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Public Transportation in Rural Areas and Tribal Communities: Guide for Service Improvements (2025)

DETAILS

222 pages | 8.5 x 11 | PAPERBACK

ISBN 978-0-309-59990-0 | DOI 10.17226/29260

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SUGGESTED CITATION

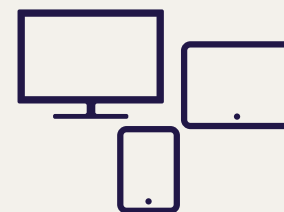
National Academies of Sciences, Engineering, and Medicine. 2025. *Public Transportation in Rural Areas and Tribal Communities: Guide for Service Improvements*. Washington, DC: The National Academies Press. <https://doi.org/10.17226/29260>.

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Research Report 1144

**National Cooperative
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Program**

TCRP

Research Report 251

**Transit Cooperative
Research Program**

Sponsored by the Federal Transit Administration

Public Transportation in Rural Areas and Tribal Communities

GUIDE FOR SERVICE IMPROVEMENTS

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**Public Transportation in Rural Areas
and Tribal Communities**

GUIDE FOR SERVICE IMPROVEMENTS

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Systematic, well-designed, and implementable research is the most effective way to solve many problems facing state department of transportation (DOT) administrators and engineers. Often, highway problems are of local or regional interest and can best be studied by state DOTs individually or in cooperation with their state universities and others. However, the accelerating growth of highway transportation results in increasingly complex problems of wide interest to highway authorities. These problems are best studied through a coordinated program of cooperative research.

Recognizing this need, the leadership of the American Association of State Highway and Transportation Officials (AASHTO) in 1962 initiated an objective national highway research program using modern scientific techniques—the National Cooperative Highway Research Program (NCHRP). NCHRP is supported on a continuing basis by funds from participating member states of AASHTO and receives the full cooperation and support of the Federal Highway Administration (FHWA), United States Department of Transportation.

The Transportation Research Board (TRB) of the National Academies of Sciences, Engineering, and Medicine was requested by AASHTO to administer the research program because of TRB's recognized objectivity and understanding of modern research practices. TRB is uniquely suited for this purpose for many reasons: TRB maintains an extensive committee structure from which authorities on any highway transportation subject may be drawn; TRB possesses avenues of communications and cooperation with federal, state, and local governmental agencies, universities, and industry; TRB's relationship to the National Academies is an insurance of objectivity; and TRB maintains a full-time staff of specialists in highway transportation matters to bring the findings of research directly to those in a position to use them.

The program is developed on the basis of research needs identified by chief administrators and other staff of the highway and transportation departments, by committees of AASHTO, and by the FHWA. Topics of the highest merit are selected by the AASHTO Special Committee on Research and Innovation (R&I), and each year R&I's recommendations are proposed to the AASHTO Board of Directors, the FHWA, and the National Academies. Research projects to address these topics are defined by NCHRP, and qualified research agencies are selected from submitted proposals. Administration and oversight of research contracts are the responsibilities of NCHRP.

The needs for highway research are many, and NCHRP can make significant contributions to solving highway transportation problems of mutual concern to many responsible groups. The program, however, is intended to complement, rather than to substitute for or duplicate, other highway research programs.

NCHRP RESEARCH REPORT 1144

Project 08-147
ISSN 2572-3766 (Print)
ISSN 2572-3774 (Online)
ISBN 978-0-309-59990-0
Library of Congress Control Number 2026930127

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TRANSIT COOPERATIVE RESEARCH PROGRAM

The nation's growth and the need to meet mobility, environmental, and energy objectives place demands on public transit systems. Current systems, some of which are old and in need of upgrading, must expand service area, increase service frequency, and improve efficiency to serve these demands. Research is necessary to solve operating problems, adapt appropriate new technologies from other industries, and introduce innovations into the transit industry. The Transit Cooperative Research Program (TCRP) serves as one of the principal means by which the transit industry can develop innovative near-term solutions to meet demands placed on it.

The need for TCRP was originally identified in *TRB Special Report 213—Research for Public Transit: New Directions*, published in 1987 and based on a study sponsored by the Urban Mass Transportation Administration—now the Federal Transit Administration (FTA). A report by the American Public Transportation Association (APTA), *Transportation 2000*, also recognized the need for local, problem-solving research. TCRP, modeled after the successful National Cooperative Highway Research Program (NCHRP), undertakes research and other technical activities in response to the needs of transit service providers. The scope of TCRP includes various transit research fields including planning, service configuration, equipment, facilities, operations, human resources, maintenance, policy, and administrative practices.

TCRP was established under FTA sponsorship in July 1992. Proposed by the U.S. Department of Transportation, TCRP was authorized as part of the Intermodal Surface Transportation Efficiency Act of 1991 (ISTEA). On May 13, 1992, a memorandum agreement outlining TCRP operating procedures was executed by the three cooperating organizations: FTA; the National Academies of Sciences, Engineering, and Medicine, acting through the Transportation Research Board (TRB); and APTA. APTA is responsible for forming the independent governing board, designated as the TCRP Oversight and Project Selection (TOPS) Commission.

Research problem statements for TCRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the TOPS Commission to formulate the research program by identifying the highest priority projects. As part of the evaluation, the TOPS Commission defines funding levels and expected products.

Once selected, each project is assigned to an expert panel appointed by TRB. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, TCRP project panels serve voluntarily without compensation.

Because research cannot have the desired effect if products fail to reach the intended audience, special emphasis is placed on disseminating TCRP results to the intended users of the research: transit agencies, service providers, and suppliers. TRB provides a series of research reports, syntheses of transit practice, and other supporting material developed by TCRP research. APTA will arrange for workshops, training aids, field visits, and other activities to ensure that results are implemented by urban and rural transit industry practitioners.

TCRP provides a forum where transit agencies can cooperatively address common operational problems. TCRP results support and complement other ongoing transit research and training programs.

TCRP RESEARCH REPORT 251

Project B-49

ISSN 2572-3782

ISBN 978-0-309-59990-0

Library of Congress Control Number 2026930127

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FOREWORD

By **Dianne S. Schwager**

Staff Officer

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NCHRP Research Report 1144/TCRP Research Report 251: Public Transportation in Rural Areas and Tribal Communities: Guide for Service Improvements provides a guide to help public transit agencies in rural and tribal communities initiate new or enhance existing public transportation services to improve mobility and accessibility, enhance performance, and consider appropriate innovations. The guide will help transit providers determine the service designs and modes that are most appropriate within their community and support implementation of those services. This guide will be of immediate use to public transportation providers and stakeholders who plan and provide public transportation services in rural areas and tribal communities throughout the United States.

Rural and tribal communities have a wide range of public transportation needs and must tailor their transit services, operations, funding strategies, and organizations to the conditions within their communities. These transit agencies have different programs and practices for overseeing, funding, and managing their services. Insufficient financial and staffing resources make it challenging for transit providers in rural areas and tribal communities to identify and meet the travel needs of the communities they serve, communicate effectively with patrons, comply with national requirements, and evaluate and adopt new technologies.

Funded jointly under NCHRP Project 08-147/TCRP Project B-49, “Improving Public Transportation in Rural Areas and Tribal Communities,” KFH Group Inc. was asked to produce a guide to (1) promote practices that are responsive to customers and aid transit providers in improving efficiency and effectiveness; (2) help transit providers better leverage and coordinate resources, comply with federal requirements, and adopt appropriate emerging technologies; and (3) address the differences among rural areas and tribal communities regarding distinguishing characteristics and how these characteristics affect public transportation services. The distinguishing characteristics include size (i.e., square miles); population density; demographics; current mobility options; economic conditions; proximity to small, medium, and large urban areas; proximity to health facilities; geography; road infrastructure; internet connectivity; funding; and weather conditions.

The guide has four parts: Part I presents new and emerging strategies and practices to enhance rural and tribal transportation services. Part II focuses on topics that support or improve the strategies and practices, which include planning for new service strategies, management and operations considerations, communications with current and prospective passengers, safety and security, technology, measuring success, funding and revenue, and accessible service. Part III addresses topics that will help rural and tribal transit agencies improve or enhance their transportation services. Part IV introduces nine case studies that

detail services or strategies implemented to improve public transportation for rural or tribal communities.

Supplemental to the guide are the full case studies and an appendix that contains the literature review. The supplemental documents are available on the National Academies Press website (nationalacademies.org/publications) by searching for *NCHRP Research Report 1144/TCRP Research Report 251: Public Transportation in Rural Areas and Tribal Communities: Guide for Service Improvements* and looking under “Resources.”

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SUMMARY

Public Transportation in Rural Areas and Tribal Communities: Guide for Service Improvements

Background

Travel needs and the potential to provide effective public transit are different in rural and tribal communities compared with more densely populated urban areas. Not only are rural areas and tribal communities different from urban areas, but they vary widely among themselves, which affects the provision of public transportation. These variations include population and population density, demographics, the local economy, distance to medical services and higher education, retail services, and employment opportunities, among other characteristics.

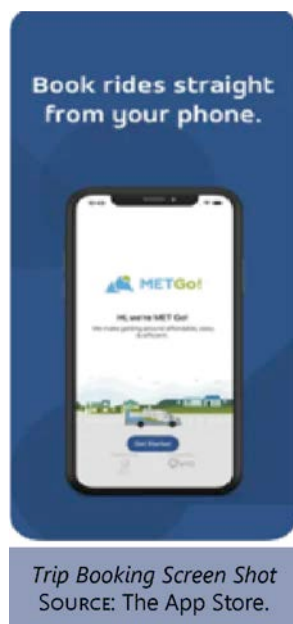
People living in rural areas and tribal communities travel predominantly by personal vehicle; however, for those lacking access to a personal vehicle, public transportation becomes critical. Many of those without such access are older adults, people with lower incomes, and people with disabilities; they disproportionately live in rural communities.

Over 84% of counties in the United States have rural or tribal public transportation services.¹ Services such as traditional fixed routes, demand responsive, deviating/flex routes, taxi-based programs, vanpools, and volunteer driver programs differ considerably by operating mode and by their providers' organizational structure, management capabilities, and effectiveness in meeting the communities' mobility needs.

Staffing and funding resources are issues that affect the transit providers' access to transit industry knowledge, including insights on new service ideas, new technologies to pilot, and approaches for working toward more effective transit for their communities. Are there new service strategies that rural and tribal transit providers should consider? What exactly is microtransit, and might it be an appropriate service design for a rural community? What technological innovations are available to improve transportation? Are there new funding resources for improving service?



2 Public Transportation in Rural Areas and Tribal Communities: Guide for Service Improvements



This Guide has been developed to fill that knowledge gap, providing rural and tribal transit providers with a resource that supports consideration of various service designs and modes that may be appropriate for their community. In addition, the Guide provides information that supports implementation of a new service design, such as planning for that new service, communications with passengers, safety and security, and ensuring effective and efficient service.

Research Process

Research efforts included the following:

- An extensive review of the literature and resources addressing public transportation in rural areas to identify issues facing rural and tribal transit providers and to seek potential strategies, practices, and initiatives to feature in the Guide.
- A comprehensive survey of rural and tribal transit providers and state departments of transportation (DOTs) to better understand key issues and challenges as well as initiatives that have been implemented to address challenges:
 - A total of 267 rural transit providers responded, which included 25 tribal transit providers.
 - A total of 39 state DOT rural program managers responded, representing 22 states.
- A compilation of effective service strategies and practices to further explore through case study research.
- Nine case studies, describing a range of service strategies and practices that improved public transportation for the respective communities.
- Research on the various topics that support transit agencies with the planning and implementation of new strategies and practices to improve public transportation.

Guidance in Four Parts

Part I starts with Chapter 2, presenting new or enhanced service strategies and operational practices that rural and tribal transit providers can consider and then pursue to improve mobility and accessibility for people in their communities. Table S-1 presents the core service design strategies that are detailed in Chapter 2 with summary information regarding when such strategies may be appropriate for a community and its transportation needs.

Chapters 3 and 4 in Part I describe additional approaches to improve mobility and accessibility: through the coordination of transportation services to serve trip needs, such as the long, multijurisdictional trips seen in rural and tribal communities (Chapter 3); and providing feeder service to the national intercity bus network, with opportunities to travel to regional population centers as well as more distant destinations (Chapter 4).

Part II includes Chapters 5 through 12, which address the following topics that support or improve the strategies and practices presented in Part I.

Planning for New Service Strategies

Planning is an important support category, necessary to pursue and implement new service strategies. Planning involves developing goals and objectives, assessing data, reaching out to customers and the community for input, and developing specific actions to implement the new service.

Management and Operations Considerations

Rural and tribal transit agencies need to address a range of topics related to managing and operating a transportation service—whether a new service or an expansion of existing service. These include such topics as hiring, training, and retaining staff, which were identified as key challenges by transit agencies that responded to the research project's survey.

Communications with Current and Prospective Passengers

Rural and tribal public transit agencies provide critical access to healthcare, employment, education, food, and other services. Accurate information and marketing of their services are essential so that current passengers can use the service and prospective passengers understand what services are available and how they can use them.

Safety and Security

Effective practices addressing transit safety and security are critical for transit providers to manage risk and ensure security, and they support implementation and operation of effective transit services.

Technology

Recent technologies have improved how public transportation services are delivered, including technologies that are becoming increasingly accessible to rural and tribal transit providers such as those that provide information on real-time bus arrivals, driver-assist or autonomous vehicle functions, mobility-on-demand apps, and security software.

Measuring Success

Evaluating transit services and measuring performance and success requires a performance evaluation process with specific goals and objectives for the transit service, defined measures to track achievement, and operating and financial data to compute the measures.

Funding and Revenue

Various federal funding programs are available and identified for rural and tribal public transit providers. Effective practices for maximizing federal funding, including the use of in-kind match, are described. Nontraditional funding sources are also identified.

Accessible Service

Accessible transit service involves meeting the requirements of federal law established by the Americans with Disabilities Act (ADA). Accessible service ensures that people with disabilities can use the transit service in their community. An overview of the ADA law is provided along with an outline of what rural and tribal transit providers need to address to ensure compliance.

Part III includes Chapters 13 through 15, which address topics that will be of interest to rural and tribal transit agencies as they navigate a path to improving or enhancing their transportation services. The three topics are state DOT role, roles of tribal governments, and compliance with federal and state requirements.

Part IV presents an introduction to the nine case studies on services or strategies implemented to improve public transportation for the rural or tribal community. These case studies demonstrate what other rural and tribal transit providers have accomplished in their quest to improve mobility and accessibility for their communities. The full case studies are available on the National Academies Press website (nationalacademies.org/publications) by searching for *NCHRP Research Report 1144/TCRP Research Report 251: Public Transportation in Rural Areas and Tribal Communities: Guide for Service Improvements* and looking under “Resources.”

Endnote

1. Mattson, J., and D. Mistry. Rural Transit Fact Book 2024, DP-325. North Dakota State University, Fargo: Upper Great Plains Transportation Institute.

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Table S-1. Service designs/modes—where to deploy.

Service Design/ Mode	Where to Consider the Service Design/Mode	Notes for Successful Implementation
Fixed-Route	• Appropriate where and when there is sufficient demand to support such a service. • Use in corridors and areas with higher population densities, where there is enough concentrated demand to be productive.	• Focus service on higher-demand trip generators (such as a large regional store, hospitals, and human service agencies) that produce peak-period job-related or program-related demand. • Focus on trip patterns in your area—origins and destinations of riders that create concentrated demand. • Consider fixed routes as one of a compendium of services in a community—they can leave persons in low-density areas without service if only fixed-route service is provided. • Less effective when routes are long, one-way loops that attempt to serve too many origins and destinations.
Flex-Route/Route Deviation	• Flex routes/route deviations are appropriate in corridors with defined routes but also pockets of demand beyond bus stop walkshed that are not far enough to warrant extending fixed-route service in these areas. • Use to serve medium-density areas with insufficient demand for traditional fixed-route, but where there is still a more pronounced demand density within a corridor.	• Appropriate where people do not live along main roads on which bus stops are not easily accessible. • Expands the reach of transit service by allowing the vehicle to deviate off the route's main road to pick up/drop off riders. • Generally, need to accept requests for deviation from all riders, and the deviation distance should be no more than $\frac{3}{4}$ mile, per FTA guidance. • No ADA paratransit obligation if deviation is for all.
Flex Routes—Checkpoint Deviation	• Because there are no defined routes, just scheduled stops, checkpoint-deviation service can be useful in areas that do not suggest a base route, especially in less dense areas where anchor destinations exist. • Flex/checkpoint deviations also serve areas with medium density but with demand insufficient for transitional fixed routes and a more dispersed population.	• A flex-route checkpoint-deviation service is suggested where there are one or more high-demand stops, such as a transit center, a big box store, a shopping center, a hospital, or a college. • Expands the reach of transit service by allowing the stops wherever there are origins or destinations that will attract sufficient ridership. • No ADA paratransit is needed/required if deviation is for all.
General Public Demand Response/ Dial-A-Ride	• Use in geographic areas during particular times/days that do not have the demand density for fixed- or flex-route transit. • Effective way to serve a large area with low and dispersed demand.	• Can be used to complement fixed-route transit or flex-route transit in less dense areas. • Also used to serve specific population subsets (e.g., older adults and people with disabilities), or for specific trip purposes such as medical trips. • Demand response service must be operated so that service equivalency is achieved for persons with disabilities.

Table S-1. (Continued).

Service Design/ Mode	Where to Consider the Service Design/Mode	Notes for Successful Implementation
ADA Complementary Paratransit	- Required if providing traditional fixed-route services. - May be possible to provide ADA complementary paratransit on fixed-route vehicles, especially when fixed-route services are provided along very long, very rural routes.	- Service should only be provided to people with disabilities who are prevented from using fixed-route transit because of their disability or disabilities. - Service must meet six service criteria, including the prohibition of capacity constraints. Transit providers must monitor service for patterns of service shortcomings, such as trip denials and late trips, among others. - Using the same vehicle to provide fixed-route and ADA paratransit can be complicated to operate because all ADA paratransit requirements must be met in serving the ADA paratransit trips.
Microtransit	- Use to expand transit service to a new area or at a new time that is unsuitable for fixed- or flex-route transit. - Can replace an underperforming fixed- or flex-route or route segment. Microtransit can typically carry between three and six passengers per hour. - May be a more effective service design in well-defined corridors or zones of operation that cannot sustain a productive fixed route, such as in small cities (including small cities in rural areas), commercial subdivisions, or strings of office parks. - Can provide more flexibility to customers than traditional fixed-route service.	- Has been used in communities of populations between 5,000 and 20,000 by rural transit providers, with service provided in a designated zone. - Focus the zone on a smaller geographic area, anchored by stops at transit connections and other major demand generators; the smaller the zone, the more productive the service. - If service equivalency cannot be achieved with on-demand service, delay service to 1- to 2-hour advance notice. - Uses smaller vehicles and mobile technology to provide dynamic routing and curb-to-curb or corner-to-corner service.
Travel Vouchers and Fare Card Subsidy Programs	- Use in conjunction with existing public and private service providers, with the riders arranging for service directly with the service provider. - Can be used to make existing transportation services more affordable for low-income riders. - Cost-effective way to target subsidies for eligible riders.	- Vouchers can be paper vouchers or scrip tickets. Each voucher or scrip ticket has a monetary value and allows riders to pay for trips on existing transportation services that participate in the program (e.g., public transit services, taxis, or other for-hire services, intercity bus carriers). - Fare cards are now more common than paper vouchers or scrip, with the transit agency adding funds to an eligible rider's card to subsidize trip-making use. - For fare cards, consider possible implications for the unbanked population. Applications using customers' fare card or bank card accounts may not be accessible to the unbanked or persons without smartphones.
Volunteer Driver Programs	- Use to meet mobility needs in areas where transit services and taxi/TNC services are limited or cost-prohibitive - Can be an effective means to accommodate unmet mobility needs of older adults and people in rural and tribal communities where transit, paratransit, and taxi/TNC services are limited or cost-prohibitive. - Cost-effective way to provide long-distance trips.	- Volunteer programs can also coexist to augment other existing transit services. - With reimbursement to volunteers for mileage at an IRS-approved rate, volunteers provide a low-cost option to meet needs, particularly for long-distance trips.

Note: TNC = transportation network company.
IRS = Internal Revenue Service.



CHAPTER 1

Introduction and Purpose of the Guide

Background

Travel needs and the potential to provide effective public transit are different in rural and tribal communities compared with more densely populated urban areas. Not only are rural areas and tribal communities different from urban areas, but they vary widely among themselves, which affects the provision of public transportation. These variations include population and population density, demographics, the local economy, distance to medical services and higher education, retail services, and employment opportunities, among other characteristics.

People living in rural areas and tribal communities travel predominantly by personal vehicle; however, for those lacking access to a personal vehicle, public transportation becomes critical. Many of those without such access include older adults, people with lower incomes, and people with disabilities; they disproportionately live in rural communities.

Over 84% of counties in the United States have rural or tribal public transportation services.¹ Services such as traditional fixed routes, demand responsive, deviating/flex routes, taxi-based programs, vanpools, and volunteer driver programs differ considerably by operating mode and by their providers' organizational structure, management capabilities, and effectiveness in meeting the communities' mobility needs.

Staffing and funding resources are issues that affect the transit providers' access to transit industry knowledge, including insights on new service ideas, new technologies to pilot, and approaches for working toward more effective transit for their communities. Are there new service strategies that rural and tribal transit providers should consider? What exactly is microtransit, and might it be an appropriate service design for a rural community? What technological innovations are available to improve transportation? Are there new funding resources for improving service?

These are among the topics explored in this Guide, which was developed to help public transit agencies in rural and tribal communities improve mobility and accessibility, enhance performance, and consider appropriate innovations.

Basic Premise of the Guide

An underlying premise of the Guide is that rural and tribal communities have a wide range of transportation needs and, to be effective and efficient, public transit providers in these communities should tailor their transit services to the conditions within these communities.

The Guide presents service design strategies and modes, including emerging nontraditional services, and practices improving efficiency and effectiveness that may be appropriate in rural

and tribal communities. Key to the success of any of these services is matching those service designs and modes with different transportation needs within a community.

Research Process

The research for the Guide was structured in two phases.

Phase 1 involved the following:

- Collection and compilation of background information to analyze issues facing rural and tribal transit providers, including a comprehensive review of written and unpublished reports and materials and the identification of potential strategies, practices, and initiatives to feature in the Guide.
- A comprehensive survey of rural and tribal transit providers and state departments of transportation (DOTs) to understand key issues and challenges:
 - A total of 267 rural transit providers responded, including 25 tribal transit providers.
 - A total of 39 state DOT rural program managers responded, representing 22 states.
- A compilation of effective service strategies and practices to further explore in Phase 2 through case study research.
- Preparation of an interim report and the interim NCHRP Project 08-147 panel meeting to present the findings of Phase 1 and receive feedback.

Phase 2 involved the following:

- Nine case studies, describing service strategies and practices that improved public transportation for the respective communities; the strategies ranged from microtransit to service coordination to improving connections from rural communities to the national intercity network.
- Research on the various topics that support transit agencies with the planning and implementation of new practices and strategies to improve public transportation. The topics include, for example, communicating with passengers, funding, safety and security, technologies, and ensuring effective/efficient service.
- Development of this Guide.

The overall purpose of the Guide is to help transit providers determine which service designs and modes are most appropriate within their community and to support implementation of those services.

Structure of the Guide

The Guide has four parts and an appendix.

Part I, *New and Emerging Strategies*, presents strategies and practices that focus on the initiation of new or enhancement of existing rural and tribal transportation services. This part contains the following chapters:

- Chapter 2: Service and Design Strategies and Practices
- Chapter 3: Mobility Management and Coordination
- Chapter 4: Regional Services and Connections to the National Network

Part II, *Support Topics*, presents Chapters 5 through 12, which address topics that support or improve the strategies and practices presented in Part I. The topics could stand on their own for public transit service and improvement; however, the Guide approaches these topics through the lens of how they support new or enhanced service strategies and what steps or actions transit providers should consider or take when implementing a service strategy.

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The chapters in Part II are as follows:

- Chapter 5: Planning for New Service Strategies
- Chapter 6: Management and Operations Considerations
- Chapter 7: Communications with Current and Prospective Passengers
- Chapter 8: Safety and Security
- Chapter 9: Technology
- Chapter 10: Measuring Success
- Chapter 11: Funding and Revenue
- Chapter 12: Accessible Service

Part III, Information Topics, includes Chapters 13 through 15, which address topics that will be of interest to rural and tribal transit agencies as they navigate a path to improving or enhancing their transportation services. The chapters in Part III include the following:

- Chapter 13: State DOT Role
- Chapter 14: Roles of Tribal Governments
- Chapter 15: Compliance with Federal and State Requirements

Part IV, Case Studies, includes Chapter 16, which provides an introduction to, and an overview of, the nine case studies of services or strategies implemented to improve public transportation for the rural or tribal community. The case studies range from a new microtransit service for a small community in central Texas to rural routes in South Dakota designed to feed the national intercity bus network, connecting rural areas with the state's largest cities as well as to destinations in other states. The full case studies are available on the National Academies Press website (nationalacademies.org/publications) by searching for *NCHRP Research Report 1144/TCRP Research Report 251: Public Transportation in Rural Areas and Tribal Communities: Guide for Service Improvements* and looking under “Resources.”

Appendix: Literature and Resources: General Guidebooks and Literature by Topic Area provides an extensive listing of the literature and resources reviewed by the research team at the start of the project, with some updates from the development of the Guide. The appendix is available on the National Academies Press website (nationalacademies.org/publications) by searching for *NCHRP Research Report 1144/TCRP Research Report 251: Public Transportation in Rural Areas and Tribal Communities: Guide for Service Improvements* and looking under “Resources.”

Endnote

1. Mattson, J., and D. Mistry. Rural Transit Fact Book 2024, DP-325. North Dakota State University, Fargo: Upper Great Plains Transportation Institute.

PART I

New and Emerging Strategies

Part I of the Guide presents service strategies and operational practices that rural and tribal transit providers can use to identify and initiate new or enhance existing transportation services with the goal of improving mobility and accessibility for people in their communities. The Guide has categorized the service strategies into three types:

Chapter 2: Service and Design Strategies and Practices presents service strategies and operations management practices aimed at improving mobility, efficiency, and effectiveness. Included in this chapter are service designs and practices that may better serve local mobility needs, such as using flex routes instead of fixed routes to expand transit's reach, which may also improve its effectiveness, and piloting an on-demand service using new mobility providers.

Chapter 3: Mobility Management and Coordination discusses management practices that offer approaches for delivering coordinated transportation services, particularly when serving older adults, people with disabilities, veterans, and low-income individuals. In rural and tribal areas, the challenges to provide mobility for these groups are often daunting, as trips are often long and may require crossing multiple jurisdictional boundaries.¹

Chapter 4: Regional Services and Connections to the National Network presents service and design strategies and related operational practices that rural and tribal transit providers can implement to address rural regional transit needs, including connections to the national intercity bus networks. Residents of rural and tribal communities benefit when they can access regional activity centers and national transportation networks with services to larger urban areas. This access may be provided directly through intercity bus services with stops in rural areas, or when their local transit providers connect to national and regional transportation networks.

CHAPTER 2

Service and Design Strategies and Practices

Introduction

Chapter 2 presents service and design strategies and practices that rural and tribal transit providers can implement to increase current and prospective customers' mobility and accessibility and improve the efficiency and effectiveness of the service. The goal is to support transit providers in considering which different service designs and modes are most appropriate for the transportation needs within their community to ensure efficient and effective service.

Objective: Matching Service Designs and Modes to Different Community and Mobility Needs

The chapter first identifies the service design types and modes and, for each, discusses how to match those service designs and modes with different transportation needs within a community (more on the planning process is presented in Part II—Chapter 5). The chapter concludes with some specific operational practices that can improve service efficiency and effectiveness in rural and tribal communities.

Service Design Strategies

Rural and tribal transit providers serve communities that have a wide range of environments, which means they must come up with a wider range of service solutions than their urban counterparts. For example, rural transit providers may serve cities with under 50,000 residents and rural counties with 150 residents; they can also serve towns that are 10 miles apart, 50 miles apart, or more, and communities on the outskirts of metropolitan areas. Rural populations tend to be older, and the percentage of people with a disability is slightly higher in rural areas.

With this variety of service areas and population needs, rural and tribal transit providers should consider how best to match the varying needs within their service area with the appropriate mode. Taking the “one-size-fits-all approach” does not result in service effectiveness or efficiency. What might be an effective and innovative approach in one area may not work in another. Rural and tribal transit providers are implementing a “family of services” to better match service types with needs.

Proper service design and operational strategies are paramount to the delivery of efficient and effective service. An improper service design often results in lower ridership and lower productivity. Applying the right service design can often improve performance at no extra cost.

The following discussions focus on where and when specific service types/modes should be considered. The modes include the following:

- Fixed-Route Service
- Flex-Transit Service

- General Public Dial-a-Ride
- ADA Complementary Paratransit
- Microtransit
- Travel Vouchers and Fare Card Programs
- Volunteer Driver Programs

Fixed-Route Service

Fixed-route transportation service operates on a predetermined route according to a set schedule. Fixed-route transit systems generally publish timetables and have designated stops where riders are picked up and dropped off. Fixed-route bus systems are the most common form of public transportation in the United States, but only 32% of rural bus systems operate fixed routes.²

Fixed routes are appropriate where and when there is sufficient demand to support such a service. With limited resources, many rural/tribal transit providers find themselves choosing between two options:

- **Maximize Efficiency.** Provide fixed routes in areas with higher population densities (origins) and high-use destinations and devote enough resources to make the service convenient via a frequent headway to attract riders. This creates a highly cost-efficient service (as measured by operating cost per passenger trip) but may leave people in lower-density areas without any transit.
- **Maximize Effectiveness.** Provide fixed routes in a wider geographic area, including areas with lower densities, so that more people have access to transit. This diverts some resources from the higher-density areas, potentially lowering the cost-efficiency of the service.

These can be opposite approaches. Some transit agencies have found that the second option, while well intended, can sometimes backfire if the service implemented consists of long, circuitous, and infrequent one-way “coverage routes” that attract limited demand because of the infrequency of service and long ride times, and because they ultimately do not get riders to their destinations in a timely manner.

When weighing or balancing the goals of the two options, transit providers should also consider that service goals can be achieved by operating other modes such as demand response service in areas not well served by fixed routes. Tactics that apply to both fixed-route service approaches include the following:

Design Fixed-Route Service with Individual Trip Patterns in Mind. An effective tactic in designing rural/tribal fixed routes is to focus on the origin and destination of individual riders. Where and when do the riders need to go to get to work? What services and shopping do they need to access? (See the example in the sidebar.)

Focus on Demand Generators. The efficiency and effectiveness of a fixed-route service can often be improved by focusing the service on higher-demand generators (such as shopping centers, hospitals, community colleges, and human service agencies) that also generate peak-period job-related or program-related demand.

Focus Fixed-Route Transit Service Where Demand Density Will Generate a Productive Service. Many rural and tribal providers operate fixed-route service in areas and at times where/when fixed-route service does fit the need, and, in particular, where there is demand density and trip patterns along a route corridor or within a concentrated network. Fixed-route service becomes less and less viable as the demand density decreases.

Add or Subtract Route Segments. Another related tactic is to add a peak-only or all-day route segment onto the end of a route if the additional demand outweighs any operational and cost shortcomings, including the possibility of having to add a bus to maintain a headway. Conversely, a transit provider can always cut a route segment altogether or during off-peak hours if the demand in that part of the transit corridor does not justify the service.

Consider Fixed-Route Transit Service as One End of a Continuum of Service Modes. If a transit provider is new in the community and the demand for public transit is unknown, it may be prudent to begin with demand response service and graduate to a fixed-route service in areas or at times when enough demand density materializes to generate a cost-efficient service. Such graduation means that the cost-per-trip favors one mode versus the other. In this case, graduation to fixed-route means that the cost-per-trip is lower than for the demand response service. However, in weighing such a decision, rural and tribal transit agencies must also consider the additional costs of infrastructure (new bus stops) and especially the ADA obligation to provide complementary paratransit service. Conversely, a transit provider can remove a fixed route or segment as suggested and replace it with a form of demand response service that is especially important for those ADA paratransit riders who may be negatively affected by the service removal.

Case Study Example: Okanogan County's TranGO

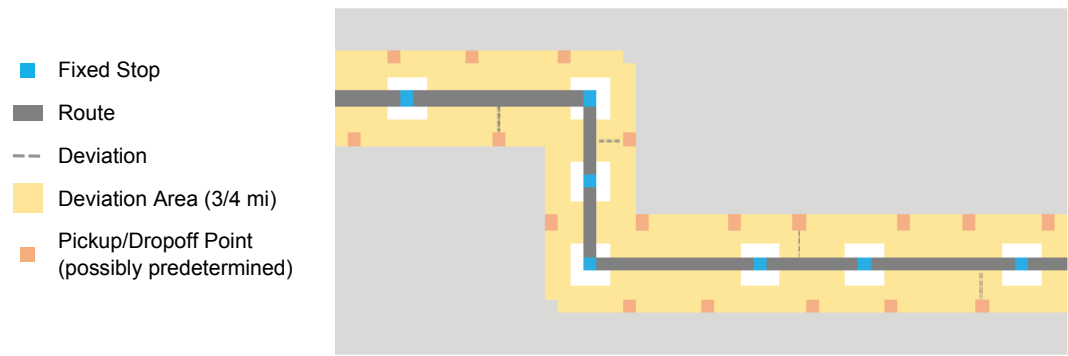
When the fixed-route design for TranGO was being planned, the design that was most favored was a hub and spoke system. But the new general manager took a different approach: To better understand the travel patterns of prospective riders, a community survey was launched. Responses suggested that a hub and spoke design would fail because it did not accommodate the trip patterns. Instead, a U-shaped model was designed with linking route segments that coordinated at different transfer points, pulsing at the transfer points every ½ hour during the peaks, and on the hour during the off-peak. The service was later expanded to start 1 hour earlier and end 1 hour later to accommodate intercity commute trips.

Flex-Transit Service

In rural and tribal communities, traditional fixed-route services may not always meet the needs of residents, particularly in communities where people do not live along main roads and bus stops are difficult to access. In these cases, communities may opt for flex services³ that pick up riders at their homes or other more favorable locations.

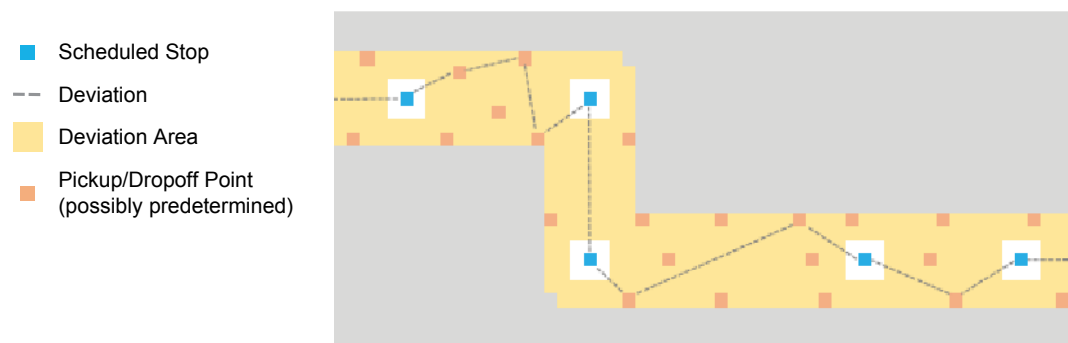
Flex-transit service is a general public hybrid service between fixed-route and demand response service and is generally provided to expand the “walkshed” typically associated with fixed-route service. The concept behind flex transit is that it can provide options for a wider base of users than traditional fixed-route service. It is common in rural areas. Two of the more common forms of flex transit in rural areas and tribal communities include (1) route-deviation service; and (2) checkpoint-deviation service, with route-deviation service working best in medium-density locations and checkpoint deviation working best in low-density areas. These two service concepts are illustrated in Figures 2.1 and 2.2, respectively.

A common element with each is that deviations from a scheduled fixed route require an advance request, which means that a transit agency needs to have a call center and a way to schedule/dispatch such requests so that the driver knows what deviations to make. Another common element is that a transit agency may charge a higher fare (a deviation surcharge) for a requested deviation.



COURTESY OF TEXAS A&M TRANSPORTATION INSTITUTE (TTI).

Figure 2.1. Route-deviation concept.



COURTESY OF TEXAS TRANSPORTATION INSTITUTE.

Figure 2.2. Point-deviation concept.

Implementing Route-Deviation Service.⁴ Route-deviation service, also called flex route or deviating route, is a general public transit service that operates mostly along an established route on an established schedule with designated stops along the route. Stops can be curb-to-curb, virtual, or physical stops. Between these stops, however, the vehicle may deviate off-route to pick up or drop off a rider who has requested the deviation within a defined off-route service area. After the pickup or drop-off, the vehicle generally returns to the route from where the vehicle exited the route (or as close to that point as possible). There are no minimum requirements for the size of the area for off-route deviations, although $\frac{3}{4}$ -mile corridors are common. Transit agencies that implement route-deviation services recognize a more dispersed demand that is just beyond a reasonable walking distance to the route/bus stop.

Policies vary on how far in advance a rider may request a pickup or a drop-off. Most systems allow same-day requests, either with a minimum of 1-hour or 2-hour notice. Some of these systems also allow next-day requests, while others require requests at least 1 day in advance. Some systems also do not require requests for off-route drop-offs, which are made known to the driver when the rider boards the vehicle.

Route-deviation services are typically provided in corridors where there are pockets of demand beyond the bus stop walkshed but not enough to warrant implementing/modifying a fixed route to these areas. Route deviation can also be used to extend service beyond the terminus of a route.

Generally, route-deviation services must accept deviation requests from all riders. While there are no minimum requirements for the corridor size where deviations are permitted,

FTA guidance states that anything less than $\frac{3}{4}$ -mile deviations *may* be considered unreasonable and hence limiting and discriminatory. In very rural areas where demand is low, transit agencies might be able to meet the complementary paratransit requirements through off-route deviations. If deviations are only available for persons with disabilities, complementary paratransit requirements must be met.

However, an off-route service area cannot be so large that deviations will consistently jeopardize the ability of the vehicle to make it to the end of the route in time to start the next scheduled route cycle. This suggests that enough slack be built into the route to accommodate a reasonable number of deviations; 10% to 15% slack is a good rule of thumb. Transit agencies may limit the number of deviations on any specific cycle to ensure an on-time start on the next cycle. Making too many deviations can be problematic if the slack in the schedule only allows a certain number of deviations per vehicle trip and the maximum number of deviations for that run has already been reached. At the same time, transit agencies need to make sure that all riders have the same opportunities to request deviations. Practices to avoid include the following:

- Not advertising the deviation aspect of the service.
- Establishing restrictive policies that limit use by persons with disabilities, such as the following:
 - Charging excessive surcharges.
 - Establishing restrictive deviation areas.
 - Limiting deviation by trip purpose.
 - Unreasonable capping of the number of deviations.

To be considered demand responsive rather than fixed-route, route-deviation services must accept deviation requests from all riders (except as noted previously) and provide nondiscriminatory service. If a transit agency were to provide a route-deviation service only at certain times or only along certain segments of the route, the transit agency would still have to provide ADA paratransit service when the route is being run as fixed-route. Otherwise, route-deviation services do not have a complementary ADA paratransit obligation because there is no prescribed fixed route. This can reduce a transit agency's operational costs.

Alternatively, a transit agency could operate **fixed-route service and ADA complementary paratransit with the same vehicle**, which would deviate from the scheduled fixed route only to serve an ADA paratransit trip. If a transit agency were to operate such a service, all requirements that would apply to ADA paratransit **must** be followed in serving ADA paratransit trips.⁵ (See the sidebar.) Included in these requirements is one related to a pattern of denials resulting from capacity constraints.

Therefore, if a transit agency chooses to operate such a commingled service, the schedule must have enough slack time in the route so as not to result in a pattern of denials that limit service for ADA paratransit riders. Such commingled service can be complicated to operate, but may be a solution for a very long, very rural route.

Implementing Checkpoint (or Point) Deviation. Checkpoint-deviation services are distinguished from route-deviation services because of one major distinction: there is no defined route, but there are scheduled stops. Between the scheduled stops, or perhaps more accurately, beyond the walkshed of those stops, the bus may pick up and drop off riders, responding to advance rider requests, within a defined area during prescribed service hours. As with route deviation, stops can be curb-to-curb, virtual, or physical stops.

Checkpoint deviation works best in low-density areas while route deviation works best in medium-density locations where there is a more pronounced demand density within the transit

In Okanogan County, Washington, TranGO provides ADA paratransit in two ways: (1) with its dial-a-ride services within the dial-a-ride zones; and (2) with its fixed-route buses (in between these dial-a-ride zones) with such buses deviating from the fixed routes to provide ADA paratransit service. This is described further in the TranGo case study in Part IV.

In Beltrami County, Minnesota, the rural transit provider added scheduled stops to its rural dial-a-ride service and, in doing so, created a checkpoint-deviation service. By adding these scheduled stops at popular destinations, the service was able to increase productivity by funneling demand.

corridor. Transit agencies typically opt to implement a checkpoint-deviation service (as distinguished from route-deviation service) in areas that are less dense and do not suggest a base route. A checkpoint-deviation service is also suggested where there are one or more high-demand stops, such as a transit center, a big box store, a shopping center, a hospital, or a college. Generally, checkpoint deviation can boost demand response service productivity by up to 50% because of walk-ons at the higher-demand scheduled stops. (See the example in the sidebar.)

Checkpoint deviation has no obligation for ADA paratransit because there is no true fixed route.

General Public Demand Response (Dial-a-Ride) Service

General public dial-a-ride is a demand responsive transportation service that allows all members of the community to request a pickup and drop-off at any location within the service area.

Operating Demand Response Service. Rural and tribal transit agencies serving large and low-density areas often choose to provide general public demand response service instead of fixed-route or flex transit, because it enables the transit agency to cost-effectively serve a larger area with a lower and more dispersed demand density. In practice, these services are often referred to as “dial-a-ride” services. Dial-a-ride services work best in areas or at times that are not conducive to fixed or flex transit, whether it is because of the demand density, road network, or topology.

As with flex transit, transit agencies implementing a dial-a-ride service must establish a call and control center that houses the four functions of reservations, scheduling, dispatching, and handling customer service calls [e.g., “where’s my ride?” calls and estimated time of arrival (ETA) requests]. While very small systems can get by putting together a schedule with a simple spreadsheet or electronic appointment book, most rural and tribal agencies support the dial-a-ride call and control center function with scheduling/dispatching software, which is also used to register customers and collect and report service data.

Most demand response services provide curb-to-curb service (noting that some systems offer door-to-door for persons with disabilities). Service is typically available on an advance-reservation and subscription basis. Advance-reservation periods are, for example, 7 to 14 days in advance.

Operating Demand Response in Different Regions. A common approach to serving a very large (e.g., multicounty) area is to use county transportation operations as contractors. It should be noted that transit agencies with different providers in different areas will need to provide a mechanism for inter-area travel, including whether this requires transfers or direct service and whether the inter-area trips suggest a zone-based fare structure. Hence, the same vehicle and driver could be used to serve different areas in this manner. Moreover, transit agencies employing this tactic have found improvements in productivity because the demand is funneled to certain days, which creates more shared rides. Where none exists, and where a rural or transit agency has limited resources, an alternative tactic is to serve different areas on different days.

In Okanogan County, Washington, TranGO has deployed dial-a-ride services in a 5-mile radius surrounding five different transit centers/transfer points along its fixed-route network. Many customers use the dial-a-ride service to go to other towns served by the network. See the TranGo Case Study in Part IV of this Guide.

Interfacing Demand Response and Fixed Transit. Rural and tribal transit agencies often provide demand response service (only) to serve an entire region, meeting transit goals. But some agencies alternatively operate demand response service that primarily serves to feed a fixed-route network. (See the example in the sidebar.)

In such instances, timed transfers are an essential ingredient to enhancing the rider experience. Moreover, some transit agencies do not charge two fares; instead, they use a free transfer system, while others simply do not charge for the demand response portion.

Implementing a Limited-Eligibility Demand Response Service. Most rural and tribal transit agencies' demand response services are available to the general public. Alternatively, some agencies offer the service to subsets of the population, such as older adults and persons with disabilities.

Migrating to or from Demand Response. Some rural transit agencies with existing flex-transit services found that switching from flex transit to a demand response service increased productivity (See the side box example.)

A demand response service can also be a cost-efficient way to test the demand for transit in a new area, and, if a concentrated demand materializes, it may suggest graduating to a fixed- or flex-route solution.

Service Equivalency. Demand response service must be operated so that service equivalency is achieved. This is an ADA requirement and key concept of nondiscrimination, requiring that service for people with disabilities be equivalent to that provided to people without disabilities. This is particularly an issue when the fleet used for demand response service is not 100% accessible (see also Chapter 12, Accessible Service).

For example, if riders without disabilities can request a trip and the response time of the vehicle is within 3 hours, then a rider with a disability who needs an accessible vehicle should expect the same response time (i.e., the accessible vehicle will arrive within 3 hours). It would not be acceptable for riders without disabilities to have that 3-hour response time, and riders using wheelchairs to have to book a day in advance for a trip because there are fewer accessible vehicles available for service. That is not equivalent.

Pulaski Area Transit in Virginia operated a route-deviation service at a productivity of 4.9 passenger trips per hour. After converting to a dial-a-ride service, the productivity increased to 7.0 passenger trips per hour. The change in service type enabled Pulaski Area Transit to serve a broader range of riders, including more general public passengers and children. A similar change was made by **Inter-County Public Transportation Authority**, a multicounty system in **northeastern North Carolina**. Here, productivity increased from 1.6 with the route-deviation service to 3.0 with the dial-a-ride service.

ADA Complementary Paratransit

Transit agencies that provide traditional fixed routes must also provide complementary paratransit for persons who, because of their disability, cannot access or use the fixed-route transit service. The requirements and obligations for ADA paratransit are detailed in FTA Circular 4710.1⁶ (https://www.transit.dot.gov/sites/fta.dot.gov/files/docs/Final_FTA_ADA_Circular_C_4710.1.pdf). See also Chapter 12, Accessible Service.

In rural and tribal communities, the provision of ADA paratransit can be similar to the demand response service models described previously, with the exception that ADA paratransit service is intended only for people with disabilities who are unable to use the transit system's fixed routes due to their disability. This distinction is relevant because some people with disabilities can use fixed-route services.

The most common service models for ADA complementary paratransit are (1) all in-house, where all daily functions are performed by the transit agency; and (2) a "turnkey" contract model where a contractor (or contractors assigned to subregions) performs the daily functions. But there are also service models where call and control center functions are performed by the transit agency while service delivery is performed by the contractor(s). Other transit agencies that use a contractor or contractors may split the call and control center functions between the transit agency and the operations contractor(s). There can be variations in which entity provides the vehicles, the facilities, the software, communications, and the in-vehicle equipment.

Providing Premium Service. While some rural and tribal transit agencies provide ADA paratransit based on the minimum requirements, others include reservations that go beyond next-day reservations and subscription service. In addition, some transit agencies provide premium service by expanding their service area beyond the required $\frac{3}{4}$ -mile transit corridors or expanding the span of service beyond fixed-route transit service hours. Note that transit agencies can charge a higher fare for premium-service trips.

Operating ADA Paratransit with Dedicated Fleet versus Non-Dedicated⁷ Service Providers. Most rural and tribal transit agencies provide ADA paratransit with a fleet of one or more dedicated vehicles, operated either in-house or by a contractor (or contractors) as mentioned. Some rural and tribal transit agencies use non-dedicated service providers (NDSPs) such as taxis, where they exist, to provide ADA paratransit service instead of using a dedicated fleet, while others use NDSPs to provide “overflow” ADA paratransit service as a way to more cost-efficiently handle a portion of their trips, but also to solve other types of operational issues. As a precondition to their involvement in providing ADA paratransit overflow service (and as part of their contract), those NDSPs must agree to meet ADA requirements, especially for their drivers. In brief, drivers need to be “trained to proficiency,” and they need to meet FTA’s drug and alcohol testing requirements.

Nontraditional Approaches to Serving ADA Paratransit Trips. Some rural and tribal transit agencies have creatively implemented nontraditional ways to provide ADA paratransit, such as implementing flex routes.

Monitoring Possible Capacity Constraints. An FTA requirement is that transit agencies regularly monitor for patterns of service shortcomings that may indicate a capacity constraint, which is not allowed. In practical terms, transit agencies must analyze their ADA paratransit data to detect whether there are any patterns of denials, late trips, missed trips, excessively long trips, or excessively high average and maximum call hold times, each of which may point to a capacity constraint. Efforts to improve the service efficiency of an ADA paratransit service may be tempered by the need to avoid such patterns.

Providing Subscription Trips (also called Standing Orders or Recurring Trips). There is nothing in the ADA that requires a transit agency to offer subscription service, but most transit agencies do. Subscription trip policies vary from agency to agency; some require a certain minimum number of times per week that the trip occurs; others offer one recurring trip per week. The general benefits of subscription trips are (1) they are a convenience for customers (who do not have to make a reservation for each trip); (2) they can reduce the call volume which can reduce call center labor costs and hold times; and (3) well-scheduled subscription trips (especially group trips going to the same location such as senior center or adult day program facility) can have a significant positive impact on the overall productivity cost-efficiency of the overall service. Frequent maintenance of the subscription trip schedules is necessary to ensure they continue to have a positive impact on productivity.

The original FTA guidance stipulated that transit agencies should not devote more than 50% of their ADA paratransit capacity to subscription trips at any given point in the service day, the underlying reason being to preserve sufficient capacity for one-time, advance-reservation trips. However, this requirement is superseded as long as there are no capacity constraints. Thus, a transit agency can exceed the 50% threshold as long as there are no capacity constraints.

Many transit agencies view the offering of subscription service as a privilege (as it is not required). Frequent cancellations of subscription service can wreak havoc on the schedule. Therefore, these transit agencies have implemented a subscription trip cancellation policy where frequent cancellations of subscription trips trigger a suspension from subscription service.

Providing Alternative Service and Programs for Persons with Intellectual/Developmental Disabilities (IDD). Many persons with IDD can qualify for ADA paratransit service. The transportation of many such individuals is often subscription in nature, with transportation to day programs and competitive work opportunities, and is generally in peak-demand periods. Some rural and tribal transit agencies have found that (1) such trips take up considerable capacity during the peaks, resulting in a capacity constraint, and (2) poor levels of on-time performance provided to these individuals on the ADA paratransit service can have an adverse impact on their day program experience much less their ability to get or hold a job. Two solutions include (1) providing vehicles to such day programs at a modest rate per month per van (that may include fuel and maintenance) and letting the transit agency provide the driver and serve the trips, and (2) contracting with a separate provider to operate the equivalent of a premium service to take the riders to/from the day programs and other locations. With either approach, transit agencies can claim these trips on their National Transit Database (NTD) reporting as long as the drivers are trained to the same proficiency as the ADA paratransit drivers.

Microtransit

An increasing number of public transit providers have begun providing an on-demand, e-hailing service for the general public. These services, called microtransit, use smaller vehicles and mobile technology to provide dynamic routing and curb-to-curb or corner-to-corner service. Customers use a smartphone application (app) to schedule and pay for a trip within a specific geofenced zone, or riders can use the telephone to book a trip.

Microtransit can be provided in different operating environments, including in small cities and communities in rural settings. While demand densities are typically lower in rural settings than in larger urban areas, microtransit operates in the same way—using app-based, real-time scheduling technology to manage trip requests and provide constantly updated real-time arrival information. By adopting this technology in the appropriate operating environment, rural public transportation providers can operate more efficient and customer-focused service in small communities.

Based on experience, microtransit typically carries between three and six passenger trips per vehicle revenue hour. It may be a more effective service design in well-defined corridors or zones of operation that cannot sustain a productive fixed route, such as in small cities (including small cities in rural areas), commercial subdivisions, or strings of office parks.

Evolution of Rural Microtransit

Microtransit services allow transit agencies to provide (or subsidize) dynamically routed on-demand service for the general public. The definition of microtransit is evolving, and there are many forms of service. For example, microtransit services can be operated with a dedicated fleet (with or without a non-dedicated overflow provider), by a transit agency, or by a contractor. Or it can be operated with one or more transportation network companies (TNCs) or taxis, operating in a non-dedicated fashion. Some microtransit services provide a choice of dedicated and non-dedicated providers.

Many in the transit industry qualify microtransit as a shared-ride service; however, there are transit agencies that have implemented what they call microtransit as an exclusive ride service; the former can be claimed in NTD reporting while the latter cannot. (To include data for NTD reports, the transit service must be shared ride.) Most microtransit services are app-driven, with backup call-in options.

Microtransit service can provide more flexibility to customers than traditional fixed-route service. Riders can individualize service by selecting both their pickup and drop-off locations,

while dynamic routing capabilities allow drivers to quickly adjust pickup locations to provide more efficient service and pick up riders within 15 to 30 minutes of the request. The hallmark of microtransit is that it is an on-demand service where general public riders can request trips for immediate fulfillment, noting that some microtransit services also allow riders to book trips in advance. But some microtransit services have an on-demand delay of sorts, requiring a 1- to 2-hour advance notice to meet certain ADA requirements related to service equivalence. Some on-demand technologies also allow walk-on riders (who have not pre-requested service at all) at high-demand locations (see information on Allowing Walk-Ons).

Variations of the microtransit design also can include different types of stops, which can vary from “curb-to-curb” stops at riders’ addresses and destinations to fixed stops at popular origins and destinations. Some on-demand technologies have the capability to identify a virtual stop, which is typically a nearby intersection (within a configurable walkshed) of a rider’s home, much like a school bus stop.

Why Implement Microtransit? Rural transit agencies are implementing microtransit services for two reasons: (1) to expand public transit to a new area or at a new time that is unsuitable for fixed-route transit service, and (2) to replace an underperforming bus route or route segment or to replace the underperforming route or route segment **when** it is underperforming.

Some rural transit agencies have also used microtransit to test the demand in a new area or at a new time. This might start as non-dedicated service if there is not enough demand to support dedicated service. If ridership grows to the point where dedicated service can be justified from a cost-per-trip perspective, the next step could be to expand fixed-route service into the area.

In Summit County, Utah, High Valley Transit (HVT) implemented a microtransit service in part to serve the higher-elevation areas that were previously unserved. HVT used all-wheel drive minivans for this service to navigate the steep grades that could not be accessed by larger vehicles, thereby expanding the reach of public transit by 25%.

Where to Implement Microtransit. Rural transit agencies are being implemented in towns and cities with populations of between 5,000 and 20,000, where fixed-route or flex transit cannot generate adequate ridership and productivity, or where traditional fixed-route buses are unable to serve an area because of the steep grades. (See sidebar.) In some rural communities, microtransit has replaced advance-reservation dial-a-ride service as an on-demand upgrade.

When to Implement Microtransit. It depends on the service context. For example, for a service expansion to new areas or replacement of routes or route segments, transit agencies are typically setting service days and hours, so they are the same as the conventional service days and hours of the fixed-route system.

But many transit agencies have implemented a microtransit service during low-demand periods, such as the middle of the day or after hours, when it is not cost-effective to operate fixed-route service. Some rural transit agencies have used microtransit to replace fixed-route segments that are not drawing sufficient ridership during low-demand periods. Other rural transit agencies have implemented microtransit systemwide after their fixed-route service has ended for the day. Some transit agencies subsidize such trips 24/7, as a way to offer mobility options at nontraditional times. For example, evening, late-night, and overnight microtransit service can help hospital employees on the graveyard shift, inventory stockers, and custodial workers to get to and from work.

Sizing the Microtransit Zone(s). Rural transit agencies tend to implement microtransit zones in smaller areas, anchored by stops at transit connections and other major demand generators. The smaller the zone, the more productive the service because trips are shorter; however,

the zone should be sized to include both origins and common destinations. If a transit agency chooses to use dedicated vehicles to provide microtransit, sizing a zone smaller than 5 to 7 square miles can keep a dedicated vehicle productive. Larger zones with more dispersed demand are more typically served with NDSPs or a mixture of dedicated and non-dedicated vehicles.

Implementing Microtransit with Dedicated Service. Providing microtransit with one or more dedicated vehicles makes the most sense (1) where a transit agency needs/desires more control over service quality; (2) where demand density and productivity are known and it makes sense from a financial perspective (e.g., where it is less expensive per trip than with an unproductive route or route segment); (3) where zones are smaller; and (4) where there is lack of NDSPs. One rule of thumb is that dedicated service makes sense where productivity of 3 to 4 passenger trips per revenue vehicle hour can be achieved. Also, a transit agency may have to supply dedicated wheelchair-accessible vehicles (WAVs) in instances where the service primarily uses NDSPs that do not have or have very few WAVs.

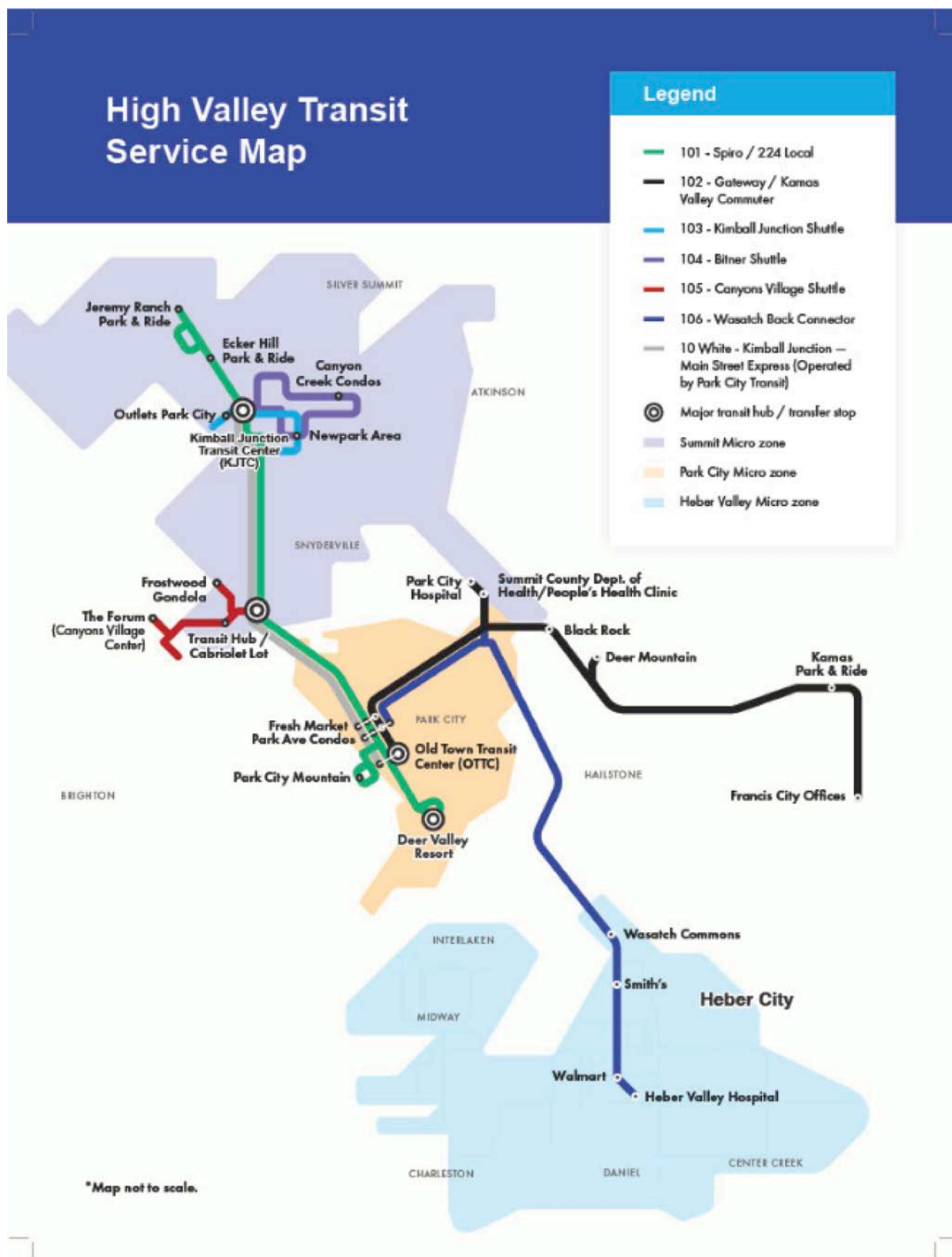
Implementing Microtransit with NDSPs. Providing microtransit with NDSPs makes sense (1) where there are taxis and TNCs already in place, although an argument can be made that “if you build it, they will come”; (2) where cost is an issue and it is reasonable to believe that the unit cost of dedicated service will likely be higher than non-dedicated service; (3) where the zone is larger (larger than 5 to 7 square miles); and (4) where the design is based on a user-side subsidy program. Using NDSPs also makes sense as an overflow partner to dedicated service or as another mobility option.

When using non-dedicated providers such as taxis and TNCs, it is important to consider whether the drivers for those providers must meet federal drug and alcohol testing.⁸ Under federal transit law, transit agencies that receive funding through FTA’s grant programs for urban as well as rural areas are required to comply with drug and alcohol testing of staff in safety-sensitive positions, including those operating, dispatching, and maintaining revenue service vehicles.

If a transit provider uses just one taxi company or other non-dedicated provider, then the FTA drug and alcohol testing requirements apply. In that case, the transit provider must ensure that the taxi company or other transportation provider complies with the testing requirements. However, if the transit provider uses two or more non-dedicated transportation providers and the riders choose which provider when booking a trip, then the “taxi exception” applies. This means that the providers of the service are not subject to the federal drug and alcohol testing requirements. (More on the taxi exception is provided in Chapter 12.)

Implementing Microtransit as a Many-to-Many or Many-to-One Service. A many-to-many service makes sense where there is a sufficient amount of intrazone travel to dispersed locations. Providing many-to-many service also makes sense as a new late-night service, late-night route replacement, or a weekend service. A many-to-many service makes sense where there is more than one major demand generator, such as a transit center or bus stop (where riders might transfer onto a fixed route) or a college, hospital, or shopping center. Limiting destinations funnels demand, which enhances productivity. Most rural transit agencies with many-to-one microtransit service implement the service to provide first-mile last-mile (FMLM) connections to a transit center hub or stop.

Implementing Inter-Area Trip-Making Policies. In the case of microtransit, zones often are nonadjacent, and therefore allowing interzonal trips in such a design is not common. Most rural transit agencies do not allow interzone trips on the microtransit service, instead using microtransit as a feeder/distributor to the fixed-route network for longer trips (see Figure 2.3). The concept is not to have the microtransit service compete with the fixed-route service. If the microtransit zones are adjacent, a rural transit agency can always allow direct service, perhaps with a higher fare or with transfers in the case of larger areas.



COURTESY OF HIGH VALLEY TRANSIT.

Figure 2.3. Microtransit as a feeder to the fixed route.

Deciding on Stop Types.⁹ Stop types can vary from service to service or even within a service. By policy, pickup and drop-off locations can be at the curb (in front of the requested location), or physical stands at anchor points, such as at bus connection points, a college/business/medical campus, or a mall or other multi-entrance building. Some microtransit services harness their supporting technology to create virtual stops that are within a configurable distance from a requested location and are usually at a nearby intersection. Virtual stops can also be set for certain geofenced areas where curb-to-curb service is a challenge or where a transit agency wishes to designate a physical stand. Some systems direct all riders to go to virtual stops. The concept behind virtual stops is to improve productivity with more shared rides and fewer stops. However, this can be problematic in certain rural areas where demand is dispersed. Transit agencies should be careful in selecting the supporting technology. Some focus solely on virtual stops while others do not provide for virtual stops at all. Considerations for many rural and tribal communities include whether there are addresses.

Setting Booking Policies.¹⁰ The defining characteristic is that riders can request service on demand (or within a very short lead time) for immediate fulfillment. As a convenience to customers, some microtransit services also allow longer-lead same-day requests, advance requests, and (less common) subscription requests. Another design element is whether to allow walk-ons at key stops versus having all riders place an on-demand request. In the case of walk-ons, drivers add in the trips as the riders get into the vehicle.

Allowing Walk-Ons. Allowing walk-ons at anchor points is great for riders because they do not need to prebook their trip, and they are perfect for timed connections at transit stations/hubs and at other major attractors that have time-sensitive demand. (See sidebar.) Drivers book trips for walk-on riders, which is a function of the technology, noting that some technologies do not have this function. The reasoning for not allowing walk-ons is that some agencies do not want drivers to have to do extra work at the anchor points, arguing that their job is challenging enough.

App versus Call-In Booking. Many transit agencies, their microtransit contractors, or service providers provide a customer-facing app or web-based booking to their riders to book trips. Many such apps provide riders with ETAs, imminent arrival notices, and the vehicle location before and after the rider is onboard. Some apps also provide a way to rate the service and provide feedback. However, in concert with Title VI requirements, transit agencies are required to provide a call-in option for customers who do not have a smartphone or do not have access to the internet. Some transit agencies use their paratransit call center for this purpose, with call center staff able to use a link to the on-demand technology to enter the caller's request.

Fare Policies. Common types of microtransit fares include flat fares or zone-based fares, where there are different zones and where a rider might be able to travel from one zone to another. Where microtransit is introduced as a new service in an area not previously served by any public transit, it is common to set the fare equivalent to a fixed-route fare. This is also the case where microtransit is exclusively used as FMLM service, noting that some transit agencies offer free fares for such a feeder/distributor service, the idea being to improve fixed-route ridership by making access easy.

In **Missouri City, Texas**, a transit agency provides microtransit service by superimposing a many-to-many service on top of checkpoint-deviation service. The checkpoint-deviation service uses two buses that shuttle between a transit center and a shopping center, picking up or dropping off walk-on passengers (allowed at the two anchor points) along the way. If no one walks onto the vehicle at an anchor point, the vehicle automatically joins the other microtransit vehicles providing many-to-many service up until it needs to be at the next anchor point.

Subsidy Policies. Common to microtransit service that relies on NDSPs such as taxis or TNCs are meter-based or technology-generated fares, which are typically distance-based, noting that TNCs can vary rates based on location and demand levels.

- With a provider-side subsidy design, there is typically a base fare and a maximum subsidy, with the riders paying any overage. For example, a \$2 base fare is paid by the rider, a transit agency-provided subsidy of up to \$13 is paid to the providers, and the rider pays any part of the fare over \$15. When using this design, it is important that the total fare, including cases where a rider pays an overage, not be different for riders who need a WAV, lest there be a service equivalency issue. Other cost control elements include limiting the number of trips per day, week, or month that a rider can take.
- With a user-side subsidy program, the transit agency matches the rider's prepayment onto a fare card or bank credit card up to a maximum subsidy per month. Then the rider uses the card to pay for allowable trips, noting that the maximum subsidy controls the number or length of the trips taken. Common match ratios range from a 1:1 match to a 4:1 match. It is important that WAV riders and other riders pay the same fare for a similar trip. In some cases, transit agencies have paid the service provider a WAV trip stipend per trip to cover the additional cost of operating a WAV, thereby allowing equivalent fares (see the section on Travel Vouchers and Fare Card Programs).

Achieving Service Equivalency. With microtransit service or with any demand responsive service provided by a transit agency, the transit agency, under the ADA, must achieve service equivalency. Seven criteria define service equivalence, but in practical terms, the most difficult to achieve and evidence is equivalent response times for riders who require the use of WAVs and those who do not need a WAV. This requirement means that the transit agency not only has to ensure that WAV service is provided by design, but also that the response time for the two different sets of riders can be tracked separately (see also Chapter 12, Accessible Service).

DART in Des Moines uses Uber and taxis for its The Flex Connect service, while also directly operating WAV service. Riders can access the service via or by calling the DART call center. Service is free and restricted to trips going to/ from the transit hubs.

In **Flagstaff, Arizona**, **Mountain Line** introduced a microtransit service in part to take advantage of slack in its ADA paratransit service. In **Greenfield, Massachusetts**, a microtransit service was similarly implemented to take advantage of its rural demand response service's extra capacity.

Providing WAV Service. There are various ways that WAV service can be provided. WAVs can be the vehicle of choice for dedicated service operated in-house or by contractors. For non-dedicated service, these are some options:

1. Many taxi companies already have or are required by municipal ordinance to have wheelchair-accessible taxicabs.
2. Transit agencies may require their provider to supply WAV service to meet service equivalency; this can include the transit agency supplying WAVs to the provider or leaving this up to the provider to supply enough WAVs.
3. TNCs, if used, are beginning to partner with drivers of WAVs; however, the supply is typically very limited (and this is often the case with user-side subsidy programs). Some transit agencies that rely primarily on NDSPs that cater to ambulatory riders either directly operate WAV service or require the TNCs to subcontract with a provider (such as a paratransit management/operations company or an NEMT company) to provide WAV service.

Coordinating/Consolidating Microtransit with Other Services. Some rural transit agencies that already provide fixed-route service (and hence ADA paratransit service as well) have coordinated, if not consolidated, their microtransit and ADA paratransit services. Where two different technologies are used to support the two services operated with separate fleets, this often involves building a software bridge between the two technologies so that the ADA paratransit dispatcher (or dispatching system) can recognize a

trip assignment solution on a microtransit vehicle. A more consolidated approach is to use the same technology and the same set of vehicles to support both services. Either approach works best where the two services overlap or are provided near each other. Drivers in either approach would need to be ADA paratransit certified. It is important in consolidated services that ADA paratransit trips be prioritized and that ADA versus microtransit service data be tracked separately. Also, microtransit can be combined with other demand response services.

Travel Vouchers and Fare Card Programs

Rural and tribal transportation can be cost-effectively provided with travel vouchers or fare cards. Both are used in conjunction with existing public and private service providers, with the riders arranging for service directly with the service provider.

Implementing Travel Vouchers.¹¹ Travel vouchers are paper vouchers, which are good for one ride. A variation of travel vouchers is scrip tickets, each of which has a monetary value. Rural and tribal transit agencies can provide or sell (at a significant discount) vouchers or scrip to eligible individuals to pay for trips with transportation providers that are willing to participate in the program. Such providers might include public transit or paratransit, taxis or private for-hire service providers, intercity bus carriers, volunteer drivers, or a friend or family member. The concept is to make existing services more affordable for the individual while also enabling the transit agency to provide service at an attractive rate.

Once a rider has acquired travel vouchers or a book of scrip tickets, the rider contacts the participating service provider to arrange service and uses the travel voucher or scrip as fare to pay for the trip. The driver then turns in the vouchers or scrip to the company office, which submits the vouchers or scrip to the transit agency as proof that the trip was taken. With some vouchers, a rider's signature and details of the trip are required. The transit agency then pays for the trip based on a negotiated rate (in the case of vouchers) or per the face value of the scrip submitted.

One of the shortcomings of travel vouchers is the propensity for fraud, which can include the duplicating of paper vouchers or scrip as well as driver-rider collusion. The former can be somewhat mitigated by the use of serial numbers and recording the serial numbers of the vouchers or scrip sold to each customer. The latter occurs when riders sell their signed vouchers to a driver at a profit, and the driver fabricates a trip not taken. This type of fraud is a little more difficult to spot unless patterns occur.

Implementing a User-Side Subsidy Program with Fare Payment Cards. Because of the fraud potential of paper travel vouchers, they have largely been replaced by fare card payments (and card readers), like those that exist with taxi meters or mobile readers such as Square devices that work with smartphones. Many rural transit agencies rely on a user-side subsidy design where they add funding to a customer's fare card or bank card account. From there, the rider arranges transportation with a provider that accepts the card. A rider will swipe the card at the beginning or end of the trip. The in-vehicle technology ties into the rider's account and decreases the account at the end of the trip. Fare card systems commonly provide more information about the service performance with data provided to evidence service equivalency, for example. Where bank cards are used, the transit agency may get more limited data about program use.

Volunteer Driver Programs

Implementing a Volunteer Driver Program. Volunteer driver programs can be an effective means to accommodate unmet mobility needs in rural and tribal communities where transit, paratransit, and taxi¹²/TNC services are limited or cost-prohibitive. Such volunteer programs can also coexist to augment other existing transit services.

With the transit agency reimbursing volunteers for mileage at an IRS-approved rate, these programs provide low-cost transportation in communities not well served by transit and can be especially appropriate for longer-distance travel, for example, to a regional hospital or dialysis facility. This is because the reimbursement rate is considerably more cost-effective than using a vehicle from a dedicated fleet or a non-dedicated fleet (if any exist in the community).

Recruiting and retaining volunteer drivers can be difficult, so a program coordinator's success may rely on ways to ease the burden on the driver. Flexibility is key, allowing a driver to pick and choose the rider the driver wishes to serve and facilitating the reimbursement process. Some drivers also like to drive on certain days, so they might ask the coordinator to fill up their manifest every Monday, for example.

Recruitment is a constant challenge although many volunteer driver program coordinators rely on the riders to recruit their own drivers (a friend or family member), and the coordinator takes it from there. A driver appreciation program can help with driver retainage.

There are ways to coordinate if not consolidate many of the administrative aspects of these programs, such as recruiting, retaining, managing, and reimbursing the volunteer drivers. Such strategies that better connect these programs will help rural and tribal transit providers improve mobility for their communities, with better leveraging and coordinating of the resources.

New technologies have also enabled ride boards for volunteer driver programs where volunteer drivers and individuals needing a ride can match up. With such technologies, volunteer drivers and riders find each other via portals to the ride board and self-match. One of the respondents to the research project's survey implemented software to improve communication between transit agency staff and volunteers while scheduling rides and completing required reports.

One of the inherent shortcomings of many volunteer driver programs is the lack of WAVs. This can be addressed by rural and transit agencies giving WAVs due to be retired to one or more volunteer drivers or organizations with volunteer driver programs.

Many of these programs have varying and sometimes conflicting parameters, such as rider eligibility or trip purposes. For example, in a given rural region, there may be volunteer driver programs that focus on older adults and veterans while others focus on the transportation of cancer patients to and from treatments or other specific trip purposes.

Practices to Enhance Service Design Efficiency and Effectiveness

In addition to the service modes and strategies presented, the research uncovered several operating practices that are being used to enhance services (applicable for both new service modes and improvements to existing services). Three of these are presented here, and Chapter 6 presents related operational/management considerations in a number of areas.

Managing Demand in Large Regional Service Areas

Regardless of whether the fixed-route, flex-route, or demand response services are provided to the general public, many rural and tribal transit agencies serving very large regions with dispersed demand implement a strategy of serving different communities on different days. By funneling travel demand to certain days, the opportunities for ridesharing increase, which

increases productivity and lowers cost-per-trip. Such a strategy has been known to double ridership and productivity.

In the rural areas around Charlottesville, Virginia, for example, Jaunt Inc. provides both rural demand response services that link outer counties to Charlottesville and other rural demand response services that provide circulator services within the outer counties. Depending on the county and service (link or circulator), service is provided six times per week, five times per week, three times per week, or two times per week. In one case, the same driver/vehicle is used to provide three times per week service on Monday, Wednesday, and Friday in one county and a two-times-per-week service on Thursday and Friday in another county.

In addition, one of the respondents to the research project's survey provides shopping trips to areas of their service on different days, thereby funneling demand to specific destinations. Yet another survey respondent limits out-of-county trips to Tuesdays and Thursdays (their lower ridership days) for medical trips; in this way, they can provide needed out-of-county trips with the existing fleet without compromising the service quality of the in-county service.

Use of Part-Time Operators

Use of part-time operators for a transit service operated with a dedicated fleet provides a cost-efficient strategy to fulfill a vehicle's daily obligations. For example, a fixed-route schedule has a 12-hour run. The schedule can be run with one full-time (8-hour) shift and one part-time (4-hour) shift. Part-time drivers can also be used for two halves of a split run.

As another approach, one rural transit system reported that it hires part-time drivers during midday to accommodate a sustained demand level during the day and fill the gap resulting from full-shift driver lunch breaks.

The lower cost stems from a lower compensation package for part-time employees. In today's current driver recruitment challenges, using part-time drivers opens employment opportunities for prospective drivers who do not want full-time positions, those who want to augment their first job, and those who may want to try out being a vehicle operator.

Maximizing Use of Subscription Service and Maintaining Subscription Trips

Another important tactic for demand response services involves efficient scheduling of subscription trips (also called standing orders). Use of subscription trips can benefit the transit agency as well as the riders, who do not have to call in for each trip.

Benefits to the transit agency accrue when the subscription trips are scheduled efficiently with the driver schedules reviewed on a routine basis to ensure efficient groupings, as these trips provide the backbone of the schedule. If subscription trips are not well scheduled, the productivity of the entire system will suffer. This is why frequent maintenance of the subscription trip schedules is so important.

Efficient subscription trip scheduling will lead to productivity improvements. Experience shows that improvements in productivity can also result from scheduling the same rider trips with the same driver (or run) from day to day. The driver knows the riders and their trips (e.g., where to position the vehicle for pickups/drop-offs, which riders need more assistance with boarding and alighting), which can lead to more efficient service.

In addition, more subscription trips translate into fewer calls with trip requests, and possibly lower labor costs for the trip reservation function, if the number of call takers can be reduced. What this means is rural and tribal communities should maximize the use of subscription trips, tempered only by leaving sufficient capacity for other one-time trips.

Summary of Strategies and Services

Improving mobility in rural and tribal communities can be accomplished in myriad ways. Possible effective strategies and services to meet this objective are summarized in Table 2.1, which summarizes where the various service strategies/modes may be appropriate and provides some notes on successful implementation.

Table 2.1. Service designs/modes—where to deploy.

Service Design/ Mode	Where to Consider the Service Design/Mode	Notes for Successful Implementation
Fixed-Route	• Appropriate where and when there is a sufficient demand to support such a service. • Use in corridors and areas with higher population densities, where there is enough concentrated demand to be productive.	• Focus service on higher-demand trip generators (such as large regional stores, hospitals, and human service agencies) that produce peak-period job-related or program-related demand. • Focus on trip patterns in your area—origins and destinations of riders that create concentrated demand. • Consider fixed routes as one of a compendium of services in a community—they can leave persons in low-density areas without service if only fixed-route service is provided. • Less effective when routes are long, one-way loops that attempt to serve too many origins and destinations.
Flex Routes/Route Deviation	• Flex routes/route deviations are appropriate in corridors with defined routes that also have pockets of demand beyond bus stop walkshed that are not far enough to warrant extending fixed-route service in these areas. • Use to serve medium-density areas with insufficient demand for traditional fixed route, but where there is still a more pronounced demand density within a corridor.	• Appropriate where people do not live along main roads on which bus stops are not easily accessible. • Expands the reach of transit service by allowing the vehicle to deviate off the route's main road to pick up/drop off riders. • Generally, need to accept requests for deviation from all riders, and the deviation distance should be no more than ¾ mile, per FTA guidance. • No ADA paratransit obligation if deviation is for all.
Flex Routes – Checkpoint Deviation	• Because there are no defined routes, just scheduled stops, checkpoint-deviation service can be useful in areas that do not suggest a base route—especially in less dense areas where anchor destinations exist. • Flex/checkpoint deviations also serve areas with medium density but demand insufficient for transitional fixed routes and a more dispersed population.	• A flex-route checkpoint-deviation service is suggested where there are one or more high-demand stops, such as a transit center, a big box store, a shopping center, a hospital, or a college. • Expands the reach of transit service by allowing the stops wherever there are origins or destinations that will attract sufficient ridership. • No ADA paratransit is needed/required if deviation is for all.
General Public Demand Response/ Dial-a-Ride	• Use in geographic areas or during particular times/days that do not have the demand density for fixed- or flex-route transit. • Effective way to serve a large area with low and dispersed demand.	• Can be used to complement fixed-route transit or flex-route transit in less dense areas. • Also used to serve specific population subsets (e.g., older adults and people with disabilities), or for specific trip purposes such as medical trips. • Demand response service must be operated so that service equivalency is achieved for persons with disabilities.

(continued on next page)

Table 2.1. (Continued).

Service Design/ Mode	Where to Consider the Service Design/Mode	Notes for Successful Implementation
ADA Complementary Paratransit	- Required if providing traditional fixed-route services. - May be possible to provide ADA complementary paratransit on fixed-route vehicles, especially when fixed-route services are provided along very long, very rural routes.	- Service should only be provided to people with disabilities who are prevented from using fixed-route transit because of their disability or disabilities. - Service must meet six service criteria, including the prohibition of capacity constraints. Transit providers must monitor service for patterns of service shortcomings, such as trip denials and late trips, among others. - Using the same vehicle to provide fixed-route and ADA paratransit can be complicated to operate because all ADA paratransit requirements must be met in serving the ADA paratransit trips.
Microtransit	- Use to expand transit service to a new area or at a new time that is unsuitable for fixed- or flex-route transit. - Can replace an underperforming fixed- or flex-route or route segment. Microtransit can typically carry between three and six passengers per hour. - May be a more effective service design in well-defined corridors or zones of operation that cannot sustain a productive fixed route, such as in small cities (including small cities in rural areas), commercial subdivisions, or strings of office parks. - Can provide more flexibility to customers than traditional fixed-route service.	- Has been used in communities with populations between 5,000 to 20,000 by rural transit providers, with service provided in a designated zone. - Focus the zone on a smaller geographic area, anchored by stops at transit connections and other major demand generators; the smaller the zone, the more productive the service. - If service equivalency cannot be achieved with on-demand service, delay service to 1- to 2-hour advance notice. - Uses smaller vehicles and mobile technology to provide dynamic routing and curb-to-curb or corner-to-corner service.
Travel Vouchers and Fare Card Subsidy Programs	- Used in conjunction with existing public and private service providers, with the riders arranging for service directly with the service provider. - Can be used to make existing transportation services more affordable for low-income riders. - Cost-effective way to target subsidies for eligible riders.	- Vouchers can be paper vouchers or scrip tickets. Each voucher or scrip ticket has a monetary value and allows riders to help pay for trips on existing transportation services that participate in the program (e.g., public transit services, taxis or other for-hire services, intercity bus carriers). - Fare cards are now more common than paper vouchers or scripts, with the transit agency adding funds to an eligible rider's card to subsidize trip-making use. - For fare cards, consider possible implications for the unbanked population. Applications using a customer's fare card or bank card account may not be accessible to the unbanked or persons without smartphones.
Volunteer Driver Programs	- Use to meet mobility needs in areas where transit services and taxi/TNC services are limited or cost-prohibitive. - Can be an effective means to accommodate unmet mobility needs in rural and tribal communities where transit, paratransit, and taxi/TNC services are limited or cost-prohibitive. - Cost-effective way to provide long-distance trips.	- Volunteer programs can also coexist to augment other existing transit services. - With reimbursement to volunteers for mileage at an IRS-approved rate, volunteers provide a low-cost option to meet needs, particularly for long-distance trips.



CHAPTER 3

Mobility Management and Coordination

Introduction

Many communities across the country are employing mobility management practices that offer innovative approaches for delivering coordinated transportation services, particularly when serving older adults, people with disabilities, veterans, and low-income individuals. In rural and tribal areas, the challenges to providing mobility for these groups are especially daunting, as trips are often long and may require crossing multiple jurisdictional boundaries. The mobility management focus on meeting individual customer needs through a range of transportation services is vital as communities balance increasing needs with funding constraints and limited resources.¹³

This chapter defines mobility management and discusses key aspects of successful programs and services. It also highlights potential funding opportunities that rural and tribal transit providers can consider when planning new or improved mobility management efforts and provides a variety of available technical assistance supports.

While the Chapter 2 presented service and design strategies to assist rural and tribal transit providers with implementing improvements that best fit their communities, mobility management offers the opportunity to fully use these services through a customer-focused approach that connects customers to their most appropriate transportation option.

Defining Mobility Management

Mobility management is an approach to designing and delivering transportation services that starts and ends with the customer. It begins with a community vision in which the entire transportation network—public transit, human service agencies, private operators, volunteer drivers, and others—works together with customers, planners, and stakeholders to deliver the transportation options that best meet the community's needs.

National Center for Mobility Management (NCMM)

The NCMM provides a variety of grants and other technical assistance for rural and tribal communities looking to implement or expand mobility management activities. These opportunities may vary from year to year and have included ready-to-launch grants, community grants, and issue-focused mobility meeting support. The NCMM website¹⁴ provides information on current opportunities and a variety of resources.

The NCMM notes that mobility management:

- Encourages innovation and flexibility to reach the “right fit” solution for customers.
- Plans for sustainability.

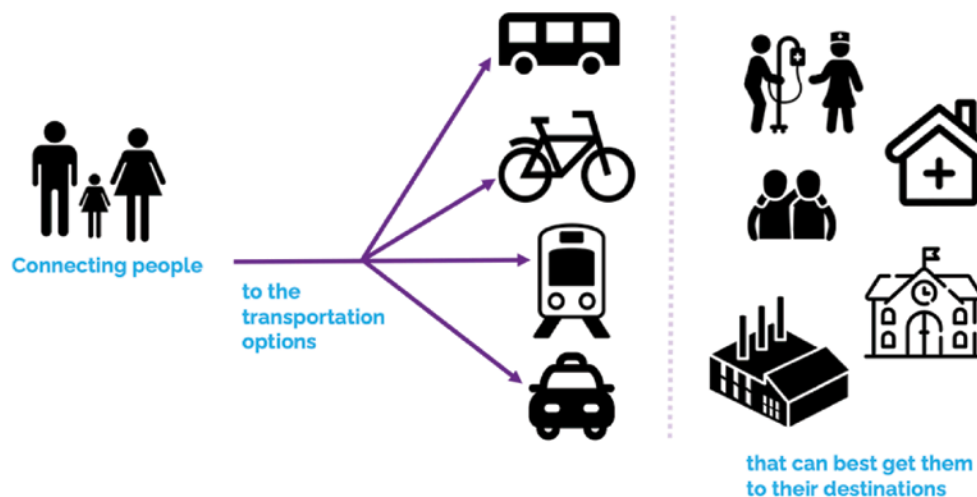
- Strives for easy access to information and referrals to assist customers in learning about and using services.
- Continually incorporates customer feedback as services are evaluated and adjusted.

NCMM also highlights that mobility management is a two-part endeavor as shown in Figures 3.1 and 3.2:

The customized approach of mobility management means no two programs are exactly alike, and a variety of entities, including rural and tribal transit providers, can serve as a mobility manager in a community. However, some components are commonly found in many mobility management programs:

- Partnerships between multiple agencies and organizations.
- A customer-driven, market-based approach that provides customers with a range of transportation options through individualized trip planning.

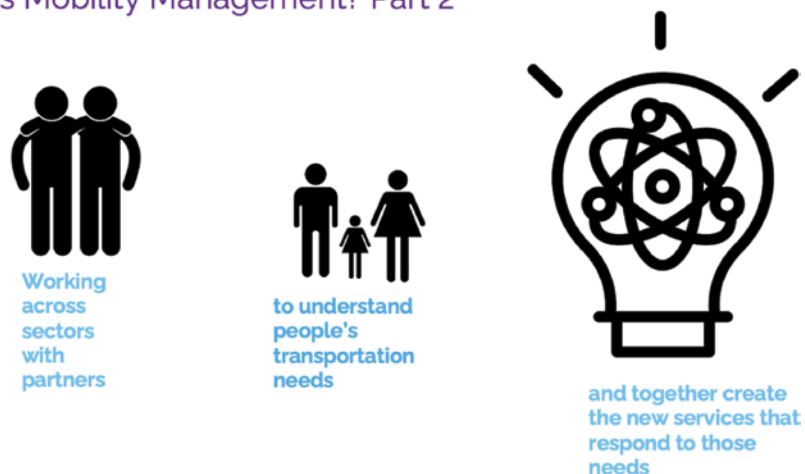
What is Mobility Management? Part 1



SOURCE: NCMM WEBSITE: <https://nationalcenterformobilitymanagement.org>.

Figure 3.1. Mobility management—Part 1.

What is Mobility Management? Part 2



SOURCE: NCMM WEBSITE: [HTTPS://NATIONALCENTERFORMOBILITYMANAGEMENT.ORG](https://nationalcenterformobilitymanagement.org).

Figure 3.2. Mobility management—Part 2.

- One-stop travel information and trip-planning centers that provide information on available transportation options and coordinate requests for transportation services. This may include the operation of a central call center for customers to gain information on available transportation options and to schedule trips.

Common Themes: Successful Mobility Management Approaches and Strategies

Research on successful mobility practices resulted in a document that can serve as a resource in the planning, implementation, and assessment of a mobility management practice.¹⁵ The document was written by the research team, and materials on the subject were drawn from that document.

The document notes that, while the customized approach of mobility management means no two programs are exactly alike, there are common components and specific strategies that are employed by successful practices that can be considered when establishing, expanding, or improving a mobility management program. While efforts through mobility management practices are often interrelated, the successful approaches and strategies are primarily segmented by four key categories detailed in the document (and shown in Figure 3.3) along with related resources and examples.

- **Community Outreach and Engagement.** Community outreach and engagement are a vital part of any successful mobility management program. Organizations with successful mobility management practices form partnerships with key community stakeholders, promote their services to the community, and lead or participate in coordinated transportation planning efforts. The document includes a stakeholder checklist with the types of organizations that can help increase support for mobility management programs.
- **Needs Assessment and Program Design.** Coupled with community outreach, successful mobility management practices are sensitive to local and regional transportation needs and build their program around these needs. There is a strong emphasis on listening to their local community, conducting ongoing needs assessments, and designing services and programs to fit the identified needs. The document includes a variety of tools and resources for conducting needs assessments and designing appropriate services and programs.
- **Program Evaluation and Assessment.** Monitoring and evaluation are important considerations for any mobility management program, especially as funding continues to be constrained for transportation projects and services. Successful mobility management practices

"Often transportation providers are looking at how can MY agency do XYZ, or how can MY agency implement coordination projects in their communities? The shift in coordination should be towards what can ALL of us do to implement a strategy. Mobility managers can help with that outside of the solo-agency thinking when it comes to coordination of projects."

Statewide Mobility Manager
Oklahoma Department
of Transportation



SOURCE: KFH GROUP 2018.

Figure 3.3. Key strategies for successful mobility management.

employ some form of program evaluation and assessment. They track program outcomes using qualitative and quantitative performance measures with established goals and objectives. The document discusses various tools and resources for assessing mobility management program outcomes.

- **Funding.** Funding strategies are the backbone supporting mobility management efforts. While successful mobility management practices explore different funding sources, often the primary ones are through FTA programs, which are discussed in the next section.

Funding and Technical Assistance to Support Mobility Management Efforts

While a variety of funding programs may support mobility management activities and may vary across the country based on state-to-state department of transportation (DOT) focus on these efforts, primary opportunities are identified as follows. Several FTA programs are listed, though it should be noted that there are competitive federal programs not listed that have provided funding in the past and could have funds allocated to them in the future. Therefore, rural and tribal transit providers are encouraged to monitor the FTA website for such opportunities.

Federal Transit Administration

Sections 5310 and 5311 Programs

Mobility management projects are eligible for capital funding under FTA's Sections 5310 and 5311 programs (providing up to an 80% federal share).¹⁶ A list of the types of activities that are eligible for mobility management capital funding can be found in the FTA circulars for each program. Because each state administers its Sections 5310 and 5311 programs differently, rural and tribal transit managers should check with their state DOT or FTA contacts for funding eligibility of mobility management projects in their area (see also Chapter 11, Funding and Revenue).

Section 5310 Program—Coordinated Planning Requirements

One of the federal requirements of the Section 5310 program is that projects selected for funding (including mobility management projects) must be included in a locally developed, coordinated public transit-human services transportation plan (coordinated transportation plan). FTA guidance defines a coordinated public transit-human service transportation plan as one that identifies the transportation needs of individuals with disabilities, older adults, and people with low incomes; provides strategies for meeting those local needs; and prioritizes transportation services and projects for funding and implementation. There are four required plan elements:

1. An assessment of available services that identifies current transportation providers (public, private, and nonprofit).
2. An assessment of transportation needs for individuals with disabilities and older adults. This assessment can be based on the experiences and perceptions of the planning partners or on more sophisticated data collection efforts and gaps in service.
3. Strategies, activities, and projects to address the identified gaps between current services and needs, and opportunities to achieve efficiencies in service delivery.
4. Priorities for implementation based on resources (from multiple program sources), time, and feasibility for implementing specific strategies and activities identified.

Innovative Coordinated Access and Mobility Pilot Program¹⁷

The Bipartisan Infrastructure Law, enacted as the Infrastructure Investment and Jobs Act in 2021, continues to provide funding for the Innovative Coordinated Access and Mobility (ICAM)

pilot program, which funds capital projects to improve coordination and enhance access and mobility to vital community services for older adults, people with disabilities, and people of low income.

The range of capital activities eligible under the ICAM program includes the following:

- Financing innovative projects that improve the coordination of transportation services, including nonemergency medical transportation services.
- Regional or statewide mobility management projects.
- Deployment of coordination technology.
- Regional or statewide projects that create or increase access to one-call/one-click centers.

Coordination with Other Federal Programs

The Coordinating Council on Access and Mobility (CCAM) is a federal interagency council that works to coordinate funding and provide expertise on human services transportation for three targeted populations: people with disabilities, older adults, and individuals of low income. The CCAM works at the federal level to improve coordination of transportation resources and to address barriers faced by states and local communities when coordinating transportation. The CCAM's mission is to issue policy recommendations and implement activities that improve the availability, accessibility, and efficiency of transportation for CCAM's targeted populations, with the vision of equal access to coordinated transportation for all Americans.

As part of the coordination efforts, CCAM encourages federal fund “braiding” for local match that allows grant recipients to use funds from one federal program to meet the match requirements of another, such as the Section 5310 program. The CCAM Federal Fund Braiding Guide provides information to potential grantees on acceptable fund braiding arrangements on transportation-related projects. The guide defines federal fund braiding for local match and examines whether federal fund braiding is allowable for 61 programs across CCAM agencies that may fund transportation. More information on CCAM and access to the braiding guide can be found at <https://www.transit.dot.gov/coordinating-council-access-and-mobility>.¹⁸ This website also provides links to multiple FTA-funded transportation technical assistance centers that offer guidance with coordination efforts.

In addition, in March 2022, FTA posted guidance clarifying coordination of human services transportation on a new transportation coordination web page. As noted by FTA, coordinated transportation involves multiple entities working together to deliver one or more components of a transportation service to increase capacity. The transportation coordination guidance aims to reduce overlap between the federal programs across nine agencies that may fund human services transportation and incentivize collaboration by clarifying eligible reporting for the NTD. This new guidance addresses several topics related to NTD reporting: definition of public transportation; paratransit; charter service; incidental use of transit assets; and trip brokering. The guidance can be found at: <https://www.transit.dot.gov/regulations-and-programs/access/ccam/about/transportation-coordination>.¹⁹

The Role of Rural and Tribal Transit Providers in Mobility Management and Coordination Efforts²⁰

With mobility needs for older adults, people with disabilities, and others continuing to increase while funding is not keeping pace, rural and tribal transit providers must seek opportunities to effectively use existing resources and to identify methods to provide more coordinated services. Through these efforts, rural and tribal transit providers may operate a mobility management

"The emphasis on a collaborative mindset is crucial. Shifting from an individual agency-centric approach to a collective strategy is a game-changer. Mobility managers indeed play a pivotal role in fostering this collaboration, breaking down silos, and expanding the reach of transportation services."

Steuben County Mobility Manager
The Institute for Human Services, Inc.
Bath, New York

program out of their organization or may join a group of stakeholders through a mobility management program facilitated by another agency in their community.

Increasing demand and shrinking resources have caused communities to think about transportation in new ways, and, through mobility management and coordination efforts, rural and tribal transit providers can be thinking more broadly than just the current services offered through their organization.

There are also a variety of state-level mobility management efforts in place across the country, so it is important for rural and tribal transit providers to connect with their state and federal funding program contacts to be aware and involved in these efforts. For example:

- The Nebraska Department of Transportation leads a statewide program that facilitates a variety of transportation projects and provides various resources to support local and regional mobility management efforts.²¹
- The Ohio Department of Transportation's Mobility Management Program supports over 40 local and regional mobility managers, providing numerous resources in their efforts to increase awareness of community transportation options and services and to ensure mobility needs are considered in planning activities.²²

Emerging and Evolving Technology²³

Rural and tribal transit providers should also stay informed on emerging and evolving technology that can be used to support mobility management and coordination efforts. As highlighted by NCMM, technologies, services, and partnerships are rapidly evolving, with the field experiencing the advent of Geographic Information System (GIS)-based tracking and monitoring of vehicles with apps that allow customers to directly interface with services and receive minute-by-minute service updates. These advances have been added to the community transportation menu piecemeal, requiring transit managers to figure out how to integrate them into existing mobility options.

PICK Transportation²⁴ is a single-branded, regional, on-demand transit system that includes four public transit agencies and 10 tribal nations. PICK Transportation is a consortium of public transit agencies (Pelivan Transit & Northeast Oklahoma Tribal Transit Consortium,²⁵ JAMM (INCA) Transit,²⁶ and KI BOIS Area Transit System²⁷) that provides public transportation for all. PICK Transportation operates during nontraditional hours, operating Monday through Friday from 5:00 p.m. to 10:00 p.m., and Saturday from 10:00 a.m. to 2:00 p.m., which allows riders to book a trip immediately from a smart phone (PICK Transportation OK app for iPhone or Google Play) or desktop and expect an ADA-compliant vehicle to arrive within minutes.

Serving rural communities in Eastern Oklahoma, PICK Transportation is a flexible, demand responsive transportation service that aims to bridge the gap with traditional public transit services. Unlike fixed-route buses, the service provides microtransit that operates on a more localized scale, using smaller vehicles such as vans or shuttles. As noted by PICK Transportation, this innovative mode of transportation leverages technology to optimize routes and schedules based on real-time demand and provides customers with convenient point-to-point transportation options. See more on this coordinated service in the case study on PICK Transportation.

At the same time, rural transit managers need to consider funding, compliance, and other requirements when incorporating these new technologies and strategies. The website of the National Center for Applied Transit Technology (N-CATT)²⁸ provides additional information on emerging technologies for transit, including mobility management.

Technology for Mobility Management: A Framework for Moving Forward

The NCMM has developed a technology typology to support mobility management efforts, with the goal to give mobility managers a framework for understanding, comparing, and evaluating the value of the many different types of technology available in the public transportation sector.

Available at <https://nationalcenterformobilitymanagement.org/technology-for-mobility-management-a-framework-for-moving-forward>, the typology covers the main technology areas for the industry, taking into account the following dimensions (National Center for Mobility Management n.d.-a):¹⁴

- **Scalability.** How well does the technology type keep up and stay relevant as an agency grows in its needs?
- **Platform Accessibility.** How much capacity is there to integrate and exchange data with other systems through a documented system-to-system interface? This type of accessibility refers to the ease of access between technology systems and should not be confused with the people-to-infrastructure meaning of accessibility commonly used in the mobility field.
- **Ease of Implementation.** Can an agency set up this category of tools themselves? If external resources are needed, what kind and how much?
- **Relative Affordability.** Roughly how affordable is the solution compared with other options?
- **Maturity.** How “fully baked” is this kind of solution? How much risk is there that a solution might not work well because the vendor has not worked out all the kinks?
- **Generalizability.** How confident can an agency be that the solution will “just work” out of the box without significant up-front investment to configure the tool or alter agency workflows? The more reliant a system is on certain conditions for it to be useful, the less generalizable it is.

Shared Mobility

While the focus of mobility management is on individual customers, rural and tribal transit managers do not want to lose sight of shared mobility through these efforts and the opportunities to use transportation resources most effectively in their communities. Information about emerging approaches to shared mobility, including a list of frequently asked questions about FTA funding and requirements for nontraditional types of services, can be found on the FTA website. The Shared-Use Mobility Center (SUMC) is another source of information about coordinating shared mobility and new technologies.



CHAPTER 4

Regional Services and Connections to the National Network

Introduction

Intercity bus services constitute an important part of the rural transportation infrastructure, providing longer-distance connections between rural and urban areas. Interest in regional services (those that fall in between local transit and intercity buses in terms of trip length) is increasing as the need for access to regional destinations grows.

This chapter presents service and design strategies and related operational practices that rural and tribal transit providers can implement to address rural regional transit needs, including connections to the national intercity bus networks. Residents of rural and tribal communities benefit when they can access regional activity centers and national transportation networks with services to larger urban areas. This access may be provided directly through intercity bus services with stops in rural areas, or when their local transit providers connect to the national and regional transportation networks. This facilitates access to destinations and services that are not available in the local community, such as major medical facilities and large employment centers. The connections also allow travel beyond the rural community to visit family and friends in distant locations across the country.

Many services that were formerly available locally are now consolidating at the regional level. Rural hospitals have been closing their doors in increasing numbers in recent years, leaving rural residents without close access to more specialized (or even basic) medical services. Necessary shopping often requires trips to more distant stores serving larger regions, and longer commutes for jobs may also drive the demand for regional services. Access to higher education facilities is another need addressed by regional and intercity bus access. However, many intercity bus riders in rural areas depend on access to the intercity bus network to maintain ties to distant family members. One of the key functions of these long-distance bus services is enabling important social connections.

There is a significant potential role for rural transit providers to develop services that enable them to become part of a connected national network that gives access across the country. The extent of the national network is not fully appreciated, but a national interline ticketing system and information system provide the potential for a connected national network. Rural residents make long-distance trips for various reasons—to reach educational facilities and for medical, military, shopping, employment, and personal business trips as well as to visit family and friends.

Because federal funding for rural intercity services is provided to the state DOTs, they play a critical role in developing and supporting services that provide intercity bus service in rural areas, or support connections to intercity bus stops in more urban areas. Rural transit providers can fill the gap and provide connections to the regional and national network with different

strategies, ranging from coordinated ticketing arrangements to actual service provision using federal grant funds.

Historical Context

At one time, all rural public transportation services were operated by private bus companies. Although bus services were originally local or regional, with individual routes linking small towns, the firms quickly developed techniques such as interline ticketing and through buses to develop larger networks and provide intercity services, linking small towns with larger urban areas. Services were connected and networks grew with the development of stations where passengers could transfer between routes, greatly increasing the mobility options by allowing users to travel to many different destinations. At stations, passengers could wait for connecting buses, get information, buy tickets, use restrooms, etc., in a climate-controlled environment, often with options for food as well.

After World War II, the increase in auto use led to service reductions and a focus on city-to-city services using major highways. Unprofitable rural services were the most likely to be discontinued. As services shifted to the interstates, many small-town stops were eliminated—particularly following the Bus Regulatory Reform Act of 1981, which eliminated federal control of routes and fares, and preempted state controls as well. Rural service continued to be affected by the bankruptcy of Continental Trailways and its takeover by Greyhound, the nationwide Greyhound strike in 1990–91, and Greyhound service restructuring in the mid-2000s. Finally, in 2020, the bus industry was severely affected by the travel restrictions of the COVID-19 pandemic, which reduced ridership to approximately 20% of pre-pandemic levels, and, more recently, Greyhound was purchased by Flixbus.

Through these developments, the least profitable services are the ones most affected—rural and small-town services. The federal policy response to the loss of rural intercity bus services has been mainly through the creation of a funding program as part of formula assistance to the states for rural public transportation. Initially, this program was a subsection of the rural public transportation program, Section 18(i), which became Section 5311(f), to provide funding to the states to support intercity bus service for rural residents. The need to connect the developing rural and small urban public transportation systems with the remaining intercity bus networks was apparent early in the history of this program. Greyhound and the Community Transportation Association of America developed the Rural Connection Program in 1989, but the actual program was severely impacted by the Greyhound strike of 1991. However, many of the approaches developed for that concept are still valid and applicable today.

One significant tool that is now available is the in-kind match provisions of the Section 5311(f) program. This allows the value of connecting unsubsidized intercity bus services to be used to meet the requirements for local operating match, rather than requiring local, state, or carrier cash match. This change in the program has increased program implementation by providing a means for states to fund rural intercity services using federal funds for operations.

Context: What to Know

For a rural public transportation entity—typically a private nonprofit, a public entity (such as a city or county), or a tribal organization—to develop and fund a service to address the need for long-distance connections requires some knowledge of the context because that can provide opportunities or constraints that will affect service design and feasibility. Federal programs, state implementation, and the intercity bus industry each have policies and procedures that will affect

what a local provider can do, and, for that reason, the research team has included a summary of the context of the federal programs and regulations, how states have implemented those programs, and the current state of the intercity bus network. Developing a rural regional service that connects to the national network may seem like a puzzle, but it is possible!

Federal Programs and Requirements

Most of the national intercity bus network is operated by private firms funded by fare revenue, but routes in rural areas and rural feeder routes and connections can be funded by FTA Section 5311(f) rural intercity bus funding. Section 5311/5311(f) funding is for transit service to residents of nonurban areas—those with under 50,000 residents.

This component of the overall FTA Section 5311 rural program requires that a minimum of 15% of a state's 5311 apportionment be used for intercity bus service unless the state's Governor certifies that there is no unmet rural intercity need. Each state contemplating such a certification must conduct a consultation process that involves the intercity carriers at least every 4 years. The 15% set-aside is a floor, not a ceiling, and Section 5311 funding can also be used to fund regional services. Section 5311(f) funding is allocated to the states. It can be used for operations, capital (such as vehicles, facilities, and technology), planning, and administration.

Other requirements of the Section 5311(f) program define intercity bus as fixed-route, fixed-schedule long-distance service using vehicles that accommodate baggage and make meaningful connections with the national network of intercity bus services. FTA guidance does not define “meaningful connection” except that this is service to the same stops with schedules coordinated to provide a convenient connection. Rural regional services can be funded with intercity funds if they make meaningful connections with intercity bus services, but can also be funded with Section 5311, state, and local funds if they do not make intercity connections—often there are rural regional needs in places with no intercity bus service, or at times that do not coincide with the remaining intercity schedules. Commuter services and charter services cannot be funded under this program. The current circular addressing the grant program²⁹ does not include a definition of commuter bus services, but the proposed update of the circular governing the program includes such a definition.

The Section 5311(f) intercity program has the same match requirements as the overall Section 5311 program. The federal share is limited to 50% of the net operating deficit of a project, or 80% of capital, planning, and administrative costs. This program has a unique option of using the value of unsubsidized connecting intercity bus service as the local match for federal operating funds. The in-kind match program allows the value of unsubsidized connecting intercity bus service to be counted as the local match for operating programs. Both the segment to be subsidized and the unsubsidized service are combined in a single project. FTA requires that the carrier providing the in-kind miles submit a letter of agreement defining the service being used (schedule, route length, etc.), and its value (frequency times length of route times fully allocated operating cost per mile). Match miles do not have to be provided in the same state. The practical effect is that many rural intercity routes can be operated without local or state cash match, if the services connect with unsubsidized services and that carrier agrees to the arrangement.

State Implementation

Intercity routes that provide connectivity to rural residents that are unprofitable to the private sector can be funded using Section 5311(f), which is provided through the state DOTs. States may identify the unmet needs through a consultation process, and either contract for services on those routes or provide grants to operators for those services. Some states solicit local/regional

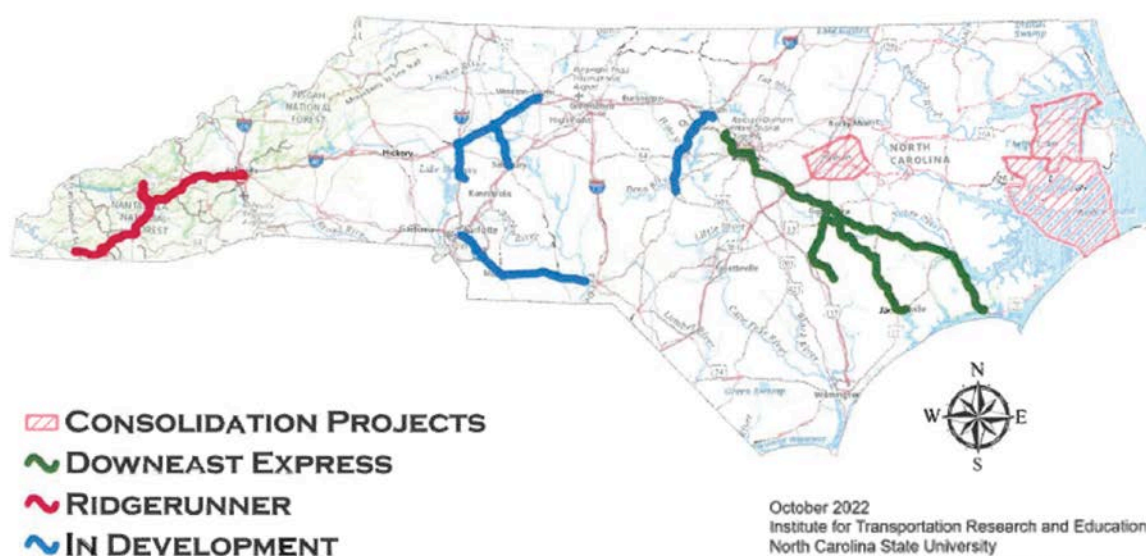
North Carolina's ConCPT CO Program

The North Carolina Department of Transportation (NCDOT) has developed a funding program for long-distance rural routes operated by three or more public transportation grantees who agree to establish formal relationships to maximize resources, gain efficiencies, and improve access to public transportation. The program calls for routes to include at least three transit systems and operate every weekday using a published schedule. The requirements call for the routes to provide access to major medical destinations and nearby regional and intercity transportation facilities such as multimodal centers and intercity bus stops. Service to other regional destinations such as community colleges can also be included. The program suggests that service should operate preferably two times per day. NCDOT will pay 50% of the net costs of the service. ConCPT funding is limited to a maximum of \$200,000 per year for each grantee involved in the coordination program. Two routes were initially funded, as shown on the map—a Downeast Express filling a gap in the intercity network, and a Ridgerunner route in the far western mountain regional of the state. The Ridgerunner has been renamed the WNC Express, and it has received ConCPT funds for FY25 and is operating on demand. The Downeast Express is operating 5 days per week, connecting Goldsboro and other rural areas with Raleigh, with many trips to medical facilities and the airport. Neither service interlines with intercity bus services. Other projects in the central part of the state were under development when COVID-19 affected the program. This program is a unique effort to use state funds to address the more regional needs while providing connections to longer-distance intercity services.

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ConCPT – Consolidation and Coordination of Public Transportation



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proposals for such services, allowing local providers to take the initiative in developing service proposals. Some states do both. Oregon DOT contracts for private intercity firms to operate several gap-filling routes, branding these services as the POINT routes; in addition, it offers grants to public and nonprofit providers for intercity services that connect with the intercity network. South Dakota provides grants to a private intercity carrier to maintain the main intercity routes and also provides grant funding to local public transit providers to operate connecting feeder routes and bus depots (see the South Dakota case study). North Carolina uses Section 5311(f) funding and in-kind match from Greyhound to contract for a statewide network of intercity routes and has been developing a new model of regional routes to feed them.³⁰

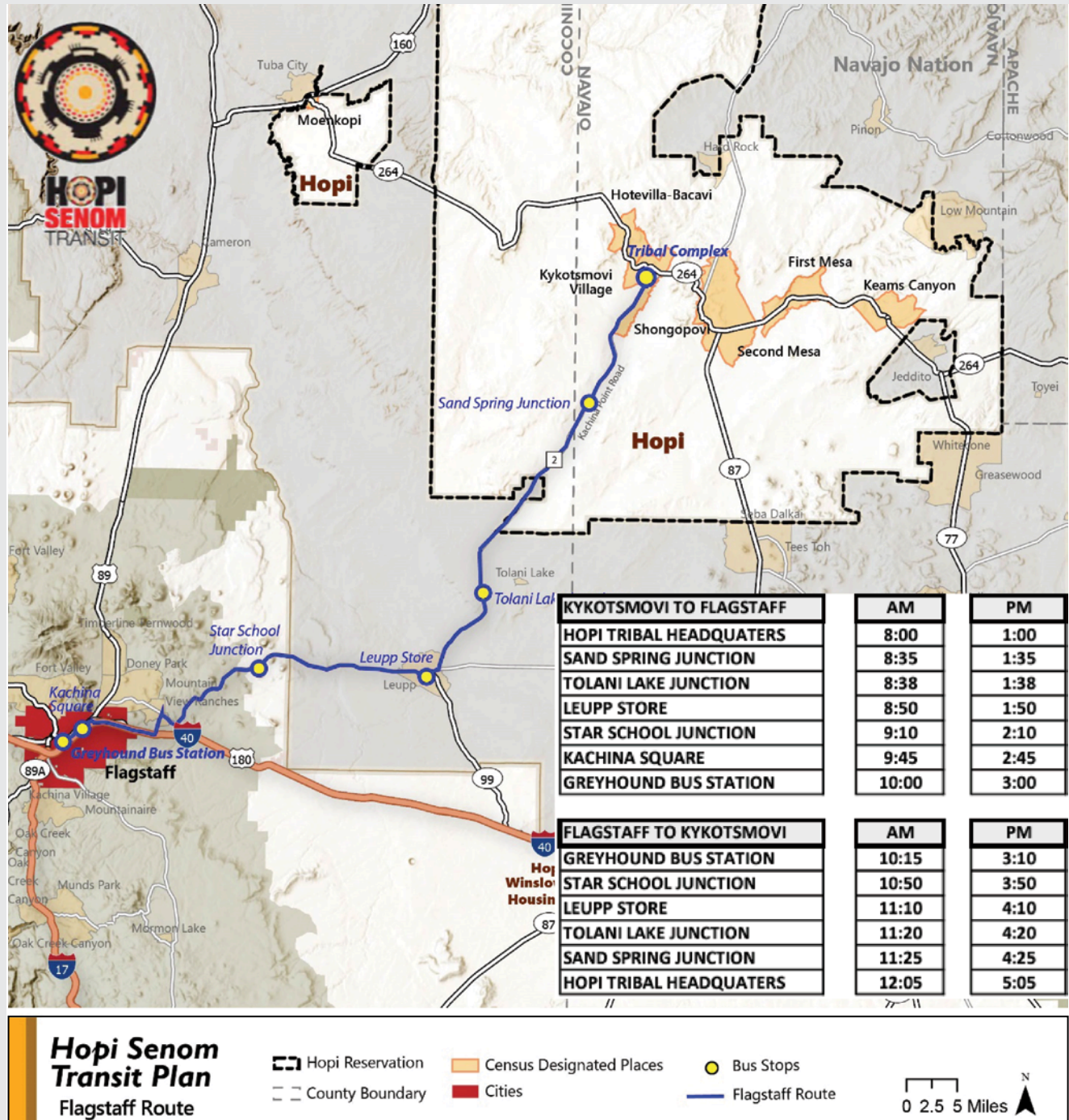
The Intercity Bus Network-Interline Ticketing

Section 5311(f) calls for projects to connect with the national network of intercity bus services, but it does not define a national network. There is an interline ticketing system that is national in scope, including Greyhound, Flixbus, and several other firms. This system also includes many of the larger regional carriers, such as Salt Lake Express, Jefferson Lines, Burlington Trailways, Adirondack Trailways/Trailways of New York/Pine Hill Trailways, Barons Bus, and others. Interline ticketing allows a passenger to buy a single ticket that can be used on multiple carriers, and the ticketing system provides information on the carriers, their connecting schedules, and where there are stops and transfer locations.

Interline ticketing depends on agreements between the carriers, allowing them to share the revenue from tickets that provide service on more than one carrier. The largest national interline ticket clearinghouse is operated through the National Bus Traffic Association, which is housed at the American Bus Association. There are other significant carriers, not part of this network, such as the firms owned by CoachUSA—Megabus, Dillons Bus, Shortline, Hudson Transit, and other regional firms such as Indian Trails and Red Coach. FTA has only stated that it is not a requirement that Section 5311(f) contractors or recipients be members of a particular association or group, but access to the greatest number of potential destinations would come from connections with the national interline network. Other independent carriers cater to ethnic communities, specialize in target markets (such as higher-fare luxury services), serve more localized regional needs, or offer service that could be considered intercity but focus on airport access. For the most part, these carriers have their own ticketing systems. There are also firms that sell tickets on multiple carriers. These are referred to as ticket aggregators or platforms, and they include sites such as Wanderu, Busbud, and Rome2Rio. They do not operate services or provide information about connections.

It should be noted that the intercity bus industry is in a state of change, and many procedures and policies regarding ticketing, interline connections, and station access are in a state of flux. The COVID-19 pandemic and its travel restrictions had a significant impact on the intercity bus industry as ridership fell to around 20% of pre-COVID-19 levels. The bus industry reduced frequencies, maintained the network, and used available federal relief funding to stay alive, but it has not fully recovered. In October 2022, Flix Mobility purchased Greyhound Lines from FirstGroup of the United Kingdom—but FirstGroup retained all the real estate (except the publicly funded Atlanta terminal). FirstGroup has sold off many of the stations and garages, requiring Greyhound, Flixbus, and their interline partners to seek new locations for stops and transfers. In general, they desire to co-locate with public transit facilities. Another aspect of the change in structure is the development of new business models. Flixbus and Ourbus are not bus companies, but technology companies planning, marketing, and selling tickets. They contract with bus firms to operate the buses, and they contract for station facilities. One impact of this is that in-kind match from one of these networks would need to be provided by the bus operator—but this is still evolving.

Hopi Senom Transit is a tribal transit provider in a rural area of Arizona. It receives Section 5311(f) funding from the Arizona DOT to operate this route, which connects its main terminal at Tribal Hall with the Greyhound station in Flagstaff. The local match is provided as in-kind from Greyhound, but the ticketing is not interlined. However, the schedules are coordinated, and passengers can make connections to and from the Greyhound services.



COURTESY OF KFH FROM *HOPI-SENOM SHORT RANGE TRANSIT PLAN, 2023*

Examples of Section 5311(f) Rural Intercity Projects

Examples of rural intercity programs and projects are identified to show the variety of approaches to providing connections to regional and national transit networks.

- South Dakota's rural transit feeder connectors to Jefferson Lines intercity services consist of three rural public transit systems that provide connections to the two Jefferson routes with fully interlined ticketing, and two rural systems that operate bus depots on the Jefferson routes.
- Colorado's Bustang/Bustang Outrider Program is a state development of an intercity and commuter intercity regional network, operated under contracts with a private firm and regional firms or nonprofits, with the state providing the buses.
- Oregon's Transit Network is a connected network of privately provided intercity bus services, state-contracted intercity bus services (the POINT routes), and connecting rural regional services.
- Virginia Breeze consists of four intercity routes connecting rural Virginia communities with the national bus network, interlined with Megabus routes in the Northeast.
- Northwest Connector consists of five rural transit operators in northwest Oregon that partner to provide a regional network of coordinated services, with co-branding, a common website, and a link to intercity services.
- Afton Express has regional commuter services connecting small towns/rural park and rides with jobs and services in Charlottesville, Virginia (operated as a route of BRITE rural regional services based in the Central Shenandoah region).
- Scenic Mississippi Regional Transit in southwest Wisconsin provides morning and evening services on four routes centered around La Crosse.
- Rural regional routes with connections to/from Greyhound stops are operated by Hopi-Senon Transit, Yuma County Transportation, and Ajo Transportation, illustrating potential options for connecting with Greyhound services.

Effective Practices Identified

Effective practices for rural and tribal transit providers that improve connections to the regional and national transportation network include the following:

Development of Regional Routes/Services. These services connect to key destinations (medical, shopping, work, education) in more distant locations outside the local service area.

Operation of Intercity Services. Services are operated by private carriers, rural public, or tribal transit entities from rural stops to connect to the regional or national intercity bus network. This facilitates connectivity to the network. Such services may be a connecting route, or, potentially, the route could become part of the intercity network.

Operation of Feeder Services. These services provide connections to intercity services by demand response or other service models. This is essentially the rural equivalent of first-mile/last-mile service, which in the rural context may be 10- or 25-mile trips.

Provision of Bus Depots/Ticket Agencies. This provision provides station facilities for intercity bus passengers, either passengers on subsidized connecting service or stops on unsubsidized intercity bus routes.

Information. Section 5311 recipients now have to provide General Transit Feed Specification (GTFS)-data for their fixed-route services as part of their annual data submission to FTA's NTD. If they operate intercity services using Section 5311(f), this data is not required, but a best practice is to provide GTFS data. Some states require this as part of their Section 5311(f) program.

Developing Rural Intercity and Regional Connections

The development of rural regional and intercity connections is a process similar to that needed for any new service or initiative. Because these services require more coordination between programs and providers, this is a process that involves several steps, and, at any one of the steps, the conditions identified and the decisions made at that point may require revisiting earlier steps (i.e., it is not a linear process). It is kind of a puzzle—the trick is identifying the end picture that is desired and fitting the pieces together.

Step One: Goal-Setting

Some providers that offer rural regional and intercity connections consider the need to offer long-distance services to major cities and the national network as part of their mission. Many rural providers already offer some type of long-distance service, such as trips for medical purposes for persons receiving nonemergency medical transportation (NEMT) under Medicaid. Some offer long-distance trips for the general public with demand responsive service, and some have more specialized long-distance trips (vanpools, commuter bus routes). Intercity connections are for the general public, with no trip restrictions.

People who use intercity bus services are generally similar to rural public transit passengers in that they cannot use a private vehicle (or prefer not to). They do not have a vehicle or have a vehicle that is not capable of a long trip. They may not have a driver's license, or their age affects their desire or ability to drive. They are part of the same pool of riders being served by rural transit, but they are also part of a broader community. One key difference is trip purpose. In almost every state and nationally, the majority of passengers using intercity bus services are traveling to visit family or friends, often for a scheduled event, such as a wedding, birthday, funeral, or reunion. They are willing to pay significant fares and endure long rides and connections; these trips are valued by the users.

There are several potential goals for intercity or regional services. Connecting with the national intercity bus network is an important potential goal, but providing connections to larger communities for other trip purposes may also be important—for example, access to medical services, shopping, work trips, access to regional or higher education, and connections to other modes. The more different markets that can be served, the more feasible the service becomes—but there are requirements linked to different funding sources.

If a rural transit provider sees a need for this kind of connection as part of the provider's mission, what are the goals? How does a rural transit provider approach such a service?

Step Two: Discovery—Gather Information

There are two initial aspects to be researched, particularly if one of the goals of the service is connecting to the national network of intercity bus services. One aspect is the intercity bus services that exist and the potential connections they offer. The other is the funding potential—how the state manages the available federal funding for rural intercity bus services.

Intercity Bus Information

Where are the nearest intercity bus stops? Who operates these services? What are the schedules? Where do they connect to other services? These are questions that greatly affect the design of potential services.

Despite the development of technology and the web, it is not as easy as it once was to identify all the intercity bus services operated in a given state or area. There is no single industry source of information similar to *Russell's Official National Motor Coach Guide*, which once provided a single source of timetable information for most scheduled intercity services, by carrier, indexed by stop. Though much service information is available in Google Transit, not all intercity carriers have developed and provided the GTFS data needed to allow potential users to discover all the available services—but it is one place to start.

However, there are two national atlases of intercity bus information based on GTFS data provided by the intercity carriers. One is the National Intercity Bus Atlas, which is being developed by the Bureau of Transportation Statistics at the U.S. DOT³¹ (see Figure 4.1). Development of this atlas is being assisted by the NCHRP Project 08-133(01), “Implementing the National Intercity Bus Atlas.” This project is underway.

Another source of data is a GTFS-based intercity bus map called MMC Network Transportation Information provided by Transcor Data Services (TDS).³² TDS supports the interline ticketing system for much of the scheduled intercity bus industry. The ticketing system includes all the data needed for GTFS mapping. This atlas includes all the participating firms and Amtrak and is updated frequently because of its relationship to the ticketing system. Another source with timetable data is the North American Bus Timetable guide to Greyhound timetables,³³ which is updated approximately every 6 months.

There are other GTFS platforms (such as Transitland³⁴) that may offer data provided by or available from intercity carriers, but they are not comprehensive. Other third-party intercity bus ticket sellers can be searched as well, such as Rome2Rio, Busbud, and Wanderu. These may also provide information on the services offered by various carriers between city pairs, but they do not offer a comprehensive map or timetables.

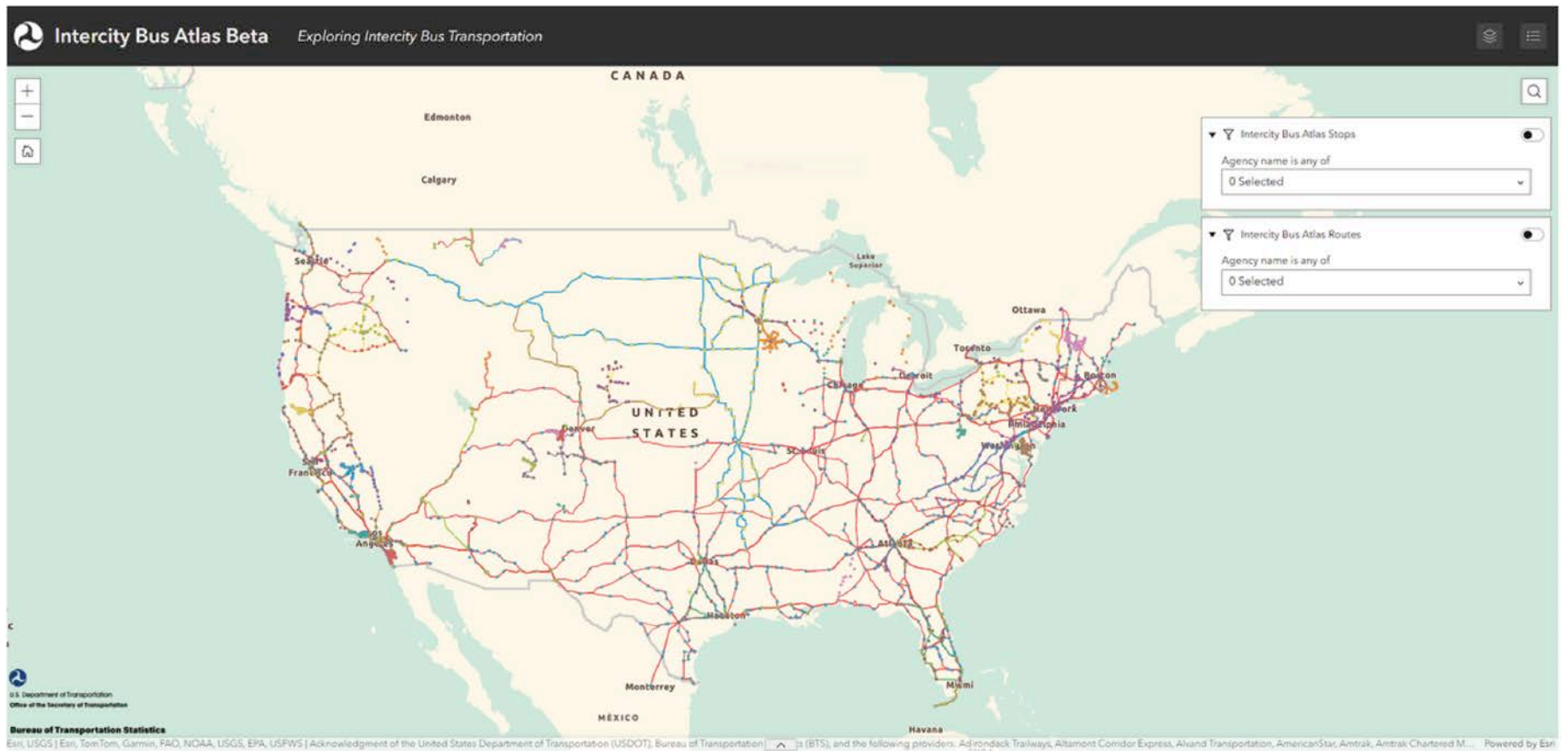
Some states have completed intercity bus studies or plans that include a comprehensive inventory of intercity bus routes and stops. It is worthwhile to contact the state DOT to see if it has conducted such a study in recent years and obtain a copy.

All those resources will assist in identifying carriers and routes. Based on that information, one can go to the individual carrier websites for more information on times and stop locations. However, carriers generally no longer provide actual timetables but only offer the user the opportunity to input hypothetical trips between endpoints as a means of obtaining detailed schedule and stop information. The national Trailways system site³⁵ provides this type of information for several carriers operating under the Trailways brand, for example.

The key information from this process is to determine the nearest intercity bus stop, whether it is served by a carrier that is part of the national system, what times buses arrive and depart, and the amenities and accessibility at that stop (is there a shelter for passengers needing to wait for connecting buses, are there restrooms, does it have access to food). It may make sense to develop a service that connects to a more distant hub with better facilities. Or it may be that there are no services provided at times that can be served and a different model of regional service is needed to address other needs (i.e., Is this an intercity feeder/connection? Or a regional service if an intercity connection is not feasible or does not address needs like work trips, medical trips, etc.?).

State Program Information

Also, at this stage in the process, it is important to obtain information about the availability of funding for intercity bus services in the state. FTA guidance on the Section 5311(f) rural intercity bus program allows for considerable flexibility in the way this program is implemented, and states vary in their approach. One source of information in each state is the FTA State Management Plan for Section 5311, which should describe the state's management of the Section 5311(f) program.



SOURCE: THE INTERCITY BUS ATLAS UNDER DEVELOPMENT BY THE BUREAU OF TRANSPORTATION STATISTICS AT THE U.S. DEPARTMENT OF TRANSPORTATION. [HTTPS://WWW.BTS.GOV/INTERCITY-BUSING/DATA-MAPS-AND-APPS](https://www.bts.gov/intercity-busing/data-maps-and-apps) (Bureau of Transportation Statistics 2024).

Figure 4.1. Screenshot of Bureau of Transportation Statistics' Intercity Bus Atlas.

Grant applications for Section 5311 may also include relevant information on funding availability and requirements. Some states include intercity bus in their overall grant applications, and others have a separate Section 5311(f) intercity bus grant application. Other states may not provide intercity bus grants but may retain the funds and contract for rural intercity bus services.

The traditional approach is to offer 5311(f) funds as part of the grant solicitation for Section 5311, with the choice of routes and services up to the applicant. The alternative model is one in which the state is the grantee, unmet needs and gaps are identified, and a request for proposals for a contractor to provide services to fill those gaps—specifying the route, frequency, stops, and so forth—is issued. Carriers bid to provide these services and are third-party contractors. The state may limit bidders to private for-profit firms with appropriate operating authority. Some states offer a mixed model—a separate grant solicitation issued by the state for particular services, not part of the general 5311 application, but including a scope of services in which the state identifies where it wants to see service. Some states also have state funding available for intercity bus services, either as match for the federal funds or to pay the entire net deficit of the intercity services.

There are other permutations in the way states implement this program. In some cases, part of the program is contracted, and part is a grant solicitation (e.g., Oregon). Some states contract for all their rural intercity services and brand them as another provider, such as Virginia Breeze, Bustang, GoBus, POINT, New Mexico Park and Ride, Travel Washington, and Vermont Translines. In these cases, there may not be any available Section 5311(f) funding for local public transit providers—but there may be an opportunity to connect with or provide stops for the state-branded network.

Step Three: Consultation

Under the Section 5311(f) guidance requirements, states contemplating certifying that there are no unmet rural intercity needs (which would allow them to reprogram the 15% intercity allocation for other rural transit purposes) must conduct a consultation process at least every 4 years. This process is intended to include an inventory of existing services, solicit input from stakeholders (including the intercity bus companies), conduct analysis, and provide a basis for deciding whether there are unmet needs. Public transit and intercity operators need to participate in this process and be as specific as possible about the unmet needs. Input can also address the way the program is managed.

As part of this process, the intercity bus operators and local/regional transit providers should be asked to provide input on the needs for intercity bus services to and from nonurban areas of the state. Transit providers (and intercity carriers) should be prepared to provide input about services that are needed, including routes or services to connect to the national network, routes that are vulnerable to being discontinued without assistance, and new or replacement routes or services to fill gaps that resulted when routes were discontinued. Needs for facilities, additional stops, and better information all need to be identified as they are eligible for FTA funding.

Step Four: Assess Feasibility—Potential Service Design

Potential services can be defined after examining the existing services and knowing where the potential connections to the intercity bus network are located and the potential connecting times. If there is no option of Section 5311(f) funding under the state's program, the requirements for a meaningful connection to the national intercity bus network may not restrict service designs; this may allow for services that arrive at the destination in the morning and leave in late afternoon, regardless of the schedules of the available intercity bus services.

Virginia Regional Transit in Culpeper, Virginia, operates a feeder route connecting its Commerce Street Hub and the Target stop with the Virginia Breeze Piedmont Express Route, which stops at the Brandy Station Park and Ride Lot approximately 6 miles away on Route 29. The local feeder/connector operates twice a day to meet both the north- and southbound Virginia Breeze services.



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Virginia Regional Transit's Culpeper County & Virginia Breeze Express Connection Service			
Stop #	Location	VRT	VA Breeze
1	Target	11:54 AM	
2	Arrival Brandy Station Park & Ride	12:05 PM	12:15 PM (North Bound)
2	Departure Brandy Station Park & Ride	12:15 PM	
1	Target	12:25 PM	
2	Arrival Brandy Station Park & Ride	12:32 PM	12:35 PM (South Bound)
2	Departure Brandy Station Park & Ride	12:35 PM	
3	Commerce Street HUB	12:50 PM	



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PIEDMONT EXPRESS		
Destination Danville (Southbound)	City/Stop	
10:15 a.m.	Washington, D.C. Union Station – Bus Deck	2:40 p.m.
11:05 a.m.	Dulles Airport Curb 2A	1:50 p.m.
11:40 a.m.	Gainesville University Boulevard Park & Ride	1:15 p.m.
12:00 p.m.	Warrenton Warrenton Park & Ride	12:50 p.m.
12:35 p.m.	Culpeper Brandy Station Park & Ride	12:15 p.m.
1:40 p.m.	Charlottesville Arlington Blvd @ Barracks Road	11:10 a.m.
2:45 p.m.	Amherst 171 S Main St.	10:05 a.m.
3:15 p.m.	Lynchburg Kemper Street Station	9:35 a.m.
3:50 p.m.	Altavista Altavista Commons Shopping Center	9:00 a.m.
4:50 p.m.	Danville Danville Transit System Hub	8:00 a.m.
	City/Stop	Destination Washington, D.C. (Northbound)

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SOURCE: Virginia Breeze Brochure, Virginia Department of Rail and Public Transportation.

Schedule Design Factors

An optimal design for a regional service or intercity connector would focus on a connection that is no more than approximately a 2-hour drive away, ideally operating in the 7:00 a.m. to 7:00 p.m. window, with a connection point (intercity bus stop) at a location that offers a good place to wait and ideally in a town that offers some other services that might address other needs (regional banks, medical services, regional shopping). The reason for the limit on distance is that there may be a need to make two round trips to provide connections to intercity bus services operating in both directions—2 hours each way for two round trips is a full 8-hour day of service. Alternatively, it may be necessary for the feeder vehicle to wait several hours to provide a return trip for passengers from the intercity coach going in the opposite direction.

In developing a schedule, there is also a need to build in enough cushion time at the connection to deal with late buses on either route—the intercity coach is likely not going to wait for the feeder services (unless the driver knows there is a connecting passenger), and the connector may have to wait for a late intercity bus to take on connecting passengers. A rural transit provider may consider other factors for the service design (e.g., can the connection trip be provided within the provider's hours when there is dispatch and backup available?).

Frequency

Another key factor in designing intercity feeder services is the overall frequency. Most intercity bus services operate daily in each direction, sometimes several times per day. Following the COVID-19 pandemic, some services were reduced to 5 days per week or even less. Before the pandemic, Greyhound (and other carriers) would not provide in-kind match for intercity feeders operating less than 5 days per week. However, with the reduction in services, there is now more flexibility on the part of the carriers, and it may be that the connecting services will not have demand that supports daily service, or even 5-day-per-week service. The frequency and schedule will need to be developed based on the expected ridership (available funding) and the schedules and input of the carrier operating the intercity service. Potential demand can be estimated based on *TCRP Research Report 147: Toolkit for Estimating Demand of Rural Intercity Bus Services*, input from the intercity carrier, historical information, and local knowledge.³⁶

Vehicles

The size of the vehicle is a function of the expected demand. Intercity bus ridership levels also peak at holidays, on Fridays and Sundays generally, and around school events. Allowance needs to be made for higher ridership at these times. At the same time, the size impacts the requirements for operating authority and the related insurance requirements. If the route is long, there may be a need for better or different seating compared with the local transit vehicles. Section 5311(f) requirements call for vehicles on funded services to be able to carry baggage and to be fully ADA wheelchair accessible. Considering these factors, there may be a need for a dedicated intercity service vehicle.

Stop Locations

Another aspect of service design concerns the stops. Unlike purely local services, there may be a need for stops at locations where passengers can wait for connecting intercity services. These may be intercity bus stops already designated and used by the intercity carrier. Or there may be a need for improved conditions. Passengers with a wait need shelter, security, a place to sit, information, restrooms, and maybe food. Because intercity bus carriers generally no longer provide package express, there is no need for lockers, storage areas, or staff to deal with freight. Another possible role for rural transit agencies is to become the bus depot and offer a waiting area, restrooms, a counter, and information for intercity bus service or for the feeder connector operated by that system. This information is also provided in this chapter.

There are several considerations in developing a rural regional service or intercity feeder. Service feasibility is a function of purpose and service design—most demand will come from combining intercity connecting passengers, commuters, and medical, shopping, and school trips on the same bus. It is very difficult to design services to do all this because of the need to schedule around the intercity bus, something that is now even more difficult with reduced intercity bus frequency.

Step Five: Working with the Intercity Carrier— Roles for the Intercity Feeder

With the background information about the existing intercity services and a conceptual service plan, it is appropriate to reach out to the intercity carrier to present the concept and discuss the potential for implementing it, including the needed agreements that address scheduling, ticketing, stop locations and terminal access, and requirements for operating authority and insurance. Information on carrier procedures and options may require changes in the service design, so this step should be viewed as a negotiation with likely iterations to the project design. A list of carrier

contacts is available from the National Rural Transit Assistance Program (RTAP). The potential ways a rural transit provider can work with an intercity carrier include the following.

Operation of Feeder Services

A rural transit operator or regional intercity bus provider can operate service that connects a nonurban area to existing intercity bus services. The information provided about how to design this service provides a basic description of the considerations involved, but several elements involve the intercity carrier with which the connection is planned.

If the feeder service is to be funded with Section 5311(f) rural intercity funding, a “meaningful connection” with the national intercity bus network is required—meaning service to the same location as the intercity bus stop with convenient schedules for connecting. Intercity carriers prefer that the feeder connector be a scheduled service, fixed-route or route-deviation, with designated stop locations. This is so that they can quote schedule information, ideally as part of their ticketing system, which lists the stops that will be served and when. In addition, with a scheduled connection, there is less chance that a passenger will be left stranded if a demand response trip has not been arranged. As noted, the intercity carriers are now more willing to have feeder connections that operate fewer than 5 days per week. The FTA Section 5311(f) regulations provide for the potential to use that funding source for “rural feeder” service, which may be a small vehicle or demand response; however, the “meaningful connection” requirement still applies.

With the meaningful connection, the feeder service would be potentially eligible for Section 5311(f) funding. However, a rural operator could offer service to the intercity bus stop as part of their overall service offering, funded out of their other program funds. The service could include demand responsive services to the intercity bus stop, or as a stop on other routes, or services not necessarily timed for a convenient connection. The service would provide for passenger access to intercity bus services. Under this scenario, the feeder carrier would go to the stop to drop off or pick up passengers and sell local tickets at their own fare levels (a terminal license might be required, see Terminal Access). This type is more likely for demand response services, but it can be part of fixed routes as well.

Depending on the nature of the agreement with the carrier, the revenue from the feeder service may be an allocation of the overall revenue of a longer-distance interline ticket, or the feeder carrier’s local fare for that service.

Interline Ticketing

If the potential service is fixed-route (or scheduled route deviation with designated stops), the rural provider should strongly consider participating in the intercity carrier’s ticketing system, offering interline tickets. A participating rural provider can provide transportation as part of a longer trip with segments provided by other intercity carriers, sharing in the revenue, with their services included in the websites of the national or regional carriers, making them effectively part of the connected national network. The revenue from the interline ticket is allocated among the providing carriers, including the feeder provider, based on mileage operated on each. The clearinghouse for doing this allocation is the National Bus Traffic Association, and the feeder provider can become a member at minimal expense if sponsored by a member intercity carrier.

The benefits of participating in interline ticketing include added ridership and revenue. Inbound passengers can discover the feeder connection, which they otherwise would have no way to know about. Local passengers can go anywhere on a single ticket, get information on connections, and purchase tickets on their phones using a credit card. If the transit provider is hoping to obtain the value of in-kind miles as match, the carriers providing the match prefer an

interline ticketing agreement because it defines the feeder connection, and the feed traffic is the main benefit to the carrier of providing in-kind match.

Another advantage for the feeder providers is operational. Now that intercity bus ticketing has capacity management systems similar to those of the airlines, the intercity carrier will know if it has passengers who need to make the connection with the feeder provider. And, through the ticketing system, the feeder provider can see that and operate only if there are passengers incoming.

Outbound passengers originating on the rural provider purchasing a ticket through the system have, in effect, reserved a seat because the ticketing systems managing capacity are holding an empty seat on the intercity bus. For connections being made in the middle of an intercity route, this is a significant advantage, as otherwise buses may be full, with no seats for persons boarding mid-route. This is also an advantage to both carriers in making sure that they are prepared for transferring passengers requiring wheelchair lifts or other ADA assistance. The rural provider knows if its passengers need ADA access and should inform the connecting intercity carrier, and the interline ticketing system can provide information on the required accessibility needs of passengers arriving on the intercity bus.

The carrier will want the feeder provider to sign an interline ticketing agreement that covers service expectations, procedures, revenue control and allocation, communication protocols, notification, insurance requirements, and stop locations.

Terminal Access

Historically, the intercity carrier would require a feeder provider to sign a terminal license, allowing the feeder provider to operate its buses in the carrier-owned intercity bus facilities. This is a time of transition in the industry, and the number of standalone, purpose-built carrier-operated intercity bus facilities is declining. For those that remain, allowing terminal access involves an agreement that addresses liability issues and the responsibility of the different parties for the cost of terminal operations, including security.

With the changes in the intercity bus industry, the potential feeder provider will need to identify the stops it intends to serve and whether a terminal license agreement with the intercity carrier is needed. It may be that they need to negotiate with an urban transit provider operating a multimodal passenger facility, or with the city or town where the stop is located if the stop is a curbside location. Many rural intercity bus stops are located in truck stops, service plazas, or restaurants, and it will require negotiation with the intercity carrier and the facility to determine whether the feeder service is under the agreement the facility has with the carrier or if the feeder provider will need a separate agreement.

Providing a Facility and Becoming a Ticket Agent

Another potential role for the rural transit provider interested in supporting rural intercity mobility is that of the agent for intercity bus firms, selling tickets and providing facilities and locations for transfers.

At one time, a rural or small-town ticket agent for an intercity bus company could generate revenue for the transit agency from the commission on intercity bus ticket sales. Although there is still some revenue from ticket sales commissions, providing a bus depot and the ticketing and information function are key support functions for retaining rural intercity bus service.

Bus companies operating services in rural areas have relied on local businesses to be their stopping points and to act as agents. These were often other businesses such as diners, restaurants, gas stations, and hotels. However, in the wake of COVID-19, finding businesses to perform

this role has become more difficult, and many businesses that did it in the past have turned over, and new owners do not wish to take on this task given the diminished revenue from ticket sales. As a consequence, many rural towns have lost all service because there is no longer an identified stop location. Carriers may be willing to add stops in towns where there is an agent and acceptable waiting conditions.

The administrative costs of operating the bus depot or transit agency are eligible expenses under Section 5311(f) from a federal perspective, though not all states have considered this possibility. South Dakota DOT allows rural transit operators to apply for funding for such expenses. Five bus agencies receive administrative funds from the state: two on the Jefferson Lines route, and an additional three on the feeder routes operated by the rural transit agencies.

Key functions of this role include the following:

- Selling tickets, including printing of e-tickets.
- Providing information to riders and assisting them with access to ticketing platforms if they have already purchased a ticket.
- Assisting passengers with baggage, ADA access, and so forth.
- Providing station facilities—indoor and outdoor waiting areas, seating, restrooms, possibly vending machines, bus bays/accessible sidewalks/pads for lifts, and signage. Often, this can be part of the rural transit providers' administrative offices.

The rural transit provider willing to see this as a needed service can take on this role as part of their portfolio of service offerings. If they are also operating a feeder route, providing the bus depot with ticketing and information at the origin point for the service will greatly assist in making the service feasible by supporting ridership in the community. If the rural provider is in a town that is located on an intercity route, taking on this role could maintain or create a stop at that location.

Operating Authority: Federal Motor Carrier Safety Administration Requirements

Intercity bus services that involve interstate commerce are regulated by the Federal Motor Carrier Administration (FMCSA) and are relevant for consideration as they affect rural intercity feeder services. United States Code (USC) Title 49 Section 13501 creates federal jurisdiction over motor carriers engaged in interstate commerce: a passenger carrier is engaged in interstate commerce when it transports passengers across state lines for compensation. Transporting passengers entirely within a state may be considered interstate commerce if the passengers begin or will finish their trip in another state. If a passenger carrier receives compensation for transporting passengers in interstate commerce, it is a for-hire carrier, and therefore must have FMCSA operating authority; obtain and file evidence of minimum levels of insurance [\$1.5 million for operating vehicles of 15 or fewer passengers (including the driver), or \$5 million for operating vehicles of 16 or more passengers (including the driver)]; and designate a process agent that can be served with court papers.³⁷ A carrier can be defined as a for-hire passenger carrier even if it is a nonprofit.

In addition, all passenger carriers must meet federal safety requirements and are subject to inspection if their vehicles are operating in interstate commerce and meet the definition of a commercial motor vehicle (CMV). A CMV has a gross vehicle weight rating (GVWR) of 10,001 pounds or more, OR is designed or used to transport nine or more passengers (including the driver) for compensation, OR is designed or used to transport 16 or more passengers (including the driver) regardless of whether the vehicle is used to transport passengers for compensation. There are also requirements for the driver to have a commercial driver's license if the vehicle is

designed to transport 16 or more passengers (including the driver), which then involves drug and alcohol program requirements. The FMCSA requirements for registration only apply if the vehicles transport nine or more passengers (including the driver) or have a GVWR over 10,000 pounds. As long as the other feeder providers operate vehicles below the capacity or weight thresholds, they do not need to register with FMCSA or carry the higher insurance levels even though they carry interstate passengers for compensation.

Compliance with these requirements is an additional step in developing a rural intercity feeder service. Many rural public or private nonprofit transit providers carry only the minimum levels of liability insurance required by their state programs, which are often less than the FMCSA levels. This results in a need to purchase additional coverage for any intercity routes to obtain the FMCSA registration. Greyhound/Flixbus requires all its rural intercity feeder providers to carry the FMCSA levels of insurance regardless of whether they operate small vehicles that do not require FMCSA registration. Other intercity carriers may differ in their requirements, but state DOTs typically require grant recipients or contractors to meet all regulatory requirements. Another concern is that the FMCSA does not accept liability insurance provided by a state insurance pool (except in Oregon), creating a need for a separate policy for an intercity route operated by a carrier in a state where the transit providers are covered by a pool.

Providing Information About Services

Taking the additional steps to make information about the services available to the public is another key step, once the service design is finalized and the relationship with the intercity network is set. If the service is fully interlined, the intercity carrier's ticketing system may generate GTFS about the service, and some states require rural subrecipients and contractors receiving intercity funding to develop and provide GTFS data for the fixed-route services. A rural transit provider operating connecting service should also include it on their website, with schedules, stop locations, and a link to the intercity carrier's website. In addition, the provider can provide the GTFS data for intercity routes or feeders on an open platform, provide it to the Bureau of Transportation Statistics, and make it available to platforms such as Google. GTFS-based data combined with other transit information results in improved marketing to the public. It allows potential users to discover service connections and be aware that communities served by the rural provider have intercity access. It is also critical to create ridership.

Step Six: Funding

Section 5311(f)

In the federal program context, the most likely source of funding for rural intercity feeder services is the Section 5311(f) program of assistance for intercity bus service. Any rural provider thinking about providing these services needs to determine how their state manages these funds. This information should be part of their FTA-required State Management Plan. Some states offer grants for intercity services and feeder services as part of the annual Section 5311 application. Some states have a separate application; some use this funding to contract with carriers to operate the services they identify to fill gaps in the statewide network; and some states do both.

If Section 5311(f) funding is available, key considerations that affect service design and the relationship to the intercity bus carrier come from requirements in the FTA program guidance. One is that the service has a meaningful connection with the national network of intercity bus services, which affects the choice of stops and the schedule. For operating assistance, typical federal match ratios require a 50% local share of the net operating deficit, and a 20% share of capital, administrative, and planning expenses applies. FTA also requires that the vehicles used have space for luggage and be fully ADA accessible.

Capital Funding

Capital funding for vehicles and facilities is also an eligible use under the FTA Section 5311(f) program guidance, but states vary in how they manage these funds, and they may have restricted it to operations. One case where capital funding may be needed is the situation of the rural public transit operator that does not have an appropriate vehicle available (luggage space, adequate seating for longer trips) for an intercity feeder and needs to have a vehicle or two to operate the service.

Section 5339(b) funding is also a possible option for vehicle capital funding through the state DOT.

In-Kind Match

A significant benefit of the FTA Section 5311(f) program is that it includes a provision for the use of in-kind match to meet the requirement for local operating match. The federal program includes statutory language, and the regulatory guidance provides for the use of the value of connecting unsubsidized bus service as the match for operating assistance. Federal transit funding for operations provides for a maximum federal share of 50% of the net operating deficit, with the remaining 50% to be provided as local match. The net operating deficit is calculated by subtracting fare revenue from the total operating cost. For most local transit, the local match is provided from local funds, generally tax revenue of one sort or another. For intercity services that serve many local jurisdictions, this is essentially impossible to arrange, leaving the state as the most likely source of local match for Section 5311(f) intercity services. However, some states have been able to provide rural intercity services largely with federal funding.

The in-kind match provisions allow a carrier to redefine an operating project to include both the subsidized segment and a connecting unsubsidized segment. One hundred percent of the value of the miles operated on the unsubsidized segment can be counted as the local match for the federal funding used to operate the subsidized segment. By careful design of projects to combine services in this way, essentially all the operating costs can be federally funded. In addition, this model allows the costs of the vehicles to be included, so there is no need to find match to purchase expensive vehicles (typically only 80% of a capital project can be federally funded).

Unlike most transit projects, because the cost of the vehicle capital (whether through depreciation charges or lease costs) is included in the fully allocated rate charged by the contractors, the state does not have to provide vehicle capital for the program for the services that are provided with in-kind match.

Carrier Provision of In-Kind Match

Carriers operating unsubsidized connecting services can provide a letter documenting the value of the connecting service. The project is defined as including both the subsidized and the unsubsidized segment, and this letter documents the unsubsidized segment in terms of the schedule, route length, miles operated, and value of the in-kind match.

Intercity carriers have requirements for providing in-kind match miles. They may require that the feeder provider join the interline ticketing system, and they will want the subsidized feeder segment to service the same stops. Carriers generally set criteria for the connecting service in terms of the maximum schedule window that can be considered a connection. Pre-COVID-19, this window was generally 2 hours. With post-COVID-19 schedule reductions, carriers may consider a longer wait acceptable—perhaps as much as 4 hours, depending on the facilities available for the connection. Intercity carriers do not generally provide in-kind match for demand response services (because demand response service cannot currently be included in ticketing systems). This is to ensure that passengers can count on connecting feeder services based on their schedule.

State Funding

Several states also provide state funding for transit projects, including intercity connections. State programs are generally unique, and a rural provider will need to reach out to their state DOT for details on the availability of funding. A state may provide some or all the local match for the Section 5311(f) program, either for operating assistance or capital. In some cases, demonstration funding may be available.

Finally, if the service that has been developed does not meet the FTA requirements for the Section 5311(f) funding, either because it does not make a meaningful connection or because it is a commuter service, it may be possible to fund it with other Section 5311 funding. In that case, the option of the in-kind match would not be available.

Fare and Other Revenue

Other sources of funding for rural intercity feeder routes include fare revenue. If the feeder service is in the interline ticketing system, this will be the prorated share of each overall ticket. If the feeder provider is also the ticket agent, there may be commission revenue on any tickets sold. Some intercity carriers pay the feeder provider per mile or per trip for feeder services. These carrier funds can be considered as revenue for the intercity feeder service. It will likely take both 5311(f) operating funds and carrier revenue to make a rural intercity route feasible.

Conclusion

Planning and implementing regional and intercity feeder services involve more considerations and include more steps than most new rural transit services. More programs and actors are involved: federal programs, state implementation of those programs, the intercity carriers, businesses at stops, federal regulators, and so forth. The requirements and conditions of each vary depending on several factors, and accommodating one requirement or condition will likely affect the design of the project and its feasibility. Developing such a project is an iterative process. The process is not linear, and there is no single best model; getting to the end service will vary with the goals, the geography, the existing services, and the availability of resources. However, the result can be an integral part of the overall program of a rural public transit's mission of providing mobility to all who need it—in this case, enabling rural residents to access the rest of the country and beyond and be part of the larger national community.

Endnotes

1. Chapter 3 on Mobility Management and Coordination is partially based on materials found in the most recent update (2024) to the National RTAPs Transit Management Toolkit (TMT) section on Coordination and Mobility Management. Members of the research team for this project contributed to or authored materials in the TMT. <https://www.nationalrtap.org/Toolkits/Transit-Managers-Toolkit/Welcome>.
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4. Rodman, W. 2022. *TCRP Synthesis 161: ADA Paratransit and Other Demand-Responsive Transportation Services in Small to Midsized Transit Agencies*. Transportation Research Board, Washington, DC. <https://doi.org/10.17226/26509>. A team member for this project, Will Rodman from the Texas A&M Transportation Institute, was the author of this synthesis report and drew from it for the discussion of route deviations and operating general public demand response services.
5. These ADA paratransit obligations are detailed in 49 CFR Part 37 Subpart F and are also covered in the FTA Circular 4710.1's Chapter 8 and 9. Also see Chapter 12, Accessible Service, in Part II of this Guide.

6. FTA. (2015). FTA Circular 4710.1: Americans with Disabilities Act Guidance. U.S. Department of Transportation, Washington, DC.
7. Dedicated service uses vehicles that are operated exclusively for the transit agency's paratransit program during a defined period of time; service is provided only to customers of the transit agency during the defined time period. Non-dedicated service uses vehicles that are not exclusively operated for the transit program. For example, a taxi company providing non-dedicated service may transport a paratransit customer on behalf of the transit agency but then is available to serve general public, private-pay passengers unaffiliated with the transit agency.
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11. Rodman, W., Berez, D., Moser, S., and Choe, J. 2016. *NCHRP Report 832: State DOTs Connecting Specialized Transportation Users and Rides, Volume 1: Research Report*. Transportation Research Board of the National Academies, Washington, DC. <https://doi.org/10.17226/23506>. A team member for this project, Will Rodman from the Texas A&M Transportation Institute, was the author of this report.
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PART II

Support Topics

Part II focuses on topics that support or improve the services and strategies presented in Part I, including guidance on implementation. These topics cover a wide range of efforts that rural and tribal transit providers should address as they consider new and emerging services and strategies.

Most of the topics could stand on their own to improve mobility and accessibility; however, the Guide approaches these topics through the lens of how they support new or enhanced service strategies and identifies steps or actions transit providers should consider or take when implementing a new or emerging service strategy.

The topics are presented in the following chapters.

Chapter 5: Planning for New Service Strategies describes planning for new services is an important support category that involves developing goals and objectives, assessing data, reaching out to customers and the community for input, and developing recommendations for service.

Chapter 6: Management and Operations Considerations addresses a range of topics that rural and tribal transit agencies face in managing and operating their transportation services, including deciding who should operate the service; selecting the appropriate vehicle; and hiring, training, and retaining staff. This last topic was identified as a key challenge by transit agencies that responded to the research project's survey.

Chapter 7: Communications with Current and Prospective Passengers describes how rural and tribal public transit agencies provide critical access to healthcare, employment, education, food, and other services. Accurate information and marketing of their services are essential so that current passengers can use the service, and prospective passengers understand what services are available and how they can use them.

Chapter 8: Safety and Security describes how effective practices addressing transit safety and security are critical for transit providers to manage risk and ensure security, and how they support the implementation and operation of effective and quality transit services.

Chapter 9: Technology addresses recent advanced technologies that have emerged to improve how public transportation services are delivered, including technologies that are becoming increasingly accessible to rural and tribal transit providers. These include technologies that provide information on real-time bus arrivals, driver-assisted vehicle functions, mobility-on-demand apps, and security software.

Chapter 10: Measuring Success concerns evaluating transit services and measuring performance and success using a performance evaluation process, with goals and objectives for the transit service as well as defined measures to track achievement and operating and financial data to compute the measures.

Chapter 11: Funding and Revenue identifies federal funding programs targeted for rural and tribal public transit. Effective practices for maximizing federal funding, including the use of in-kind match, are described. Nontraditional funding sources are also identified.

Chapter 12: Accessible Service for transit involves meeting the requirements of federal law established by the Americans with Disabilities Act (ADA). Accessible service also ensures that people with disabilities can use the transit service in their community. This chapter provides an overview of the ADA law and outlines what rural and tribal transit providers need to address to ensure compliance with ADA regulations.

CHAPTER 5

Planning for New Service Strategies

Introduction

Effective planning supports rural and tribal transit providers with the implementation of new and enhanced services to improve mobility and accessibility. Chapter 2 discusses new and emerging service strategies that can help rural and tribal transit providers increase their current and prospective customers' mobility and accessibility and improve service and cost efficiency. That chapter also provides some guidance for when the various strategies and practices may be appropriate or beneficial for a community. This chapter discusses the planning process involved in matching those strategies and practices for a transit system and its community.

The planning process is a cooperative, reflective, and generative opportunity to address and anticipate transportation challenges and issues facing rural and tribal community members and entails collecting and compiling relevant data and information, developing and considering strategies and alternatives, pursuing meaningful public engagement, and making recommendations for implementation. As a result, it is vital to make well-informed, data-driven, and performance-based decisions about how to spend limited transit funds, and it is crucial to the development and delivery of new and enhanced rural and tribal public transportation services.

What Is Involved?

Planning is the first step in the iterative and cyclical process of planning, programming, implementing, and evaluating transit. The planning process is an opportunity to set the strategic direction of transit services with the establishment of a vision, mission, and goals for the community's new and enhanced public transportation services, and to analyze ways to prioritize investments and accomplish targets.

The framework for performance-based planning and programming is illustrated in Figure 5.1 and summarized by the following key questions:

- Where do we want to go?
- How are we going to get there?
- What will it take?
- How did we do?

General Transit Planning Process

The Transportation Planning Capacity Building (TPCB) program¹ is jointly administered by the U.S. DOT's FHWA and FTA and provides training, technical assistance, and support relating to transportation planning efforts. TPCB developed an informational publication called



SOURCE: *THE TRANSPORTATION PLANNING PROCESS BRIEFING BOOK* (TPCB 2018), U.S. DEPARTMENT OF TRANSPORTATION, FEDERAL HIGHWAY ADMINISTRATION AND FEDERAL TRANSIT ADMINISTRATION.

Figure 5.1. Framework for performance-based planning and programming.

*The Transportation Planning Process Briefing Book (Briefing Book)*² that includes an overview of transportation planning as well as major policy and planning issues to consider.

As summarized in the *Briefing Book*, transportation planning entails the following:

- Engaging with stakeholders and the general public.
- Establishing a shared vision and goals.
- Monitoring existing conditions in relation to performance goals.
- Forecasting changes in population, employment, land use, and major corridors.
- Identifying needs by developing performance measures and targets.
- Analyzing improvement strategies and assessing the tradeoffs of alternatives.
- Developing both short-range programs and long-range plans for capital, management, and operations.
- Estimating the impact of transportation investments and improvements in relation to performance goals and other considerations (e.g., air quality, congestion, etc.).
- Developing a financial plan for implementation and sustainable programming.

The National Rural Transit Assistance Program (RTAP) developed a *Transit Manager's Toolkit*³ that serves as a guidebook and reference manual for new transit managers and includes an overview of information relating to strategic and service planning for rural transit. National RTAP also developed several technical briefs highlighting critical aspects of the planning process, including *Getting Started: Creating a Vision and Strategy for Community Transit* and *Developing, Designing, & Delivering Community Transportation Service*.⁴

Additional practical guidance about the planning process is provided in the modules developed for the Ohio DOT's Active Transportation Academy⁵ and the report on *Best Practices in Transit*

Service Planning compiled by the University of South Florida's Center for Urban Transportation Research for Florida DOT's Research Center.⁶

These sources provide comprehensive and practical guidance on the planning process, including both strategic planning (setting the stage) and service planning.

1. Strategic Planning
 - Developing Vision and Mission Statements
 - Developing Goals, Objectives, and Performance Measures
 - Adopting an Appropriate Governance Structure
 - Establishing Meaningful Opportunities for Community Stakeholders
2. Service Planning
 - Analyzing Needs
 - Existing Sociodemographic, Employment, Geographic and Land Use, and Service Data
 - Comparing with Peer Communities and Services
 - Convening Public Meetings and Forums
 - Surveying Passengers and Community Members
 - Conducting Focus Groups and Interviews
3. Identifying Resources
 - Financial (Federal, State, and Local Government, Private, Nonprofit)
 - Capital (Facilities, Vehicles)
 - Human (Operators, Administrators, Services)
4. Considering Service Options
 - Fixed Service
 - Flexible Service
5. Selecting Service Design and Policy
 - Parameters for Service Area, Hours, Fares
 - Policies, Procedures, and Responsibilities of the Transit System and Riders

Resources for Transit Planning

For rural areas, transportation planning processes are facilitated through the state, with the National Rural Transportation Assistance Program (National RTAP) [49 USC 5311(b)(3)], which provides funding to assist in training, technical assistance, and other support services for transit operators in nonurbanized areas.

Eligible activities through the National RTAP include planning, capital, operations, job access, reverse commute projects, and the acquisition of public transportation services. Transportation planning processes for tribal communities are supported by the Public Transportation on Indian Reservations Program (also called the Tribal Transit Program) [49 USC 5311(c)(1)(A)], which is a set-aside from the Formula Grants for Rural Areas program. Eligible activities include planning, capital, operations, administrative expenses, and the acquisition of public transportation services. Federally recognized Native American tribes are also eligible recipients of funds under the Formula Grants for Rural Areas program.

FTA provides funding for two entities that offer resources to support the transportation planning process in rural, small urban, and tribal communities. The National RTAP offers training and technical assistance resources to the rural transit industry, facilitates the development of networks, promotes coordination of services, and provides a database of information about the rural transit industry. Meanwhile, the National Center for Applied Transit Technology provides “practical, replicable resources” to aid in the application of “technological solutions and innovations.”

Planning Process for New Service Strategies or Designs

The planning steps aimed at implementing new service strategies or designs parallel the general performance-based planning process outlined. Four core tasks define the process:

1. Gain relevant knowledge.
2. Pursue meaningful engagement.
3. Develop appropriate recommendations.
4. Pursue robust implementation.

This planning process can directly support the adoption and establishment of new and emerging service strategies. The determination and selection of an appropriate public transportation service design is a critical contribution to the planning process and should be aligned with the community's vision and mission for transit service, oriented toward realistic and achievable performance targets, and based on the analysis of needs, identification of resources, and consideration of service options. See Example of Service Planning, showing how the City of Battle Creek, Michigan, transit staff led the planning process toward implementation of new countywide microtransit service.

The range of environments across rural and tribal communities means that a range of service solutions should be considered, and what may be effective and successful in one area may not work well in another. The planning process should inform and empower rural and tribal communities to adopt the most appropriate service design and improve service efficiency and effectiveness. It should address the tradeoffs between service that prioritizes coverage and ridership (frequency) and support making an informed determination.

For example, the needs analysis should help rural and tribal communities avoid providing fixed-route service in low-density/low-ridership areas where a flexible service would be less expensive and at least as effective, if not more so. The needs analysis can help to identify spatial patterns in population and employment density in relation to roadway conditions and major trip generators and inform the determination of whether a rural and tribal community is a good candidate for a service enhancement, such as the replacement of a low-performing fixed-route or demand response service with microtransit.

Concept of Productivity Versus Mobility

Key to the selection of a community-based service strategy is agreement and buy-in on goals for service changes. To understand when a particular service design is appropriate, the community must first agree on a shared vision for transit. Does the community want to maximize mobility for all in the community, or does it want to maximize productivity? Both goals can be achieved, but they may conflict with each other. The selection of service designs and strategies can be driven by two objectives: (1) maximizing productivity (and reducing costs) and (2) providing mobility for the community as a whole.

To decide which is more important, and, ultimately, how to balance these two goals, it is important to first understand the concept of system productivity and how it relates to cost, especially for demand response services operated with dedicated vehicles. With most of the underlying cost for such services being hourly-based, productivity—measured as passenger trips per revenue vehicle hour (RVH)—is a measure that can be used to convert an hourly cost to a cost per trip (cost/RVH divided by trips/RVH = cost/trip). Thus, the higher the productivity, the lower the cost per trip.

Lowering the cost may enable a transit agency to provide more trips for the budget available. For example, going from one passenger trip/RVH to two passenger trips/RVH cuts the operating

cost per trip in half and allows the vehicle to transport twice as many riders in the same period of time. However, in the process, the system may have to reduce or restrict services to lower-density areas where services are less productive.

Performance measures provide an opportunity to monitor progress toward and the achievement of realistic and achievable goals established during the planning process and ensure transportation decisions are data-driven and evidence-based. This relates to the important concept of productivity in the development and delivery of new and enhanced rural and tribal public transportation services; the planning process is where decisions are made about service design and operational strategies that directly affect key performance measures for service efficiency and effectiveness.

Effective Planning Practices in Rural and Tribal Communities

Effective planning practices in rural and tribal areas can use the resources provided by National RTAP, including the various toolkits, to help with the planning function. Planning strategies that rural and tribal transit providers could employ include the following:

- Engage the community when planning transit services.
- Use a pilot to test a new service strategy or service.
- Coordinate with other organizations serving your riders, particularly those riders who are older adults, have disabilities, and/or low incomes, to better understand the community's mobility needs.
- Where appropriate, consider consolidating with neighboring transit systems to help realize economies of scale, which, among other benefits, may allow more staff resources for the planning function.
- Plan and pilot the service.

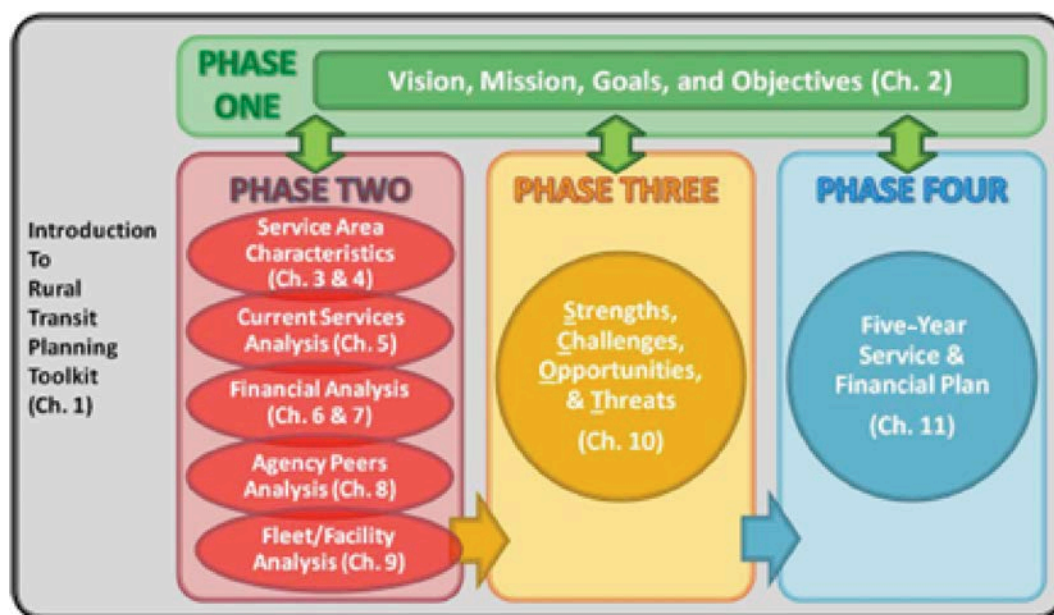
As a crossover with the technology topic, several new technologies could be effective as transit planning tools, including the use of route-planning software to redesign the routes.

The research conducted for the Guide found numerous reports about planning, including planning for rural areas. The following two are particularly useful.

- The *Transit Development Plan Guidebook*,⁷ developed by the Oregon DOT, provides best practices and steps for preparing transit development plans (TDPs). According to the guidebook, TDPs describe transit provider goals and identify needs and strategies to achieve them over a 20-year horizon or other specified time frame. A TDP can inform and help integrate transit needs into transportation system plan (TSP) updates and other planning processes.
- The *Toolkit for Rural Transit Planning: Five-Year Operations and Financial Plan*,⁸ developed by Edrington et al., gives rural transit agencies a resource for short-range planning. It provides a step-by-step overview of the rural transit planning process, including tools for developing vision, mission, goals, and objectives; capturing a service area snapshot; projecting a future service area; assessing transit service types, costs, and funding; benchmarking against peers; assessing assets; conducting a strengths, challenges, opportunities, and threats (SCOT) analysis; and developing operations and financial plans. The toolkit provides an information-rich resource with tables, maps, and performance data to effectively communicate with stakeholders and partners⁹ (see Figure 5.2).

Example of Service Planning

An example of an effective planning practice can be found in Battle Creek Transit's planning for countywide transit service.



SOURCE: EDRINGTON ET AL. 2012.

Figure 5.2. *Four-phase approach to planning.*

Background

The Calhoun County Coordinated Mobility Pilot (publicly branded as BCGo) is a countywide microtransit service operating throughout Calhoun County, Michigan, serving a population of approximately 133,000 across a 718-square-mile service area (see Figure 5.3). The pilot initially launched in March 2021 for 12 months with \$395,505 in Service Initiatives Grant funding from the Michigan DOT (MDOT). Since then, MDOT has provided an additional \$800,000 in continuation funding to operate the pilot for a total of 36 months.

The BCGo pilot is the result of a comprehensive, multiphase planning process to improve existing service in Battle Creek (the largest town in Calhoun County, with a population of 52,721 in the 2020 Decennial Census) and introduce new service to parts of Calhoun County that were previously unserved by public transportation. BCGo builds on the findings of the 2020 Calhoun County Transit Study and is part of a strategy to build public support for a millage and the



COURTESY OF BATTLE CREEK TRANSIT.

Figure 5.3. *BCGo serves passengers.*

establishment of a countywide transit authority. The 2020 Calhoun County Transit Study was an important milestone in the effort to establish countywide service. It was completed by a consulting team comprising staff from Foursquare Integrated Transportation Planning Inc. and Monahan Mobility Consulting, LLC. The 2020 Calhoun County Transit Study included demand estimates for new countywide service to ensure efficient and effective service design, and BCGo pilot staff continues to set goals for service improvement in Phase II of the pilot, including those related to better coordination of dispatching and service across the relatively large service area.

The Planning Process

City of Battle Creek Transit staff, including the Transit Director and Mobility Manager, played an integral role in leading the planning process toward countywide transit service, including managing consultants for the 2020 Calhoun Transit Study, securing state funding for the BCGo pilot, and managing the contracted microtransit services with Liftango. In addition, they are playing an ongoing leadership role in the evaluation of the pilot's performance, which has entailed the collection and evaluation of pilot performance metrics, including measures of productivity, customer satisfaction, and rider quality-of-life impacts.

The 2020 Calhoun County Transit Study and BCGo pilot demonstrate the vital role of the planning process in building support for new and enhanced transit services, as well as ensuring their successful implementation. In particular, the 2020 Calhoun County Transit Study provided an opportunity to identify the vision, goals, and objectives of countywide transit service, consider governance alternatives, and evaluate existing conditions and unmet needs. The study included service, implementation, and financial plans that equipped Battle Creek Transit staff with the information to successfully request state funding for the BCGo pilot, itself an important opportunity to ground-test the plans and build further support toward a countywide transit authority.

Costs

The 2020 Calhoun County Transit Study was funded jointly by the FTA, Calhoun County, Battle Creek Community Foundation, Battle Creek Unlimited, and additional stakeholders within the project area for approximately \$175,000.

The BCGo pilot has been funded by the MDOT for a combined total of \$1,195,505 over 36 months.

Multiple members of the Battle Creek Transit staff devote a portion of their staff time to support the operation of the BCGo pilot, including dispatchers who book trips and take payments for customers who do not have access to a smartphone or choose to make a reservation over the phone; maintenance staff who maintain vehicles and ensure they are in safe condition; and administrative staff (including the Transit Director and Mobility Manager) who support the day-to-day operation of the pilot, provide technology support and customization of the mobile app, analyze data, engage in conversations regarding the success and lessons learned, and work on a plan (vision) for the service once the pilot phase is complete.

Lessons Learned

One of the most significant and ongoing challenges for the BCGo pilot has been a driver shortage, which has constrained the pilot to two vehicles in service and led to a significant number of unmet trip requests. Transit workforce challenges are widespread throughout the United States and have received increasing attention as more transit agencies limit or cut service due to

labor shortages. Devoting additional focus in the planning process to transit workforce issues may ensure that new and enhanced transit strategies are deployed without (or with fewer) labor constraints.

Working with the Liftango technology platform has allowed for a high level of service customization and access to backend data, but it also required an ongoing investment in Battle Creek Transit staff time. Smaller communities with fewer staffing resources may decide that an off-the-shelf service platform may be a better fit given their needs and staffing capacity.

CHAPTER 6

Management and Operations Considerations

Introduction

The management and operation of transit service involve myriad components, from hiring and training staff to developing customer information and policies to maintaining vehicles and putting service on the street. This chapter presents guidance on management and operational issues to help rural transit providers meet some of the challenges they might encounter when implementing new or enhanced service strategies or practices.

The chapter does not attempt to cover all of the components of managing and operating new service strategies, though some of these are addressed in the remainder of Part II. Instead, the chapter focuses on key management and operational issues identified as important through the research project's survey.

At the onset, it has to be acknowledged that transit providers in rural and tribal communities are often short-staffed, with the manager wearing many hats. The boundary lines among the various functions of providing transit service (administration, planning, finance, operations, and maintenance) may be blurred or may not exist. Compounding this, there may be a dearth of institutional knowledge as the bulk of the institutional knowledge often leaves with the previous manager.

Managing and Operating New Transit Service Strategies

Planning or implementing a new transit service strategy requires many decisions, tasks, and activities. Some of these relate to the type of service strategy to be implemented; others are important regardless of the service strategy. Some are relevant only if the service is operated directly by the transit agency.

Part II contains chapters devoted to those subjects. Chapter 6 delves into other management decisions that are specific to the implementation of new service strategies, namely the following:

- How to decide who operates the service—whether the service will be operated in-house, by a contractor, or involve a transportation network company (TNC)/taxi (non-dedicated) provider.
- How to hire, train, and retain adequate and competent staff.
- How to select vehicles and vehicle storage/staging appropriate to a new service design.
- How to monitor the new services on the street, collect appropriate data, and report the data.
- How to establish or adjust the customer-service functions and set customer-facing service policies.

Deciding Who Operates the Transit Service

One of the key decisions to be made by the transit agency's governing board is whether day-to-day service should be operated directly by the transit agency or whether a contractor should be hired to manage and operate the daily service. Many rural and tribal transit agencies directly operate their transit services. Some agencies contract for service, and some rural and tribal transit agencies do both. Existing arrangements for who operates the service may need to be changed (or at least examined) as new service models or strategies are introduced.

While many rural transit and tribal services are operated in-house or by one contractor, transit agencies often need to be creative and try options beyond their basic service model. This includes using non-dedicated services, such as those provided by taxis or TNCs.

The service models that describe the division of day-to-day responsibilities can vary from transit agency to transit agency, serving different rural and tribal communities. The most common service models are (1) all in-house, where all daily functions are performed by the transit agency; and (2) a turnkey contract model, where a contractor (or contractors assigned to subregions) perform all the daily functions.

There are also service models where call and control center functions are performed by the transit agency and service delivery is performed by the contractor(s), and others where call and control center functions may be split between the transit agency and the operations contractor(s). In addition, where a contractor or contractors are used, there can be variations in which entity provides the vehicles, the facilities, the software, communications, and the in-vehicle equipment.

Factors to consider when deciding whether to operate the service directly or use a contractor relate to the following:

- Cost for the particular service.
- Availability of staff and the level of expertise within the transit agency to operate the service.
- Control over day-to-day operations.
- State or local laws that may be an issue.
- The transit agency's longer-range planning strategies.

Use of a contractor does not relieve the transit agency from meeting requirements that convey with federal and/or state grant funds used for the service. The transit agency is responsible for closely monitoring service to ensure the contract requirements are being met. This requires staff to ensure compliance through monitoring and management of the contractor.

Generally, a major advantage of operating service directly is control over day-to-day service. With the use of a contractor, responsibility for day-to-day service rests with the contractor, which includes hiring and training drivers and other operational staff; managing and operating the service day-to-day; vehicle maintenance (in many cases); and service monitoring and operational reporting. The transit agency, however, retains control over planning, policy development, marketing and public information, service evaluation, and typically customer complaints and feedback.

An advantage of using a contractor may be lower operating costs than providing service directly, as experience has shown that labor costs for contractors tend to be lower than direct transit agency operations. While transit contractors are often private companies or organizations, they can be other public entities as well.

For example, the Golden Crescent Regional Planning Commission (GCRPC), based in Victoria, Texas, directly operates a rural demand response service in two of the counties in its region. GCRPC contracts out to other county governments or human service agencies to provide rural demand response service in the other six counties of the region. In some rural regions, this can

involve a pass-through of funding to the remote municipal operators. Elsewhere, the financial transaction involves invoicing monthly for services provided.

Using Non-Dedicated Service

Using non-dedicated service can be an effective strategy to meet varying demand levels or to supplement regularly scheduled dedicated transit service.

Non-dedicated service uses vehicles that are not exclusively operated by the transit agency. A taxi company is a good example of paratransit service, transporting an eligible paratransit customer on behalf of the transit agency. Then the taxi company is available to serve the general public, private-pay passengers unaffiliated with the transit agency. Dedicated service uses vehicles that are operated exclusively for the transit agency, typically its paratransit program, during a defined period of time; service is provided only to customers of the transit agency during that time.

Taxis and TNCs may not be prevalent in rural and tribal communities, and so the opportunities to use their non-dedicated services as an operational strategy may be limited. However, both taxis and TNCs have been known to expand their service area to some rural areas if there is a demand for their services. Often, this is contingent on a transit agency having a program that will guarantee at least some trips for them. Indeed, many transit agencies have subscribed to the concept that if we build it, they will come, and this can work.

Other potential sources of non-dedicated service provision include NEMT providers that are involved in providing rural Medicaid and hospital-sponsored trips.

Using a Contractor¹⁰

Independent rural transit districts typically operate their own service. However, when a city, county, or council of governments wants to initiate or expand public transportation in their area but has no transit experience, using a contractor may be a strategy to be considered, at least initially.

Providing certain transit agency assets to a contractor may also be a cost-effective strategy. The assets typically involved with transit operations include the following:

- Facilities that house the management, administrative, and support teams; call and control center; operations and maintenance functions; heavy maintenance equipment; and the vehicles themselves (revenue and nonrevenue).
- Scheduling and dispatching software, customer-facing technologies, and related hardware.
- Telephone system and all hardware used in the call and control center.
- Vehicle communication system and related equipment.

Rural and tribal transit agencies that provide vehicles, software, and other equipment to their contractor(s) and allow their contractor(s) to operate service out of transit agency facilities can use FTA capital funding (at an 80/20 percent match versus operating funding at a 50/50 percent match).

Improving Competition for Procurement of a Service Provider

By providing the key assets for transit operations to the contractor, the transit agency creates a more level playing field for the procurement of service providers as compared with procurements when proposers are required to provide these assets. For example, a local provider may have fully amortized its facility long ago or via a long-term lease and has a cost advantage over another interested nonlocal provider who must find and pay for a facility.

Conversely, requiring contractors to provide their own facility may drive up the cost because the cost of a short-term lease (matching the contract term) will likely be more expensive. Moreover, the options for locations available to proposers may be suboptimal, whereas a transit agency can take its time to find the optimal location and facility.

Rural and tribal transit agencies that provide their own vehicles to the service providers under contract generally (1) want direct control over the type of vehicle and the replacement/retirement schedule; (2) can transfer vehicles to a new contractor when a contract is not renewed; and (3) can shift vehicles between or among multiple contractors if the amount of work each contractor performs changes. Transit agencies requiring contractors to provide their own vehicles are generally less interested in the administrative chores associated with fleet procurement and maintenance.

Similar to vehicle provision, a rural or tribal transit agency may wish to directly license demand response scheduling software because it wishes to have direct control over how the software works (including the control of scheduling algorithmic parameters and desired modifications) and can take advantage of the 80/20 percent split of capital funding used to license the software. It can also continue to license the software and provide it to an incoming turnkey contractor while not having to endure the transfer of data from one system to another (should the incoming contractor wish to use another software system).

Providing vehicles and the software scheduling system also provides some consistency as a new contractor is retained, and there is no need to convert data from one system to another. As important, the transit agency directly controls which assets are obtained.

If transit agencies wish to require their contractors to provide their own facility, there are two tactics to help minimize the cost of such facility leases. These tactics are not mutually exclusive:

- Transit agencies can lengthen the contract base term and the number of option years, thereby enabling service providers to enter into longer (and therefore less expensive) leases.
- Transit agencies can require their contractor to place the transit agency in the first position to assume a facility lease if the contractor's contract ends prematurely or should the lease period exceed the term of the contract. This allows the service provider to take a chance on a longer, less expensive lease because they have an out if, for example, the transit agency decides not to exercise the option years because of poor performance. Moreover, in this event and with such a provision, the transit agency has the right of first refusal to assume the lease and can then sublease the facility to the incoming contractor.

Providing Vehicles to the Service Contractor

A transit agency may face challenges in providing vehicles to its contractor if the vehicles have problems or if the seating configurations hinder productivity and scheduling flexibility. These challenges may be addressed by involving the contractor in the vehicle procurement decisions as well as decisions on fleet mix (i.e., using different types of vehicles in the dedicated fleet).

A second type of challenge may occur when the provider's contract comes to an end. During the lame-duck period, proactive maintenance may subside. Moreover, the transition of vehicles from the old to the new contractor can be fraught with disagreement about which entity covers the cost of required and needed maintenance. To avoid these conflicts, clear standards and financial obligations need to be included in the transition section of each contract.

Providing Scheduling Software to the Service Contractor

Providing the scheduling software can sometimes result in a case of finger-pointing (i.e., the contractor attributing its performance to the scheduling software or the scheduling parameters

as controlled by the transit agency). Transit agencies need to be careful about allowing a turnkey contractor access to the scheduling parameters without any oversight, as contractors have been known to alter these parameters to maximize their financial position. On the other hand, transit agencies should involve the turnkey contractor in all proposed modifications and the analysis of the results.

Hiring, Training, and Retaining Staff

Whether or not a new service is operated directly by the transit agency (often referred to as in-house service) or contracted, the hiring, training, and retention of operating staff are key management functions.

One of the primary challenges articulated in the research project's survey was the shortage of vehicle operators or drivers, including drivers with a commercial driver's license (CDL). This challenge started nationally in urban and rural areas before the COVID-19 pandemic and became particularly acute during the pandemic and beyond. While the country is easing out of the pandemic, hiring and retaining drivers remain a concern. As one survey respondent said, "We've been forced to suspend some services due to the shortage of CDL drivers."

Much of the research addressing the hiring and training of transportation staff was done before the COVID-19 pandemic and focused primarily on urban transit. However, it is important to acknowledge that, while exacerbated by the recent public health emergency, current transit workforce issues are not entirely new.

Information focused on staffing issues for rural transit is available through the National RTAP Transit Manager's Toolkit. This resource provides guidance and suggested practices on a range of topics, which include the following:

- Wages and Benefits
 - Ensure that the transit agency is a competitive employer. Consider the cost of living in the transit agency's region and typical wage and benefit packages offered by employers in the area to help establish that the transit agency is competitive with other employers.
- Recruiting
 - Appeal to job seekers interested in giving back to their community, which may draw interest from retirees and millennials.
 - Look for job seekers who have a positive attitude and like working with people.
 - Provide support in securing a CDL license where needed.
- Training
 - Ensure compliance with federal training requirements. For example, is a CDL required to operate the transit agency's vehicles? What about drug and alcohol testing?
 - The ADA requires that drivers and other transit agency staff be trained to proficiency, as appropriate to their position, to provide appropriate assistance to individuals with disabilities, and to treat them respectfully and courteously.
- Refresher Training
 - Reinforce safety and security measures.
 - Provide information on new policies and procedures.
- Retention
 - Create an employee recognition and incentive program.
 - Develop a mentoring program.
 - Provide ongoing training to help staff improve, grow, and stay engaged.

An APTA report,¹¹ concerning the workforce shortage, provided several recommendations to address the problem.

- Increase compensation.
- Improve worker schedules.
- Create a positive work environment.
- Demonstrate a clear path to promotions or raises.
- Ensure worker safety.
- Provide ongoing training.
- Increase hiring practice effectiveness.

APTA has also published a toolkit¹² to help transit agencies assess their workforce situation and develop strategies for enhancement and to improve outcomes.¹³ Strategies are provided in the following topic areas: recruiting; hiring; training and onboarding; working conditions and policies; and transit agency culture and morale. A sample of these strategies follows:

Under recruiting, strategies include the following:

- Cultivate, evaluate, and strengthen the referral network.
- Promote accessibility in the application process.

Under hiring, suggestions include the following:

- Hold mass hiring events.
- Audit and streamline the hiring process.
- Evaluate testing requirements, which can become roadblocks.

For training and onboarding, suggested strategies include the following:

- Expand on-the-road training.
- Develop accessible and flexible training.

For working conditions and policies, the toolkit lists several practices, such as the following:

- Reevaluate worker schedules and consider schedules that promote flexibility.
- Expand full-time work.
- Incentivize shift distribution through pay.

The transit agency's culture and morale can:

- Help with employee retention—keeping the staff that you have.
- Develop a clear pathway for growth, with career ladders and efforts to motivate staff to pursue professional development.
- Consider developing programs for engagement, recognition, and acknowledgment.

A 2019 study in North Carolina found that workforce challenges were more pronounced in rural areas and for smaller transit systems, with more unfilled positions the farther a transit agency was from an urbanized area.¹⁴ The study also found that higher pay leads to lower vacancies, and that reliable hours and pay attract workers.

Selecting Appropriate Fleet of Vehicles and Vehicle Staging

As transit providers implement new services, selecting the right vehicle for transit operations requires numerous decisions, including the following:

- Matching vehicles to the service(s) (e.g., size and seating capacity/configuration)
- Fuel/power source

- Vehicle staging
- Infrastructure and facilities in place (or improvements required) for the successful operation and maintenance of various vehicle types

Additional issues related to vehicles include the following:

- Experience with the COVID-19 pandemic may be another factor that transit agencies consider: should the vehicles be larger than previously acquired to allow for driver shielding and empty seats between onboard passengers?
- Alternative fuels and motive power are also factors to consider. However, will rural and tribal transit providers be able to support new fuel types and vehicle technologies and equipment that are state of the art, while being hundreds of miles from maintenance centers that conduct major repairs?
- An additional consideration is that a vehicle is more than a conveyance for a transit agency—vehicles are the face of the transit service. Each vehicle is the out-front image of the service. To a significant extent, the community's impression of the service is based on the appearance and look of the transit provider's vehicles.
- Facilities are another issue and constitute the next major capital consideration after vehicles. From maintenance and operations facilities to bus stops, shelters, and transfer facilities, all these are necessary for a properly functioning transit service. Some transit providers that are attached to local governments may use existing government administrative and maintenance facilities, but many providers must make decisions about vehicles and address facility needs.
- Maintenance facilities should be evaluated separately, as there are other factors at play. A recent study among Texas rural and small urban transit systems indicated that transit agencies with in-house maintenance reduced their maintenance costs vehicle downtime as well.

Vehicles for Fixed-Route Service

Rural and tribal transit agencies may need different types of fixed-route vehicles depending on the terrain encountered on individual routes. For example, Hopi Senom Transit on the Hopi Reservation in Arizona has fixed routes that routinely carry 25 or more riders on good roads. Medium-duty or large cutaway buses that can carry 25 to 30 riders are most appropriate for these routes. Hopi Senom Transit also has a route that climbs up mesas in the service area at the rate of 700 feet in 2 miles, with switchback turns. Out of necessity, smaller, more nimble vehicles with lower capacities are required on this route.

The Right Fixed-Route Vehicle for the Need

In most cases, transit agencies opt for larger rather than smaller vehicles. While there are niches for smaller vehicles (see the example of Hopi Senom Transit previously mentioned), larger vehicles are usually more flexible in their use and can eliminate the need for a second bus. When using FTA funds, the added local capital cost of the larger vehicles is minimal. While the operating costs are higher, the added flexibility can also reduce the number of vehicles needed as service expands.

Bus Typologies

There are generally three types of fixed-route buses to consider. While general cost information is introduced here, rural and tribal transit agencies should check prices regularly. It should also be understood that, with an 80% federal match, the cost for a medium-duty bus may be about \$20,000 to \$30,000 more per bus than a light-duty cutaway. For the additional cost, a transit

agency would get vehicles that could last twice as long, provide a more comfortable ride, and have a greater capacity or more room for social distancing.

Cutaway—Small Bus

Cutaway chassis vehicles are smaller than buses and usually have a high floor. These vehicles customarily have a seating capacity of eight to 30 seats, and their size can vary significantly from 15 to 30 feet long. These vehicles have a 5- to 7-year life as a frontline vehicle, less if used in heavy-duty service. These small buses are used in a variety of applications, such as feeder buses, dial-a-ride, and ADA paratransit service, and on lightly traveled fixed routes. All buses must have lifts or be low-floor with ramps. Depending on size and configuration, these vehicles are in the lowest price range.

Medium-Duty Transit Coach

Medium-duty low-floor buses, typically 30 feet in length, are practical for use in some rural, small city, and commuter service operations. These buses are designed to last up to 10 years and allow for a standard bus configuration without the cost of a heavy-duty bus. Medium-duty transit coaches seat 20 to 25 passengers and can typically transport two to six people using wheelchairs. Important for long routes, these buses are more comfortable than cutaways. These vehicles are typically in the mid-price range.

Low-Floor Heavy-Duty Bus

This vehicle typology is useful for transit agencies needing large-capacity vehicles to meet demand. Rural or tribal systems would only use low-floor heavy-duty buses when operating a high-density commuter service. Most larger transit systems use low-floor heavy-duty buses for regular fixed-route service. These buses are generally 30 to 40 feet in length and are designed to last 12 years in heavy-duty service. The low floor and wide door allow for rapid boarding and alighting. These vehicles seat 30 to 40 passengers, with additional room for standing. They are in the highest price range.

The choice of vehicles will, to a large degree, be determined by funding availability and capacity requirements. Some areas may require a small cutaway bus, while others could justify a heavy-duty bus under the right conditions. Where possible and feasible, transit agencies should typically purchase larger vehicles if a choice is available.

Vehicles for Demand Response Service—Nonmicrotransit

Historically, rural and tribal transit agencies have been subject to state vehicle procurements where they have little control over the types of vehicles procured. While often difficult for rural and tribal transit systems to achieve, uniform fleets provide flexibility to schedulers and dispatchers while minimizing the breadth and expense of a parts inventory.

However, some transit agencies may need larger-capacity WAVs to accommodate increased numbers of trips during peak periods or for large group trips (e.g., trips to senior centers or human service agencies). In some cases, this need may result in the transit agency acquiring a uniform fleet of larger vehicles when such larger vehicles are needed only at specific times. The transit agency, then, may be unnecessarily spending more on purchasing, operating, and maintaining larger vehicles.

A mixed fleet, with larger and smaller vehicles, some WAVs and some not, can reduce both capital costs and operating costs, as smaller vehicles tend to be less expensive.

Additionally, the use of smaller vehicles enables transit agencies to operate vehicles with non-CDL drivers. Because non-CDL drivers typically have lower wages than CDL drivers, recruitment, training, and operating costs are reduced.

Many rural transit agencies consider these reasons and observations of the percentage of trips requiring WAVs—typically 20% to 25% of overall trips—when stratifying their fleet with WAVs of varied sizes and configurations. They can also replace or augment their fleet with sedans or non-accessible vehicles.

In addition, supply chain issues that were particularly acute during the pandemic have resulted in delays for transit agencies receiving more common types of paratransit vehicles (e.g., large and midsize accessible cutaways); expanding the fleet mix to include non-WAV vehicles shortens the procurement cycle.

Vehicles for Microtransit

Microtransit vehicles are typically built on a smaller vehicle chassis, allowing for more maneuverability in tight areas. These are low-floor, ramp-equipped vans with a capacity of 8 to 10 passengers with two wheelchair positions. While it is optimal to have smaller vehicles for microtransit service, it is not a prerequisite for starting service. Many small cities and towns have narrow streets that sometimes require larger transit vehicles to back up. Rural and tribal transit agencies will want to make sure that the microtransit technology used eliminates or minimizes backing up.

As an example, a Capital Area Rural Transportation System (CARTS) microtransit vehicle is depicted in Figure 6.1. This vehicle has a narrower profile (80 in.) than a typical cutaway bus (94 in.) and can maneuver better in some rural environments. These vehicles can also be ramp-equipped for faster and easier boarding for people using wheelchairs or who have trouble walking up steps. These vehicles cost about \$100,000 each.

Drivers interviewed during other microtransit projects have mentioned that the routing algorithm of some microtransit technology platforms does not always consider how large a cutaway bus is, and using the smaller passenger vans minimizes the risk of damaging parked cars and could increase the service's efficiency.¹⁵



PHOTO COURTESY: KFH GROUP INC.

Figure 6.1. CARTS NOW vehicle.

Alternative Fuel Vehicles

Significant improvements have been made to electric and hybrid buses. Their success has been proven in urban areas. Unfortunately, the options for rural and tribal transit agencies are more limited.

Issues that transit agencies in rural and more isolated areas face with electric and hybrid vehicles include the following:

- Alternative fuel vehicles require an infrastructure and staff to support them. This is a major commitment that many rural and tribal systems cannot make on their own. Specialized maintenance equipment and appropriately skilled and trained technicians are necessary to maintain the vehicles. To ensure there are enough vehicles to make the infrastructure investment cost-effective, a local government-wide commitment to the use of alternative fuel vehicles is needed, so that costs for the infrastructure and specialized technicians can be shared.
- Hybrid buses work best in stop-and-go traffic that recharges their batteries. Hybrid buses have few, if any, advantages over long-distance routes with few stops, which are typical in many rural areas. Further, like all alternative fuel vehicles, these buses also require specialized maintenance.

Diesel Vehicles

Diesel vehicles are still the most practical alternative for many rural and tribal transit agencies. For most remote areas, diesel buses will continue to be the most practical propulsion method for the next 5 to 10 years.

A word of caution: There are often grant opportunities available for hybrid or electric buses that offer a lower or no match. Rural and tribal transit systems must use caution when considering their options because, without the required support infrastructure and staff, these vehicles will have little to no effectiveness and will be a drain on the transit agency.

Vehicle Storage and Staging Facilities

Many rural and tribal transit agencies serving large and sometimes multicounty regions stage their vehicles at satellite parking areas or at the drivers' homes in the areas that the drivers serve daily. The following are examples:

- Jaunt, a public transportation system, serves a six-county region in central Virginia and stages 24 of its 102 demand response vehicle fleet in remote locations. A dozen of these vehicles are stationed at one remote county's public works facility, where Jaunt also has access to one maintenance bay and a nearby car-wash facility. The remaining 12 are parked at drivers' residences or a nearby parking facility.
- Garrett County Transit in western Maryland uses a senior center (located 25 miles from the main facility) to park five of its transit vehicles.
- Inter-County Public Transportation Authority (ICPTA) is a multicounty system in northeastern North Carolina. ICPTA reports that some of the hiring decisions made by its operator are influenced by where the applicant lives, as the agency does encourage some of its operators to take the transit vehicle home at night to reduce deadheading.

Using remote locations for parking vehicles has increased as more and more rural transit providers use software that can transmit driver manifests to in-vehicle tablets or mobile data terminals (MDTs) and are no longer using paper manifests.

The concept behind the remote vehicle storage strategy is to reduce the number of daily deadheading miles that otherwise would be necessary if the vehicles were all to be parked centrally.

The additional deadheading adds unnecessary driver pay hours, and in the era of driver recruitment issues, requiring drivers to commute to the central facility and then to deadhead back out to the region where they live adds unnecessary time to their day, the commute portion of it being unpaid. The unnecessary deadhead also adds wear and tear to the vehicles.

How a vehicle is fueled is central to this strategy. If vehicles are fueled at gas stations with fuel cards, a park-out strategy works well. If vehicles must be fueled at the central facility, time must be made for the vehicles to come into that facility. If the vehicles are used to link outer-county riders to the area where the central facility is located, daily fueling can take place at that time. However, for vehicles that provide circulator service in an outer region and rarely come into the central area, fuel cards can be a better solution.

As another strategy, a survey respondent stations buses at strategic places of high use instead of having the vehicle return to the central depot.

As part of the remote vehicle storage option, it is important that a transit agency's maintenance technicians have access to vehicles for preventive maintenance activities and as-needed repairs. Therefore, vehicle swaps need to happen at opportune times that coincide with such activities. ICPTA also reported that using remote vehicle storage locations limits the amount of face time that a driver has with supervisors, and this can present challenges for monitoring. An additional concern is security, as many remote parking facilities are not secured; thus, a safe location becomes a consideration for a satellite parking location.

Monitoring New Services on the Street

The initiation of a new service strategy or service may require a transit system to introduce new or additional monitoring efforts and to collect and review new operating data.

Daily Operations—Monitoring Service

Whether the transit service is operated in-house or contracted, monitoring service on the street is important. This allows for firsthand observation of the service, including, for example, observations of drivers and their performance as well as schedule adherence. Some of the observations should be discreet so that drivers do not know they are being monitored.

Many rural systems do not have road supervisory staff; therefore, office and management staff can be trained as road supervisors and be deployed on an as-needed or “in-the-area” basis.

With directly operated service, the transit agency, where possible, should ensure some level of street supervision periodically. The frequency may depend on various factors, such as whether there have been complaints by riders, whether time checks for the schedule are needed, whether there are new drivers who need evaluation, and when there are accidents or incidents.

For contracted service, the transit agency should require that the contractor provide street supervision, but the transit agency should also do some of its own street supervision. This gives the transit agency insights into its service that are not possible by reviewing operational reports.

Data Collection and Reporting

The ongoing collection and reporting of operational data are needed for various reasons. Transit providers need a way to determine whether the new services meet the goals they set. For example, how data is collected may need to change (particularly if the new services are not

under the direct control or supervision of transit staff). Further, new measures of success may be needed for new operating modes.

Recipients of 5311 funding are required to collect and report financial and operating data on an annual basis for the National Transit Database (NTD). The NTD provides a reporting manual¹⁶ with instructions on the reports. States may also have certain reporting requirements. Maryland, for example, requires its 5311 recipients to provide quarterly operating reports.

As important as meeting federal and, in some cases, state reporting requirements, rural and tribal transit agencies need to collect and report operating data on a routine basis to assess service performance, take corrective actions if needed, and monitor where the transit agency is achieving its goals and objectives (see Chapter 10 for more on performance evaluation).

Customer Service

One of the primary goals of implementing new service strategies is to better serve the needs of the community. In rural and tribal areas, this often involves meeting the mobility needs of people who are dependent on the transit system to access essential services. Moving or adding to a new service strategy will enhance the mobility of customers. Changes may need to be made to the customer services function, and customer-facing service policies may need to be changed or created. Part II, Chapter 7, Communications with Current and Prospective Passengers, in the Guide provides more information on this subject.

Comments, Complaints, and Commendations

Rural and tribal transit agencies need to establish a customer-service function so that riders can provide input—good, bad, or indifferent. Through the transit agency’s website or by other means familiar to the community, information on methods that riders or others can use to provide feedback should be provided. Because many rural and tribal communities struggle with internet access, any customer-service program or process should include a phone option as well as the agency website option. Policies and procedures should be established for the transit agency’s response to the feedback, including the expected response time.

When the transit agency implements a new service strategy, particularly if implemented as a pilot, it may be useful to survey riders to understand their use of the new service and comments they may have. That feedback can help the transit agency understand if changes or fine-tuning should be considered.

Customer-Facing Service Policies

Service policies are important. They provide a framework of transit service for the riding public as well as for the transit agency staff. Service policies spell out the operating hours and days, the fare structure, and any other information that the public needs for taking trips. The transit agency staff also needs that information. Drivers, for example, need to know the fare structure and what sort of fare media can be used. Many rural and tribal transit agencies have “How to Ride” guides that include all the information the riding public needs to know.

Certain policies are required. As an example, the ADA requires that riders be able to travel with a service animal. ADA regulations further specify what questions can be asked by the transit agency if there are concerns that the animal is not a service animal for a rider with a disability.

Additionally, some policies support safe and effective transit service. A code of conduct (or rules of behavior) lets riders know what is expected of them while riding transit, including whether there are consequences for noncompliance, and what the consequences are.

Other policies may be specific to the mode of service. For fixed-route service, as one example, some transit agencies require strollers to be folded before boarding, while others may specify a certain place on the bus for strollers to remain unfolded.

For paratransit service, several policies support improved paratransit. This includes a no-show policy, which helps prevent operational issues related to no-shows. If the paratransit service functions as the transit agency's ADA paratransit, the transit agency needs to ensure that its no-show policy adheres to FTA guidance.



CHAPTER 7

Communications with Current and Prospective Passengers

Introduction

Rural and tribal public transportation agencies provide critical access to healthcare, employment, education, food, and other services. Up-to-date communications and marketing of their activities and operations are essential to ensure that current passengers may effectively continue to use the service and that prospective passengers understand what services are available and how they can use them.

Communications targeted toward current passengers include informative messaging about current transit operations, schedules, fares, ride scheduling, upcoming events, and other changes to normal operations. Communications with prospective passengers use methods that improve understanding of services and their benefits, including fare structure, hours and days of service, and any eligibility requirements.

Since access to public transportation is often a crucial link, ensuring that information regarding the service remains up-to-date and accessible for all current and prospective users is important. This includes establishing and maintaining easy communication methods for those without access to the internet or a smartphone, those with limited English proficiency, and people with disabilities.

Increased marketing efforts, including travel training, social media, community engagement, and branding, may result in increased ridership and a reduced misconception that rural transit services are only for older adults or people with disabilities.

What Is Involved?

In rural and tribal communities, strategies to disseminate information vary widely. Communication practices or strategies identified in the research project's literature review and surveys of public transit agencies and state DOTs include the following:

- Printed materials (flyers or brochures).
- Transit agency websites.
- Social media.
- Email newsletters.
- Printed newsletters.
- Newspaper ads.
- Signage.
- Vehicle wraps.
- Telephone systems.

- Real-time communication technologies (traveler information systems or arrival prediction systems).
- Rider and community surveys.
- Outreach events (at community meetings and events).
- Travel training.

The National Rural Transit Assistance Program (National RTAP) provides a marketing toolkit¹⁷ with guidance, tools, and successful examples to help develop transportation marketing materials. When developing a successful communication strategy, the following components should be considered:

- The audience receiving the communication, and where they are located.
- The content, format, and accessibility of communication.
- The timing and frequency of communication.
- Communication dissemination media and who has access to that media.

A mixture of communication types is recommended in a successful public transit communication strategy because some options, like social media, may not be accessible to some current or potential users. Providing materials in Spanish or other languages commonly spoken in the system's coverage area or having available staff or volunteers who are fluent in those languages can help increase accessibility. See the example later in this chapter of the Southwest Area Regional Transit District and its efforts to improve accessibility through multilingual communications and considerations for passengers with disabilities.

Building trust is key to successful engagement with a rural community. It can be built by providing timely and accurate information on public transit agency operations and by actively listening to community members' feedback. One method of obtaining such feedback and determining whether the system is meeting the public's expectations is through current user and community member surveys.

Social Media

Social media, including Facebook, X (Twitter), Instagram, and LinkedIn, is increasingly used as a tool for outreach, education, engagement, and sharing information. Social media communication tends to fall into five categories:

- Timely updates: Real-time information for transit riders.
- Public information: Providing information on the services available, fares, etc.
- Citizen engagement: Connecting with riders and non-riders.
- Service updates: Updating the public when changes to normal operations occur.
- Employee recognition.

Several lessons learned on using social media were identified:

- Social media is not very effective at reaching older adults and low-income communities.
- Social media should be integrated with traditional communication methods.
- Social media requires staff time and effort to keep profiles maintained and up-to-date.
- Social media can provide a transit agency with customer feedback, which may be negative. Listen to this feedback.
- Social media can help create connections with riders.

Branding

Consistent branding helps unify a transit system and improve name recognition, which has been shown to increase new ridership and reduce confusion surrounding the transit system.

Branding is also particularly helpful in reducing the perception that a transit system is for older adults only, which may be an issue in communities where the public transit system began as a senior-based service.

The main components of branding include the following:

- Name: Something short and easy to say and remember.
- Logo: Should be representative of the name.
- Vehicle Graphics: The bus is a rolling billboard.
- Bus Stop Signage: Universal bus symbols, logo, phone number, website.
- Facilities: Bus stop shelters and benches.

An example of branding for a new transit service is provided in Mitchell County Transportation—Pine Line.

How to Support New and Emerging Service Strategies

Communication is a key component of a successful transit service, especially one that is new and emerging. When a service is in the process of launching or as its parameters change over time, getting the word out is important. Some key components of communication outreach include the following:

- Develop a Communications Plan. Develop a plan that works for the target community. What communication channels do people in the community use (e.g., social media, radio), especially the community of riders you are looking to serve currently and in the future?
- Engage with the Local Community. Building a strong local community board will help with everything from outreach to engagement to funding opportunities. Examples of partners include the following:
 - City or county officials.
 - Service providers (including human services).
 - Funders.
 - Local businesses.
 - Riders.
 - Other notable community members.
- Identify Key Community Events. These events allow transit service providers to advertise and engage with current or prospective riders and become a part of the community.
 - Examples include local fairs, community meetings, and farmers' markets.
- Design for Accessibility. Accessibility resources and services are important from the beginning: from designing to meet the requirements of the ADA to developing products in the most spoken community languages.
- Recognize the Importance of Branding. Transit service branding must be relatable to its community. Branding should let people know what the service is.
- Budget for Communications. Make sure that the transit service has money in both the planning and first-year funding to complete quality communications. Funding is often limited, so be creative about the who, what, when, where, and how of the communications.
- Engage in Self-Reflection and Analysis. This is a necessary process used to reframe marketing efforts.
- Do Not Reinvent the Wheel. Excellent resources already exist, like the National RTAP Marketing Toolkit and rural providers with expertise in transit communications. Reach out, connect, engage, and build the rural transit network.

Examples of Outreach and Communications

Southwest Area Regional Transit District

The Southwest Area Regional Transit District (SWART), in southwest Texas, began service in 1981 as part of a local community action agency. It became a standalone system in 2012.

As of 2023, SWART serves eight counties in the Hill Country region of Texas down to the border of Mexico, with a headquarters in the town of Uvalde. With a fleet of 55 vehicles, SWART primarily operates on-demand service with two fixed routes in the community of Eagle Pass.

Ridership is highest among older adults and people with disabilities; however, post-COVID-19, the focus has shifted to transit for economic and community development.

SWART has put a focus on communication as a key component of the success of its services. Best practices in their outreach methods include the following:

- A variety of outreach methods. Social media, websites, email blasts, newspapers, and flyers.
- Bilingual communications. Communications materials are provided in English and Spanish; the majority of SWART employees are bilingual.
- ADA-accessible communication. Considering vision- and hearing-impaired riders' need to access information, SWART provides large-print schedules and marketing materials.
- Community outreach. Attending community events such as food pantries, health fairs, and adult education.
- Mobility managers. SWART has full-time mobility managers whose job is to reach out and connect with current and future riders, and ensure they have access to the resources they need to ride the bus.
- A stakeholder committee. Creating a diverse group of community members to help build support and get the word out (e.g., mental health providers, the faith-based community, law enforcement, workforce development, business community).
- Building a rural transit network. SWART doesn't reinvent the wheel; it reaches out and learns from others who are providing similar services.

In the end, when it comes to communication, do more than you think necessary.

SWART is a great example of a rural transit provider that has grown and succeeded over time through strong community partnerships, growing and adapting services to community changes, and networking with other rural transit providers across the country. For more information, see Transit District, Middle Rio Grande, TX. | SWART at paseoswart.org.¹⁸



Mitchell County Transportation—Pine Line

Mitchell County Transportation in North Carolina implemented the Pine Line after receiving a grant from the National RTAP in 2021. The Pine Line is a deviated fixed-route service that complements the county's existing demand response transit by providing regular, free service to the area of Spruce Pine, North Carolina (population: 2,332).

The Pine Line provides access from several neighborhoods and housing complexes to Walmart, Mayland Community College, shopping and dining opportunities, and recreational opportunities in the area. The Pine Line has had great success using a variety of marketing efforts and community connections, which resulted in several riders using the system on day one. The system has since seen around 46 boardings per day.



When implementing the Pine Line, key components of the Mitchell County marketing strategy included the following:

- Creating branding to use on all marketing materials, which made the Pine Line bus distinct from the countywide demand response service. The branding featured a recognizable local icon, a spruce pine—the town’s namesake.
- Using community connections.
 - The Pine Line had the support of a Transportation Advisory Board (TAB), which had members from human services organizations, the local newspaper, and the local economic development council. One of the TAB members helped translate marketing materials into Spanish for the local Hispanic population.
 - The Mitchell County Transportation Director and local chief of police put boots on the ground and raised awareness for the Pine Line by handing out flyers and answering questions.
 - A local DJ rode the Pine Line and interviewed passengers to promote the system on a popular local radio station.
- Using low- or no-cost marketing, including attending local events and festivals, local newspaper and radio, printed marketing materials (e.g., flyers), Facebook, and QR codes, which enable users to view a Google Map with the Pine Line stops and bus schedule.

The Pine Line is a great example of rural community members coming together to support a transit program. The Director of Mitchell County Transportation found that word of mouth was key to the success of the system, noting that once people saw the benefits of the Pine Line, they began to believe in what the transit system was doing for community members and advocated for the system. To learn more about the Pine Line, see: <https://www.mitchellcountync.gov/departments/transportation/>.¹⁹

CHAPTER 8

Safety and Security

Introduction

Transit safety and security concerns affect passengers, vehicle operators, other transit workers, vehicles and equipment, public and private property, homeland security-related threats, and the traveling public. Effective practices addressing transit safety and security are critical for transit providers to manage risk and ensure security, and they support the implementation and operation of effective and quality transit services.

Arguably, the most important priorities for a transit agency are safety and security; however, these are two of the many responsibilities involved in providing transit in rural and tribal communities. As such, transit managers in these communities may be challenged to maintain the emphasis and quality that is needed for safety and security.

What Is Involved?

The FTA requires that transit agency operators receiving Section 5307 funds and located in an urbanized area with a population of 50,000 or more develop a Public Transportation Agency Safety Plan (PTASP). Since rural transit agencies do not operate in an urbanized area, a PTASP is not required. Tribal transit agencies are also not required to develop an agency safety plan. However, the FTA indicated that it would continue to evaluate the safety risk posed by small operators to determine the need for future regulation.²⁰

Although agency safety plans are not currently required for rural and tribal transit agencies, there are aspects of safety and security planning that are recommended for all transit agencies, no matter the size. Safety planning should be modeled after the safety management systems (SMS) framework.

SMS is a term that is used to describe a comprehensive, collaborative approach to managing safety where management and labor work together to control risk more effectively. It is an approach where detection and correction of safety problems occur earlier, data is shared and analyzed across the transit agency, and safety performance is measured more precisely.²¹

Having a safety and security plan akin to a PTASP is recommended for all transit agencies, including those operating in rural and tribal communities. Security should be part of the plan. Some elements to include in the transit agency safety and security plan are as follows:

- Safety and Security Management Policy. Define the safety and security goals.
- Performance Measures and Targets. Define safety and security performance measures and targets.

- **Communication Plan.** Determine how employees, contractors, volunteers, and leaders will be informed of the plan.
- **Organizational Chart.** Define who is responsible for safety and security management.
- **Reporting Program.** Determine how safety and security events will be reported.
- **Risk Management Plan.** Determine how to identify hazards and hazard consequences, assess risk, and identify mitigation strategies.
- **Performance Monitoring Measurement Plan.** Determine how safety and security event data will be collected and analyzed.
- **Change Management Plan.** Consider how changes may introduce new hazards or impact safety and security performance.
- **Continuous Improvement Plan.** Determine the process to track safety and security performance (progress toward targets).
- **Training Program.** Determine who will be required to participate in training (operators, employees, contractors, volunteers), what training will be required, when the training will be required, and how often retraining will be required.

Planning or Implementing New Transit Service

Implementation of a new or enhanced service strategy should include a review of a transit provider's safety and security plan and consideration of what changes might be needed. Will the service extend into a new, farther-out part of the service area? If so, are there new operational issues to address that might affect safety and security? Are new and different transit vehicles being deployed? If so, should vehicle operator training be expanded to ensure operators understand what is new and different about the vehicles? These issues fall under the change management element listed and should be addressed by transit providers, with updates to documentation of the transit provider's safety and security plan.

When considering what changes to safety and security might be needed, the following guidance is offered for operational safety and transit security.

Safety

Understanding the elements of a safe bus transit system is crucial for an SMS framework. Vehicles, infrastructure, operators, passengers, and the traveling public all have a role to play in the framework. A safe system includes all elements: from the maintenance and operation of the bus, to the loading and unloading of passengers, to the actions of pedestrians, bicyclists, and drivers.

Safety on the Road

The U.S. Department of Transportation has adopted a Safe System Approach as the guiding paradigm to address roadway safety. The principles of a Safety System Approach are helpful in the context of a rural and tribal transit bus system and should be a way to think about safety. These principles include²² the following:

- **Death and Serious Injuries Are Unacceptable.** A Safe System Approach prioritizes the elimination of crashes that result in death and serious injuries.
- **Humans Make Mistakes.** People will inevitably make mistakes and decisions that can lead to or contribute to crashes, but the transportation system can be designed and operated to accommodate certain types and levels of human mistakes and avoid death and serious injuries when a crash occurs.
- **Humans Are Vulnerable.** Human bodies have physical limits for tolerating crash forces before death or serious injury occurs; therefore, it is crucial to design and operate a transportation system that is human-centric and accommodates physical human vulnerabilities.

- **Responsibility Is Shared.** All stakeholders—including government at all levels, industry, non-profit/advocacy, researchers, and the general public—are vital to preventing fatalities and serious injuries on our roadways.
- **Safety Is Proactive.** Proactive tools should be used to identify and address safety issues in the transportation system, rather than wait for crashes to occur and react afterward.
- **Redundancy Is Crucial.** Reducing risks requires that all parts of the transportation system be strengthened so that if one part fails, the other parts still protect people.

Some types of crashes that tend to be more common include the following:

- **Fixed Object:** The bus strikes a fixed object.
- **Sideswipe:** The bus sideswipes another vehicle, or another vehicle sideswipes the bus.
- **Rear End:** The bus is rear-ended or the bus rear-ends others.
- **Right Angle:** The bus strikes a vehicle at a right angle, or another vehicle strikes the bus at a right angle.

Other types of crashes are crucial to avoid because they are likely to be more severe. These include bus crashes with pedestrians, bicyclists, and other vulnerable road users. Being aware of the actions that bus operators should practice and those that should be avoided is critical. For example, backing the bus is something that should be avoided. If it cannot be avoided, the get out and look (GOAL) approach should be taken, or a second person should assist with the backing procedure.

Another example is addressing the lack of visibility around obscured spots, such as the ones the bus frame creates (A-pillar and B-pillar). Rocking and rolling (leaning left and right) in the seat to see around pillars is a method used to train operators to look for people or objects that may be hidden. For many transit agencies, bus crashes occur when the servicing bus stops. Training and retraining are recommended to address these and other safety concerns.

A Focus on the Infrastructure

Partnering with roadway owners to ensure that the infrastructure meets the needs of the transit agency is critical to having a safe system. For rural and tribal roadways, a basic need is a lane width wide enough for transit vehicles.

Buses typically need a minimum of 11-foot-wide travel lanes to operate. If curb and gutter or other physical barriers are present, 12-foot-wide travel lanes are preferred. Wide shoulders or bicycle lanes are needed for bicyclists and can be used by pedestrians if sidewalks are not present.

Shoulders that are 10 feet wide or more give a space for the bus operator to pull out of the travel lane completely when picking up or dropping off passengers. Roadway signs and signals, along with markings and other infrastructure like pavement surface and lighting, must be installed and maintained. Addressing visibility during dark lighting conditions and poor weather situations is important for bus safety.

Pedestrian Safety

Pedestrian safety is an important component of transit because every trip begins and ends as a pedestrian trip. The FHWA sponsored a study on pedestrian safety in tribal communities, *Pedestrian Safety in Native America*.²³ This is an important topic as Native Americans have the highest rates of pedestrian injury and death per capita of any racial or ethnic group in the United States.

Native American communities require community-specific injury prevention intervention activities for community-specific pedestrian safety problems. Moreover, tribal communities may lack coherent inter- and intra-community transit, which leaves only walking, cycling, or driving

as the choices for transportation. Also, individuals living below the poverty line mitigate high transportation costs by using public transit, walking, or cycling. Successful strategies identified in the FHWA study included education and media-based interventions, law enforcement interventions, child education, and pedestrian facility improvements.

Protecting the Health of Transit Provider Staff and Passengers

The health of bus operators was an issue before the pandemic due to the physical demands, environmental factors, and stress related to their jobs. The COVID-19 pandemic emphasized the need to protect the health of vehicle operators as well as passengers. This is part of ensuring a safe transit service.

Lessons were learned during the pandemic, including the importance of sanitizing vehicles. Not only did this help protect operators and passengers, but informing the public about that process gave the community greater willingness to use public transit as the pandemic waned. The pandemic also demonstrated the need for public transit for workers in essential occupations and people with ongoing health issues, such as those with kidney disease on dialysis, who do not have private transportation.

Safety Considerations

Some overarching safety considerations include the following:

- Consider the first/last mile for passengers in rural and tribal areas. Work with roadway owners to request and ensure that infrastructure is available for these passengers to walk or bike to/from their destinations. Accommodations, such as sidewalks, bicycle facilities (bike lanes or wide shoulders), and established crosswalks, should be present.
- Investigate crashes and record safety incidents, and analyze data to understand and address patterns of safety concerns.
- Have a system where bus operators, passengers, and others can report safety concerns. Investigate these concerns and respond to the person, informing them of planned actions.
- Regularly train bus operators and those who have been involved in crashes to ensure that they understand the safe practices to incorporate. Basic first aid and defensive driving are also important.
- Incorporate wellness programs that provide operators with resources to assist with health issues, such as avoiding drug and alcohol abuse, addressing sleep disorders, and encouraging physical activity and a balanced diet.
- Regular safety meetings are important to share concerns and discuss safety training and tips for safe driving practices. Regularly focusing on a particular crash type or safety concern at different times throughout the year can increase awareness.

For rural and tribal transit agencies, the agency safety plans, if developed, will guide safety management. At its core, having an SMS where hazards are identified, reported, mitigated or avoided, and evaluated is key to managing risk.

Transit Security

There is clear evidence that transit security is a complex issue. Transit crime and perceived security play an important role in transit use and ridership. Ridership levels could be more than 10% higher if people felt more secure when traveling and waiting at transit facilities. Certain groups may be more susceptible to feelings of insecurity. Crime is a particular issue raised by transit agencies as operator assaults have increased. Tribal communities have expressed the need to address safety for their operators.

*TCRP Report 180: Policing and Security Practices for Small- and Medium-Sized Public Transit Systems*²⁴ acknowledges that small- to medium-sized transit operations' security needs and demands are significantly different from those of larger transit operations.

This report identifies security risks that transit operations face, and it is further contextualized by identifying the security environment inherent to small- and medium-sized transit operations.

The report discusses the five stages of protection activity—prevention, mitigation, preparedness, response, and recovery—and the identified potential security countermeasures that could be used at small- and medium-sized public transit systems.

The following strategies focus on minimizing security concerns:

- CPTED. Crime Prevention Through Environmental Design is a strategy for preventing and deterring crime based on the theory that effective design and the use of the built environment can lead to a reduction in crime incidence and fear and involves the following:
 - Access control (fencing, walls).
 - Surveillance (visibility—naturally provided and/or with camera systems).
 - Territoriality (creating a sense of ownership).
 - Maintenance (having a clean and tidy setting).
 - Activity support (having local events or land use that attracts legitimate users).
- Target Hardening. Deters criminal activity with stronger doors, windows, locks, and alarms.
- Technology. Systems for surveillance, detection, lighting, and communication.
- Policing. Security presence.
- Training for Targeted Staff. Such as de-escalation training and conflict management.
- Public outreach and education. Increasing awareness by providing information on the agency website, printed materials, interior vehicle advertising, social media, and announcements.
- Policy. For example, allowing nighttime courtesy stops.

Addressing security and perceived safety in transit environments is multifaceted and layered. Feelings of insecurity are challenging and could be quite costly for transit agencies to address. One example pertains to concerns about people who are experiencing homelessness and/or mental health issues. Transit agencies, no matter the size, have seen increased pressure on the system to accommodate and address the needs of this population. Transit agency property is sometimes seen as a place for sitting or sleeping, likely increasing the unease and feelings of insecurity that passengers experience. At least one transit agency respondent to the research project's survey reported that homelessness and drug use among passengers were ongoing issues. This transit agency also noted that when one homeless rider boarded the bus, others got off.

Impairment/intoxication is another common concern of bus operators and passengers. Working with partners in social services is important for assisting people in these situations.

Careful consideration of security and safety must be given to locations where people board and alight vehicles. Passengers need to wait away from fast-moving motor vehicle traffic and have shelters to protect them from intense sun, rain, or other harsh weather. The locations should include lighting for nighttime use.

Role of Communications

At a basic level, rural and tribal transit agency operators and dispatchers need to be able to communicate so that concerns can be reported and addressed as quickly as possible. In rural areas and tribal lands, law enforcement and other types of assistance may be many miles away, so communication is key.

Technology may be appropriate to enhance the lines of communication and can be used to identify, evaluate, and address security issues. Lighting at bus stops and inside the vehicle should be considered.

Mobile applications can play a role. Mobile apps allow passengers to use their mobile devices, including phones, tablets, or watches, to report incidents to transit agency authorities, such as suspicious activities, disruptive behavior, and safety or maintenance issues, safely and conveniently.

Basic surveillance systems use fixed-view CCTV cameras within facilities such as bus stations, vehicles, and parking lots. Also, incident reporting systems are recommended to document and review data to address concerns.

Transit agencies across the United States have adopted different models of security and law enforcement presence. Smaller and rural transit agencies may set up partnerships or have good communication channels with law enforcement officers at the state, county, municipal, or tribal government level to respond to and assist with security concerns.

CHAPTER 9

Technology

Introduction

Over the past three decades, advanced technologies have emerged to improve public transportation service delivery. These technologies are becoming increasingly accessible to rural and tribal transit operators and range from mobility-on-demand mobile applications—including cloud-based software-as-a-service (SaaS)²⁵ such as self-service ride booking and mobile fare payment—to real-time bus arrival information, to service planning and operations software applications.

These technologies can benefit rural and tribal transit providers that are attempting to initiate new and enhance existing rural and tribal public transportation services, just as they can benefit urban operators. Since many transit services in rural areas and tribal communities include demand responsive features, technologies that assist in real-time communication and vehicle dispatching are of particular interest.

As transit agencies across the nation face driver and transit labor shortages on top of shortfalls in transportation funding, intelligent transportation systems (ITS) and other advanced technologies have the potential to solve, and possibly aggravate, challenges depending on their application. This chapter is intended to inform agencies serving rural and tribal areas in making decisions around adopting technology and implementing strategies that support new and improved transit services.

Several technologies are frequently discussed in the press for their use or potential uses in urban contexts, but these technologies have yet to find an established role in rural and tribal transit services. Autonomous vehicles are an example. While innovation is ongoing, more established driver support features [Society of Automotive Engineers (SAE) Levels 1 and 2] like adaptive cruise control, lane centering, and braking assist may be of more interest in some rural areas.

The ability to evaluate and adopt newer technologies can be constrained by factors such as limited financial and staff resources; limited availability of internet and cellphone network connectivity; long distances; and rates of internet access, access to computers, and adoption of smartphones among riders. Such access is trending upward but remains lower in rural areas, influenced by demographic factors such as age, income, and educational attainment.²⁶ These constraints are especially prominent in the most rural areas, where low population density, long routes, and distance from major metropolitan areas can be challenging to manage. Despite these considerations, experience shows rural transit operators are actively seeking to take advantage of technology to improve all aspects of their public transportation services.

This chapter begins by summarizing core technologies that are fundamental to developing and implementing other advanced systems. The chapter identifies and describes technology

applications and how they apply to various service strategies that support transit services. Note that some of the technologies described are likely beyond what a typical rural or tribal transit provider might adopt in the near term, such as autonomous vehicles. Nonetheless, these more advanced technological options are included in the discussion.

What Is Involved?

ITS refers to innovative technologies that enhance informed decision-making by different stakeholders in transportation, from customers to planners and operators.²⁷ While new and emerging technologies have the potential to augment the quality and reliability of transit services, many of these functions rely on a series of fundamental technologies. This section provides an overview of these core technologies and how they are necessary for positioning transit providers to implement ITS.

Vehicle Communications

The communications infrastructure within a transit system is used to communicate text, voice, data, or video within vehicles, between vehicles, and with off-board systems. The type of technology adopted may be constrained by the communications infrastructure available.

Equipment on board buses must be interconnected through a vehicle area network (VAN), through wired (Ethernet) or wireless (WLAN) transmission. External communications from the vehicle can be established through satellite phones, smartphones, cellular-enabled tablets, or basic radios and Global Positioning System (GPS).

GPS

GPS is the most common technology for locating vehicles and does not require internet or cellular connectivity. GPS devices are relatively inexpensive to procure, operate, and maintain, and function at the core of automatic vehicle location (AVL) systems.

The key considerations for any geolocating system are latency and update rate. The update (or polling) rate is the frequency at which the device calculates and transmits its location, while latency refers to the delay in time between when the data is first polled and when it appears to the end user, either in a trip-planning application or a dispatcher's screen. To deliver high-precision vehicle-location data, latency should be low, and the update rate should be 30 seconds or less. Many systems default to 10-second polling rates, which means that a vehicle's location is queried every 10 seconds.

Standalone GPS systems have the added benefit of maintaining data transmission without cellular connectivity, which can be spotty in some rural or densely forested areas.

External Communications

Connectivity to data centers and platforms beyond the bus is necessary for location sharing, real-time information, and data transfer. Operating data from buses is valuable for dispatchers and maintenance purposes. Drivers and dispatchers need to communicate while vehicles are operating, which can take place over the phone, messaging, or radio.

Two-way radios are still the most common and familiar technology for drivers and dispatchers to communicate about emergencies, detours, and driver sign-ins and sign-outs, especially in areas without cellular reception. Radio setups can be procured as part of the bus or as part of computer-aided dispatch (CAD)/AVL systems. Many agencies have opted for the integration of radio system controls with tablets and other driver interfaces. However, radio communication is not a medium for transmitting data, which typically relies on cellular communications.

Cellular connectivity is the dominant industry approach for connecting buses to off-board data centers. Some state DOTs negotiate master service agreements with data providers and allow transit agencies to use the same contract. This reduces the administrative and procurement burden on the transit agency. The limitations of cellular connectivity and infrastructure can guide agencies to design the most continuous service in their areas.

Vehicle Internal Connectivity

There are a variety of ways to interconnect onboard technology components through a VAN, which can be established through Ethernet cables, cellular modems, or Wi-Fi internet. Components that many agencies look to integrate include the farebox, headsigns, announcement systems, driver tablet interfaces, and passenger counters.

The SAE developed the Transit Communications Interface Profiles (TCIP) as the standard for facilitating onboard hardware communications. Some agencies define the TCIP protocols they expect vendors to use and comply with in technology procurements; this approach is particularly common for agencies with existing technologies installed.

CAD/AVL

AVL is a system that uses in-vehicle computers to transmit GPS data so that dispatchers can send instructions to drivers and track bus locations and movements in real time. CAD consists of a series of hardware and software components that collect AVL information and communicate with off-board systems, allowing dispatchers to have vehicle information on hand in real time. Both CAD and AVL require wireless data communications to communicate data between vehicles and backend systems.

CAD/AVL systems are essential technologies that enable many functions and services for both transit operators and passengers. CAD/AVL systems allow dispatchers to visually assess transit services in real time and respond to detours and unexpected incidents. Data collected from AVL systems helps providers pinpoint where delays are taking place and design efficient bus routes and schedules.

Onboard the vehicle, CAD/AVL packages can be integrated with headsigns, stop annunciators, and other passenger information systems. AVL systems also integrate with silent alarm and video monitoring technologies, which enhance safety and security in vehicles. AVL systems also enable real-time bus information and service alerts for transit riders not yet onboard.

For many rural and tribal agencies, demand responsive services have long been an attractive service model to meet travel demand across a large geographic area. CAD/AVL is a particularly important technology for agencies seeking to implement dynamic demand response service models. However, agencies may not see efficiency improvements commensurate with the cost and effort of implementing CAD/AVL unless they are seeking to use it for broader functions.

Vehicle logic units (VLUs) and mobile data terminals (MDTs) serve as the driver interface with CAD/AVL systems and can provide centralization of bus hardware functions.

Vehicle Logic Unit

The VLU is the onboard computer that manages communications between the vehicle and a central dispatch system and serves as the interface to the vehicle radio. It is the central data processing and storage device for a vehicle and can automate the processing of AVL data as well as data from other on-vehicle hardware.

Mobile Data Terminal and Mobile Digital Computers

An MDT is a portable interface that enables drivers to access information visually. MDTs can exist as ruggedized tablets, laptops, or other customized hardware interfaces onboard transit

vehicles. Drivers can use the MDT to program the headsign, manage the electronic farebox, and receive real-time directions from dispatch on how to navigate detours and disruptions for fixed-route service and the trip manifest for demand response service. MDTs can also present dashboards on key performance indicators, such as on-time schedule adherence, fare transactions, and vehicle maintenance condition.

Typically, an MDT communicates with a backend computer stored in a cabinet elsewhere on the bus, which includes a VLU in some vendors' implementations. MDTs have increasingly been deployed as fully functional computers, hence the more recent term of mobile digital computers; however, "MDT" remains the popular term referring to these devices.

VLU and MDT Implementation Considerations

Many CAD/AVL providers design proprietary VLUs to provide unique capabilities that differentiate their product. If implemented well, this approach can bring benefits to a transit provider but also ties the agency's fleet to a central, proprietary piece of infrastructure. If the CAD/AVL vendor were to change, replacing the VLU could threaten interoperability with other components. Alternatives to proprietary VLUs include commercially available computers, programmed with open data standards, which are similarly stored in a cabinet on the bus. Unfortunately, such an approach remains inaccessible to many transit agencies due to the high level of technical knowledge and expertise needed to configure a custom computer unit for a fleet.

Mobile Applications

Mobile applications improve access to transportation services and allow passengers to plan trips or access real-time transportation information. Applications are beneficial for transit users and transit providers and offer several functionalities that use transportation technology and include the following:

- **Trip and Journey-Planning Applications.** These applications provide passengers with public transit directions, offering alternative routes, transfers, and schedules. Trip-planning applications may operate in real time but can also present static timetables.
- **Ticketing and Fare Payment Applications.** These applications allow transit users to purchase digital tickets for transit services. Applications may also facilitate fare payment on or near the transit vehicle and management of virtual and nonvirtual fare payment cards.
- **Real-Time Information Applications.** These applications communicate real-time transit vehicle information, including locations and departure and arrival times. Real-time applications can deliver service alerts and other transit information.
- **Integrated Mobility Applications.** These applications are comprehensive mobile applications that provide trip-planning functionality across multiple modes of transportation for a seamless user experience.

Considerations for agencies in developing their mobile application strategies include the following:

- **Accommodating Multiple Functionalities in One Platform.** Riders are less likely to download or use multiple apps for the same transportation service. A singular application that unifies some or all of the most important services (e.g., trip planning, fare payment, real-time

information) can be more convenient for riders than multiple single-purpose apps and streamlines data collection and analysis to better understand rider behavior.

- **Leveraging Existing Applications.** Creating or procuring a new app that replicates the same services provided by others may require more staff (in-house and contracted/managed) and financial resources, particularly for rural and tribal agencies.
- **Coordinating and Collaborating on Applications Across Agencies.** A multiagency application can spread a cost burden more manageably. Coordinating with other agencies can also offer a common platform for integrated ticketing and trip-planning across multiple modes, agencies, and service areas.

Automated Fare Collection

Automated fare collection (AFC) describes any method of fare collection in which fares are not paid directly to a transit officer in exchange for entry to a vehicle. Used by transit agencies beginning in the 1960s, AFC includes technologies like electronic fare gates, ticket vending machines, magnetic stripe and smart cards, standalone validators, and, most recently, contactless payment terminals and mobile wallet readers. These technologies de-emphasize the role of cash and individual tickets in fare payment and ticketing. Regardless of technology, a backend fare payment system provided by the vendor is required to facilitate electronic payments.

Automating fare collection and ticket validation can allow agencies to reduce their staff needs, reduce fare processing costs, increase efficiency, reduce fraud, and enhance rider convenience.

General Transit Feed Specification

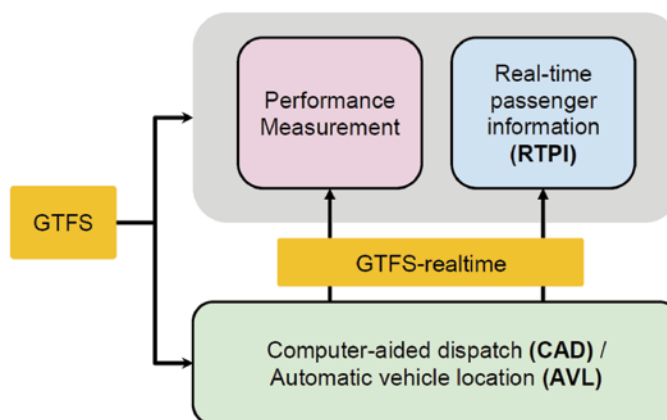
GTFS is a standardized data format for public transportation schedules and geographic information. The standardized format allows data to be published and shared. Published GTFS feeds can be used by a variety of third-party applications for planning, reporting, and passenger information.

The core specification is GTFS Static (or GTFS Schedule), which captures fundamental information about routes, schedules, fares, and stops. GTFS Static is used in applications that visualize journeys and routes. As an accessible, text-based Open Standard, it has allowed many agencies and third-party GTFS consumers to innovate extensions to the core specifications. For example, GTFS Realtime is an extension of GTFS that integrates vehicle-location data (from AVL) and service alerts to provide live information on the transit system, allowing the data to support real-time tracking and trip- and journey-planning applications.

Recently, several data vendors have collaborated to develop GTFS-Flex, which provides booking rules and real-time service information that accounts for route deviations, detours, and demand response transit. Larger agencies have developed and maintained their own extensions to suit their data needs.²⁸ The California Integrated Travel Project (Cal-ITP) is further developing an Operational Data Standard (ODS) based on GTFS, incorporating operational elements such as deadheading and layover facilities.

Adopting GTFS provides numerous transportation technologies, such as real-time passenger information and a variety of service strategies with a common dataset for route, stop, and schedule information. It also improves regional and interagency coordination by facilitating data sharing between agencies. Figure 9.1 illustrates how GTFS relates to CAD/AVL systems and feeds information to transit technology applications.

There are many resources available to agencies for building GTFS feeds. State DOTs can procure master contracts to provide and maintain GTFS for agencies, alleviating the procurement



SOURCE: Aaron ANTRIM, TRILLIUM SOLUTIONS, REAL-TIME TRANSIT INFORMATION: BENEFITS, TECHNOLOGIES, COMPONENTS, EXPERIENCES, AND RECOMMENDATIONS; FROM WHITE PAPER FOR ODOT. [HTTPS://WWW.OREGON.GOV/ODOT/RPTD/RPTD%20DOCUMENT%20LIBRARY/REAL-TIME-WHITE-PAPER.PDF](https://www.oregon.gov/ODOT/RPTD/RPTD%20DOCUMENT%20LIBRARY/REAL-TIME-WHITE-PAPER.PDF). SEE ALSO ODOT, TRILLIUM SOLUTIONS, AND IBI GROUP. "ON BUS HARDWARE, SOFTWARE, STANDARDS & INTEROPERABILITY." STATE LIBRARY OF OREGON, 2021. [HTTPS://WWW.OREGON.GOV/ODOT/RPTD/RPTD%20DOCUMENT%20LIBRARY/BUS-HARDWARE-SOFTWARE-STANDARDS-INTEROPERABILITY.PDF](https://www.oregon.gov/ODOT/RPTD/RPTD%20DOCUMENT%20LIBRARY/BUS-HARDWARE-SOFTWARE-STANDARDS-INTEROPERABILITY.PDF).

Figure 9.1. Relationship of GTFS to CAD/AVL systems and transit technology applications.

and administrative burden on rural and tribal transit providers. The State of Oregon has one such contract with Trillium Consulting to develop GTFS feeds for agencies across the state, as described later in this chapter. Some CAD/AVL vendors can provide this service. The National RTAP provides a free-to-use GTFS builder tool as well as instructional support, which can help agencies train staff and build capacity in managing and maintaining GTFS (see <https://www.nationalrtap.org/Technology-Tools/GTFS-BUILDER>).²⁹

In 2023, the FTA began requiring all transit agencies that receive federal funding to provide a GTFS-Static dataset of fixed-route services, as part of NTD reporting. This applies to all agencies, including “full,” “reduced,” “tribal,” and “rural” reporters. Recognizing that only 35% of NTD reporting agencies have an officially published and maintained GTFS feed, the FTA is also providing additional technical assistance to agencies for establishing these datasets.

Geographic Information System

GIS is computer software that is used to store, manage, and visualize spatial and geographical data for map creation and analysis.³⁰ GIS allows agencies to manage transit-specific geographic data, such as routes and bus stops, and visualize it with other data, such as demographics or travel patterns. GIS data enables interactive maps and integrates with CAD systems to support public transit route-planning and location-based tools used for real-time passenger information and customer-service functions.³¹

GIS software is accessible via various platforms, including ESRI ArcGIS, which is commercial software with licensing fees, and QGIS, which is open-source and freely available. Considerations in selecting commercial or open-source software include cost, in-house staff expertise, and resources available to rural and smaller transit agencies.

Transit agencies may have access to GIS expertise and resources through other local government agencies, such as counties or councils of governments. The U.S. Department of the Interior’s Branch of Geospatial Support provides Tribal governments with GIS software, training, and technical

assistance. Web-based operations software can also provide agencies without access to GIS with the capability to manage and visualize their data.

Procurement Considerations

Some key principles and themes related to the contracting and procurement processes were identified from various peer agency and technology vendor interviews. These best practices can help staff at rural and tribal agencies tailor their requests and mitigate risk.

- Before developing an RFP, proactively reach out to vendors and peer agencies to fact-find and understand what outcomes are desired.
- Choose the right tools for your community and ridership, not just what is popular in the industry (outcome-oriented procurement).
- Explore joint procurement efforts with neighboring agencies for economies of scale.
- Find opportunities to incorporate technology procurement with future fleet purchases or obtain technology through fleet procurement contracts.
- Prioritize procuring vehicle communications, CAD/AVL systems, and GTFS as part of any technology adoption effort.
- Design procurement scope with emphasis on a standards-based, commercially accessible system of technology over proprietary hardware and connections.
- Scope for an integrated package with one primary technology vendor who is responsible for subcontracts and/or technology integration.

The key challenges facing transit agencies include staff capacity, internal technical knowledge, and the cost of adoption. Writing procurement scopes and specifications can be complex and costly if agencies are unsure of what to look for. As such, some of the best practices identified are around information gathering and taking an outcome-oriented approach to procurement. This means listing and categorizing the agency's desired goals and outcomes for technology and asking vendors how their solutions can help the agency holistically achieve defined goals.

This is in contrast with many standard procurement processes, where contracts emphasize checking off various requirements with specific pieces of technology or being overly prescriptive in the specifications of individual hardware components. Requirements can also be right-sized to community needs; for example, one agency found the live map provided by their CAD/AVL provider was adequate for their riders' needs, and they did not require specific arrival time predictions.

The effectiveness of fundamental, backend systems is a greater priority than trying to establish a multitude of public-facing technologies at once. Consistent vehicle connectivity will enable high-quality data and information to be transferred within and between transit vehicles, powering CAD/AVL and real-time information. With CAD/AVL systems, rural and tribal agencies should evaluate the following to better understand their own needs:

- Likelihood of implementing peripheral technologies such as automated passenger counters (APCs) and interior multimedia displays.
- Adoption of CAD/AVL for microtransit or other demand responsive services.
- Scale and size of the agency, and availability of in-house technical expertise.
- Risk tolerance for system modularity and technological interoperability.

Bundling technology procurement under one contract can be beneficial, with a lead vendor taking responsibility for the integration and interoperability of multiple technology components. While reliance on a single prime vendor can come with risks, adopting a standards-based approach when scoping the contract can help mitigate these factors. A key example is driver interfaces such as VLUs and MDTs, which are often the integration point of CAD/AVL systems. Selecting a solution

that uses readily available commercial tablets and computers instead of proprietary equipment from CAD/AVL vendors is one strategy for minimizing risk. To make CAD/AVL systems more accessible to rural and tribal agencies, state DOTs can provide technical assistance and advocate for open-source standards and modular components.

Open-source standards are also valuable in ensuring different technologies are compatible with each other and accessible to agency staff and planners.³² The Transit ODS is an initiative pioneered by Cal-ITP³³ and supported by various private vendors to maintain and define standards for transit operations information. It builds on GTFS but tailors the specifications for operational needs. It is a significant milestone in moving SaaS vendors toward technology compatibility and can be a useful specification to reference in developing RFPs and procurement scopes.

Rural and tribal agencies can partner with each other to take advantage of market economies of scale and create efficiencies in procurement. Agencies can also work with state DOTs to identify any master services agreements (MSAs) or overarching contracts that have already been procured and explore opportunities to contract from the MSAs. Research conducted by Trillium Consulting for the Oregon Department of Transportation (ODOT) identified that the current technology marketplace best serves transit systems that operate 1,000 or more vehicles and have greater internal technical capacity and staff to select across multiple vendors and contracts.³⁴ Taking a joint procurement approach can alleviate the administrative and legal burdens on agencies with limited staff. For example, ODOT has a master contract (with Trillium Solutions) that transit agencies across the state can leverage to maintain their GTFS feeds. The FTA maintains a joint procurement clearinghouse for the purchase of rolling stock for the same reasons, but it is only available to federal grant recipients. State DOTs can take a leadership role in connecting transit agencies and empowering joint procurement practices.

How Technology Can Enhance or Support New and Emerging Services and Strategies

Technology can serve an important role in supporting new as well as emerging services and strategies that rural and tribal transit providers may pursue. This section discusses various technology applications and how they can be applied to various service enhancements and strategies (see Table 9.1).

Trip-Planning and Media Applications

Website

A transit agency website is often the first source of information for potential riders. Websites convey information about the routes and schedules of services offered, fares and payment, agency rules, policies, and procedures. They also include administrative and planning details, such as transportation plans or programs, agency job postings, contact details, and news about services.

A passenger-focused transit website includes three main attributes: passenger experience and ride information; articulation of the transit agency brand; and backend maintenance of the website.³⁵

- Passenger experience and ride information should be oriented to passenger usability. Administrative information can be provided, but it should be secondary. Elements can include interactive route maps with real-time vehicle tracking and/or static route maps with published schedules. Trip-planning and service information should be as easy as possible to find and understand, minimizing the number of clicks/taps needed.
- Agency brand articulation prompts rider recognition of agency logos and other marketing elements used on vehicles, bus stop signs, and other informational materials.

Table 9.1. Various technology applications by type.

Technology		Type
Core Technologies		
	Vehicle Communications	Software
	CAD/AVL	Software
	Mobile Applications	Software
	Automated Fare Collection	Software
	GTFS Static and Real-Time	Software
	GIS	Software
Technologies to Support Transit Services and Strategies		
Trip-Planning and Media Applications	Website	Rider Info
	Real-time Trip-Planning and Route Information Mobile Application	Rider Info
	Demand Response Self-Service Booking	Rider Info
Software	Service Planning and Operations	Software
	Fleet Management	Software
	Automated Dispatch	Software
Information Displays	Real-Time Information Screens	Hardware
	Destination Signs (headsigns)	Hardware
	Interior Digital Displays	Hardware
Electronic Fare Payment	Automated Fare Collection	Operations
	Integrated Ticketing	Operations
Safety and Security	Data Management System	Hardware
	Silent Alarms	Hardware
	Bus Stops	Hardware
	Cameras (in-vehicle and stop/station)	Hardware
In-Vehicle Technology	Automated Passenger Counters (APCs)	Operations
	Automatic Voice Annunciator (AVA) Systems	Rider Info
Vehicles and Fleet	Driver Support (SAE Level 1 or Level 2)	Operations
	Autonomous Vehicles	Operations
	Zero-Emission Vehicles	Operations

- Backend website infrastructure supports the display of GTFS and other real-time data so that the passenger can get the most up-to-date information, and the website can adapt more seamlessly to changes in service.

Websites should be accessible by passengers using web browsers on computers and mobile devices, preferably without downloading mobile applications. Websites should also be accessible to passengers with disabilities.

Rural transit agencies may have issues maintaining a website of this type because it requires either a contracted website designer or an in-house web designer adept at creating a transit-focused website. For this reason, some rural and tribal agencies may have websites that may be difficult to navigate or are not optimized for different devices.

Website Coordinating Five Transit Agencies' Service—NW Connector, Oregon

The NW Connector website developed for an alliance of five transit agencies across northwest Oregon shows daily route schedules for a variety of services operated by the Alliance's five agencies. The website offers a static and interactive map for all agencies; integrated trip planning across the agencies; schedules and other information for each agency individually; information on visitor passes valid for travel on all the agencies' services; and trip ideas for riders to explore through a series of updated service area maps. It is accessible on both mobile and web devices. The shared effort spread the cost among the agencies and provides a unified view of service in northwest Oregon.

The screenshot displays the NW Connector website. At the top, there is a search bar and the NW Connector logo with the tagline "PUBLIC TRANSIT ACROSS NW OREGON". Navigation buttons for "SCHEDULES, ROUTES, FARES" and "MORE" are visible. Below the navigation bar, four main service areas are highlighted: "How to Ride" (FAQs, Policies, Accessibility, and Rider Guides), "Connections" (Connecting Services: POINT, Amtrak, and more), "Data and Apps" (Third-party Apps and GTFS data for developers), and "Contact Us" (Questions, Service Information, and Feedback).

The main content area features a large map of northwest Oregon showing transit routes. To the right of the map is a "PLAN A TRIP" section with input fields for "Starting Address", "Ending Address", and "11/19/2023", along with a "Time" field and a "GET DIRECTIONS" button. Below this is a "GET YOUR PASS" section with text about 3 and 7 day all-you-can-ride passes and a "GET YOUR PASS" button. Further down is an "EXPLORE TRIP IDEAS" section with text about browsing hikes, sight-seeing opportunities, and entertainment accessible by transit, and a "SEE TRIP IDEAS" button.

The footer contains the NW Connector logo and description, followed by logos and names of the five member agencies: Tillamook County Transportation District, Columbia County Rider, Lincoln County Transit, Benton Area Transit, and Sunset Empire Transportation District.

SOURCE: NW Connector.

Real-Time Trip-Planning and Route Information Mobile Applications

Real-time trip-planning mobile applications provide transit riders with up-to-the-minute route information, including service alerts and the ability to view vehicle locations in real time. Applications can also share live updates on traffic conditions, service delays, and disruptions, and provide other in-app alerts. This information helps transit riders make informed decisions and adjust their travel plans accordingly.

Transit agencies can publish their information for use in an app procured by the agency (such as through its CAD/AVL vendor) or by a state DOT. Transit agencies can also publish their real-time information as open data to the cloud, making it available to riders using a variety of commonly used transit-focused trip-planning mobile applications, such as Transit and Moovit, and widely used multimodal trip planners, such as Google Maps. Partnering with an application provider such as Transit or Moovit can also provide agencies with a more customized presence as well as access to anonymized data from the platform.

Service Strategy	Support of Service
Fixed-Route	Passengers can see real-time delays and disruptions to departure and arrival times, including whether schedule changes may impact transfers. Trip-planning applications can also provide alternative routes and modes of travel to the destination.
Demand Response	Trip-planning applications can communicate flexible-route departure and arrival times and changes to routes. For on-demand transit, trip-planning applications should incorporate the ability to request a ride from the app.
Fare Policy and Programs	Trip-planning applications can host user profiles for passengers to log in and request and access travel voucher/fare card subsidy programs.

Real-time mobile applications generally require access to a mobile phone. This may limit user adoption and accessibility, so alternative access must be provided for those without such access.

Implementing real-time trip-planning mobile applications presents challenges in data accuracy and integration with technical infrastructure. Inaccurate data can lead to incorrect trip information, which makes service unreliable for riders. Infrastructure required to support real-time applications includes core technologies such as wireless networks and AVL capabilities on vehicles to provide backend systems with information and feed the mobile application with real-time data.

State DOTs Can Help Transit Agencies Provide Mobile Applications—Vermont Agency of Transportation (VTrans) and Minnesota Department of Transportation (MnDOT)

State DOTs are well-positioned to enter public-private partnerships to help rural agencies develop and maintain trip-planning mobile applications, with the added benefit of enabling cross-region rural trip planning. Two examples highlight these efforts:

- Vtrans conducted a project to develop a trip planner wholly owned by the State of Vermont to provide rural transit trip planning without ongoing

(continued on next page)

State DOTs Can Help Transit Agencies Provide Mobile Applications—Vermont Agency of Transportation (VTrans) and Minnesota Department of Transportation (MnDOT) (*Continued*)

license fees (<https://www.connectingcommuters.org/travel-apps/trip-planner/>). The tool was built on open-source Open Trip Planner software. The project, which included public-private partnerships, helped expand the capabilities of GTFS-flex standards and enabled use of the technology for all public transit providers in the state.

- MnDOT launched a project (<https://mntransitplanner.com/#/>) to help residents in rural areas plan, book, and pay for multimodal trips. MnDOT worked with Transit, Cambridge Systematics, and Trillium Solutions to launch a pilot project providing customer experience upgrades that might otherwise be difficult for small agencies to procure individually, such as GTFS and online payment and trip booking. The pilot ran through April 2024.

SOURCES: Federal Transit Administration, Vermont Agency of Transportation (VTrans) Flexible Trip Planner (Report 0150), 2020. <https://www.transit.dot.gov/research-innovation/vermont-agency-transportation-vtrans-flexible-trip-planner-report-0150>. Carol Schweiger, "Open Source Software and Open Data – What They Are and How to Use Them," National Center for Applied Transit Technology (N-CATT), October 2020.

MnDOT, "MnDOT launches first multimodal trip planner to focus on rural transportation services," March 1, 2023. <https://www.dot.state.mn.us/news/2023/03/01-statewide-Multimodeltrip.html>. Transit App. "Four Transit Projects Putting the 'State' in State-of-the-Art," August 23, 2023. <https://blog.transitapp.com/four-transit-projects-putting-the-state-in-state-of-the-art/>.

Mobile App for Paratransit Service—Franklin Regional Transit Authority, Greenfield, Massachusetts

Franklin Regional Transit Authority's Access Program offers passengers the ability to schedule ride services through its mobile application. Passengers can select pickup and drop-off locations, and the application will send a push notification when the ride is en route. The notification provides the bus ID number, and passengers can enter the application to view the bus location in real-time.

SOURCE: FRTA. FRTA Access Program, October 5, 2022. <http://frta.org/getting-around/frta-access-program>.

Demand Response Self-Service Booking

On-demand mobility enables customers to request a trip from a service provider without requiring a reservation. This is achieved through a digital platform to match customer requests for trips to vehicles available to provide the trip. These services include the agency's dispatch staff (which uses a computer desktop concierge platform to make trip requests on behalf of the customer), texting trip requests to the digital platform, or using a touchscreen kiosk at a key pickup point.³⁶ Demand responsive booking can integrate various modes of transportation, such as buses, trains, TNCs, and even walking or biking, and makes it easy for users to plan multimodal trips.

Service Strategy	Support of Service
Demand Responsive and On-Demand Microtransit	Demand response services pick up passengers at requested addresses and allow customers to make reservations with a smartphone, potentially allowing trips to be made with a shorter advance reservation period.
ADA Paratransit	Allows ADA paratransit customers to make reservations with a smartphone and provides information on trip arrival.

Self-Service Web Portal for Demand Response Trip Booking—Link Transit, Wenatchee, Washington

Link Transit uses a self-service web portal from EcoLane to allow riders to book trips in advance on its demand responsive services, freeing up dispatchers and enabling 18% of trips to be same-day requests, 71% of which are picked up within 30 minutes. The self-service portal works with EcoLane software (see Software section in this chapter) to automate scheduling and dispatch of the trips.

SOURCE: Link Transit Case Study, EcoLane. [https://www.ecolane.com/hubfs/Offeres/ecolane-linktransit-casestudy-v2%20\(1\).pdf](https://www.ecolane.com/hubfs/Offeres/ecolane-linktransit-casestudy-v2%20(1).pdf).

Software

Service Planning and Operations

Transit agencies can use service planning and operations software to design and manage aspects of their operations, such as route planning, scheduling, fleet planning, and fare structure. The software available ranges from multipurpose and comprehensive capabilities to more specialized and focused on specific aspects of service and operations, such as run cutting and blocking (i.e., assigning specific runs to individual operators, and grouping trips or runs with similar start and end times, respectively).

Both general and specialized software have advantages and disadvantages depending on the service strategy. Some contract service operators or technology hardware vendors have software offerings that may be bundled and could be suitable for a transit provider's needs. A critical function of any scheduling software is to generate a GTFS feed for an agency.

Service Strategy	Support of Service
Fixed-Route	Service planning and operations software helps transit agencies with scenario planning and cost analysis to improve efficiency and coverage.
Demand Response	Software can optimize the allocation of on-demand vehicles based on ridership and demographics. Passenger requests for vehicles can be valuable input for software to understand demand.
ADA Paratransit	Software can support the planning of pickup and drop-off locations based on need and demand from communities that use paratransit services.
Fare Policy and Programs	Software can incorporate demographic and ridership data to focus service in communities or areas with transportation needs.

Examples of software include Remix by VIA, Optibus, HASTUS, Trapeze, TransLoc, DoubleMap, Samsara, EcoLane, Passio, and Swiftly.

Service planning and operations software requires accurate data input to operate effectively. The infrastructure in rural communities may need to be upgraded to support data collection and communication to provide accurate data input to the software. Rural agencies may also lack on-site technical expertise and should consider costs for training, operations and ongoing management, and maintenance.

Planning Microtransit—LYNX, Capital Area Transit System, Baker, Louisiana

LYNX by Capital Area Transit System (CATS) is a microtransit service launched in June 2022 to complement a fixed route and provide residents in the north suburbs of Baton Rouge access to jobs, education, healthcare, and essential services. CATS used Remix—VIA’s route-planning and scheduling software—to plan service areas and communicate to potential passengers their transit options before and after implementation of LYNX, including common pickup and drop-off locations.

SOURCE: Metro Magazine. New Mobility. Jan. 9, 2023. <https://www.metro-magazine.com/10189330/in-this-suburb-microtransit-is-the-key-to-expanding-opportunities-for-residents>.

Fleet Management

Fleet management is the process of planning, operating, and maintaining the vehicles required for the transit service and travel needs of a community. Aspects of fleet management include vehicle procurement and maintenance, fuel management, and fleet tracking. Effective fleet management considers passenger capacity and resource management in response to demand for transportation. Examples of software that handle these functions include Fleetio, Clever Devices’ SmartYard, Geotab, Webfleet, Samsara, and Passio Technologies.

Fleet Maintenance & Management Software—Kayak Public Transit, Confederated Tribes of the Umatilla Indian Reservation

Kayak Public Transit, operated by the Confederated Tribes of the Umatilla Indian Reservation, provides rural regional public transportation across four counties in Northeastern Oregon and Southeastern Washington. Kayak implemented Fleetio as a fleet maintenance and management software to track fuel, mileage, and work history on their vehicles and quickly address maintenance needs. Kayak’s use of fleet management software also fulfilled performance and compliance reporting to external auditors.

SOURCE: Fleetio. Kayak Public Transit. <https://www.fleetio.com/resources/case-studies/kayak>.

Transit agencies in rural communities typically have limited staff; therefore, for fleet management software to be effective, it should be easy to use, comprehensive in features, and have reporting functionalities. With more functions and an intuitive user interface, smaller agencies can reduce costs and simplify reporting for compliance. Rural agencies may not have sufficient staffing within their agencies and may outsource their fleet management. Conversely, if the software is simple and comprehensive, it may be more feasible and cost-effective for fleet management to stay within the transit agency.

Service Strategy	Support of Service
Fixed-Route	Fleet management software can track vehicles in real time and track performance and operations. It can also manage vehicle life cycle and maintenance. Software can also support compliance with local, state, and federal standards.
Demand Response	Software can help optimize fleet size and type, and track statistics useful in allocating and reporting costs.
ADA Paratransit	Software can optimize fleet size and type in response to demand for paratransit vehicles.
Regional and Interagency Connections	Software can improve management and communication between different transit agencies to optimize regionwide fleet inventory and resources.

Automated Dispatch

CAD/AVL is one of the core technologies identified earlier in the chapter, and CAD is the computer-aided dispatch element of that technology.

Dispatch operations is a typical staff function that includes tracking vehicles as they enter and go out of service; fielding calls from customers; communicating with drivers, which includes relaying information about incidents, missed stops, and trip or transfer requests; and scheduling demand response and paratransit trips.

Software provided by a CAD/AVL vendor (or integrated with those capabilities) supports and optimizes these traditional transit operations functions by allowing dispatchers to track vehicles in real time and more easily communicate with drivers.

Dispatch software may include or overlap with scheduling software for tasks such as managing driver and vehicle assignments. CAD software enables dispatchers to optimize driver and vehicle assignments and can include self-service trip-booking capabilities for demand response services. Examples of software vendors include GMV Synchromatics, Trapeze, REI HUB, TripSpark, Ecolane, Samsara, Optibus, and Swiftly.

Service Strategy	Support of Service
Fixed-Route	CAD software can coordinate dispatch and driver communications, provide real-time information about vehicles, and capture performance metrics for internal assessments and external audits.
ADA Paratransit	Software can provide a platform for passengers to request, confirm, and cancel trips. It may also be a platform to communicate with drivers regarding route deviations and schedule changes. This can help improve service operations and passenger satisfaction.
Regional and Interagency Connections	If a region has multiple transit agencies or transit providers, software can potentially integrate across providers.

Rural communities implementing automated dispatch software should consider route complexity, hardware and other software compatibility, and ease of participation by drivers and passengers. The complexity of routes and interagency connections of a region may affect the inputs required for automated dispatch software to function effectively. Smaller communities may not have the hardware, software, or technical infrastructure in place to support compatibility. Implementation should also consider training needs for drivers to interact with vehicle-based components of the software (i.e., MDT) (see the following examples of CAD/AVL).

Improving Service with CAD/AVL—Palo Verde Valley Transit Agency, Blythe, California

Palo Verde Valley Transit Agency (PVVTA) provides transit service to the City of Blythe, California. PVVTA operates nine ADA-compliant 18-passenger buses and six support vehicles, all equipped with GMV Syncromatics' dispatch and vehicle tracking software. Since implementing GMV Syncromatics in 2023, PVVTA has improved passenger and fare logging, real-time vehicle location tracking, and fleet maintenance. Tablets installed on buses allow operators to log passenger counts and fare payment/pass type as passengers board the vehicle.

Passengers can view arrival times and vehicle locations in real time on www.ridepvvta.com; those who cannot access the website may still call PVVTA dispatch. Operators can communicate to dispatch directly through text messaging capabilities on the tablets. Real-time tracking of support vehicles, including service trucks and utility vehicles, helps staff monitor and optimize how they are deployed. For smaller transit agencies such as PVVTA, this software helped streamline transit operations by capturing more accurate data and reducing manual work by staff. Lessons learned from the implementation include factoring in training time for staff; identifying strategies to serve passengers who may not have internet or smartphone access; and updating infrastructure such as bus stop signs to communicate the availability of new capabilities.

SOURCE: PVVTA, Interview, October 24, 2023.

Upgrading CAD/AVL System—Cascades East Transit, Bend, Oregon

Cascades East Transit (CET) secured funding to replace its legacy CAD/AVL system, which was cumbersome and resulted in staff scheduling demand response trips manually. CET took 1 year to develop a scope of work and start the bid process. It ultimately selected Passio as a single contractor, using Passio's system for CAD/AVL capabilities on its fixed-route service. CET viewed upgrading its CAD/AVL system as foundational, before rolling out other public-facing capabilities. Since the agency has limited staff, it opted for basic tools aligned with its needs. For example, Passio's native live map has met the needs of CET's riders; CET did not implement arrival predictions. Drivers' tablets include a button for counting boardings and alightings, which CET uses in lieu of APC.

CET's contract with Passio includes partnering with EcoLane for a separate system to manage its demand response operations, which comingle Medicaid NEMT trips on rural dial-a-ride; the software helps CET effectively track different funding types. The shift to dynamic scheduling of demand response trips was initially challenging for veteran drivers but was ultimately well received by its schedulers and makes it easier for new drivers. The agency has tuned the system to balance efficiency (which has increased its trips per revenue hour) with meeting requested appointment times.

Since Passio can also manage scheduling for deviated route services, there is some overlap in capabilities between the two systems. CET maintains flexibility by installing both applications on 25% of its fleet. CET uses EcoLane for its Community Connector route serving the Confederated Tribes of Warm Springs Reservation, to maximize control for deviated service provided within Warm Springs. Conversely, it plans to use Passio for the flex route service it plans to introduce in the city of Redmond.

SOURCE: CET, Interview, September 26, 2023.

Information Displays

Real-Time Information Screens

Real-time information screens provide travelers with current information on vehicle arrival times, delays, and real-time service alerts. They can be implemented at transit centers and stations as well as at bus stops. These screens come in various types and formats, ranging from traditional LED signs to eco-friendly options like solar-powered or battery-powered electronic paper (e-paper) displays.

While LED displays have been traditionally used, many agencies have moved to energy-efficient e-paper (also known as electronic ink or e-ink) and solar technologies that do not require a connection to the electrical grid. In applications where high-visibility, low-power consumption, and readability in various lighting conditions are crucial, e-paper displays are an increasingly popular choice for transit agencies. Real-time information screens are commonly found at train and BRT stations, but agencies are now working with e-paper display vendors to more widely extend their use to bus stops.

Service Strategy	Support of Service
Fixed-Route Transit Service	E-paper or solar-powered arrival screens, as opposed to LED displays, eliminate the need for electric wiring, which makes it more cost-effective to deploy at individual bus stops.
Demand Response Service	Displays at fixed stops inform travelers of estimated vehicle arrival times.

To fully leverage the capabilities of e-paper displays, integration with CAD and AVL systems is necessary to enable the delivery of accurate real-time information about transit schedules and vehicle arrivals.

Bus Stop Displays—Sonoma County Transit, California

To improve customer service and streamline transfers, Sonoma County Transit (SCT) invested in battery-powered bus stop displays and partnered with vendor Connectpoint. “With our focus on improving customer service, solar-powered Connectpoint Digital Bus Stops have proven to be reliable and efficient to deploy, especially in rural areas where access to power is limited or too costly,” noted Bryan Albee, Transit Systems Manager, Sonoma County Transit.

Their digital bus stops using e-paper displays offer real-time departure and schedule information as well as service or emergency alerts. The signs can be retrofitted to existing bus stops or shelters and can be managed remotely via a system provided by the vendor.

SOURCE: Sonoma County Transit Invests in Sustainable ConnectPoint® Digital Bus Stops. *Connectpoint*, January 28, 2021. <https://connectpoint.com/sonoma-county-transit-invests-in-sustainable-connectpoint-digital-bus-stops/>.



For effective remote management and updates, e-paper displays require a reliable wireless (cellular or Wi-Fi) or Ethernet connection. This connectivity ensures that content can be remotely managed and refreshed, keeping the displayed information up-to-date and relevant. Where wireless access is limited, rural agencies may need to explore alternative solutions or invest in improving their connectivity infrastructure to fully leverage the benefits of e-paper and/or solar displays.

Destination Signs

Destination signs are electronic signage above the windshield that display the route number and destination of the vehicle to riders waiting for an approaching bus. A similar sign can be installed on the side next to the front door and on the rear of the bus for greater visibility.

Signs can also provide basic information about the service and pertinent information on events, advertisements, and notices to the user and the public. These systems can be controlled manually by the operator, typically by selecting options programmed from a centralized data feed.

Service Strategy	Support of Service
Fixed-Route Transit Service	Destination signs are an essential element of passenger information for fixed-route service.
Demand Response Service	Destination signs provide awareness of the type of service provided by a vehicle and can help facilitate the use of common vehicles for different services.

The controls for the destination signs can be integrated with CAD and AVL technology and other digital networks to automate sign operations. Integration with CAD and AVL is preferred, but it is not necessary to leverage the capabilities fully. In a rural context, it can be manageable for operators to manually switch the signage at the end of the route or the beginning of their shift. There is a range of options in destination signage, from liquid crystal display (LCD) screens to full-color LEDs, and basic monochromatic designs (see an example in Figure 9.2).



COURTESY OF YAMHILL COUNTY TRANSIT.

Figure 9.2. *Example destination sign.*

Interior Digital Displays

Onboard digital displays are integrated LCD or e-paper displays on vehicles used to provide information to users onboard the vehicle. The primary use of these displays is to update riders on the location and destination of the vehicle; these systems are often integrated with voice annunciators to better serve customers with vision impairments. In some cases, they can be integrated with camera systems to provide awareness of safety and security on the bus. A secondary use of the displays can be to promote, inform, and engage with riders on the vehicle, including general announcements and other nontransit-specific community information.

Service Strategy	Support of Service
Fixed-Route Transit Service	Interior displays require a small increase in operation and maintenance costs and require ongoing effort to ensure content is being updated.
Demand Response Service	Displays can be used to help users connect with resources in the community and provide relevant information about the transit service.

The controls for the destination signs can be integrated with CAD and AVL technology, and other digital networks to automate the sign operations. Integration with CAD and AVL is needed to provide location and destination information. In a rural context, a wireless connection at a maintenance facility is recommended to more easily manage the signs remotely. Several vendors are available; ensuring integration with the CAD/AVL vendor and other components is an important consideration.

Figure 9.3 shows an example from Yamhill County Transit in Oregon.

Electronic Fare Payment

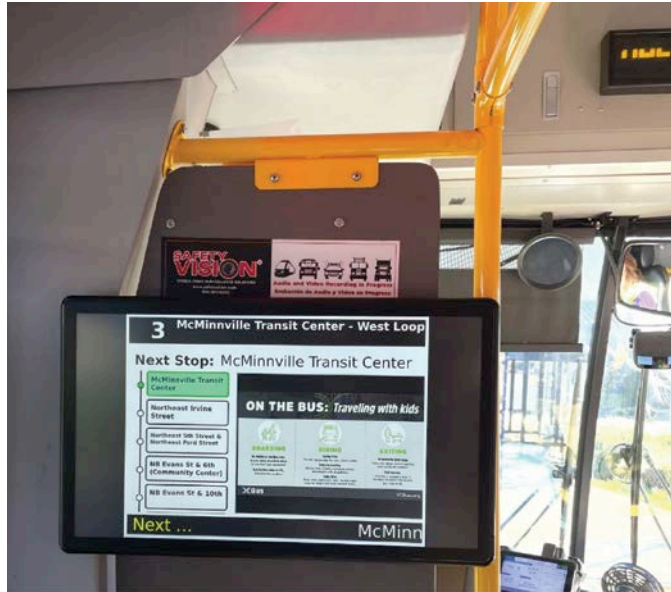
Automated Fare Collection

AFC describes any method of fare collection in which fares are not paid directly to a transit officer in exchange for entry to a vehicle. AFC can include a backend payment system provided by a vendor; ticket vending machines, automated fare collectors, and validators; and digital fare media. Digital fare media options can include transit smart cards with RFID (radio-frequency identification) chips, bank cards enabled for contactless payments, and mobile ticketing (including both mobile applications and payment using a linked debit or credit card).

The convenience of contactless and mobile payment fare payment technology can help agencies attract new riders to transit. Electronic fares can also help agencies increase efficiency, reduce fraud, and increase safety for operators on rural routes (due to no/reduced cash onboard).

Significantly, electronic fare payment reduces costs for processing cash fares, which can represent a significant share of fare collections.³⁷ It also enables easier administration of fare pass programs and discounted fares, including low-income or senior citizen passes, fare capping (i.e., limiting fares charged beyond a given number of rides per day or per month), and fare reciprocity agreements. Electronic fare payment can also provide agencies with anonymized information on transit ridership patterns that can be used to analyze and optimize service.

Challenges in adopting AFC technology include buses and bus stops that lack the space or infrastructure to host ticket/card vending machines and certain kinds of validators and fare collectors. Accommodating different fare collection types is important to riders who may lack access to banking or smartphone connectivity and can pay only with cash.³⁸ This concern appears more prevalent in rural contexts where comparatively fewer businesses have adopted smart wallet and cashless technology. However, electronic fare payment can make it easier for agencies to implement the types of discount programs described.



COURTESY OF NELSON\NYGAARD.

Figure 9.3. Interior multimedia display, Yamhill County Transit, Oregon.

Integrated fareboxes can accept cash and cashless payment types and transit smart cards, although accepting cash adds to the complexity and expense of the fareboxes.³⁹ Mobile ticketing that does not require onboard validation (such as displaying a ticket to the driver) is a mobile ticketing strategy that agencies can implement with low to minimal upfront cost (e.g., some vendors have only a transaction-based cost). Mobile fare payment applications may either be part of a vendor's app or be integrated into a more general-purpose transit mobile device app (such as Transit or Moovit). Another source of information on system and payment technology is the National Center for Applied Transit Technology (N-CATT) New Fare Payment Systems and Payment Technology Guidebook.⁴⁰

Pilot Smart Card Program—Yuma County Intergovernmental Public Transportation Authority (YCIPTA), Arizona

In 2016, the American Public Transportation Association (APTA) operated a 90-day pilot program in Yuma County to modify the smart card program and allow for online ordering and recharging of cards. This pilot was part of an effort to develop a standards-based system that could be more easily adopted by small and rural agencies. Because Yuma County does not fall under the scope of any metropolitan regional contactless payment system, there was debate over the efficacy of investing in new smart card infrastructure for a small system isolated from major cities. The pilot was deemed successful, however, and the modifications made by APTA were permanently integrated into YCIPTA's fare collection system. Yuma County Transit serves both a highly agricultural and a multicultural region, including several tribal communities. According to the County's 2021 Transit Plan, the County is nearly two-thirds Hispanic or Latino.

Pilot Smart Card Program—Yuma County Intergovernmental Public Transportation Authority (YCIPTA), Arizona (*Continued*)

The pilot identified best practices to overcome barriers to passenger adoption given the County's rural character and demographics, including providing point-of-sale terminals at retail establishments, offering limited-use cards, and offering larger fare discounts.

SOURCE: Allen, W. E., and Murray, R. D. 2016. Transit IDEA Project 79: Implementation of Smart Card Automatic Fare Collection (AFC) Technology in Small Transit Agencies for Standards Development. <https://onlinepubs.trb.org/onlinepubs/IDEA/FinalReports/Transit/Transit79.pdf>.

Service Strategy	Support of Service
Fixed-Route Transit Service	AFC improves efficiency and allows for multiple payment types.
Demand Response Service	AFC improves efficiency and allows for multiple payment types, improving ease of fare payment on vehicles, online, and in advance of trips.
ADA Paratransit	AFC improves efficiency and allows for multiple payment types, though contactless payment types are less commonly accepted on paratransit vehicles, partially due to lower adoption by an often older clientele.
Mobility Management and Coordination	AFC allows riders to travel between different modes and transit services more easily.
Fare Policy and Programs	AFC can standardize payment types and fares across modes.

Integrated Ticketing

Integrated ticketing builds on the foundation of AFC by allowing riders to travel across multiple modes or transit types on the same smart ticket or smart card. It allows travelers to connect to different modes within the same transit system (a typical application among urban agencies) or

Service Strategy	Support of Service
Fixed-Route Transit Service	Integrated ticketing allows riders to transfer between transit modes and service types with ease and facilitates longer trips on transit.
Demand Response Service	Integrated ticketing between fixed-route and demand response services encourages riders to use both modes.
Regional and Interagency Connections	Integrated ticketing across multiple regions and agency service areas allows riders to travel longer distances on a single transit ticket or payment type.
Mobility Management and Coordination	Integrated ticketing encourages multimodal transit and facilitates trip planning.
Fare Policy and Programs	Integrated ticketing can standardize payment types and fares across modes.

Coordinating Fares—15 Transit Agencies Across Ohio and Kentucky

EZ Fare combines 15 transit agencies across Ohio and Kentucky for riders' ease of transfer. Rather than managing tickets and apps across multiple agencies, EZ Fare allows riders to access ticketing information for all agencies on one application. Some of these transit agencies cover rural areas of the state otherwise unconnected to urban areas like Cincinnati or Columbus. Using AFC technology, value can be added to transit passes through the app and then scanned as barcodes by participating agencies for validation. The tap-and-ride system also allows riders to validate their transit pass as they board and as they exit a vehicle, regardless of which agency provides transit, with fares capped to prevent overcharging.

SOURCE: Transit blog. <https://blog.transitapp.com/four-transit-projects-putting-the-state-in-state-of-the-art/> and <https://www.neoride.org/post/projects-ezfare>.

Mobile and Integrated Ticketing Applications in Oregon

Columbia Area Transit (CAT), Sunset Empire Transit District (SETD), Tillamook County Transportation District (TCTD), Lincoln County Transit, Benton Area Transit—Several agencies serving northwest Oregon have adopted Token Transit as a common mobile ticketing platform that can facilitate connecting trips across these agencies along the Columbia River and Pacific Coast.

- Approximately 70% of CAT rides are paid through digital fares on Token Transit or through the Gorge Pass, which is digital or physical. There is a small transaction fee subtracted from CAT fares every time someone purchases a pass digitally. Initially, CAT implemented fares with a different provider, but it was able to easily transfer the fare collection and ticketing process with minimal upfront costs.
- SETD has implemented fare capping and programs for low-income riders, including reduced fare passes offered through the Token Transit platform. SETD's transit offices will set up in-person cash transit accounts for underbanked and unbanked individuals. Token Transit could ultimately allow people to travel on services operated by the five providers in the Northwest Connector alliance using the NWConnector Pass.

Cascades East Transit (Bend, Oregon)—CET has continued to operate fare-free following the COVID-19 pandemic, with the exception of its recreational services. Drivers can use a handheld reader to quickly validate QR codes on tickets purchased using the UMO mobile app. CET has used this contactless payment system from Cubic (previously Touchpass) for the past 6 years, which has also been adopted by several other providers in Oregon, making it possible for people to use the same system in multiple parts of the state. CET has also been able to make unused readers available to intercity service providers operating in its service area in Central Oregon, including the Central Oregon Breeze and Pacific Crest Bus Lines, enabling fare reciprocity.

SOURCES: Columbia Area Transit, Interview, October 7, 2022.

SETD, Personal Communication; September 30, 2022. Oregon DOT, Profile of Tillamook County Transportation District, <https://www.oregon.gov/odot/RPTD/RPTD%20Document%20Library/TCTD-NWOTA-Profile.pdf>; Cascades East Transit, Interview, September 26, 2023.

services offered by different transit providers using a common ticketing and payment interface. Rural transit systems and agencies are more likely to see a benefit from enabling riders to conveniently travel across service areas.

Safety and Security

Transit vehicle security systems include video cameras, silent alarms, and AVL. These technologies support transit agencies in monitoring and responding to situations onboard vehicles

and along transit routes and can be used in combination to pinpoint a vehicle's location during an incident and provide real-time assistance or record a comprehensive view of an incident.⁴¹

Security systems include surveillance, sensor, and alarm technology. Surveillance may use video or audio feed, and many systems also include recording capability. Sensor technology includes metal detectors and motion sensors. Alarms may be either driver-activated or user-activated.

Security systems may be located onboard vehicles, in public areas such as stops or transit stations, or in nonpublic areas, including maintenance yards, garages, or inside buildings. AVL technology allows for monitoring of vehicle location, which can result in reduced response times.⁴²

Data Management System

Data management systems, also called safety management systems, rely on data from various sources to ensure rider and operator safety. These systems integrate camera, alarm, and other vehicle information to identify, assess, and mitigate incidents while they are happening and then use records from each incident to monitor, measure, and evaluate how an agency responds to incidents.

The FTA requires that each urban agency implement a transit safety management plan and report incidents and safety events annually. This requirement was designed to promote a culture of safety within transit agencies. Such a plan is not currently required for nonurban agencies, but elements can be considered best practices and may be required by state DOTs.⁴³

Silent Alarms

Silent alarms, also called operator-activated panic buttons, are used by operators in-vehicle to alert authorities to the need for emergency assistance without alerting passengers.⁴⁴ Alarm responses can be configured in several ways depending on a system's safety protocol. Alerts can be sent directly to authorities for immediate assistance or to control centers that will then decide the appropriate response and contact the necessary authorities. In many systems, alarms also activate an exterior display so that passersby are aware of potentially dangerous situations on the bus and can also act or alert authorities.

When combined with AVL data and camera surveillance systems, silent alarms can ensure quick and targeted emergency action. AVL allows authorities to quickly locate vehicles. In Denver, studies found that the introduction of AVL and silent alarms in 1992 led to a 33% decrease in passenger assaults because police were able to intercede earlier.⁴⁵ Live surveillance systems allow dispatch to communicate the issue at hand to authorities so that they can be adequately prepared to respond. At Boston's MBTA, when an operator triggers a silent alarm, dispatch center operators immediately connect to the video surveillance systems to describe situations to responders.⁴⁶

In large transit systems, silent alarms may be triggered multiple times a day, though only a few situations per week will require emergency assistance or outside intervention. Through repeated events, larger systems have practiced identifying and responding to incidents and emergencies and have a clear and well-practiced safety protocol. In smaller transit services, operators may be less familiar with silent alarms and following the subsequent safety protocol.

Additionally, emergency response times are significantly slower in rural areas due to both a shortage of responders and longer distances between locations.⁴⁷ While silent alarms can still be useful, such operators may have to trigger panic buttons earlier to receive emergency assistance promptly. It is possible, though, that this could result in false alarms, misuse, or improper reactions.

Bus Stops

Several factors impact the safety and security of a bus stop or transit station. Lighting is a significant design element affecting safety at a bus stop. Proper lighting deters criminal activity and preserves rider safety.⁴⁸ Stops should be sufficiently lit from above to ensure riders are visible to

Service Strategy	Support of Service
Fixed-Route Transit Service	Silent alarms support fixed-route services by alerting responders to emergencies on vehicles.
Demand Response Service	Silent alarms support demand response service by alerting responders to emergencies on vehicles, even when the vehicles are not following fixed, more trafficked routes.
ADA Paratransit	Silent alarms support ADA paratransit by offering faster emergency response to riders with disabilities who may be more vulnerable to injury or incident on vehicles.

drivers and other passersby. Most stop and station lights use shelter- or pole-mounted solar LEDs to provide the industry standard of 1.5- to 2-foot-candles if a hard-wired electrical connection is not available.

Low-vision riders often report difficulties being noticed by bus drivers at night and may feel unsafe waiting at dark stops, in addition to not knowing if the bus will stop for them. Devices like “Spin to be Seen” use reflective materials to alert drivers of riders waiting at stops.

To increase transit network legibility for low-vision riders, stops and stations can also install plaques with Braille labels, tactile paving at boarding and alighting areas, or radio-frequency or Wi-Fi sensors to improve wayfinding and prevent accidents. Large transit agencies like the New York MTA have also piloted apps to translate visual wayfinding signage into auditory cues for low-vision riders (NaviLens).

Seating at stops increases comfort and stop accessibility, especially for older adults, riders with disabilities, and riders traveling with young children. Some areas with a significant unhoused population opt to remove benches or install architecture such as slanted benches, metal dividers, wave-design seating, or sonic Mosquito alarms to discourage sleeping or loitering at stops and inside bus shelters. This can be a point of contention in discussions of stop/station safety regarding which population is prioritized when considering the design of bus stops.

Public perception of a bus network plays a considerable role in its ridership. In regions where public transit is not a dominant travel mode, bus stops and stations are visible transit infrastructure and serve as a marker of service quality for non-riders.⁴⁹ Transit safety and potential rider accident prevention are also key in rural communities where emergency response services might not be timely due to distance and a lack of responders.

Rural areas typically have more stringent local light pollution regulations in addition to state-wide regulations to preserve night skies. These regulations can supersede the need for brightly lit stops and stations. The distance between stops in rural systems, especially for intercity buses, also makes wayfinding assistance technologies between stops harder to implement, which may mean transit becomes difficult for some low-vision riders.

Service Strategy	Support of Service
Fixed-Route Transit Service	Bus stops and stations are the primary boarding/alighting destinations for fixed-route service, and safe and well-designed stops and stations improve accessibility and riders’ perception of safety and security on transit.
Demand Response Service	This can also apply to fixed stops for demand response service, like a point deviation service.

Cameras

Transit agencies use in-vehicle and stop/station camera technology to increase safety and security of bus drivers and passengers and to build more confidence in their service. Surveillance systems reduce fraudulent claims, deter crime and vandalism, protect drivers, and reduce collisions.

In demand response transit systems that do not follow a fixed route, cameras are often the arbiter of rider complaints. Camera footage can show whether drivers arrived at the proper location for rider pickup and followed proper protocol on arrival.

Vendors offer network cameras and onboard surveillance. Older systems and buses may operate analog VCR technology, but this is increasingly uncommon as more systems are using digital video recorders (DVRs). Cameras on buses are commonly paired with a DVR that continually records and stores feeds from multiple viewpoints. The number of cameras and possible angles is limited by DVR capacity. The DVR computer is stored in the same cabinet as the VLU.

DVRs are often not included in CAD/AVL procurements, so drivers may still have to activate alerts or alarms themselves. In advanced CAD/AVL systems, DVRs can automatically detect emergencies, such as vehicle accidents or driver assaults, and alert dispatch.

Buses are subject to shock and vibration, have rapidly shifting light conditions, and have minimal storage space, all of which complicate camera installation and security. There are also debates over civil liberties compromised by audio-enabled cameras because most riders are unaware that camera technology on transit vehicles may also include audio recording.

Rural transit systems may have limited staff capacity to regularly review or continuously monitor camera footage, making video footage less effective for timely emergency response. Footage is still useful in resolving complaints or gathering additional information for onboard incidents.

Service Strategy	Support of Service
Fixed-Route Transit Service	Cameras record incidents and rider and operator behaviors for reporting, so that incidents can be properly reported and responded to.
Demand Response Service	Cameras record incidents but also resolve disputes about arrival and drop-off times, no-shows, and other common rider complaints.
ADA Paratransit	Cameras record incidents but also resolve disputes about arrival and drop-off times, no-shows, and other common rider complaints. They are also important for monitoring driver behavior and compliance with safety protocols when serving riders with disabilities, many of whom require sensitivity and operator assistance.

Surveillance Camera System—Fresno County Rural Transit Agency, California

Starting in 2015, the Fresno County Rural Transit Agency began a thorough upgrade of their surveillance camera system to address rider and vehicle safety concerns. While buses are en route, a combination of in-vehicle and dashboard cameras monitor riders, drivers, and traffic conditions. Stop/station and maintenance yard cameras round out surveillance. Both camera types allow the agency to monitor footage live and record feeds for later viewing. The cameras are said to increase riders' perception of safety at bus shelters, especially in the most rural communities where some common bus stops and routes are less visible to passersby.

SOURCE: Fresno County Rural Transit Agency, Resolution 2018-06. https://www.ruraltransit.org/wp-content/uploads/2018/06/FCRTA-6.28.18-Board-Agenda_Part2.pdf.

In-Vehicle Technology

Automated Passenger Counters

APCs are electronic scanner devices that can be installed near vehicle doorways to count boarding and alighting activity. APCs are used to conduct in-person surveys and ride check counts. When paired with CAD/AVL systems, APCs can provide boarding activity by stop location, which is valuable for planners. During the COVID-19 pandemic, APCs were used to estimate bus crowding and capacity for transit riders to meet social-distancing requirements, a function that remains important post-pandemic.

Service Strategy	Support of Service
Fixed-Route Service	APCs are easy to implement on fixed-route systems, where defined stop locations help produce cleaner data. APCs can help agencies understand fixed-route service performance, capacity constraints, and the highest ridership locations.
Demand Response Service	APCs are typically less applicable to demand response service because dispatch systems track pickup and drop-off locations.

APCs have technological limitations. Depending on the design, the infrared laser scanner may count any movement passing through it, which can include obstructions or vehicle operators’ entrances and exits. Newer generations of APCs are more capable of differentiating between people and objects (such as bicycles). Regardless, the FTA requires APC data interpretation to undergo certification for accuracy to be used in NTD reporting. This typically means cross-checking APC data against manual surveys or alternative software solutions. Some APCs now come with an internal camera for count verification purposes. The NTD Full Reporting Policy Manual provides a checklist for the certification process.

Electronic alternatives to APCs also exist in the form of video counting, where CCTV can be installed to monitor doorways, and software algorithms can then process the footage to count people. One best practice for agencies looking to procure APCs is to be specific in RFPs when defining accuracy requirements. Managing and cleaning APC data also requires staff time that may be beyond the capacity of rural agencies.

The cost of installing, integrating, and using small numbers of APCs and support hardware may not be justified depending on a transit system’s ridership and fleet size. At scale, the automation of ridership counting can greatly benefit an agency in labor hour savings; however, costs for implementation can range from \$1,000 to \$5,000 per vehicle, with an additional \$5,000 to \$10,000 for the requisite software licenses to interpret the data.⁵⁰ While not strictly required, additional work is often needed to integrate APCs with MDTs and CAD/AVL systems. Integrations can minimize driver interaction and maximize APC functions such as collecting real-time information and establishing rich databases of historical data. In procurement, agencies should ensure that vendors have a proven record of supporting agencies in achieving NTD certification and that their accuracy is quantifiable.

Overall, adoption of APCs among rural and tribal agencies has been low because of costs, technology, and staff limitations. Although dated, a 2010 nationwide survey of transit agencies receiving Section 5311 funds found that only 2% had APCs in use. This was a total of 10 agencies, from 451 responses across 45 states. Findings from that survey also reported that only 15% of respondents indicated any expectation of adopting APCs by 2015.⁵¹

Automatic Voice Annunciator Systems

An automatic voice annunciation (AVA) system typically consists of hardware components such as an in-vehicle computer and speaker system combined with software that has text-to-speech

capabilities and can store stop-level geographic information. Under U.S. DOT ADA regulations, fixed-route public transit services must provide announcements at transfer points, major intersections, and destinations. An AVA system has become a popular way to standardize this information and remove the task of making announcements from drivers.

Stop announcements are typically made when the transit vehicle reaches a certain distance from the stop (approach announcement), and then again on arrival (arrive announcement). Some transit systems also have exterior vehicle speakers for the benefit of passengers waiting at the bus stop.

Service Strategy	Support of Service
Fixed-Route Service	Except for detours, stop locations on fixed-route services are predetermined, which makes automation of stop announcements a relatively straightforward process. The AVA system improves accessibility and the availability of trip information for riders.
Demand Response Service	The applicability of an AVA system to demand response services would depend on the service design. Because automation requires defined stop locations, services that are point-to-point or have pre-defined destinations would be better suited to this technology.
Regional Connections	The AVA system is particularly beneficial in making transfers and connections known, especially for riders who have vision-based disabilities. Rural agencies often operate intercity routes, which would likely intersect with other transit providers.

The automated system requires real-time geolocation and route information to announce stops at the appropriate juncture. As such, integration with CAD/AVL systems is assumed, and an AVA system is often procured as part of a CAD/AVL package. Clarity and consistency in reading stop names are important for riders, and agencies should look to standardize stop information when establishing an AVA system. The CAD/AVL and AVA system vendor should also have an established and consistently updated audio library for easy text-to-speech conversions. Some software platforms allow custom announcements to be added at specific stops, such as to announce landmarks, destinations, and other routes for transferring.

Recently, text-to-speech fields have been added to the GTFS standard, which allows the AVA system to draw data from the GTFS feed rather than the vehicle's AVL in isolation. Using GTFS has many benefits, for example, it makes the AVA system consistent with other passenger information systems reliant on GTFS, such as mobile trip-planning information. With the introduction of the GTFS-flex extension, the AVA system can adapt to detours and route deviations and continue to provide stop announcements.

For rural and tribal transit agencies, the AVA system is complementary to CAD/AVL technology, assuming that fleet vehicles have onboard speaker systems. Having a GTFS feed is a plus, which can enable automated stop announcements for both fixed and demand response services. On demand response vehicles, announcements can be especially helpful for low-vision or older riders. Automation of demand response stop locations is best conducted through a GTFS-flex extension integration, or at least by having defined pickup and drop-off points with preprogrammed names.

Vehicles and Fleet

Within this category, technologies include driver support, autonomous vehicles, and zero-emission vehicles. The leading opportunities for rural and tribal transit providers in the short term are likely to include integrating driver support technologies into vehicle procurements and transitioning to zero-emission vehicles.

Driver Support

Driver support is a spectrum of assistance options designated by the SAE, ranging from no assistance (Level 0) to minor safety and driver comfort assistance to full vehicle automation (Level 5)(see Figure 9.4). SAE Level 1 and Level 2 both provide partial support but require drivers to be active and always have their hands on the wheel. Level 1 technologies like adaptive cruise control (ACC) and lane assistance provide continuous assistance with acceleration, braking, or steering. Level 2 support entails multiple Level 1 support mechanisms being used in tandem and the use of advanced driver assistance systems (ADAS) in certain, usually more dangerous, situations. This can include acceleration, braking, and steering support, all of which can be over-ridden by the driver.

SAE Level-1 support is available in some Sprinter passenger vans and Ford and Chevrolet 15-passenger van models, usually in the form of adaptive cruise, brake, or lane control. Cutaway buses and chassis-on-body minibuses can also be equipped with integrated Level-1 and Level-2 assistance technologies, with intelligent ACC being by far the most common. Most vehicles on the road, however, still operate at SAE Level 0 (no automation).

Although less-trafficked, rural roads could serve as opportunities to experiment with driver-supported vehicles, rural roads also provide distinct challenges. Roads may have low visibility and limited sight distances and can be unpaved or poorly paved, with limited or no lane markings. These environments pose a challenge because vehicles with any level of automation need to “see” a considerable distance ahead (twice their braking distance) to safely drive on highways.⁵² Slow-moving and large vehicles like tractors, farm equipment or transport vehicles, or large freight and grain trucks can also confuse assisted acceleration technology. Additionally, there is a larger risk of animals appearing suddenly on the road in rural contexts. Autonomous vehicle technology is currently attempting to address these situations.⁵³

Service Strategy	Support of Service
Fixed-Route Transit Service	Driver support can improve safety for riders and operators.
Demand Response Service	Driver support can improve safety for riders and operators, especially along less-trafficked roads with less predictable obstacles.
ADA Paratransit	Driver support can improve safety for riders and operators, which is especially important when riders are secured in wheelchairs and/or are more vulnerable to injury or likely to be injured in accidents.

Autonomous Vehicles

Autonomous vehicles operate at higher levels of SAE automation (SAE Levels 3 through 5). The autonomous vehicles on the market for public transit usage are predominantly SAE Level 4, or high driving automation. These vehicles require no assistance from drivers and are completely responsible for all driving and navigational tasks. Because they do not receive driving assistance from human drivers or operators, these vehicles are best suited to specific, geographically limited service areas and defined routes, such as airport shuttle services, small fixed-route bus services, intracity taxis, and intracity freight or delivery services.

SAE Level-4 vehicles have been deployed in various pilot studies in the United States and Europe.⁵⁴ These range from driverless robotaxis operated via ride-hailing apps in San Francisco to the Autonomous e-ATAK bus deployed by Michigan State University (MSU) in November 2021, which became the largest autonomous bus on U.S. roadways. Designed for MSU students and staff, the electric and autonomous bus completes its 2.5-mile run daily, though the university



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	SAE LEVEL 0™	SAE LEVEL 1™	SAE LEVEL 2™	SAE LEVEL 3™	SAE LEVEL 4™	SAE LEVEL 5™
What does the human in the driver's seat have to do?	You are driving whenever these driver support features are engaged – even if your feet are off the pedals and you are not steering			You are not driving when these automated driving features are engaged – even if you are seated in "the driver's seat"		
	You must constantly supervise these support features; you must steer, brake or accelerate as needed to maintain safety			When the feature requests, you must drive	These automated driving features will not require you to take over driving	

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	These are driver support features			These are automated driving features		
What do these features do?	These features are limited to providing warnings and momentary assistance	These features provide steering OR brake/acceleration support to the driver	These features provide steering AND brake/acceleration support to the driver	These features can drive the vehicle under limited conditions and will not operate unless all required conditions are met	This feature can drive the vehicle under all conditions	
Example Features	• automatic emergency braking • blind spot warning • lane departure warning	• lane centering OR • adaptive cruise control	• lane centering AND • adaptive cruise control at the same time	• traffic jam chauffeur	• local driverless taxi • pedals/steering wheel may or may not be installed	• same as level 4, but feature can drive everywhere in all conditions

SOURCE: SOCIETY OF AUTOMOTIVE ENGINEERS. [HTTPS://WWW.SAE.ORG/STANDARDS/CONTENT/J3016_202104/](https://www.sae.org/standards/content/j3016_202104/).

Figure 9.4. Levels of driving automation.

ensures a licensed bus driver remains onboard as an added precaution. However, these technologies remain out of reach even for most urban transit agencies as they are prohibitively expensive and financially risky, given the few examples of autonomous transit services currently operational.

Once developed, tested in different contexts, and expanded, Level-4 vehicles offer unique possibilities to small towns with small workforces or working-age populations. Prolonged post-pandemic driver shortages have affected most public transit agencies, with rural agencies experiencing some of the most severe shortages as rural populations age. Autonomous vehicles could allow rural agencies to respond to and rebound from driver shortages without cutting necessary service. This can only occur once autonomous vehicles become available at different scales and price points, however.

Many rural areas rely on demand response service and various specialized services to serve many trip needs. Autonomous vehicles are not an ideal fit for this kind of service. Between the boarding and alighting assistance typically needed for older riders and riders with disabilities and the need for flexible pickup and drop-off points, autonomous vehicles cannot provide an adequate level of service without an operator present on the vehicle. Agencies could look to hire support workers and staff rather than drivers with commercial driving licenses, which would help alleviate labor challenges for paratransit operators. However, there are many complexities around the staff member/support worker's ability to override the vehicle's autonomous functions, and training around this remains relatively untested.

Service Strategy	Support of Service
Fixed-Route Service	The consistent nature of fixed-route service is more conducive to autonomous vehicle programming. However, support staff would likely be needed to provide assistance.
Demand Response Service	Microtransit services of multiple sizes (individual taxis and larger vehicles) could use autonomous vehicles to expand services, but support staff would likely be needed to provide assistance.

Zero-Emission Vehicles

Unlike most transit vehicles in service today, zero-emission vehicles (ZEVs) use electricity or alternative fuel cells to power a battery instead of relying on internal combustion engines and nonrenewable fuels. These vehicles produce no exhaust beyond water vapor and oxygen, no pollutants, and no particulate matter.

Considerations for the two primary types of ZEVs are summarized as follows:⁵⁵

- **Battery-Electric Vehicles (BEVs).** BEVs exclusively use stored chemical energy in batteries without another propulsion source. The batteries are recharged by plugging into an electric power source or through conductive or wireless chargers. The majority of zero-emission vehicles in use are battery-electric ones due to the lower production, fuel, and maintenance costs. They represent most of the recent growth in both full-size and small electric vehicle (EV) bus adoption in the United States.⁵⁶ Vehicles currently have a range of 100 to 150 miles on a single charge, although some new vehicles coming onto the market have a range of 200 to 300 miles or more. To date, ZEVs have not been applicable to many rural contexts due to topography and infrastructure. BEVs can easily traverse hills, but driving uphill consumes a significant amount of energy, requiring vehicles to regularly recharge. This can prove especially difficult for rural transit systems with longer routes and distances between stops and bus depots.
- **Fuel Cell-Electric Vehicles (FCEVs).** Powered by an electrochemical reaction to convert compressed hydrogen gas to electricity, FCEVs emit only water vapor and heat. These vehicles

are not financially viable in many transit systems due to the higher upfront cost of fuel, vehicles, and maintenance [which can be similar to compressed natural gas (CNG) vehicles] along with access to a hydrogen fuel supply suitable for buses. Once vehicle costs are comparable to diesel or BEVs, the longer range of FCEVs may be better suited for long-distance trips typically operated by many rural and tribal providers; the refilling time for fuel cells is also significantly shorter than the charging times for BEVs. In addition, the Federal Bipartisan Infrastructure Law is providing funds to ramp up clean hydrogen fueling infrastructure.⁵⁷

For both types of EV technology, the cost of buses and charging or fueling systems may limit adoption.

Additional Resources

- Aamodt, A., K. Cory, and K. Coney. Electrifying Transit: A Guidebook for Implementing Battery Electric Buses. National Renewable Energy Laboratory. Washington, DC. 2021.⁵⁸
- Linscott, M., and A. Posner. *TCRP Research Report 219: Guidebook for Deploying Zero-Emission Transit Buses*. Transportation Research Board, Washington, DC. 2021.⁵⁹
- U.S. Department of Transportation. 2023. Charging Forward: A Toolkit for Planning and Funding Rural Electric Mobility Infrastructure.⁶⁰

From Diesel to Clean Diesel—Lane Transit District, Eugene, Oregon

While not technically zero-emission, there are energy types that can reduce vehicle emissions. Clean diesel vehicles use fuel derived from commercial byproducts like cooking oils to be used in traditional diesel engines. Lane Transit District in Eugene, Oregon, switched from diesel to a renewable diesel fuel source—R99 derived from fats and oil byproducts—so it could continue using existing vehicles while reducing particulate emissions by 61% and introducing no new carbon into the atmosphere. This could be a stopgap solution for smaller transit agencies such as those serving rural and tribal communities, without the funding to invest in BEVs or FCEVs and their charging and fuel infrastructure but do want to explore carbon reduction strategies.

SOURCE: Lane Transit District (LTD). 2025. <https://www.ltd.org/latest-news/ltd-switches-to-renewable-diesel/>.

Service Strategy	Support of Service
Fixed-Route Transit Service	Transit agencies can greatly reduce their emissions by implementing zero-emission vehicles for fixed-route services because they are typically the largest part of an agency's service.
Demand Response Service	Implementing zero-emission vehicles for demand response services may be a cost-effective strategy to reduce emissions because microtransit and deviation vehicles are typically smaller.



CHAPTER 10

Measuring Success

Introduction

Measuring the success of transit service requires a performance-measurement framework that provides a process for evaluating the service and determining whether the service is meeting its goals and objectives, and whether changes and improvements are suggested or needed.

What Is Involved?

Measuring the success of rural and tribal transit involves an understanding of the goals and objectives of the transit provider for its service, identification of measures that can describe how well the transit provider is meeting those goals and objectives, and the availability of data to calculate those measures. Each transit provider needs to establish its goals, determine specific objectives within those goals, and identify appropriate performance measures and targets. Such measurements support a transit provider's implementation of new or enhanced transportation services, allowing the transit agency to understand the impacts of the services and to take corrective actions if needed.

Measures most frequently used by rural and tribal transit agencies include those that are more easily tracked and calculated, such as ridership, trips per mile or per hour, operating cost per passenger trip (or trip subsidy), and operating cost per mile or per hour.

Increased ridership and efficient use of transportation funds are important for all rural and tribal transit agencies. These measures are useful for transit agencies to understand. It is also important for agencies to be aware of their comparability to other peer transit agencies.

However, transit agencies often have other goals that are not measured by these metrics and may even conflict with some goals. Performance measures that address only a few of the transit agency's goals will not provide a complete picture of how well it is performing and its value to the community.

Besides the efficient use of transportation funds, the FTA Section 5311 program also includes goals such as enhancing access in rural areas to health care, shopping, education, employment, public services, and recreation; improving mobility for all, including older adults, individuals with disabilities, and low-income individuals; increasing the availability of transportation options; and encouraging mobility management. Individual transit agencies may have identified additional goals for their service, including tribal providers who may have a unique set of goals, as the more common performance measures do not necessarily show how well agencies are meeting these other goals.

The transit literature includes various resources devoted to the topic of transit performance measurement that are useful for transit providers in developing and implementing their own performance-measurement process. While many focus on urban transit, there are several that are aimed at rural providers.

Literature Provides Useful Guidance

Two “how-to” reports are particularly useful for rural transit providers. They offer metrics that may be used, define a process for assessing the performance and impacts of rural transit, and offer support for telling your story:

- Performance Evaluation for Rural Transit Systems.⁶¹
- How to Assess Your Impact and Tell Your Story, Technical Brief.⁶²

TCRP Report 136: Guidebook for Rural Demand-Response Transportation: Measuring, Assessing, and Improving Performance includes how to measure rural transit performance and strategies for improving rural transit.⁶³

An important first step in the measurement process identified in various studies is to define goals and objectives or establish values to be achieved by the transit service. Goals should be easy to understand and measurable, so progress toward achieving those goals can be understood. However, performance measures are often chosen not because of their relationship to the transit agency’s goals and values but because of ease of measurement, availability of data, or other reasons.

A study of rural transit systems in North Carolina found that the performance measures the agencies used were not well-connected to their stated values or mission statements.⁶⁴ For example, most transit agencies included customer service as one of their values, but only about one-quarter of them included metrics related to customer focus. The study authors concluded that the lack of internal consistency could be due to a few reasons, including unawareness of the importance of establishing metrics consistent with their goals and values, difficulty in creating appropriate metrics, or perceived pressure to include metrics they thought the state wanted to see.

While not focused solely on rural transit, *TCRP Report 88: A Guidebook for Developing a Transit Performance-Measurement System* contains useful information about linking goals and objectives with performance measures.⁶⁵ Examples of measures for small transit systems and rural areas, including one from a small transit system with both fixed-route and paratransit service, are described. The small transit agency found that data on boardings and alightings, which were collected every 2 to 3 years, were particularly useful for assessing performance.

Traditional metrics typically address efficiency and effectiveness. These include measures such as ridership, trips per vehicle revenue mile (VRM) or vehicle revenue hour (VRH), operating cost per passenger trip, operating cost per VRM or VRH, and on-time performance, and some rural agencies use customer satisfaction surveys. These are key performance measures, along with safety metrics such as safety incidents per 100,000 vehicle miles. However, a more diverse set of metrics is needed to evaluate outcomes and success in achieving the transit agency’s goals.

Less Traditional Measures

The use of mobility indicators is suggested in one study as a way to evaluate the success and outcomes of rural transit. This study, done in Mississippi, concluded that performance measurement should include mobility indicators because of the unique environment in which rural transit operates.⁶⁶ These indicators include trip and passenger characteristics, which show how the rural transit provider is meeting the transportation needs of different types of rural residents.

This could include the number of trips by purpose. The mobility indicators can also be used for marketing purposes.

Other less traditional measures to assess the impact of rural transit include the rate of use by older adults or people with disabilities, percentage of medical trips or trips to congregate meal sites, number of employment trips, number of individuals using the service for independent living, and passenger feedback on service quality.

Rural transit agencies may need to focus on outcomes that show how the rural transit system affects the community's overall quality of life.

Livability is a term used to describe the quality of life in a community. The extent to which rural transit improves livability is an important indicator of success. According to Edrington et al. (n.d.),⁶⁷ performance measures that can be used to show how rural transit contributes to livability include the following nontraditional metrics:

- Local operating investment per operating expense
- Ridership per index of transit need population
- Revenue vehicles per total rural land area
- Percentage of workers who did not drive alone to work
- Household income after transportation plus housing expenses
- Ridership per developed land area

Peer Comparisons

Performance metrics are useful for transit agencies for tracking performance and progress over time, but they can also be used for comparison with other similar transit systems. However, each rural system operates in its own environment, and differences in various uncontrollable factors will influence the performance of the system and should be recognized.

A more sophisticated evaluation of the outcomes within a community can be obtained through a cost-benefit analysis. Research for the Minnesota DOT documented ways that transit benefits riders as well as communities and the larger society. The researchers developed measurement techniques, collected survey data to estimate benefits, and developed a tool to allow rural transit agencies in the state to estimate the benefits of their services and calculate benefit-cost ratios (Mattson et al. 2020⁶⁸; Mattson and Peterson 2021⁶⁹).

How to Support New and Emerging Service Strategies

Transit providers implement new or enhanced services because they recognize a need. These services or programs are introduced with specific goals to address those needs. Evaluation is an important part of the process for improving service delivery and achieving goals. Program evaluation is referred to as a systematic method for collecting, analyzing, and using data to examine the effectiveness and efficiency of programs and to contribute to continuous improvement. Such an evaluation helps organizations better understand their performance and guides them in making better decisions so they can achieve their goals.

Transit agencies first need to establish clear goals and objectives for new or enhanced services, and they need a way to determine whether they have accomplished those goals. As identified in the literature and by respondents to the research project's survey, there are many types of measures and data that transit agencies can collect and analyze to show if they are meeting their goals.

Common operational measures to collect include ridership, VRM, VRH, and operating costs. These data must be reported to the NTD for transit agencies receiving Section 5311 funding. Efficiency measures can be calculated based on these data, such as trips per revenue mile, trips per

revenue hour, cost per trip, cost per revenue mile, and cost per revenue hour. Other measures that address quality of service include on-time performance and trip denials. Measuring trip denials is important for understanding the amount of demand that is not being met due to a lack of capacity. Safety can be measured based on the number of incidents per 100,000 vehicle miles.

Measuring the impact on the community requires additional nontraditional measures. Data on the number of trips per purpose shows how the service is improving access to different activities or destinations. For example, the number of health care trips shows how the service is improving access to health care and positively impacting health and quality of life in the community. The number of work trips shows how the service helps people maintain employment and helps businesses attract labor. The number of shopping trips shows how the service supports local businesses. See the example of how the Comanche Nation Transit focused on new measures of success following their experience during the COVID-19 pandemic.

Data on the characteristics of users shows the impact the service has on older adults, people with disabilities, low-income individuals, and others who lack private transportation. Transit agencies that serve many of these individuals can improve their access to health care, work, and other activities. Data on user characteristics and trip purposes can be collected through rider surveys and evaluated by looking at common destinations.

Because the goal of new or enhanced services is to address needs within the community, feedback from community members is important for measuring progress toward this goal. This could include not just quantitative data, but also qualitative data based on comments received. The method for collecting this information should be systematic, such as annual customer surveys or regular meetings with stakeholders and the community. Annual rider surveys help identify service gaps, rider needs or desires, rider satisfaction, and problems. Transit agencies often use the information collected from these surveys to improve their services and achieve better outcomes. For example, one transit agency commented in the research project's survey that they received several complaints from riders about needing to schedule rides days ahead for simple trips such as going to the bank or buying groceries. In response, the transit provider started an in-town, same-day service, and their ridership tripled within a year.

Data collected through this evaluation process can have multiple purposes. First, it shows if the transit agency is meeting its goals. Second, it provides feedback on how the transit system can improve its service to achieve better outcomes. Finally, it provides information that can be used for promoting services in the community.

Example of Measuring Success

Comanche Nation Transit

Background

Comanche Nation Transit is a tribal transit service providing transportation services in several communities in southwest Oklahoma. They provide a demand response service, open to the public, 5 days a week. Services are headquartered in the towns of Lawton/Fort Sill and Anadarko, and services are also provided to several surrounding communities.

Setting Goals, Measuring Outcomes, and Responding to Needs

Comanche Nation Transit follows a process of identifying needs, setting goals, observing their progress toward meeting those goals, and responding to needs in the community to continually improve service. The COVID-19 pandemic played a large role in altering those goals and expectations.

Comanche Nation Transit's general goal is to better assist clients and meet the needs of the community. One way they measure this is the VRM of service. Before the pandemic, they had a target goal for VRM of service. To achieve that goal, they made some changes to the way they scheduled clients and their service hours, and they focused on purchasing ADA-accessible vehicles.

As a result of these changes, they saw a dramatic increase in dialysis trips, and medical trips in general increased. They were able to exceed their VRM goal, and the large increase in trips showed that there was a need in the community that previously was not being met. The data showed that the service was successful, measured by metrics such as vehicle miles, ridership, medical appointments and work trips made, and the number of new clients, but another result that showed that the service was having a positive impact in the community was that they were being talked about as “the Comanche Bus, the one that helps get people to work and medical appointments.” Community satisfaction is an important measure of success.

When the COVID-19 pandemic hit, circumstances changed, and achieving their original goals became more challenging. Agency staff brainstormed how the pandemic could affect their goals and how they could respond. They decided to refocus their goals. While service was limited, the agency focused on getting people to appointments and work, which involved teaching dispatchers how to better schedule trips and expanding service in one rural area. They did not meet their original goal, but they did better than anticipated because of the changes they made. As the pandemic subsided, service picked up again, and, in 2023, they not only surpassed their goal for VRM, they recorded the most VRM in the transit system's history.

Comanche Nation Transit tracks VRM and the number of clients served per day as a way of measuring how well they are meeting the needs of the community. They also conduct outreach within the community to understand how satisfied clients are with the service and where there are additional needs. This outreach includes talking with riders to understand their satisfaction with the service and engaging with the community to learn if people can make the trips they want or need to make.

Lessons Learned

One of the lessons learned from the pandemic is to be flexible and have the willingness to learn and try new ideas. They found that success can be measured by how often your service is requested and how people talk about your system in the community. Goals should be specific, measurable, achievable, realistic, and time-bound, and, once those goals are achieved, a transit agency should not settle but should set more goals to continuously improve and make a bigger impact. While Comanche Nation Transit uses data to measure success at achieving goals, as a tribal transit system, their success is ultimately demonstrated by seeing their clients being able to go where they want and need to go. They can see this by being engaged in their community. They found it is important to build relationships within the community, to market and communicate your service, and to go to or set up community meetings.

CHAPTER 11

Funding and Revenue

Introduction

Rural and tribal transit providers need adequate funding to support their operations and remain sustainable over the long run. No rural or tribal transit system can operate effectively without adequate funding and likely needs even more ingenuity and flexibility than urban transit providers to identify and effectively leverage revenue from federal, state, and local sources.

Rural and tribal transit providers often rely on revenue sources from multiple grantors and sponsors at the federal, state, and local levels (see Figure 11.1), and each may have varying rules, regulations, and reporting requirements. The two main sources of funding are local (41% of revenues) and federal (40% of revenues), with state funding accounting for just less than 20% of rural transit revenues.

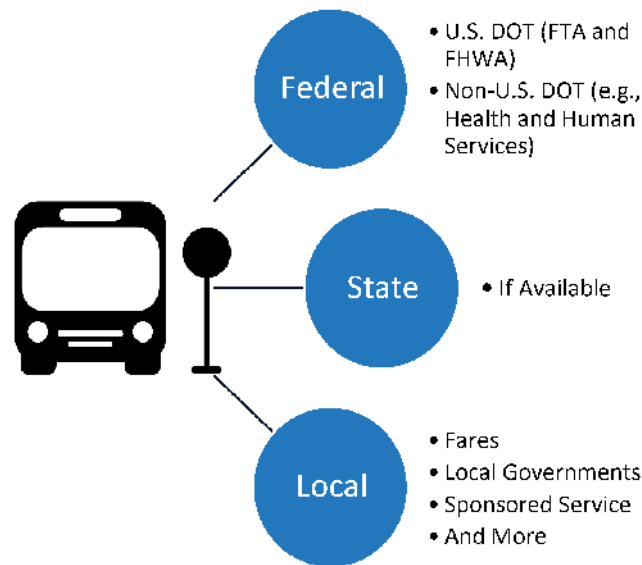
Although federal funding is a significant source of support for rural transit systems, state and local funding—both from local governments and revenues generated directly by transit systems (sponsorships and advertising) and fares—is also a critical part of the rural transit revenue picture, with local funding being a particularly important component.

For rural transit systems to succeed, identifying adequate and reliable sources of local revenue is pivotal. Not only does local revenue support expenses, but it also provides a much-needed nonfederal share of federal grants. Last, state revenues are also an important part of the picture. Most states (except Alabama, Hawaii, Nevada, and Utah⁷⁰) provide some form of state funds for public transit, with varying levels of funding and types of transit projects supported.

Some rural transit systems have had success in generating sponsorship or advertising revenue for their systems. Major retailers have been known to sponsor service and, at times, fund service at their locations. In one small city (with a population of 25,000) in Texas, the system generated \$150,000 in sponsorships. In Michigan, a large retailer purchases transit service open to the public. In Idaho, a casino supports the rural transit system with local funding.

This chapter provides basic information about sources of funding for rural and tribal public transit, with a focus on how to best acquire, use, and leverage funding to implement a new transit service or service strategy. This chapter is organized as follows:

- Under the heading What Is Involved?, federal funding programs for rural and tribal public transit are described, including those provided by the U.S. DOT as well as other federal departments.
- Under How to Support New and Emerging Service Strategies—A Focus on Funding, effective practices for maximizing federal funding are explained, and other funding sources and strategies are discussed.



SOURCE: ADAPTED FROM TEXAS A&M TRANSPORTATION INSTITUTE
MAKING DOLLARS AND SENSE OF TRANSIT FINANCE (2023).⁷¹

Figure 11.1. Sources of revenue to support public transit.

This chapter does not provide much information on state funding for transit because of the significant variety of revenue streams, grant programs, and funding allocation methodologies across states.

What Is Involved?

Federal funding is a significant resource for rural and tribal public transit and comes in many different forms, including grant programs administered by the FTA, grant programs outside of the FTA but still inside the U.S. DOT, and other non-U.S. DOT federal programs that support transportation.

Federal funding for rural areas primarily comes from the FTA through the Section 5311 program (Formula Grants for Rural Areas), which includes several subprograms, such as Rural Formula Funding, Tribal Program, Appalachian Program, and National RTAP.

In addition to Section 5311, FTA also apportions funds to rural areas through the Section 5310 program (Enhanced Mobility of Seniors and Individuals with Disabilities) and the Section 5339 program (Bus and Bus Facilities).

However, the world of federal funding for rural and tribal public transit can be daunting with all the different programs, regulations, and guidance. To help guide rural and tribal transit providers, this section is divided into the following topics:

- Foundational Concepts for Federal Funding Programs
- FTA Grant Programs
- Other U.S. DOT Programs
- Non-U.S. DOT Programs

Foundational Concepts for Federal Funding Programs

Before looking at specific federal grant programs that support public transit, it is important to understand a few foundational concepts related to federal funding programs.

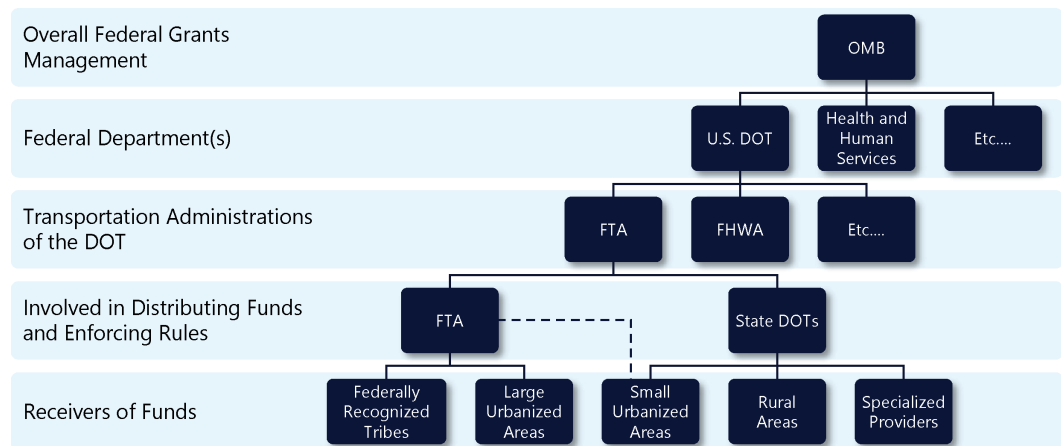
First, when dealing with any type of funding, especially federal, it is essential to understand the regulatory framework guiding the receipt and use of those funds. In the regulatory framework surrounding transit, there are three levels: law, regulation, and official guidance/policies, as shown in Table 11.1. All three levels of the regulatory framework could exist at the federal, state, and local levels, depending on a transit provider's operating context.

A second foundational concept for federal funding programs for transit is understanding the entities engaged in transit regulation and federal funding. Figure 11.2 displays a snapshot of the chain of command regarding FTA grant programs. Other federal departments beyond the U.S. DOT support transportation; however, to show these in the figure would create too much detail.

As shown in Figure 11.2, the Office of Management and Budget (OMB) is responsible for overall management of federal grants. OMB sets the regulations for federal awardees, and the bulk of OMB's rules are documented in 2 CFR 200. Federal departments like the U.S. DOT, the Department of Health and Human Services, and others are the executive departments of the federal government that are the administrators of federal grant programs that are authorized by law.

Table 11.1. Transit funding regulatory framework.

Type of Document	What It Is	Sources	Federal Example
Law	The law that has been adopted by the appropriate governing bodies.	Federal: United States Code (USC) State: Your state's code of law Local: County or city code	49 USC 5311: Formula Grants for Rural Areas. Contains the laws governing the Section 5311 program.
Regulation	Rules established for enacting law by executive/administering departments. These are not the same as the law, but they are the rules to be followed based on the administering department's execution of the law. Regulations have authority because they are based on a law.	Federal: Code of Federal Regulations (CFR) State: Your state's administrative code or code of regulations	2 CFR 200, also known as the Super Circular. Contains the rules governing federal grant awards. 49 CFR 604, Charter Service. Contains the specifics of the rule that restricts recipients of FTA grants from operating charter service. The authority for this rule comes from 49 USC 5323(d).
Official Guidance/Policies	More detailed instructions from an entity (e.g., FTA) for implementing its programs and following its rules. Guidance has authority because it comes from the entity administering the grant program, which follows established laws and regulations.	Federal: FTA Circulars State: DOT policies, state management plans, and other state plans	- FTA Circular 5010.1E, Award Management Requirements. Contains general requirements for recipients of FTA grants. - FTA Circular 9040.1G, Formula Grants for Rural Areas: Program Guidance and Application Instructions. Contains specific requirements for recipients of Section 5311 grant funds.



Notes: OMB is not technically a direct supervisor of federal departments; however, OMB does have a supervisory role. The dotted line between FTA and small urbanized areas signifies the fact that most small, urbanized areas are direct recipients of FTA urban funding.

SOURCE: TEXAS A&M TRANSPORTATION INSTITUTE MAKING DOLLARS AND SENSE OF TRANSIT FINANCE WORKSHOP (2023).

Figure 11.2. Agencies involved in transit regulation and funding.

U.S. DOT consists of several administrations, including the FTA and the FHWA. These administrations, including the FTA, are responsible for direct administration and oversight of specific U.S. DOT grant programs. However, the FTA does not work alone—state DOTs distribute grant funds and enforce rules for some grant programs and service areas; specifically, state DOTs are responsible for oversight and funds apportioned for small urbanized and rural areas (e.g., Section 5311).

Therefore, depending on the type of area and transit provider, as well as the specific federal grant program, a transit provider may have multiple oversight and funding entities to be aware of and adhere to. For example, a typical rural transit agency receiving Section 5311 funds will need to know and follow at least the following:

- Their applicable local laws and regulations.
- State DOT laws, regulations, and policies regarding the allocation and use of Section 5311 funds.
- FTA regulations and policies regarding the use of Section 5311 funds and those generally applicable to all FTA grant programs.⁷²

A third foundational concept is that there are two types of grants: (1) formula and (2) competitive. Formula grants are those where some type of formula (usually contained in law or regulation) defines how funds will be distributed to recipients. Formula funding is the most consistent and reliable source of funding for transit. Competitive grants (sometimes called discretionary grants) are those where applicants compete for funding. Transit providers submit applications to the funding agency, and the funding agency decides which applicants are awarded, usually following some form of scoring or evaluation criteria. Competitive grants can be an excellent source of revenue and are often used for low-frequency, higher-cost projects (e.g., building a transit facility) but can also provide much-needed funding to rural and tribal systems for more traditional, ongoing needs (e.g., replacing aging vehicles).

A fourth foundational concept is that federal grant programs support specific activities. Often called eligible activities, these activities supported by the grant typically fall into four main categories:

1. **Operating assistance.** Operational costs (less operating revenues like fares).
2. **Capital.** Procuring or improving capital assets like vehicles and facilities.

3. **Administrative.** General administration costs like management salaries, liability insurance premiums, and office supplies.
4. **Planning.** Studies, peer reviews, and the exchange of information, feasibility assessments, and other technical activities.

Each of these grant activity categories contains a significant number of activity line items (ALIs), which are the specific cost types in a grant budget. The wording for an ALI depends on the granting agency (e.g., FTA or the state DOT). However, to provide some examples, the capital grant activity could include the following:

- Preventive maintenance
- Procuring rolling stock
- Capital cost of purchased transportation

Understanding what grant activities are supported by each grant is crucial to being able to fully leverage federal funding and to ensure a transit agency is only seeking reimbursement of allowable costs under the specific grant. For example, if a grant only supports capital expenses, then seeking reimbursement for operating expenses under that grant is not allowed.

Another foundational concept associated with federal funding is the concept of nonfederal share. Virtually all federal grant programs require that some portion of a project supported by a federal grant also be supported by nonfederal funds (often called match or local match). Federal grant programs typically express the requirement for nonfederal share as the percentage of the project cost that is not covered by the grant's maximum federal share. For example, most capital activities under FTA grants have an 80% maximum federal share. That means that the remainder of revenue for the capital activities funded by the grant has to come from nonfederal sources (e.g., state funds and local contributions).

Consider the example of buying a transit vehicle. In this example, the vehicle costs \$200,000. If the transit agency is using an FTA grant to buy that vehicle, the maximum federal share of the vehicle expense is 80% of \$200,000, or \$160,000.⁷³ That means the transit provider will need to supply the remaining \$40,000 (20%) with nonfederal funds (e.g., a grant from the state DOT).

The maximum federal share varies across grant programs and grant activity but generally follows the values shown in Table 11.2.

Operating assistance is limited to a 50% federal share of net operating expenses, which means that fare revenue and other sources of program income must be applied to operating expenses first before federal operating assistance is sought. Federal operating assistance can only be applied to the operational expenses that remain after fares are applied.

Table 11.2. Typical maximum federal shares under FTA grants by grant activity.

FTA Grant Program	Operating Assistance	Capital	Administrative	Planning
Section 5311	50%	80%	80%	80%
Section 5310	50%	80%	N/A	N/A
Section 5339	N/A	80%	N/A	N/A

Note: The values in the table are representative of typical federal share maximums and can serve as a rule of thumb to aid in general understanding. Certain exceptions and restrictions exist, however, and will be discussed in the FTA Grant Programs section. Also, if a transit agency is receiving an FTA grant as a subrecipient (e.g., under a state DOT), the entity passing the FTA funds to your transit agency can set restrictions on the types of activities supported. For example, some states do not allow Section 5311 subrecipients to seek an 80% federal share of administrative costs—all costs must be reported and reimbursed as operating.

Another foundational concept is that not all costs a transit provider might incur are allowable to be paid for (i.e., reimbursed) by a federal grant. A full discussion of allowable and nonallowable costs is beyond the scope of this Guide; however, the main point is that, for a cost to be able to be reimbursed by a federal grant, the following must apply:

- It must be a reasonable and necessary cost for the provision of transit service.
- It must be a cost that aligns with a grant activity and ALI supported by the specific federal grant.
- It must not be expressly nonallowed as defined in law and/or regulation (e.g., 2 CFR 200.420 *et seq.*).

Chapter 5 of National RTAP's *Fundamental Financial Management for Rural Transit Providers*⁷⁴ contains a more robust discussion of cost allowability under an FTA award.

A final foundational concept regarding federal funding for transit is that using federal funds requires compliance with other federal requirements in addition to those discussed so far. For example, any recipient of federal funding for transit may also need to adhere to requirements for Title VI, the ADA, drug and alcohol testing, and NTD (and perhaps state) reporting.

These requirements are usually expressly defined in a grant agreement, and you should carefully read and make sure you understand the requirements.

FTA Grant Programs

This section provides an overview of the main FTA grant programs that support rural and tribal public transit: Section 5311, Section 5310, and Section 5339.⁷⁵ For each of these grant programs, the purpose of the program, typical recipients of the grant program (including which entities are involved in getting the funding to transit providers), eligible activities, federal share maximums, and any other special information relevant to rural and tribal transit providers are described. The latest FTA circulars and links for access to the grant programs are also noted.

It should be noted that specific states are eligible for sliding-scale federal share maximums as outlined in the Section 5311 circular. The federal share maximums presented in this chapter ignore those sliding-scale rates.

Note that the information contained in this section is not a replacement for and is not intended to supersede information, guidance, or policy of federal or state agencies. Readers are encouraged to consult FTA circulars and state documents on their own and contact their FTA or state DOT representative for additional clarity and guidance on all these grant programs.

Section 5311: Formula Grants for Rural Areas

Information for this section was obtained from several sources, including FTA's Section 5311 web page.⁷⁶

The Section 5311 program includes several subprograms:

- National Rural Transit Assistance Program (National RTAP)
- Appalachian Development Public Transportation Assistance Program (ADTAP)
- Tribal Transit Program
- Rural Formula Program, which also includes the Intercity Bus Program

The following information is generally applicable to the rural formula program unless stated otherwise.

Recipients. Rural areas and federally recognized tribes. Under the rural formula program, FTA apportions funds to states that then allocate funds to subrecipients. Typically, state DOTs are the entities that allocate funds to rural providers in the state (including to tribal operators),

and state DOTs have their own methods for determining how to allocate funds. Federally recognized tribes are also eligible to receive funds directly from FTA under the Tribal Transit Program, which is a subprogram of Section 5311.

Eligible Activities and Federal Share Maximums. Table 11.3 displays the eligible activities and federal share maximums for the Section 5311 rural formula program.

Rural transit providers should contact their state DOT to get more information about how their state manages and allocates Section 5311 funds received from the FTA.

The Tribal Transit Program contains two subprograms:

1. **Formula Program.** Distributed to federally recognized tribes providing public transportation on tribal lands.
2. **Discretionary Program.** Distributed competitively. Tribes must respond to an FTA call for projects.

Under the Tribal Transit Program (TTP) formula program, federal funds can be 100% of project costs (no local match required). The TTP discretionary program has a federal share maximum of 90% (except for planning grants).

States eligible for ADTAP may use the funds for operating assistance (50% maximum federal share), capital projects (80% maximum federal share), and planning activities.

For more information, see FTA Circular 9040.1H⁷⁷ (or the latest version), Formula Grants for Rural Areas: Program Guidance and Application Instructions, and the FTA Section 5311 website.⁷⁸

Section 5310: Enhanced Mobility of Seniors and Individuals with Disabilities

Information for this section was obtained from several sources, including FTA's Section 5310 web page.

Recipients. Urbanized and rural areas can include nonprofit organizations providing transportation. FTA apportions Section 5310 funds to designated recipients in large urbanized areas (UZAs).

Table 11.3. Section 5311 rural formula program grant activities.

Grant Activity	Federal Share Maximum(s)	Details/Notes
Operating Assistance	50% of net operating expenditures	—
Capital	• 80% generally. • 85% vehicles for ADA or Clean Air Act (CAA) compliance. • 90% of vehicle-related equipment is related to compliance with ADA and CAA. • 90% for projects related to bicycles.	Certain operational expenses are reimbursable as capital expenses, including ^a • Preventive maintenance of vehicles and facilities. • Mobility management. • Capital cost of contracting. • ADA paratransit operations (Up to 10 percent of awarded funds without restriction. Up to 20% of awarded funds under certain circumstances).^b
Administrative^c	80%	—
Planning^c	80%	

^a Some states may restrict the use of federal capital assistance for these activities.

^b See 49 USC 5302(4)(I).

^c Some states may not allow the use of federal assistance for these activities.

FTA apportions Section 5310 funds for rural areas and small UZAs to states. Typically, state DOTs then allocate their received Section 5310 funds for rural and small UZAs using their own formula-based and/or competitive processes.

Eligible Activities and Federal Share Maximums. Under the Section 5310 program, projects are divided into traditional projects and nontraditional projects.

At least 55% of Section 5310 funds must be used on traditional projects. Traditional projects are capital projects planned, designed, and carried out to meet the needs of older adults and people with disabilities when public transportation is insufficient, unavailable, or inappropriate, for example:

- Buses and vans.
- Wheelchair lifts, ramps, and securement devices.
- Transit-related information technology systems, including scheduling/routing/one-call systems.
- Mobility management programs.
- Acquisition of transportation services under a contract, lease, or other arrangement.⁷⁹

The remaining 45% of Section 5310 may be used for either traditional projects or nontraditional projects. Nontraditional projects are capital or operating projects that exceed the requirements of the ADA, improve access to fixed-route service, and decrease reliance by individuals with disabilities on ADA complementary paratransit, or alternatives to public transportation to assist older adults and individuals with disabilities, for example:

- Travel training.
- Volunteer driver programs.
- Construction of an accessible path to a bus stop, including curb cuts, sidewalks, accessible pedestrian signals, or other accessible features.
- Improvement of signage or wayfinding technology.
- Incremental cost of providing same-day service or door-to-door service.
- Purchase of vehicles to support new accessible taxi, ridesharing, and/or vanpooling programs.
- Mobility management programs.

Projects being funded with Section 5310 funds must be included in a regional public transit and human services transportation plan.

Table 11.4 displays the eligible activities and federal share maximums for the Section 5310 program.

For more information, see FTA Circular 9070.1G⁸⁰ (or the latest version), Enhanced Mobility of Seniors and Individuals with Disabilities Program Guidance and Application Instructions, and the FTA Section 5310 website.⁸¹

Section 5339: Grants for Buses and Bus Facilities

Information for this section was obtained from several sources, including FTA's Section 5339 web page.⁸²

The Section 5339 program contains several subprograms:

- **Formula Program:** FTA apportions funds by formula to recipients of Section 5307 and Section 5311 funds.
- **Bus and Bus Facilities Competitive:** Funding is competitive and is for capital expenses related to procuring or improving buses or bus facilities.
- **Low- or No-Emissions Competitive:** Also called the "Low-No" program, funding is competitive and is for capital expenses related to procuring low- or no-emission buses and related facilities.

Table 11.4. Section 5310 program grant activities.

Grant Activity	Federal Share Maximum(s)	Details/Notes
Operating Assistance	50% of net operating expenditures.	Only for eligible Section 5310 nontraditional projects (e.g., costs for exceeding the requirements of the ADA).
Capital	80% generally. 85% vehicles for ADA or Clean Air Act (CAA) compliance. 90% of vehicle-related equipment is related to compliance with ADA and CAA. 90% for projects related to bicycles.	Certain operational expenses are reimbursable as capital expenses, including^a - Preventive maintenance of Section 5310-funded vehicles. - Mobility management. - Capital cost of contracting. - Costs that support ADA paratransit (restrictions apply).^b
Administrative	Not applicable in most cases.	Administrative costs for voucher or volunteer programs are an eligible expense.
Planning	Not applicable in most cases.	Statewide planning benefiting rural areas may be funded at a 100% federal share.

^a Some states may restrict the use of federal capital assistance for these activities.

^b Expenses for ADA paratransit must be for eligible subrecipients and projects in a coordinated plan (not simply supporting a transit agency's required ADA paratransit service).

Recipients. State or local governmental entities and direct recipients that operate fixed-route bus service.⁸³ FTA apportions funds to large UZAs, and designated recipients in large UZAs apply for and receive funds directly from FTA. FTA apportions funds for small UZAs and rural areas to states. Typically, state DOTs are the entities that receive the funds and then allocate or award Section 5339 funds to rural providers in the state.

Eligible Activities and Federal Share Maximums. Eligible activities vary slightly by Section 5339 subprogram; however, they include capital projects to:

- Replace, rehabilitate, and purchase buses, vans, and related equipment.
- Construct bus-related facilities, including technological changes or innovations to modify low- or no-emission vehicles or facilities.

Table 11.5 displays the eligible activities and federal share maximums for the Section 5339 program.

For more information, see FTA Circular 5100.1A⁸⁴ (or the latest version), Bus and Bus Facilities Program: Guidance and Application Instructions; the Section 5339 fact sheet; and the FTA Section 5339 website.⁸⁵

Other FTA Programs

Although not a significant portion of federal funding for rural and tribal areas, these additional FTA programs are worth mentioning.

- **Ferry Service for Rural Communities.** A competitive grant program authorized under the Infrastructure Investment and Jobs Act (IIJA) Section 71103.⁸⁶ Funds are awarded to states and can be used for capital and operating assistance for a ferry service that operated a regular schedule at any time during the 5-year period ending March 1, 2020, and served not less than two rural areas located more than 50 sailing miles apart. This program could help support existing ferry service in rural areas.

Table 11.5. Section 5339 program grant activities.

Grant Activity	Federal Share Maximum(s)	Details/Notes
Operating Assistance	Not applicable.	Not allowed under Section 5339.
Capital	• 80% generally. • 85% vehicles for ADA or Clean Air Act (CAA) compliance. • 90% of vehicle-related equipment is related to compliance with ADA and CAA. • 90% for projects related to bicycles.	• Applicants for the low/no program can propose partnerships with other entities, satisfying the competitive procurement requirements under 49 USC 5325. • Applicants for the competitive programs must submit a zero-emission fleet transition plan with their applications for projects related to zero-emission buses. • Five percent of all grants for competitive grants related to zero-emission vehicles or related infrastructure can be used for workforce development activities unless the applicant certifies that less is needed to carry out its zero-emission fleet transition plan.
Administrative	Not applicable.	Not allowed under Section 5339.
Planning	Not applicable.	Not allowed under Section 5339.

- **Areas of Persistent Poverty Program.** A competitive grant program supporting planning, engineering, or development of technical or financial plans to assist areas of persistent poverty. The federal share is no less than 90%. This program could be helpful to rural and tribal areas hoping to conduct studies for new transit routes; engineering for transit facilities and improvements to existing facilities; innovative technologies; planning for low- or no-emission buses; planning for a new bus facility or intermodal center that supports transit services; integrated fare collections systems; or coordinated public transit and human service transportation plans to improve transit service in an Area of Persistent Poverty or Historically Disadvantaged Community, or to provide new service such as transportation for services to address the opioid epidemic.⁸⁷
- **Innovative Coordinated Access and Mobility Grants.** A competitive grant program that seeks “to improve access to public transportation by building partnerships among health, transportation, and other service providers,”⁸⁸ with a specific focus on better “coordination of transportation services and non-emergency medical transportation for older adults, people with disabilities, and people of low-income.”
- **Mobility, Access, and Transportation Insecurity Program.** A competitive grant program administered by the University of Minnesota that aims to fund projects that solve real-world problems like housing, health, and access to opportunity through innovative solutions that integrate transportation into a solution that meets the needs of low-income individuals and/or communities of need.⁸⁹

Other U.S. DOT Programs

This section provides a summary of other U.S. DOT funding programs that support public transit in rural and tribal areas but that are not administered solely by the FTA.

FHWA Programs with Flexible Funding. Several federal grant programs under the FHWA allow funds to be used or flexed for public transportation. However, it is important to note that, because these programs are administered by the FHWA, which apportions funds to states, rural and tribal transit providers need to be actively engaged in the statewide, regional, or metropolitan planning activities that typically drive the decision-making process for these programs.

Congestion Mitigation and Air Quality (CMAQ) Program (23 USC 149). CMAQ provides funding to states and local governments for transportation projects to help meet the requirements of the CAA by improving air quality and reducing congestion (<https://www.fhwa.dot.gov/safetealu/factsheets/cmaq.htm>).⁹⁰ CMAQ funds are traditionally used to support capital projects, for example:

- EVs and charging stations.
- Diesel engine replacements and retrofits.
- Transit improvements.
- Bicycle and pedestrian facilities.
- Shared micromobility projects.

However, CMAQ funds can also be used for operating assistance of public transit under certain conditions.

CMAQ funds are apportioned by the FHWA to states, which then distribute funds according to their specific policies. Rural and tribal transit providers that operate in nonattainment areas (i.e., areas that do not meet the air quality requirements outlined in the CAA) may be able to leverage CMAQ funds for expanded and improved transit services. CMAQ funds have also been used by transit agencies to support their vanpool programs.

National Highway Performance Program (23 USC 119). Funds from the National Highway Performance Program (NHPP) can be used for transit capital projects⁹¹ if the following criteria are met:

- A transit project is in the same corridor as, and in proximity to, a fully access-controlled highway designated as a part of the National Highway System.
- The transit construction or improvements will reduce delays or produce travel time savings on the fully access-controlled highway and improve regional traffic flow.
- The construction or improvements are more cost-effective, as determined by benefit-cost analysis, than an improvement to the fully access-controlled highway.

NHPP funds are apportioned by the FHWA to states, which then distribute funds according to their specific policies. Given the restrictions on the use of NHPP funds for public transit, this program is not a likely source of federal funding for rural and tribal transit providers.

Surface Transportation Block Grant Program (23 USC 133). The Surface Transportation Block Grant (STBG) Program provides the greatest flexibility in the use of funds (<https://www.fhwa.dot.gov/bipartisan-infrastructure-law/stbg.cfm>).⁹² The funds may be used as:

- Capital funding for public transportation capital improvements, carpool and vanpool projects, fringe and corridor parking facilities, bicycle and pedestrian facilities, public sidewalk improvements to comply with the ADA, and intercity or intracity bus terminals and bus facilities.
- Funding for planning for surface transportation planning activities; wetland mitigation; transit research, development, and technology transfer; and environmental analysis.
- Funding for other eligible projects, including transit safety improvements and most transportation control measures. Transportation control measures are strategies intended to reduce vehicle use or improve traffic flow. These measures include improved public transit, traffic flow improvements and HOV lanes, shared ride services, pedestrian/bicycle facilities, and flexible work schedules.

STBG funds are apportioned by FHWA to states, and the apportionments to states come with some specific requirements for use in certain areas:

- 55% of a state's STBG apportionment is obligated to these areas in proportion to their relative shares of the state's population:
 - Urbanized areas with populations greater than 200,000.
 - Urbanized areas with populations of at least 50,000 but no more than 200,000.

- Urban areas with populations of at least 5,000 but no more than 49,999.
- Areas with populations of less than 5,000.
- The remaining 45% may be obligated to any area of the state.

Rural and tribal transit providers may be able to use these funds to help support public transit improvements and expansions.

Non-U.S. DOT Programs

There are also federal programs outside the U.S. DOT that provide funding to support transportation projects. Most of the funds for these programs flow from the relevant federal department (e.g., the U.S. Department of Health and Human Services) to a state body charged with administering the federal grants (e.g., a state department of health and human services).

The Coordinating Council on Access and Mobility (CCAM) has identified 130 federal programs that can provide funding for human services transportation for people with disabilities, older adults, and/or individuals with low income. These 130 programs are documented in the CCAM Program Inventory and available on CCAM's website.

Examples include the following:

- Supplemental Nutrition Assistance Program (SNAP), Employment and Training Program, Food and Nutrition Service, U.S. Department of Agriculture.
- Vocational Rehabilitation Grants, Rehabilitation Services Administration, U.S. Department of Education.
- Medical Transportation Program (Medicaid) for Non-Emergency Medical Transportation, Centers for Medicare and Medicaid Services, U.S. Department of Health and Human Services.

One might ask how to be able to leverage these programs for public transportation, and, largely, it depends on the program. In some cases, rural and tribal transportation providers may be able to apply directly to the administering agency for specific transit projects and programs. In other cases, rural and tribal transportation providers may be able to operate as contractors, providing purchased transportation services to the state or local entities responsible for administering the federal programs.

When transit providers are receiving non-U.S. DOT revenue for providing human services transportation, they may not only be able to benefit from the increased ridership and service productivity but also may be able to use that revenue as nonfederal share for their FTA grants.

The topic of operating sponsored human services transportation and using non-U.S. DOT federal funds as match to FTA grants is discussed in more detail in the following sections.

How to Support New and Emerging Service Strategies— A Focus on Funding Effective Practices for Maximizing Federal Funding

Although there are federal grant programs to help support public transit in rural and tribal areas, maximizing the use of federal funding remains a significant challenge. This section contains effective practices for maximizing federal funding. The practices are grouped into two types:

1. Increasing federal revenue available to support transit projects and programs (i.e., finding more federal funding).
2. Better leveraging available federal funding, including the following:
 - Maximizing the federal share of transit projects and programs.
 - Generating more local match for federal grants.

Increasing Federal Revenue Available to Support Transit Projects and Programs

One approach to maximizing federal revenue to support rural and tribal transit projects and programs is to find more funding. Although there are many potential approaches, several are discussed as follows:

- Pursuing competitive opportunities.
- Participating in statewide, regional, and metropolitan planning agencies.
- Operating a federally sponsored service.
- Understanding funding formulas and leveraging them to your advantage.

Pursuing Competitive Opportunities

The previous section titled Federal Funding Programs for Rural and Tribal Public Transit highlighted the federal funding programs that support rural and tribal public transit. Although some of those programs are formula-driven, others are competitive (e.g., the Section 5339 competitive program) and require the submittal of an application specific to the grant program that explains the project and why the project would be a good candidate for the grant program.

It may be difficult to be fully aware of and prepared for federal competitive grant opportunities. However, here are a few pointers to help:

- Subscribe to the FTA's email updates to be made aware of notices of funding opportunity (NOFOs) and other FTA news. One can subscribe by visiting the FTA's website and finding the Subscribe to Email Updates (or similar) button or link.
- Attend FTA and other national, statewide, and regional transit association webinars and/or state DOT webinars and meetings that discuss funding opportunities. (Especially after a new transportation authorization bill or annual appropriations bill is passed.) Transit agencies should note upcoming opportunities that seem to fit the agency's needs as well as the region's needs. Get stakeholders together early, preferably before the NOFO is released, allowing a transit agency to determine whether there is a feasible project and if there is interest from stakeholders.
- Consider making a person within the transit agency the "grant authority," that is, the person responsible for tracking upcoming and released NOFOs and preparing the transit agency for the application process.
- Have studies, documents, and data at the ready that help to demonstrate the need for the project or program. For example, if there is interest in seeking funding for bus stops and shelters, try to have a study done that documents that customers want bus stops and shelters, how many bus stops and shelters are needed, and so forth.

When a NOFO is released:

- Read over the NOFO carefully to understand the following:
 - Critical deadlines.
 - The purpose of the NOFO (what type of projects is it trying to fund).
 - Evaluation criteria for selecting projects.
 - Understand any funding information, like:
 - Overall amount available.
 - Minimum, typical, or maximum project size.
 - Restrictions on project selection (e.g., only two per state or a minimum amount for rural or urban areas).
 - Nonfederal share requirements and acceptable forms of match.
 - Application requirements (e.g., supplemental forms, letters of commitment or recommendation, cost-benefit analyses).

- Attend the associated webinar to get a layperson's understanding of the funding opportunity and to ask questions.
- The NOFO may be worth pursuing if:
 - The transit agency has a project idea that aligns well with the NOFO's purpose and evaluation criteria.
 - The transit agency has enough stakeholders from other entities interested in supporting the project (e.g., local governments, planning organizations, local associations, and businesses).
 - There is enough data or documents on hand to help demonstrate the need.
- If the decision is to apply for the grant, get the stakeholders together to move the application forward
- Before submitting the application, ensure that the final application demonstrates how the project aligns with the NOFO's purpose and criteria.

As with all competitive funding opportunities, there are usually more losers than winners; however, the act of compiling the information for the project can help solidify the project idea and cement stakeholder support, even if the project is not selected for funding. If the project is not selected, and there is an opportunity to receive feedback from the granting agency, doing so will provide feedback on what improvements and changes might have made a difference. There may be another chance to submit the project for competitive funding.

Participating in Statewide, Regional, and Metropolitan Planning Agencies

When focused on the day-to-day operations of public transit, it may be difficult to carve out time for participating in a transportation committee or attending meetings that are not directly related to public transit. However, there is significant value in being in the room when transportation funding decisions are made. As was discussed in the section Other U.S. DOT Programs, there are several federal transportation funding programs with a lot of money that are not specifically focused on public transit but that can be used to fund public transit projects and programs (e.g., the STBG program). However, transit providers are not always directly involved in the decision-making process for how those funds get spent, meaning that decision-makers may not be thinking about how transit could be part of the solution to help them meet their congestion reduction, environmental, or improved mobility goals.

Therefore, it is important to identify the entities in the region that are involved in the decision-making process for programs like CMAQ, NHPP, STBG, and other transportation funding programs, and to make known and visible the services, value, and needs of the transit agency. A transit agency should consider attending board meetings of rural planning organizations, metropolitan planning organizations, and councils of governments. Consider volunteering to be on a technical advisory committee or to perform some other role that can provide opportunities to voice the value and needs of public transit.

It is always better to be in the room where decisions are made. Although it may take some time, with a consistent and clear message, it may be possible to obtain more federal funding to support transit.

Operating Federally Sponsored Service

Another approach to increased federal revenue for public transit is to operate human services transportation for non-U.S. DOT federal programs. A few of these programs were mentioned in the Non-U.S. DOT Programs section. One common program is operating NEMT under the Department of Health and Human Services' Medicaid program. Because there are so many different federal programs that support human services transportation (130 according to CCAM), this chapter refers to operating transportation for these programs as operating federally sponsored service.

Sponsored service refers to service funded by a non-U.S. DOT entity, where the non-U.S. DOT entity is paying for the cost of transportation for their clients or beneficiaries under a contract with the transit provider. As discussed previously, if a transit provider is operating a sponsored service, it usually is as a contractor to a local or state human services agency rather than as a recipient of federal funds. One of the benefits of operating federally sponsored service is that the revenue from the sponsored service can typically be claimed as local match to FTA grants, including Section 5311, 5310, and 5307 (urban formula funds). Revenue from sponsored service could reduce or even potentially eliminate the need to raise additional nonfederal share for the FTA grants.

The complexities surrounding operating federally sponsored service are many, and a full discussion is beyond the scope of this Guide; however, readers are encouraged to stay tuned to CCAM and to review CCAM's currently available documents, including the following:

- CCAM's Program Inventory⁹³ lists the federal funding programs that support human services transportation.
- CCAM's Cost-Sharing Policy Statement describes general policies and guidelines for sharing (i.e., allocating) the costs of transportation when multiple federal programs are paying for the transportation.⁹⁴
- CCAM's Federal Fund Braiding Guide describes guidelines for using federal funds as match to other federal grants.⁹⁵

In addition, Chapter 3 of National RTAP's *Advanced Topics in Financial Management for Rural Transit Providers*⁹⁶ discusses guidelines for pricing and operating sponsored service and may be helpful for transit agencies seeking to start, increase, or improve their sponsored service operations.

Understanding Funding Formulas and Leveraging Them to Your Advantage

One last practice is to understand and leverage funding allocation formulas for formula programs. For rural and tribal providers, this includes not only understanding FTA's funding formulas for Section 5311, Tribal Transit, and ADTAP; Section 5310; and Section 5339; but also understanding how funds apportioned to the state DOT are then allocated to rural and tribal providers in the state.

States are quite different in their approaches. For example, in Virginia, the Department of Rail and Public Transportation (DRPT) sets aside some Section 5311 funds it receives for its statewide capital program that pools funds from Section 5311, Section 5339, and other federal programs, and then awards the remaining Section 5311 funds as operating assistance grants to rural providers in the state. Assuming funding is available, rural providers receive up to 50% of their fiscal year's operating budget as a Section 5311 operating assistance grant. DRPT does not allow any operational expenses to be reimbursed at the capital federal share rate (e.g., preventive maintenance or administrative), even though grant activities are recognized as allowable under the Section 5311 program.

In Texas, the Public Transportation Division (PTN) of the Texas Department of Transportation (TxDOT) sets aside some Section 5311 funds it receives for a statewide discretionary program and then allocates the remaining funds using a formula based on a rural provider's historical Section 5311 award, population, land area, vehicle miles, and low-income population. PTN allows rural providers to use their Section 5311 award for all applicable grant activities, including allowing rural providers, to seek the full allowable federal share (e.g., 80% for preventive maintenance, 80% of the capital cost of contracting, and 80% of administrative costs). PTN awards

Section 5339 funds to rural providers using a formula based on total vehicle miles. Rural providers benefit from knowing how funds are distributed and can:

- Work to maximize their award, calculated by a formula, if applicable.
- Work to maximize the federal share of their costs by categorizing costs into higher federal share grant activities (if allowed and following consistent cost accounting). This strategy is discussed in more detail in the Maximizing Federal Share of Transit Projects and Programs section.
- Understand the potential funding implications of making service changes (e.g., reducing service, adding new service, expanding service area) and of changes resulting from demographic shifts.

Better Leveraging of Federal Revenue

Another way to maximize federal revenue supporting transit projects and programs is to work to leverage the federal revenue already available. This strategy includes two main practices:

1. Maximizing the federal share of transit projects and programs.
2. Generating more local match for federal grants.

Maximizing Federal Share of Transit Projects and Programs

This practice is mainly only relevant to transit providers that are having difficulty spending down their federal grants due either to having a relatively small transit program compared with the grant award or to having limited local match. This section is written from the perspective of traditional FTA grant programs (i.e., Section 5311); however, the general concepts discussed apply to any federal grant that has different federal share maximums for different grant activities.

As discussed in the FTA Grant Programs section, FTA grants have different allowable grant activities, and these different grant activities have different federal share maximums (e.g., 80% federal share for capital expenses but 50% federal share for operating assistance). If a transit agency is having difficulty spending down the FTA grants or coming up with an adequate local share, it may benefit the transit agency to make sure that expenses are classified into the grant activities that have the highest federal share rates (i.e., capital and administrative). Note that state DOT rules may restrict grant recipients from using other grant activities, as is the case in Virginia.

Here are some expenses that do not sound like capital expenses but are allowed (if permitted in the transit agency's state) to be treated under the capital grant activity at 80% federal share in the Section 5311 program:

- Preventive maintenance, including maintenance of vehicles and facilities. The maintenance costs do not have to be solely related to your preventive maintenance program. All costs that are classified under the vehicle maintenance and facility maintenance functions in the Uniform System of Accounts (USOA) can be claimed as preventive maintenance expenses and reimbursed at an 80% federal share.
- Extended warranties that do not exceed industry standards.
- Computer hardware and software.
- Lease of equipment and facilities (when a lease is more cost-effective than purchase).
- The capital portion of costs for service provided under a contract (also called the capital cost of contracting).
- The introduction of new technology into public transportation.
- ADA paratransit service (not to exceed 10% of your Section 5311 award).⁹⁷
- Mobility management.
- Associated capital maintenance, including equipment, tires, tubes, and materials.⁹⁸

Also, under the Section 5311 program, some project administrative expenses can be reimbursed, if allowed by the state, as an administrative expense at an 80% federal share. Examples include the following:

- General administrative expenses (e.g., salaries of management staff).
- Marketing expenses.
- Insurance premiums or self-insurance reserve payments.
- Office supplies.
- Rentals of facilities and equipment.
- Standard overhead rates.
- Administering a drug and alcohol testing program.
- Administrative costs for promoting and coordinating ridesharing (assuming the activity is part of a coordinated public transportation program).

FTA circulars have the full details of grant activities and federal share maximums, and readers are encouraged to consult the relevant FTA Circular for full details.

One last option worth adding is exploring the use of non-FTA grant programs like CMAQ or STBG to help fund transit improvements. These programs generally allow for an 80% federal share regardless of the type of expense—operating or capital.

Generating More Local Match for Federal Grants

It can be difficult, especially for rural and tribal transit providers, to have enough local match to leverage the full value of their federal grant(s). However, there are a few effective practices aimed at increasing local match, including the following:

- Taking advantage of in-kind match.
- Using volunteers.
- Generating more revenue locally.

Another way to generate more local match for federal grants is to operate a federally sponsored service, which could allow the transit provider to claim the revenue from the sponsored service as match. This strategy is discussed in the Operating Federally Sponsored Service section.

Taking Advantage of In-Kind Match

In-kind match is something (a service or a good) needed by a transit provider but received by a transit provider (typically from a third party) without the transit provider incurring any expense. NCHRP Project 20-65/Task 75, “Baseline Research on Allowable In-Kind and Local Match Sources,” produced a guidebook titled *Use of In-Kind as Match for Federal Transit Administration Awards*.⁹⁹ This guidebook provides the most recent and comprehensive guidance regarding the use of in-kind match to FTA grants.

It is important to note that some states may restrict the use of in-kind match. If a transit agency is already not using all the agency’s available grants and no additional match funds are needed, claiming in-kind match does not necessarily add any financial benefit. However, if the transit agency is having difficulty coming up with enough match to draw down the full FTA grant that is available (and using in-kind match is allowed in the state), then using in-kind match may be an effective tool.

There are several different types of potential in-kind match, including the following:

- Labor
- Rental space
- Land and buildings

- Equipment
- Goods and services
- Indirect costs
- Intercity bus feeder service
- Vanpool credits
- Travel expenses

There are varied requirements and procedures for each type of in-kind match that are beyond the scope of this Guide; however, a few general guidelines are provided as follows.

- The in-kind goods or services must meet the same requirements as costs eligible for reimbursement under the FTA grant. The in-kind goods or services must be part of the project budget and must fall under a grant activity supported by the grant.
- The in-kind goods or services can only be claimed as match in a single award (the in-kind goods or services cannot be claimed in multiple federal awards).
- The in-kind goods or services must be counted as both an expense and a revenue in the grant budget (e.g., if you need a good or service, it costs \$X. That good or service is received in-kind at a value of \$X).
- The in-kind goods or services must be fully documented.

Here are a few examples of transit systems that use in-kind match, found in the NCHRP Project 20-65/Task 75 guidebook:

- Missoula Ravalli Transportation Management Association (MR TMA) has a long-term lease with the University of Montana for land on which FTA funds were used to build a transit facility. The lease is for \$1/year for 20 years; the local match is the net present value of the appraised value of the 20-year lease.
- Lehmi County Economic Development Association (in Salmon, Idaho) receives Lehmi County-provided insurance and maintenance service at no cost to the agency—this is used as in-kind match.
- West Virginia Department of Transportation Division on Public Transit uses unsubsidized connecting intercity bus service operated by Greyhound Lines as in-kind match for intercity feeder service.
- Potomac Rappahannock Transportation Commission uses the capital cost of vanpool acquisitions and leases by private providers of vanpools as credits for capital projects. More details of this complex item are in the NCHRP Project 20-65/Task 75 guidebook.
- Some local Missouri newspapers print bus schedules at no cost to OATS Transit in Missouri, allowing OATS Transit to claim these services as in-kind match.

The following section focuses on the topic of using volunteers, which is a potential source of in-kind match; however, for more guidance on using in-kind match, readers are encouraged to check out Chapter 6 of National RTAP *Fundamental Financial Management for Rural Transit Providers*¹⁰⁰ and the NCHRP Project 20-65/Task 75 guidebook¹⁰¹ on in-kind match.

Using Volunteers

Using volunteers is both a cost management and a local match strategy, because volunteer labor typically costs less than staff labor, and the net cost savings of using volunteer labor can potentially be claimed as in-kind local match to FTA grants. One of the most common ways to use volunteers is to have volunteer drivers.

Volunteer drivers are individuals who donate their time to operate vehicles that provide public transportation service. Many volunteer driver programs began as providers of transportation service for older adults; however, volunteer driver programs currently serve other ridership markets as well. Transit systems can also use volunteers to perform other much-needed transit functions.

OATS Transit in Missouri serves as an interesting case study for using volunteers.¹⁰² OATS Transit is a private nonprofit corporation providing public transit, serving 87 out of 114 of Missouri's counties. As a Section 5311 subrecipient with about \$10 million in operating expenses, OATS can account for about 20% of its local match from in-kind services by leveraging a volunteer network comprising around 500 volunteers that perform vital transit functions, including the following:

- Reservations
- Passenger assistance
- Marketing and community relations

Volunteers track their time spent on OATS tasks, and OATS uses a third party to calculate the hourly value of the volunteers' time.

The NCHRP Project 20-65/Task 75 guidebook on in-kind match contains many more case studies on transit providers that use volunteers as a cost management tool and also as a way to generate more local match.

Finally, the CTAA hosts the National Volunteer Transportation Center, which contains many resources focused on volunteer driver programs.

Generating More Revenue Locally

Although in-kind match may enable a rural or tribal transit provider to leverage more match without having to have the cash on hand, another way to generate more local match is to raise more local cash revenue. This increased local revenue can come from a variety of sources, including taxes or fees dedicated to transit, contributions from local governments, contributions or donations from nongovernmental entities, and also from revenues directly or indirectly related to the provision of transit service (e.g., advertising or selling maintenance services to third parties).

Such revenue can be considered external or internal.

External Revenue

External revenue refers to contributions or payments from local organizations received to offset transit expenses. It is a direct payment for specific riders' trips or in exchange for other goods and services provided by the transit agency. Increasing external revenue can often be a challenge, especially if the increase in revenue for transit comes at the expense of other nontransit programs or is funded by an increased tax or fee on individuals or businesses. Examples of organizations that might be able and willing to contribute to support transit include the following:

- Municipal, county, or other local governments (often from general funds and not dedicated to transit).
- Local organizations focused on human services, workforce development, or education (e.g., hospitals, colleges and universities, health care centers, and training centers).
- Businesses and business development organizations.

However, the key challenge is how to convince these organizations to help fund transit service. Although material on this topic could fill an entire guide, a few general guidelines are worth mentioning:

- **Communicating Value.** Transit providers need to understand how to effectively communicate the value of the transit service they provide or want to provide. Help organizations realize the full benefits of public transit in terms of economic, business development, mobility, health, and environmental benefits—whether benefit types resonate most with the potential funding organizations. Although national facts and figures may help in this narrative (e.g., APTA states

that every \$1 invested in public transit generates \$5 in economic returns), locally relevant facts, figures, and anecdotes are usually more poignant.

- **Transparency and Accountability.** A transit agency should continually maintain the highest standards of transparency and accountability in all aspects of the transit operation. Be transparent about costs, sources of revenue, service quality, and service challenges. Conduct all operations (financial and service) with the highest level of accountability and control. There is nothing that undermines the credibility of a transit system more than misusing funds, having safety incidents, or even having the appearance of lacking necessary financial and operational controls.
- **Customer Service.** Continually maintain the highest level of customer service possible. This includes dealings with riders as well as external stakeholders. Maintain an agency-wide culture of customer service where positive and respectful relationships and communications are expected and encouraged at all times.

Internal Revenue

Internal revenue refers to payments received by the transit provider in exchange for providing transportation to specific beneficiaries or in exchange for providing other goods and services to third parties.

Examples of potential sources of internal revenue include the following:

- Operating sponsored service
- Advertising
- Concessions
- Fees for transit ID cards
- Selling maintenance services
- Selling energy sources (e.g., if the transit agency generates its own hydrogen)
- Renting out vehicles and facilities
- Operating approved charter service¹⁰³

Transit agencies often have more control over how they can generate internal revenue because transit providers are not looking for “contributions”; they are selling a service to interested parties. Although revenue streams from these sources may be small (e.g., transit advertising is not typically a large revenue stream, especially for rural and tribal transit providers), revenue adds up. And, if a transit provider is having trouble generating enough local match for FTA grants, a little bit more local revenue can make a big difference in the amount of federal funding that can be drawn down.

It’s important to highlight that FTA guidelines outline what sources of revenue are eligible to be used as match to an FTA grant, and, although most of the sources previously listed are typically allowable as match, transit providers must remember that, for any source to be eligible as match, that source must be described in the grant budget and approved by the funding agency, whether it is the state DOT or FTA.

CHAPTER 12

Accessible Service

Introduction

Accessible transit service involves meeting the requirements of federal law established by the ADA. Accessible transit service also means that people with disabilities can effectively use the public transit service in their local community and get to and from where they need to go.

In rural and tribal communities, many people rely on their cars for transportation, but for those without access to private transportation, public transportation becomes critical. Many of those without such access are people with disabilities, older adults, and people with low incomes—individuals who disproportionately live in rural communities.

Transit providers in rural and tribal communities that are considering a new transit service, expanding existing service with a new strategy, or implementing a technology enhancement such as an app to book trips need to ensure that such actions comply with the ADA and meet the law's relevant requirements so that their services are accessible to community residents with disabilities.

Transit services in rural and tribal communities vary considerably and include demand responsive services, deviating/flex routes, traditional fixed routes, and taxi-based programs. New and innovative services and strategies are also being considered, such as microtransit and on-demand services using TNCs. ADA requirements include those that apply to all transit services and also include requirements specific to transit service modes. Transit providers need to understand what requirements apply to their specific services and where to go for the detailed information and support that may be necessary.

The ADA

Background

The Americans with Disabilities Act (ADA) of 1990 is a federal law providing a comprehensive framework and approach for ending discrimination against persons with disabilities. The law's five titles address employment, public services, and public accommodations and services provided by both public and private entities.

Public transportation is addressed in Title II and Title III, with Title II covering public entities and Title III covering private entities. Title III differentiates between private entities providing transportation as a primary business, such as taxis, and those providing transportation as a secondary business, such as a hotel and its shuttle service for guests.

After the passage of the law, the U.S. DOT adopted regulations implementing the transportation-related requirements of the ADA, with the FTA responsible for ensuring compliance by public transportation agencies. The regulations were written so that compliance with the ADA

regulations is a condition of compliance with Section 504 of the Rehabilitation Act of 1973, the federal legislation preceding the ADA that prohibits entities receiving federal funds from discriminating against people with disabilities.

Relevant ADA regulations for transit agencies, including those receiving Section 5311 funds and including transit agencies in tribal communities that receive funding from the Tribal Transit Program [Section 5311(c)] are codified in 49 CFR Part 37 Transportation Services for Individuals with Disabilities (ADA) and 49 CFR Part 38 Americans with Disabilities Act (ADA) Accessibility Specifications for Transportation Vehicles.

The ADA regulations and guidance that is periodically issued by the FTA provide a framework for public transportation agencies to follow in planning and operating services accessible to people with disabilities and to ensure ADA compliance.

Several important factors regarding the ADA are as follows:

- The regulations apply whether or not the public transit provider receives federal funding.
- If the public transit provider uses a private entity to provide transportation through a contract or other arrangement or relationship (including a grant, subgrant, or cooperative agreement), the private entity must follow the regulations that apply to the public transit provider as if the public transit provider were directly operating the service. In other words, the private entity is “standing in the shoes” of the public transit provider.
- The stand-in-the-shoes requirements apply to private organizations that receive Section 5311 funds, whether through a subgrant directly from a state agency or through a subrecipient that then enters an agreement with a private contractor for service.

Planning or Implementing New Transit Service

When a rural or tribal transit provider begins planning or is soon to implement a new transit service or strategy, ADA requirements must be considered. This is important not only to ensure compliance with federal law but also to ensure all residents in the community, including those with disabilities, can use the new service.

It is also important to recognize that certain regulations apply to all types of public transit services, for example, training requirements for transit agency staff. Other regulations apply only to a specific service mode, for example, fixed-route or demand response. The regulations are outlined in the following section, with basic information provided. For more in-depth information and guidance regarding the ADA regulations, two very useful resources (the first one is developed specifically for rural transit services) should be consulted:

- National RTAP’s ADA Toolkit,¹⁰⁴ published in 2021
- FTA’s Circular 4710.1 Americans with Disabilities Act Guidance,¹⁰⁵ published in 2015

The following ADA regulations are also valuable resources:

- 49 CFR Part 37: Transportation Services for Individuals with Disabilities¹⁰⁶
- 49 CFR Part 39: Americans with Disabilities Act (ADA) Accessibility Specifications for Transportation Vehicles¹⁰⁷

Materials in this chapter were based on these documents.

ADA Regulations

This section describes general requirements that apply to all transit service types.

Nondiscrimination

This is the fundamental and overarching requirement of the law: transit providers cannot discriminate against people with disabilities.

A few examples of what this means:

- A transit provider cannot, because of the disability, deny an individual with a disability the opportunity to use the public transit system if that individual can use that service.
- A transit provider cannot require a rider with a disability to use a seat belt or shoulder harness unless other riders on the same vehicle are required to do so.
- A rider with a disability cannot be required to travel with an attendant.
- An extra fare or fee cannot be charged for riders with disabilities, including a rider using a wheelchair.

Note that the ADA states that it is not discrimination to refuse transit service to a rider with disabilities when that rider engages in violent, seriously disruptive, or illegal conduct, or becomes a direct threat to the health or safety of others. However, this does not include a rider with a disability who appears in such a way or has involuntary behavior that may offend, annoy, or inconvenience other riders or the transit provider's staff. That latter rider cannot be refused service.

Provision of Service Requirements

Maintenance of Accessibility Features

Transit providers must keep the accessibility features of their vehicles and facilities in operating condition so they can be used by riders with disabilities. Should an accessibility feature be damaged or become inoperable, that feature must be repaired promptly.

When an accessible feature is not operable, a transit provider must take reasonable steps to ensure that riders with disabilities who would otherwise use the feature can be effectively served.

Vehicle Lifts and Ramps. Transit providers must ensure that the lifts and/or ramps on their vehicles are in operating order to serve riders who need to board with the lift or ramp.

Note that pre-trip lift cycling is not a regulatory requirement; however, it is considered a good practice across the transit industry and has been adopted by many transit providers, including those operating in rural environments.

If a vehicle driver finds that the lift or ramp is not working while in service, the driver must report the failure as soon as possible.

If a spare vehicle is available, the vehicle with the inoperative lift must be taken out of service before the beginning of the vehicle's next day of service, and the lift or ramp must be repaired before the vehicle returns to service.

If there is no spare vehicle available and if removing the vehicle from service reduces the transportation service for the community, the transit provider may keep the vehicle with the inoperative lift or ramp in service for no more than 5 days, where the service area population is 50,000 or less. When the service area population is more than 50,000, the vehicle may stay in service for no more than 3 days. After the stipulated period, the vehicle must be taken out of service until the lift or ramp is repaired.

A transit provider must serve a rider using a wheelchair if the rider's wheelchair meets the definition of a wheelchair and can be accommodated by the lift or ramp and fits within the vehicle's securement area.

According to the ADA, a wheelchair is defined as a mobility aid belonging to any class of three- or more-wheeled devices, usable indoors, and designed or modified for and used by individuals with mobility impairments, whether operated manually or powered.

Transit agencies may decline to carry a wheelchair/occupant if the combined weight exceeds that of the lift specifications or if carriage of the wheelchair is demonstrated to be inconsistent with "legitimate safety requirements."

Transit vehicles that are ADA compliant per the base specifications in 49 CFR Part 38 will be able to accommodate, at a minimum, an occupied wheelchair weighing up to 600 pounds and measuring 30 inches in width and 48 inches in length (formerly known as a “common wheelchair”).

Note that some wheelchairs with their occupant weigh more than 600 pounds, and some vehicles have lifts designed to handle up to 800 pounds. The transit provider must serve the rider who, in combination with their wheelchair, weighs more than 600 pounds but no more than 800 pounds.

A transit provider may deny transportation if carrying the wheelchair and its occupant would be inconsistent with legitimate safety requirements, as when, for example, the wheelchair is so large it would block an aisle or interfere with the safe evacuation of passengers in an emergency.

The vehicle driver is not required to permit wheelchairs to ride in places other than designated securement locations, but an individual using a wheelchair cannot be denied transportation because the device cannot be secured or restrained satisfactorily by the vehicle’s securement system.

The driver may recommend, but cannot require, that a rider using a wheelchair transfer to a vehicle seat.

The driver must assist individuals with disabilities who need or request assistance with the use of securement systems, ramps, and lifts. If this requires a vehicle driver to leave their seat, they must do so.

Transit providers must permit individuals with disabilities who do not use wheelchairs, including standees, to use a vehicle’s lift or ramp to enter the vehicle.

Transit providers must also allow riders using wheelchairs, when using the lift (or ramp), to board facing forward or backward.

Other Service Requirements

- **Service Animals.** Transit providers must allow a rider to be accompanied by a service animal. The official definition of a service animal is “any guide dog, signal dog, or other animal individually trained to work or perform tasks for an individual with a disability, including, but not limited to, guiding individuals with impaired vision, alerting individuals with impaired hearing to intruders or sounds, providing minimal protection or rescue work, pulling a wheelchair, or fetching dropped items” (49 CFR §37.3). If a transit provider has any questions as to whether a service animal is, in fact, a service animal, two questions may be asked of the rider:
 - Is the animal a service animal required because of a disability?
 - What work or task has the animal been trained to perform?

A transit provider may not require a special harness or special ID for the animal, nor may the driver ask about the rider’s disability.

- **Transit Service Information.** Transit providers must provide information about their service in formats that their riders can use. This includes providing written material in accessible formats, having accessible websites, and ensuring alternatives to audio communications.
- **Oxygen Supplies.** Transit providers cannot refuse to serve a rider with a disability from traveling with a respirator or portable oxygen supply as long as the device is properly secured.
- **Boarding and Alighting.** Transit providers must give riders adequate time to board and deboard.

Reasonable Modification

Public transit providers are required to make reasonable modifications to their policies, practices, and procedures when the modifications are needed to avoid discrimination and to ensure

that the providers' programs and services are accessible to people with disabilities. Transit providers are to establish a process that allows riders to request a modification and that allows the transit provider to consider whether the request is reasonable. When possible, such requests are to be made in advance of when they are needed by the rider.

Training

The ADA requires that transit providers ensure their staff are “trained to proficiency,” as needed for their position. This means that staff members, once trained as appropriate for their position, can consistently and reliably operate and use accessibility features, provide appropriate assistance to riders with disabilities, and treat the riders courteously and respectfully.

Complaint Process

Transit providers are required to identify a staff member who is responsible for coordinating compliance with the ADA. Transit providers are also required to establish a process for riders to register alleged violations of the ADA. Many transit agencies title this employee the “ADA Coordinator.”

The process for registering an ADA complaint must be advertised, such as through the transit provider's website. Transit providers that receive federal funding must retain the details of all ADA complaints for 1 year and then a summary of the ADA complaints for 5 years.

Fixed-Route Service

Buses and Vans

The ADA requires that transit providers acquire vehicles that are accessible to people with disabilities, including those who use wheelchairs, to provide fixed-route service. This includes all new vehicles and, with rare exceptions, any used or remanufactured vehicles.

ADA regulations spell out what a transit agency must do if it intends to acquire an inaccessible used vehicle or an inaccessible remanufactured vehicle for fixed-route service; refer to the National RTAP ADA Toolkit: “Fixed Route Bus Requirements” and FTA's Circular, ADA Guidance, Chapter 4: “Vehicle Acquisition and Specifications.”

Operating Service

Lift or Ramp Becomes Inoperable. If the vehicle's lift or ramp becomes inoperable while in service, and the schedule shows that the next accessible vehicle on the route is more than 30 minutes away, the transit provider must promptly provide alternative transportation to riders who are not able to use the vehicle because its lift does not work.

Note that, for a ramp failure, the vehicle driver may be instructed to deploy the ramp manually, so alternative transportation may not be needed.

Stop Announcements. The ADA requires that transit providers announce bus stops so that riders know of upcoming stops. Announcements are to be made at least at transfer points with other routes or other major destination points and at intervals along the route so that riders with visual disabilities or other disabilities can orient themselves to their location.

Stop announcements can be provided by the transit driver or by an automated stop announcement system on the vehicle. When the vehicle has an onboard public address system (required for vehicles over 22 feet in length), the driver can use that to announce stops.

Access to Priority Seating and Wheelchair Securement Area. Transit drivers must ask riders sitting in the priority seating area or in fold-down seats over the wheelchair securement

area to move if a rider with a disability needs to use the priority seating or the securement area. Note that the driver is required to ask a rider to move, but is not required to enforce that request if the person refuses to do so.

Flag Stop Service. Rural transit providers operating fixed-route service may offer “flag stop” service, that is, an individual standing along a fixed route may wave or signal to the vehicle’s driver as it approaches to stop so the individual can board. For such a service, the vehicle driver typically has the discretion to stop for a pickup, as well as a drop-off, only where it is safe to do so.

While a route may offer flag stops, the route is still a fixed route, with the requirement for ADA paratransit, which is important, as flagging the driver may be difficult for some riders with disabilities. For example, a rider with a vision disability may not recognize the approaching bus until it is too late to signal the driver for a stop.

ADA Complementary Paratransit

Transit providers that operate fixed-route service are required to also provide ADA complementary paratransit service for riders who are not able to use the fixed-route service because of their disability. Note that ADA complementary paratransit is often referred to as ADA paratransit, without the formal term complementary.

ADA regulations for paratransit are elaborate. A transit provider implementing new fixed-route service with the requirement to provide ADA paratransit should refer to detailed guidance that can be found in the following:

- National RTAP ADA Toolkit, pages 27–39.¹⁰⁸
- FTA Circular 4710.1 Americans with Disabilities Act Guidance, Chapters 8 and 9.¹⁰⁹

Major elements of ADA paratransit include the following:

- **Origin to Destination Service.** This means that the transit provider must ensure that an ADA paratransit-eligible rider can get from their origin location to their destination location. In practice, a transit provider can determine, as a policy, that their service is “door-to-door” or “curb-to-curb.” However, if “curb-to-curb,” the transit provider must offer “door-to-door” service for those riders who, because of disability, are not able to get to and from the curb.
- **Six Service Criteria.** ADA paratransit must meet six criteria, which frame its comparability to the fixed-route service:
 - Operate in the same service area as the fixed-route system, defined as a $\frac{3}{4}$ -mile corridor on either side of bus routes and around rail stations.
 - Operate during the same days and hours as the fixed-route service.
 - Have a comparable response time as fixed-route, defined as accommodating trip requests for a particular day during normal business hours on the previous day (i.e., next-day service).
 - Have comparable fares to fixed-route, defined as fares that are no more than twice the base, nondiscounted adult fare for fixed-route service.
 - Meet requests for any trip purpose; that is, there can be no trip purpose restrictions or priorities.
 - Operate without capacity constraints, meaning no waiting lists, trip caps, or patterns and practices of a substantial number of trip denials, untimely pickups, or excessively long trips (onboard ride times). Excessive hold times for telephone access are also defined as a capacity constraint.
- **Eligibility for Service.** The following items apply to eligibility for service.
 - ADA paratransit is intended for those individuals whose disability prevents the use of the fixed-route service. The key question is, does the individual have the functional mobility skills to use fixed-route service? It does not depend on a medical diagnosis or type of disability.

The ADA regulations provide specific details regarding eligibility for ADA paratransit service, with eligibility conferred within three categories: those whose disability prevents the use of the fixed-route service at all times, those whose disabilities allow the use of fixed-route service at some times (e.g., only during daylight hours), and those with a disability that is temporary and therefore unable to use fixed-route during the temporary period.

- Transit providers are to establish a process for determining eligibility, with regulations specifying how that process is to be implemented. Among other requirements of the eligibility determination process, individuals determined eligible are to be given some type of documentation that they are ADA paratransit eligible.
- **Public Participation.** The ADA requires that transit providers have a mechanism for individuals with disabilities to participate in the development and assessment of the service. This may be an ongoing advisory group that meets periodically, or it could be meetings or workshops that meet at specific times. If some changes or revisions alter the ADA paratransit service, then individuals with disabilities must be consulted.
- **Policies and Procedures.** Transit providers should establish and publicize policies and procedures that govern their ADA paratransit service. For example, is the policy that drivers provide door-to-door or curb-to-curb at pickups and drop-offs? Policies and procedures frame how the service is to be operated and what the riders should expect.

Route Deviation Service

Route deviation is a service variation that can be an effective way to stretch limited transit resources to serve a low population-density area. Route deviation is also known as deviated route, deviated fixed-route, flexible-route, or flex-route service.

It is important to recognize that route deviation is recognized by the ADA as either fixed-route or demand response. There is no “route deviation” service defined by the ADA. The distinction between whether route deviation service is considered fixed-route or demand response depends on who can request a deviation off the fixed route. If requests for a deviation from the route can be made by any rider, the ADA considers the service to be demand response, and the service must comply with requirements for demand response service.

If a request for a deviation from the route can be made only by riders with disabilities, then the route deviation service is considered to be fixed-route, which means the service must comply with requirements for fixed-route, including ADA paratransit.

Note that the ADA regulations do allow commingling or combining ADA paratransit and fixed-route riders on the same vehicle so that the route deviation service operates as fixed-route with ADA paratransit provided to serve riders determined eligible for ADA paratransit. In this way, the same vehicle serves riders boarding at bus stops and also deviates to pick up and drop off ADA paratransit riders. However, this can be complicated and may be difficult to operate because the ADA paratransit service must comply with ADA paratransit’s six service criteria, including the prohibition of capacity constraints. For example, if there are multiple requests from ADA paratransit riders for a deviation, it will be difficult for the vehicle to keep to its fixed schedule. Commingled service may be an option for a very long rural route, but the service must be monitored to ensure compliance with ADA regulations.

Demand Response Service

Buses and Vans

Transit providers operating demand response service must acquire accessible vehicles. Inaccessible vehicles may be acquired only when the demand response service, when viewed

in its entirety, provides equivalent service to riders with disabilities, including those who use wheelchairs.

Providing an equivalent level of service means that the demand response service operated for riders with disabilities, including those who use wheelchairs, is equivalent to the demand response service provided to other riders according to seven criteria:

The criteria are as follows:¹¹⁰

- Response time
- Fares
- Geographic service area
- Hours and days of service
- Restrictions or priorities based on trip purpose
- Availability of information and reservation capability
- Any constraints on capacity or service availability

For example, if general public riders can request a trip and the response time of the vehicle is within 3 hours, then a rider with a disability should expect the same response time (i.e., the accessible vehicle will arrive within 3 hours). It would not be acceptable for general public riders to have that 3-hour response time, and riders using wheelchairs to book a day in advance for a trip. That is not equivalent.

A transit provider must be able to demonstrate this equivalence, according to the seven criteria, before any acquisition of an inaccessible vehicle. The standard FTA certifications and assurances associated with federal grant funds contain a certification of equivalent service form. Depending on the requirements of the state program administering federal grants, this form is to be submitted before acquiring an inaccessible vehicle or be available on request, such as during a triennial review or other compliance review done by the FTA or the state DOT.

Operating Service

Demand response services for the general public do not require ADA complementary paratransit. But demand response services for the general public must provide equivalent service to riders with disabilities, including those who use wheelchairs, to meet ADA's fundamental requirement that prohibits discrimination against people with disabilities.

Service Equivalency. Service equivalency is measured by the seven criteria listed previously. It is important to understand that these seven criteria are different from the six service criteria for ADA paratransit. The former criteria compare the demand response service provided to riders with disabilities, including those who use wheelchairs, to the demand response service provided to other riders. The comparison must show that the service is equivalent. The latter criteria—for ADA paratransit—are used to determine the comparability of the paratransit service to the fixed-route service.

It is also important to note that FTA guidance on service equivalency makes it clear that there are various types of demand response service, such as those operated as taxi subsidy programs and route deviation, and those for subsets of the general public, including people with disabilities and older adults. The service equivalency requirements apply to these various types of demand response service. FTA's guidance for taxi subsidy programs, for example, discusses service equivalency for such programs: taxi subsidy programs provided by transit agencies using FTA funding are subject to the equivalent service requirements. The guidance specifically addresses the response time criterion: "transit agencies operating taxi subsidy programs must monitor response times for riders with disabilities, including those who use wheelchairs, to ensure these riders experience comparable response times to other riders," according to the FTA Circular with ADA guidance.¹¹¹

Commingling Demand Response Service and ADA Paratransit. Rural transit agencies that provide a demand response service and that must also provide ADA paratransit may want to combine the two services rather than operate two separate services.

Combining or commingling the two services can be done, but the transit agency must ensure that the ADA paratransit component of the service meets ADA paratransit requirements. To do so, the transit provider should differentiate the ADA paratransit riders from other eligible riders through their IDs or some other method, and then monitor the service to ensure compliance and that there are no capacity constraints. For example, the transit agency could list all trip request denials for a sampled period and then assess the denials to ensure they did not disproportionately go to ADA paratransit-eligible riders.

Contracting with Taxi Companies and TNCs

Accessible Service

Transit providers may contract with taxi companies and/or TNCs (such as Uber and Lyft) to provide demand response service. In such cases, the transit provider must ensure that service equivalency is achieved, so that riders with disabilities, including those who use wheelchairs, receive the same level of service as other riders according to the seven service equivalency criteria listed previously.

Taxi companies and TNCs may not operate any accessible vehicles, so the transit provider must find and provide accessible service. There are options. The transit provider could identify and include as a provider a taxi company that has some accessible vehicles (some taxi companies do operate at least some accessible vehicles); find another provider that operates accessible vehicles, such as an NEMT company; or the transit provider could operate the accessible component with its own vehicles.

FTA's Drug and Alcohol Testing

Another issue when a transit agency plans to work with taxi companies and TNCs is drug and alcohol testing.

Under federal transit law, transit agencies that receive funding through FTA's grant programs for urban as well as rural areas are required to comply with drug and alcohol testing of staff in safety-sensitive positions, including those operating, dispatching, and maintaining revenue service vehicles.

The requirements also apply to the contractors and subcontractors of transit agencies. The arrangements for contractors and subcontractors may be formal, such as a written contract, or an informal arrangement "that reflects an ongoing relationship between the parties."¹¹²

If a transit provider uses a taxi company and/or a TNC for their demand response service and the transit provider schedules the trips to that company (or companies if more than one contractor is used), the taxi company and/or TNC must comply with the drug and alcohol testing requirements.

On the other hand, if there is more than one company operating service for the transit agency and the rider chooses which company to book a trip with (in this case, the transit provider is not scheduling trips for the companies), then the drug and alcohol testing requirements do not apply. This is because the riders have a choice of companies (whether taxi companies or TNCs) to use for their trips. Rider choice is the key determinant that FTA's taxi exception applies to.

The taxi exception was initially developed to address transit agencies' use of multiple taxi companies, such as for taxi voucher programs, recognizing the practical difficulty of trying to

administer a drug and alcohol testing program for multiple companies. For the taxi exemption to apply, there must be at least two companies that riders can choose from for trip-making.

More and more transit agencies are considering the use of TNCs to provide transportation services, and the FTA has published useful information and guidance on their use. This includes guidance not only on drug and alcohol testing but also compliance with the ADA.¹¹³

Reservations Capability

When a transit agency uses TNCs to provide demand response service trips—for microtransit or another on-demand service—riders request trips through a smartphone app and/or online. For these services, the transit agency must ensure that there is a call-in option so that individuals who do not have a smartphone or access to the internet have access to the transportation service. The transit provider can provide the call-in option, using a link to the TNC's technology to enter the rider's trip request. Or the transit provider could require a call-in option in its agreement with the TNC. Note that some of the national TNCs provide a call-in option on their own.

Fares

Fare payment is another consideration when using TNCs. To pay for a trip on Uber or Lyft, a rider needs a credit or debit card. But what about riders who do not have such a card? Transit providers generally suggest that those riders use a pre-paid debit or cash card, which can be obtained at various retail locations such as drug stores and convenience stores.

Microtransit Service

Transit providers in rural areas are increasingly considering microtransit as a service strategy. It can be used in different ways, such as to provide or expand transit to new areas or times, to test the demand for transit, and to replace underperforming fixed-route service to improve productivity and reduce cost per trip.

Microtransit is essentially a modern version of advanced reservation dial-a-ride service, with trips provided on demand using new technology. Riders typically request trips through an app on their smartphone.

Microtransit can be operated by the transit agency, by the transit agency's paratransit contractor, and/or with taxis and/or TNCs.

As is required for demand response service, microtransit service must provide equivalent service to riders with disabilities, including those who use wheelchairs; microtransit is a demand response service.

A straightforward way to ensure equivalency for accessible service operations is to ensure that the fleet is 100% accessible. If the transit provider uses taxis or TNCs that do not operate any accessible vehicles, the transit provider must find and include a provider with accessible vehicles, or the transit provider could operate the accessible component of the service itself.

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PART III

Information Topics

Part III of the Guide includes Chapters 13 through 15, which address topics that may be of interest to rural and tribal transit agencies as they navigate a path to improving or enhancing their transportation services.

Chapter 13: State DOT Role provides information about the role of state DOTs in supporting rural and tribal transit providers. Not only does this include the allocation of federal funds to providers, but it also provides training opportunities and ensures compliance with federal grant requirements.

Chapter 14: Roles of Tribal Governments presents information about selected topics related to tribal transit, including the role of tribal governments in supporting public transportation in tribal and nontribal communities. Key topics addressed include governance, funding and compliance, tribal and nontribal systems working together, and operations and planning.

Chapter 15: Compliance with Federal and State Requirements addresses the important responsibility that rural and tribal transit providers have for managing their projects and operating their services in compliance with federal and state regulations. The chapter outlines the underpinnings of these regulations.

CHAPTER 13

State DOT Role

Introduction

State DOTs have a significant role for rural and tribal transit providers not only in the allocation of federal funds to providers but also in their support in developing, improving, and sustaining rural and tribal transit services.

This chapter provides supplemental information about selected topics related to the role of state DOTs in supporting rural and tribal transit providers (see also Chapter 11, Funding and Revenue, in Part II and Chapter 14, Roles of Tribal Governments, in Part III).

Funding and Oversight

The 2014 GAO Report to the Committee on Banking, Housing, and Urban Affairs, U.S. Senate, titled *Public Transportation: Federal Role Key to Rural and Tribal Transit*,¹ describes a review of FTA's Formula Grants for Rural Areas (Section 5311) Program, which is FTA's only funding program dedicated to rural and tribal transit. The review evaluates funding, oversight, and support mechanisms for the rural transit program and overviews relevant challenges faced by tribal and rural transit providers. The report is based on interviews with 15 state transportation agencies (subsequently referred to as state DOTs), 30 rural and tribal transit providers, FTA staff, and industry organizations, as well as an analysis of relevant NTD data.

As stated in the 2014 GAO report, oversight of rural transit program funding (covering compliance and reporting) is provided at two levels. FTA generally provides oversight of state DOTs (the grantees), who, in turn, provide oversight of rural and tribal transit providers (the subgrantees). FTA requires states to develop state management plans that describe how they will award funds to subgrantees and how they will administer their role in the rural transit program (among other outcomes).² FTA conducts state management reviews (covering compliance) every 3 years to assess each state's administration and implementation of the rural transit program.

Available research on the topic of the effectiveness of state management plans or best practices for developing and administering them is limited, but one relevant research effort is documented in a 2011 paper entitled *Section 5310 Transportation State Management Plans: A Baseline Review* and published in the *Journal of Public Transportation*, Volume 14, No. 2.³ That research effort assessed state management plans that were adopted no later than 2005. The state management plan elements reviewed included definitions of rider eligibility (pertinent to Section 5310 funding), requirements for documenting need, coordination requirements and practices, progress toward federal goals, vehicle accessibility requirements, and geographic/coverage goals. The authors offered the following recommendations for state DOTs:

- In every funding cycle, review state regulations and practices to assess if they are improving the integration of public transit systems. Regulations and practices that are too rigid might disincentivize innovation and hinder evolution.
- Focus on identifying where funding investments can be made that will improve the integration of public transit systems.
- Establish the expectation that subgrantees provide transit service as part of a larger, integrated transportation system.

In 2019, FTA conducted a workshop that described and provided guidance for addressing the most common deficiencies in state management plans. This guidance is available at <https://www.transit.dot.gov/sites/fta.dot.gov/files/docs/regulations-and-guidance/safety/133551/fta%E2%80%99s-state-management-review-top-10-findings-workshop.pdf>.⁴

More information about FTA state management reviews is available from FTA at <https://www.transit.dot.gov/regulations-and-guidance/program-oversight/state-management-oversight>.⁵

The 2014 GAO report found that, overall, funding rural and tribal transit services can be challenging. While seven of 30 interviewed providers indicated that state and local funds were adequate to support most of their operations, 16 of the 30 surveyed providers reported difficulties in finding local match (from local or state sources) for rural transit program funds. Eleven of 15 state DOTs and 22 of the 30 providers stated that the rural transit program was “essential to sustaining transit in their communities.”

The providers also “frequently indicated that without rural transit program funds, transit service in their area would be reduced or, in some cases, eliminated.” Some providers reported creating dedicated local tax funding to provide local match.

The authors of the 2014 GAO report also found that four of the 15 state DOTs provided state funding for rural transit that exceeded what was provided through the federal rural transit program. Specifically, “the state of Michigan provided about one-third of the total operating expenditures in fiscal year 2012, according to state transportation agencies officials.”

The authors of the 2014 GAO report did not identify specific transit providers who created dedicated local tax funding to provide local match for rural transit program funds. *TCRP Synthesis 154: Innovative Practices for Transit Planning at Small to Mid-Sized Agencies*⁶ provides case studies of small and mid-sized transit providers that have successfully obtained local tax funding for transit.

The 2014 report *Enhancing Transit Service in Rural Areas and Native American Tribal Communities: Potential Mechanisms to Improve Funding and Service* by the Mineta Institute⁷ references surveys conducted by Nevada DOT in 2008 and 2011 to “identify and document rural transit services, needs, and challenges.” The surveys identified inadequate funding as the major constraint on the adequacy of rural and tribal transit services in the state. The goal of the 2014 report was to identify funding sources to supplement the FTA funding administered by the state.

The surveys described in the 2014 Mineta report indicated that “inadequate funding was the primary reason that limited the most desired services in rural and tribal communities.” The surveys also showed that “most of the providers mainly used federal and state traditional [DOT] funding programs as well as local transportation agency funds.”

Potential alternative federal funding sources identified by the 2014 Mineta report authors for Nevada transit providers include the Department of Health and Human Services, the Department of Housing and Urban Development, the Department of Veterans Affairs, the Department of the Interior, the Department of Education, the Department of Agriculture, and the Department of Labor. A challenge associated with accessing these alternative sources is the large staff effort that

may be needed to investigate the funding programs and assess their applicability to local needs. The report notes that investigating alternative funding programs can be “overwhelming” for providers.

The 2014 Mineta report authors identified transit operational strategies “that providers can implement to more efficiently utilize available funding and transit resources.” These strategies include the following:

- Reducing trip lengths and travel times (e.g., by grouping trips geographically or relying on alternatives such as vanpooling).
- Providing flexible transit services (e.g., flex routes), which might be more efficient in some contexts.
- Enhancing coordination of services with providers who have overlapping or adjacent service areas (e.g., modifying how transfers are managed).
- Applying technology (e.g., to eliminate or reduce trips and to monitor service delivery).

Relationships Between State DOTs and Tribal Transit Providers

The 2014 GAO report found that relationships between tribal transit providers and state DOTs can create challenges. The authors of the report stated that “the availability of rural transit program funds to subrecipients of the state transportation agency varied depending on the nature of the tribe’s relationship with the state.” Accordingly, “some tribes reported a preference for working with their local state government as a subrecipient, whereas others strongly preferred working directly with the federal government because of strained tribe-state relations.”

A 2018 webinar presented by the Build America Transportation Investment Center (BATIC) Institute⁸ identified the following challenges associated with coordination between state DOTs and tribal governments:

- Negotiating political conflicts between tribes and states, which can interfere with finding solutions that benefit all stakeholders.
- Waiving sovereign immunity.
- Entities not being familiar with mechanisms such as 202(a)(9) transfer agreements.
- Limited tribal input on state plans.
- Limited number of working relationships with state DOT leaders.
- Outreach efforts from state DOTs to tribes limited to NEPA requirements.
- Lack of collaboration.

A 2021 article by the Brookings Institute⁹ suggests that relationships between tribal governments and state DOTs might be improved if there is a better understanding of the funding challenges faced by the former (e.g., restrictions on tribes’ use of bonds as a funding mechanism). The article identifies state-tribal tax compacts as an opportunity for coordination but notes that such agreements have disadvantages as well as advantages.

The 2018 BATIC webinar identifies principles in the U.S. DOT Tribal Consultation Plan. These principles suggest actions that might be useful in state DOT practice:

- Proactively improve and maintain partnerships with tribal governments.
- Acknowledge and support tribes’ self-government, self-determination, and tribal sovereignty.
- Improve existing programs.
- Provide technical assistance in a timely way.
- Build administrative capacity.

Illustrative state DOT resources and practices for managing state DOT involvement in tribal transportation are described in Table 13.1. More on this subject is provided in the Guide’s next

Table 13.1. State DOT involvement in tribal transportation.

State DOT	Description of Rural/Tribal Transportation Resources and Practices
Arizona (ADOT)	The ADOT Tribal Consultation Policy was adopted on September 1, 2006, and is regularly updated. The policy requires ADOT to "work with the Tribes, Communities, and Native Nations to consult and coordinate on State and Tribal transportation in missions and goals." The ADOT website¹⁰ was developed "to support those efforts [required by the Tribal Consultation Policy] and to improve state-tribal intergovernmental relations through resource information sharing. It is designed to be a central location for state-tribal transportation partnerships, projects, activities, groups, links, and other related information." As part of multimodal planning studies, ADOT collaborates with tribes through the ADOT Planning Assistance for Rural Areas Program. The training page of the above-linked website provides an online training course to assist ADOT staff in working with tribal governments.¹¹ The workshop is accompanied by a downloadable handbook.¹²
Wisconsin (WisDOT)	The website https://wisconsindot.gov/Pages/doing-bus/civil-rights/tribalaffairs/default.aspx¹³ describes WisDOT's Tribal Affairs initiative, which "works exclusively on tribal relations and all transportation-related issues affecting tribal communities." Programs administered under the Tribal Affairs initiative include the Elderly Transit Program (https://wisconsindot.gov/Pages/doing-bus/local-gov/astnce-pgms/transit/tribal-eld.aspx),¹⁴ the Inter-Tribal Task Force (https://wisconsindot.gov/Pages/doing-bus/civil-rights/tribalaffairs/ittf.aspx),¹⁵ and the annual Tribal Transportation Conference (https://wisconsindot.gov/Pages/doing-bus/civil-rights/tribalaffairs/transp-conf.aspx).¹⁶
Alaska	The web page at https://dot.alaska.gov/cvlrts/tribal.shtml¹⁷ provides information about tribal transportation policies and resources. Included is an administrative order¹⁸ about coordination between the state and tribal governments.
Montana	The web page https://tribalnations.mt.gov/¹⁹ is the home page of the Governor's Office of Indian Affairs. The Office was established in 1951 by the Montana legislature to facilitate effective tribal-state communications with special attention to the discussion and resolution of issues that Indian peoples face regarding their unique political status and as full citizens of the state of Montana. The Office of Indian Affairs serves as a liaison between the state and the Tribes and promotes economic development, environmental protection, education, support for social services, and enduring goodwill. The page provides reports (e.g., the annual Tribal Relations Report), data, resources (including a tribal relations handbook), and a tribal directory. The tribal relations handbook is available at https://tribalnations.mt.gov/_docs/Tribal-Relations-Handbook.pdf.²⁰
North Dakota	North Dakota's long-range transportation plan (https://www.dot.nd.gov/sites/www/files/documents/ExecutiveSummary_July2021.pdf)²¹ provides limited information about tribal transportation in North Dakota.
Missouri (MoDOT)	The MoDOT web page at https://www.modot.org/tribal-consultation²² provides information about MoDOT tribal consultation activities in the context of historic preservation.

Table 13.1. (Continued).

State DOT	Description of Rural/Tribal Transportation Resources and Practices
California DOT (Caltrans)	This web page²³ provides information about Caltrans' Native American Liaison Branch (NALB), which is a program that "works to help California Native American communities identify and implement transportation safety solutions that will lead to improved safety for all—motorists, passengers, bicyclists, and pedestrians" (https://www.brookings.edu/articles/separate-but-unequal-how-tribes-unlike-states-face-major-hurdles-to-access-the-most-basic-public-finance-tools/).²⁴ The NALB is part of Caltrans' Division of Transportation Planning. In addition to identifying and implementing transportation safety improvements, the NALB "serves as liaison among tribes, Caltrans, federal agencies, other state agencies, and local and regional agencies for the purpose of helping to establish and maintain government-to-government working relationships with tribes and to provide information, training, and facilitation services related to issues affecting Native American communities." Each Caltrans district has a District Native American Coordinator and liaisons to coordinate with tribal governments. Caltrans maintains a local development review guidance document that includes content about coordination and consultation with tribes. This document is available at https://dot.ca.gov/-/media/dot-media/programs/transportation-planning/documents/office-of-smart-mobility-and-climate-change/administrative-draft-tribalguidance-ldigr-remediated-a11y.pdf.²⁵ The document applies to planning studies as well.

SOURCE: STATE DOT WEBSITES, ACCESSED FEBRUARY 2022.

Notes: Not all the information in the table is transit-specific. Some of the state DOTs may have programs and initiatives that are not listed.

chapter, Roles of Tribal Governments. The randomly selected sample of states in Table 13.1 shows that state DOTs vary greatly in the level of resources they dedicate to tribal transportation and in the extent of practices they undertake with respect to planning for, coordinating with, and supporting tribal transportation.

Training and Technical Support

The 2014 GAO report found that useful training is available to transit providers on topics such as funding, to an extent. The authors of the report stated that providers generally found value in training offered by FTA, but "a few noted that FTA-sponsored webinars were sometimes full or that a lack of local or state travel funding prohibited participation in training." Given the expanded availability and acceptance of online training since 2014, the lack of access to training provided by FTA, National RTAP, and others as a result of funding might not be a pressing concern at this point. However, providing financial support for training (e.g., in the form of travel assistance or subsidized registration costs) or providing training directly is something that state DOTs can consider.

Blume et al. (2021) provides case studies of small and mid-sized transit providers with innovative approaches to obtain cost-effective training in *TCRP Synthesis 154: Innovative Practices for Transit Planning at Small to Mid-Sized Agencies*.²⁶

Data and Reporting

The 2014 GAO report noted that state DOTs report data to the rural NTD on behalf of their subgrantees. The report did not identify any positive or negative examples of how state DOTs accomplish this. However, the research team's experience supporting state DOTs with such reporting indicates that a review of state DOTs' practices and procedures for obtaining, verifying, and reporting data to the rural NTD might be informative.

Coordination Assistance

Coordinating rural and tribal transit services can be challenging. In the 2014 GAO report,²⁷ 12 of the 15 state DOTs and 11 of the 30 providers identified coordinating transit services as a challenge. The authors stated that, in 2012 research, "we found that state and local efforts to coordinate transportation services—such as coordinating councils, call centers, and regional planning—can be hindered due to challenges such as insufficient federal leadership, changes to state policies, and limited financial resources." The authors provided examples of coordination challenges:

- The Navajo Nation includes portions of three states and receives funds from each. However, buses purchased with a given state's funding by Navajo Nation Transit Service are sometimes required to be used only within that state. This can create service inefficiencies.
- Three transit providers noted that their states' switching to statewide brokerages for Medicaid-funded NEMT services made coordinating transit services more challenging.

The authors also provided examples of coordination successes:

- State funding of regional transit services in Washington linked separate transit systems and facilitated their coordination. This resulted in higher ridership and effectively expanded service areas.
- Multiple transit provider staff have stated that creating personal relationships with staff of proximate transit providers improves coordination.

The 2014 Mineta report identifies coordination between providers as a potential challenge. While coordination could provide additional resources and opportunities, coordination could also result in "service expansion geographically as well as increased service hours," and a need to figure out how to address service gaps and service duplication, given pressure to "spend local funds locally." Coordination might also require providers to reconsider topics such as insurance and driver drug and alcohol testing.

Hiring and Retaining Bus Drivers

The 2014 GAO report found that hiring and retaining bus drivers can be challenging. Twenty-four of the 30 rural and tribal transit providers interviewed reported that it was difficult to hire and retain qualified drivers. Some providers reported attempting to solve this problem by offering driver training as part of employment. This same issue was reported in this research project's surveys—the survey of state DOTs and transit providers.

The Transit Workforce Center reports that apprenticeship programs for "frontline" transit workers, such as bus drivers, can generate a positive return on investment.²⁸ Apprenticeship programs positively affect bus driver hiring and retention, according to a 2019 article by the Eno Center for Transportation.²⁹ Multiple sources of funding (e.g., through a state department of labor) are available to support apprenticeship programs. To be eligible for funding, apprenticeship programs must be registered by a state or federal agency. State DOTs can assist transit providers that are trying to establish eligible apprenticeship programs.

 CHAPTER 14

Roles of Tribal Governments

Introduction

There are many similarities between tribal and nontribal transit systems. However, the issues and complexities that come with managing or engaging with a tribal transit agency and the transit needs of tribal communities are unique. It is important to recognize that tribes are sovereign nations. (The term “tribes” as it is used in this chapter is inclusive of Native Americans and Alaska Natives.) Tribes are not a subset of federal, state, or county governments.

Like many forms of government, tribal governments focus on the role of their governing bodies; councils (including tribal councils); staff turnover; effective tribal government support; relationship with the federal, state, and local governments; funding; finding qualified employees; and building and maintaining adequate facilities. Further, most tribes have their own culture, which may affect how they govern, plan, and work with other entities.

Some tribal communities are proximate to urban areas, some are suburban, while others may be rural or extremely isolated. Some tribes join into regional public transportation service, while others operate their own transit service. The operation and service design concepts are the same as they are for any other rural system with similar geography and demographics. The rules and guidelines still apply to tribal transit.

The U.S. Census identifies 324 federally recognized American Indian reservations (Facts for Features: American Indian and Alaska Native Heritage Month: November 2022. U.S. Census),³⁰ and all are on the same level as the nation on the standard hierarchy of census geographic entities³¹ (a display of the relationships between legal, administrative, and statistical boundaries maintained by the U.S. Census Bureau). Some of the geographic intricacies that tribal transit systems encounter are also explained in the U.S. Census website publication, *Understanding Geographic Relationships: American Indian Areas*: “Federal American Indian Areas and off-reservation trust lands do not need to fall within any other geography, like states or counties. In fact, they often cross state and county lines.”³²

To provide transit, some tribes participate in regional services, either coordinating with or contracting for such services, while others operate their own transit services. With tribal systems, as with any other rural transit system, there is a wide range of services and methods of operating the services. The service design concepts and planning for tribal transit services are generally the same as they are for any other rural system with similar geography and demographics. Most rules and guidelines that are relevant for any transit system, especially rural systems, apply to tribal transit.

As with many transit systems, funding for tribal transit can be a challenge, and a variety of sources may be used to ensure adequate funding for the desired service. While tribal governments have access to most traditional transit funding sources (FTA Programs for Tribal Recipients),³³

they also have access to a few specialized sources through the Public Transportation on Indian Reservations Program (Tribal Transit Program or TTP).³⁴

It is important to note that tribal governments do not have certain funding sources that other local governments have, such as property taxes. Tribes may find local funding to start a transit service, but it is not always reliable or sustainable. Recognition of tribal transit's complexity is becoming more evident in the policies and processes of federal and state agencies.

This chapter provides information about selected topics related to tribal transit, including the role of tribal governments in supporting public transportation in tribal and nontribal communities. Key topics include the following:

- Governance
- Funding and Compliance
- Tribal and Nontribal Systems Working Together
- Operations and Planning

Note that this chapter includes considerable information that is contained in the National RTAP Tribal Transit section of the Transit Manager's Toolkit,³⁵ published in 2024 and available at <https://www.nationalrtap.org/Toolkits/Transit-Managers-Toolkit/Tribal-Transit/Welcome>.

Governance

Understanding the relationships between tribal governments and transit stakeholders at the federal, state, and local levels is important. The sovereignty of tribes directly affects how tribal governments and transit funding sources interact with each other.

Tribal sovereignty is the inherent power of individual tribes to create and enforce laws within jurisdictional boundaries and to apply laws to activities and people within their jurisdiction. Tribal band autonomy refers to the idea that tribes are unique. There should be no expectation that any two tribal governments will function in the same manner.

The significance of tribal sovereignty and the impact it has on a tribe's ability to provide necessary transportation services for its community cannot be overstated. The operation of a public transit program produces a complex set of relationships within tribal government and between the tribe, with agencies at the federal, state, and local levels, with the tribal and nontribal public, and with numerous vendors and consultants.

As of 2024, the U.S. Department of the Interior's (DOI's) Bureau of Indian Affairs (BIA) lists 574 sovereign Tribal Nations (variously called Tribes, Nations, Bands, Pueblos, communities, and Native villages) with a formal nation-to-nation relationship with the U.S. government. These tribal governments are legally defined as "federally recognized Tribes." Federally recognized Native American Tribes or Alaska Native villages, groups, or communities are identified by the U.S. DOI BIA's annual Federal Register listing of Indian Entities Recognized by and Eligible to Receive Services from the United States Bureau of Indian Affairs.³⁶ A tribe must be federally recognized to be eligible for federal transportation funds (see also Chapter 11, Funding and Revenue, in Part II, and Chapter 13, State DOT Role, in Part III).

Tribal Self-Governance and Self-Determination. Tribal Self-Governance refers to the authority of tribes to self-administer certain federal programs and services for their tribal members. This authority comes from three major pieces of legislation:

- The Indian Self-Determination and Education Assistance Act of 1975 (ISDEAA), as amended, 25 USC 5301 et seq.
- The Tribal Self-Governance Act of 1994, 25 USC 5362 et seq.

- The Practical Reforms and Other Goals to Reinforce the Effectiveness of Self-Governance and Self-Determination for Indian Tribes Act of 2019 (The PROGRESS Act), Public Law 116-180, Summary: Public Law No: 116-180 (10/21/2020).

ISDEAA describes self-determination as the ability of tribal governments to determine how to build and sustain their own communities. As of 2024, the DOI's Office of the Assistant Secretary - Indian Affairs (AS-IA) Office of Self-Governance reports that Self-Governance Tribes represent approximately 50% of all federally recognized tribes nationwide.

More information about the Tribal Self-Governance Program is available at the DOI AS-IA's Office of Self-Governance website.³⁷

Additional insight into the organization of today's tribal governments, including the governing bodies of Alaska Native communities, is available from the National Congress of American Indians report: Tribal Nations & the United States: An Introduction, Updated February 2020.³⁸ BIA's map of the regions Indian Lands of Federally Recognized Tribes of the United States³⁹ offers a clear visual representation of the Tribal Nations and is updated regularly.

More information and resources about government-to-government relationships between federally recognized tribes, the U.S. Government, and individual state governments are available in the Tribal Transit Section of National RTAP's Transit Manager's Toolkit under Governance.

Funding and Compliance

The requirements around administration and funding for tribal transit programs can vary based on how a tribe decides to access its share of the Public Transportation on Indian Reservations (Tribal Transit) Program (TTP)⁴⁰ federal funds. For example, a tribal transit system that operates under the sponsorship of a tribal government that has chosen to access funds directly from the federal government through a Tribal Transportation Program Agreement (TTPA)⁴¹ with the FHWA will have different administrative requirements than a tribal transit system that operates under a Tribal Transportation Self-Governance Program (TTSGP)⁴² compact with the U.S. DOT.

The following section describes the relationships between federal funding agencies and tribal transit programs and provides information and resources about federal and state funding sources and the types of funding contracts that are possible.

Administration and Funding

When the Infrastructure Investment and Jobs Act (IIJA), also known as the Bipartisan Infrastructure Law (BIL), was signed into law on November 15, 2022, it continued the TTP with no changes to the formula allocations designed to fund planning, capital, and operating assistance for tribal public transit services. The purpose of the TTP also remained the same: to provide safe and adequate transportation and public road access to and within Indian reservations, tribal lands, and Alaska Native Village communities. However, in direct response to the agency directives associated with the federal administration's New Actions to Support Indian Country and Native Communities, the BIA, the FTA, and the FHWA were among the many federal agencies that have updated information on their websites to provide tribal communities and Native people with clearer direction and guidance on how to obtain available funding through the TTP and other transit-related resources. In direct response to the agency directives associated with the federal administration's New Actions to Support Indian Country and Native Communities, the BIA, the FTA, and the FHWA are among the many federal agencies that have updated information on their websites to provide Tribal communities and Native people with clearer direction

and guidance on how to obtain available funding through the TTP and other transit-related resources. See BIA's press release: Interior Department Strengthens Tribal Consultation Policies and Procedures⁴³ (December 1, 2022).

FHWA. As an agency within the U.S. DOT, FHWA is responsible for program administration of the Federal Lands and Tribal Transportation Program (FLTTP)⁴⁴ and provides oversight for direct funding agreements with federally recognized tribes. The tribal lands portion of the FLTTP is managed by the FHWA's Office of Tribal Transportation. This office jointly manages and administers the TTP under a memorandum of agreement between FHWA and the U.S. DOI's BIA.

"The Federal Highway Administration's (FHWA) Office of Federal Lands Highway and the Bureau of Indian Affairs (BIA) provide funding and support for addressing the transportation needs of the 574 federally recognized tribes. The FHWA's Office of Tribal Transportation provides stewardship and oversight for direct funding agreements with 135 of the federally recognized tribes and provides support for all FHWA activities affecting Tribal transportation. The BIA provides support for the remaining tribes."⁴⁵

FTA. A primary role of the FTA is to provide financial and technical assistance to local public transit systems, including rural and tribal transit systems. The FTA also provides regional tribal liaisons⁴⁶ with the explicit purpose of supporting tribal transit.

Federal funding for tribal transit in rural areas comes mainly from the FTA through Section 5311, Rural Areas Formula Program, which includes several subprograms, including the following:

- Tribal Transit Competitive Program, 49 USC Section 5311(c)(1).⁴⁷
- Tribal Transit Formula Grants - 5311(c)(1)(B) | FTA (dot.gov).⁴⁸
- Appalachian Development Public Transportation Assistance Program (ADPTAP), 49 USC Section 5311(c)(3). As of this writing, this program is not accepting new submissions. ADPTAP is in the BIL, and ADPTAP FY 2023 funding⁴⁹ was published at <https://www.transit.dot.gov/funding/apportionments/table-9-fy-2023-section-5311-and-section-5340-rural-area-formula>.
- Rural Transportation Assistance Program (RTAP), 49 USC Section 5311(b)(3).⁵⁰

In addition to the main federal funding programs for tribal transit in rural areas, several support programs are available for federally recognized tribes providing public transportation services. Three of the most significant programs are the TTSGP, the Public Transportation on Indian Reservations TTP Program, and the Public Transportation on Indian Reservations Tribal Transit Competitive Program. The latter two programs are set-asides from Section 5311.

TTSGP. The purpose of the TTSGP is to provide tribes and tribal organizations with greater control, flexibility, and decision-making authority over U.S. DOT funds used to carry out TTPs, functions, services, and activities (PFSAs) in tribal communities. A tribe is eligible to participate in the Program if:

1. The tribe requests participation in the Program by resolution or other official action by the governing body of the tribe; and
2. The U.S. DOT determines, based on the evidence submitted by the tribe, that, over the 3 most recent fiscal years, the tribe has demonstrated financial stability, financial management capability, and transportation program management capability per regulatory criteria.

The following funds may be included in a funding agreement through the TTSGP:

- Tribal Transportation Program funds
- Transit Program funds

- Discretionary or competitive grant funds
- Title 23 (Federal-aid Highway) funds
- Section 5311 funds awarded to a state

Once a tribe is a part of the TTSGP, it would have day-to-day responsibilities for ensuring financial and procurement standards are in place and being followed, and for records management and property management standards. More information about the TTSGP may be found on the U.S. DOT Tribal Transportation Self-Governance Program web page.⁵¹

TTP. The TTP was established in 2013 under the Moving Ahead for Progress in the 21st Century Act (MAP-21) (2012). The TTP program provides two types of funding: formula program funds [Tribal Transit Formula Grants—5311(c)(1)(B)] and competitive program funds [Tribal Transit Competitive Program 49 USC 5311(c)(1)(A)]. Before MAP-21, TTP was known as the Indian Reservation Roads Program. In 2021, the BIL continued TTP with no changes to the formula allocations for tribal public transit services. BIA and FHWA jointly administer the distribution of TTP funds.

The TTP is a set-aside of the Formula Grants for Rural Areas Program (Section 5311) and has two main components:

- TTP formula funding is apportioned to tribes who report transit data to the NTD (<https://www.transit.dot.gov/funding/grants/tribal-transit-formula-grants-5311c1b>). Census data is also used under the formula program.
- TTP competitive funding is made available in a Notice of Funding Opportunity each year (<https://www.transit.dot.gov/tribal-transit>).

The FHWA's Tribal Transportation Program Data Report for 2021, Attachment A: TTP Program Summary,⁵² and the FHWA OTT Webinar: TTP-101 Tribal Transportation Program Overview⁵³ provide useful descriptions of the contracting mechanisms available to eligible TTP recipients. Tribes with a TTPA may receive their funding in advance directly from FHWA and are responsible for managing their TTP funds. The FHWA is responsible for the stewardship and oversight of the provided funds and ensures that the program is carried out in accordance with the contract and statutory and regulatory requirements.

Four additional contracting mechanisms are available for tribes that opt to work directly with BIA, which include the following:

1. **BIA Direct Service Agreements.** In consultation, the BIA is responsible for carrying out all aspects of TTP on behalf of the tribe, including delivery, stewardship, and oversight. In FY 2021, 22 tribes were reported as having Direct Service Agreements with BIA for their TTP.
2. **BIA PL93-638/Self-Determination Contracts.** Contracts are developed between BIA and a tribe in accordance with the ISDEAA, Public Law 93-638. Generally, each contract is for a specific project or activity. The tribe is provided TTP funding on a contract-by-contract basis from BIA and is responsible for carrying out the work identified. The BIA is responsible for the stewardship and oversight of the provided funds and ensures that the program is carried out in accordance with the contract and statutory and regulatory requirements. In FY 2021, 174 tribes were reported as having self-determination contracts.
3. **Self-Governance Compacts.** Compacts are developed between the tribe and the DOI's Office of Self-Governance. The tribe receives its funding in advance from the DOI and manages its TTP in a way that is similar to FHWA Program Agreements. However, these compacts include certain protections and other options afforded to them in accordance with the ISDEAA. The DOI is responsible for the stewardship and oversight of the provided funds and ensures that the program is carried out in accordance with the compact and statutory and regulatory requirements. In FY 2021, 38 tribes were reported as having self-governance compacts.

4. **BIA Government-to-Government (G2G) Agreements.** A tribe operating under a G2G Agreement receives its funding in advance from BIA and is responsible for managing its TTP. The BIA is responsible for the stewardship and oversight of the provided funds and ensures that the program is carried out in accordance with statutory and regulatory requirements. In FY 2021, 214 tribes were reported as having G2G Agreements.

More detailed descriptions of the primary federal programs and grants that are available to tribal transit providers and the relationships between the various funding sources are available in the Tribal Transit Section of National RTAP's Transit Manager's Toolkit under Tribal Administration and Funding.⁵⁴

The U.S. DOT/FHWA's Tribal Transportation Delivery Guide – A Guide for Tribes with an FHWA Tribal Transit Agreement, March 2023 explains how the federal agencies (and departments) must work cooperatively together in a G2G relationship with tribes to administer the TTP.

The FTA's Funding Opportunities for Tribal Entities website contains the most current information on all facets of working with the FTA to obtain funding through numerous FTA programs for which federally recognized Native American Tribes or Alaska Native villages, groups, or communities, as identified by the U.S. DOI's BIA are eligible as direct recipients and subrecipients (state pass-through).

Compliance and Technical Assistance

A federally recognized tribe may choose to access its share of Public Transportation on Indian Reservations Tribal Transit Program TTP federal funds directly from the federal government through a TTPA with the FHWA or a TTSGP compact with the U.S. DOT. Tribes which are not federally recognized remain eligible to apply to the state(s) they reside in as subrecipients under the state's apportionment for Section 5310, 5311, or 5307 funding.

- **Tribes as Subrecipients.** If the tribe receives funding through the Section 5311 program, it is often the case that the state is the FTA direct recipient, and the tribe is a subrecipient of the state. As a subrecipient of the state, the tribe has the same grant program administration requirements and responsibilities as other tribal and nontribal Section 5311 subrecipients. Tribal subrecipients enter into written agreements with the state that indicate the terms and conditions of assistance for their project, and the state submits certifications, assurances, and NTD data to the federal government on their behalf.
- **Tribes as Direct Recipients.** Federally recognized Native American Tribes or Alaska Native villages, groups, or communities are eligible direct recipients under the Public Transportation on Indian Reservations Program (tribal transit program or TTP) in Section 5311(c)(1). Federally recognized tribes may also elect to become direct recipients under Section 5311 in addition to being subrecipients under a state 5311 program. For direct TTP recipients, tribal requirements vary in specific ways from those under the Section 5311 program.

Tribal Transit Technical Assistance Assessments. While the FTA does not conduct oversight reviews of tribal recipients, it does conduct Tribal Transit Technical Assistance Assessments designed to provide technical assistance to tribes that are direct recipients of TTP funding. Tribal transit managers who are recipients of TTP funds are encouraged to review and use the FTA's Tribal Transit Program Assessment Manual Fiscal Year 2023⁵⁵ as a resource for ensuring a basic understanding of FTA's legal and regulatory requirements. Written in an easy-to-follow question-and-answer format, the manual lists the FTA's basic expectations and requirements for TTP recipients and contains detailed explanations of the actions required by TTP recipients to

ensure compliance. Areas of review include legal, award management and project management, financial management and NTD, transit asset management, satisfactory continuing control, maintenance, procurement, charter service, procurement, federal nondiscrimination requirements (civil rights programs), ADA, and other program requirements.

According to the FTA's Tribal Governments website, the FTA identifies tribal recipients, on an annual basis, that could benefit from an assessment. Geographic diversity, overall recipient coverage, and independent requests from tribes are considered. An assessment consists of a desk review before the visit with assistance from FTA Region staff, an on-site visit with tailored technical assistance, and follow-up assistance on an as-needed basis for up to 90 days following the visit. To indicate interest in a site visit, tribal recipients are to notify their regional tribal liaison.

Tribal and Nontribal Systems Working Together

There are numerous benefits to be gained by tribal and nontribal systems working together. Many tribes operate transportation services that are open to the general public as well as to tribal members.

Tribal and nontribal transit systems serve transportation needs that could be met through initiating relationships with other local transit services. There is a wide variety of different tribal governments and tribal transit systems, and the purpose of tribal systems may be different. Some are strictly tribal-focused. Other tribal systems serve the greater community, providing trips to both tribal and nontribal residents. Still, the success of a tribal transit system often means overcoming issues related to vast service areas and low-population densities, service reliability, and cost. The development of partnerships with other tribal and nontribal systems can help overcome these issues.

These relationships often require enhanced communication, collaboration, coordination, connectivity, and consolidation, which will result in increased opportunities for the entire community. Working together involves considerable communication between the transit provider, the tribe, the state DOT, and the FTA.

Benefits of Working with Tribes and Tribal Transit Programs

There are real benefits to be realized in the process of engaging with tribes and tribal transit programs. Understanding the value of working together is particularly important when rural and tribal communities are experiencing gaps or overlaps in transit coverage. The overlap or duplication might be between two tribal systems or between a tribal and nontribal system.

Communication

Sharing information about respective rural and tribal transit programs and services is the first step to helping more people access transportation to health care, employment, education, and other basic needs. However, communication goes both ways, so receiving information from the tribe or rural system is just as important as sharing information. Some examples of how a tribal transit system can benefit from open communication with other rural transit systems are the following:

- Obtaining tribal members' input on how local transit services could result in more effectively meeting community transportation needs and reducing transit system gaps and overlaps.
- Cross-cultural, historical, and political exchange, mutual respect, and dialogue may nurture relationships and refine intergovernmental communication.

Collaboration and Coordination

Collaboration and coordination are particularly important in challenging service areas with little in the way of transit resources. Tribal transit systems, tribal systems, and nontribal systems working together afford opportunities to increase service and ridership. Here are some examples:

- Writing grants collaboratively is a major selling point; two tribes or tribes and nontribal systems working together make a strong statement.
- Improving transit system sustainability and longevity by sharing costs and pooling financial resources for service provision (e.g., partnerships can reduce administration and training costs to both rural and tribal transit systems).
- Cross-jurisdictional coordination and engagement to enhance regional transit and grow collaborative systems with shared system investments.
- Shared maintenance resources can go a long way to keeping funds local and within the tribal entities.
- Eliminating duplicative services reduces costs and allows for other improvements.
- Increased access to grant funds (i.e., rural transit grant applications become more competitive when funding requests are for service provision to a broader population).
- Coordination of support functions like driver training, maintenance, group procurements, and other nonthreatening functions is to everyone's benefit.
- Providing a firm foundation for additional future intergovernmental partnerships to achieve numerous common multidisciplinary goals.

Connectivity, Consolidation, and Operations

The ability of tribal and nontribal systems to connect and work together on regional goals offers opportunities for all systems to improve their operations. Here are some examples:

- Connections to the intercity bus and rail network allow residents to travel across the country.
- Added transit system connections to cultural, historical, and tourist sites.
- Allowing riders to travel beyond jurisdictional lines as needed, seamlessly increasing access for all riders.
- Working together on maintenance and training allows opportunities for specialty work that small systems cannot afford.
- Potential to administer the tribal portion of a rural regional transit system, if the tribe is too small to take on transit administration or desires tribal transit service provision but prefers not to self-administer the service.
- Eliminating costly duplication of services, where possible, enables greater service and ridership for residents and visitors.

For more thoughts and ideas on how to best engage with tribal governments and tribal transit systems, the reader is encouraged to review *Tribal and Non-Tribal Systems Engaging*⁵⁶ in the Tribal Transit Section of National RTAP's Transit Manager's Toolkit.

Overcoming Operational Challenges in Tribal Areas

The BIA describes the size of Indian land areas: "There are approximately 326 Indian land areas in the U.S. administered as federal Indian reservations (i.e., reservations, pueblos, rancherias, missions, villages, communities, etc.). The largest is the 16 million-acre Navajo Nation Reservation located in Arizona, New Mexico, and Utah. The smallest is a 1.32-acre parcel in California where the Pit River Tribe's cemetery is located. Many of the smaller reservations are less than 1,000 acres."

Challenges in Coordinating Services. Tribal transit systems often, but not always, cover large areas of service, have long travel distances, and provide service from hubs that are secluded

from other communities in their region. They are responsible for transporting tribal citizens and often other residents to neighboring communities and urban centers to provide access to employment, medical services, educational opportunities, goods, and services. Tribal transit providers typically have lower ridership numbers and higher mileage per vehicle than their urban counterparts. Working to identify, address, and alleviate these transit planning and operating factors can prove challenging.

Service Coordination Issues and Approaches. Given the challenges that come with operating a transit system with low-density populations, long travel distances, and dispersed destinations, it may be difficult to understand how best to maximize the use of scarce resources. Buses may not have full occupancy or may sit idle for periods of the day. Existing routes may duplicate those run by a neighboring service provider. In addition, tribal transit systems often face challenges due to limited staffing, administrators who are new to the field of transportation, and smaller operating budgets. Tribal transit systems can realize mutual benefits by partnering with other tribal and nontribal systems. The North Central Regional Transit District (NCRTD) case study included in this report tells the story of how a regional transit district helped smaller Pueblos feel comfortable in coordinating services with larger city and county resources (see the NCRTD case study in Chapter 16, Part IV of the Guide).

Tribes Working with States, Counties, and Local Governments

Transportation services for older adults, tribal elders, people with disabilities, those with lower incomes, and others with limited transportation options in tribal communities are supported by numerous transportation and human service programs. Many states are trying to facilitate the coordination of these transportation programs, working at the local, regional, and statewide levels. Tribal transit providers will want to be aware of the efforts in their states and of federal programs to better understand how they can benefit from involvement.

The Coordinating Council on Access and Mobility (CCAM) is a federal interagency council convened by FTA that coordinates funding and provides expertise on human services transportation for people with disabilities, older adults, and individuals of low income. Tribal-specific resources are listed in the CCAM inventory,⁵⁷ such as the TTP, Section 5311 Tribal Transit Program, Indian Employment Assistance, Indian Schools - Student Transportation, Native American Employment and Training, and Special Diabetes Program for Indians (SDPI).

Tribal governments play a central role in the success of individual tribal transit systems. While the transit needs in many tribal communities may be similar, the policies and processes that govern individual tribes and tribal communities may be very different. As will be described later in this section, great care should be taken to ensure that tribal leaders and government officials are made aware of and are invited to participate in every step of any transit project that may have tribal implications. In addition, nontribal entities desiring to provide services to or coordinate with existing tribal transit services should be encouraged to take the time to learn how to interact appropriately with local tribal government and officials.

Most state DOTs have programs in place to work with tribal governments on federally funded transit projects using, for example, Section 5311 and Section 5311(c)(1)(b). With limited federal, state, and tribal funds available, proposals featuring cooperative arrangements to serve the transit needs of an area are encouraged; funding proposals are evaluated based on the avoidance of duplicative or parallel transit activities.

Working relationships between state and tribal governments vary from poor to very good. Many state DOTs have a tribal liaison, so if there are plans to work with a tribal government/transit system, a first step should be checking with the state DOT for help in developing a relationship

with the tribes. Further, most tribal transit systems participate in their state or regional transit associations, and the FTA has Regional Tribal Liaisons that can also help connect nontribal transit systems with tribal systems.

In addition, state DOTs, through their relationships with federal funding sources, have a mandated responsibility to establish meaningful consultation processes with tribes when a project is determined to have tribal implications. Tribes, counties, and other local governments may also find it helpful to establish such processes for tribal consultation and joint meetings for local transportation decision-making. For additional information about the interaction between states and tribal governments, please see Chapter 13, State DOT Role.

What follows is an adaptation of the Fundamental Principles and Best Practices for Tribal Consultation Processes as recommended by the White House Council on Native American Affairs.⁵⁸ The Tribal Transit Section of National RTAP's Transit Manager's Toolkit also includes this guidance.

Fundamental Principles for the Development of Tribal Consultation Policies⁵⁹

- Tribal consultation policies should strive for uniformity, which includes using common definitions, processes, and time frames for purposes of consistency, ease of tribal use, and promoting a whole-of-government approach.
- Tribes are sovereign nations with self-governance and self-determination rights.
- Tribes have a two-way Nation-to-Nation relationship with the United States.
- Tribal consultation may be initiated at the request of a Tribal government.
- Tribal treaties are substantive federal law and do not derive from other federal statutes.
- Treaties, like the U.S. Constitution and federal statutes, are the supreme law of the land.
- Tribal treaties are binding legal agreements between or among two or more sovereign nations.
- Tribal consultation should be meaningful. A meaningful consultation is, among other inclusive practices: carried out in a timely, efficient, and responsive manner; transparent and predictable; accessible, reasonable, flexible, and fair; founded in the principles of good faith and respectful of the sovereignty of Native American Tribes; and includes reasonable accommodation (e.g., changing of timelines, project parameters), where appropriate.
- Information obtained from tribes should be given meaningful consideration, and tribal consultation should strive for consensus or a mutually desired outcome.
- Include both nontribal and tribal officials with decision-making authorities regarding the proposed policy or action that has tribal implications.
- Consider conducting the consultation in a manner that prioritizes the participation of official tribal government leaders.
- Be familiar with tribally enacted consultation laws, statutes, policies, or protocols and, as appropriate, develop mutually agreed upon consultation protocols tailored to the consultation standards of individual Tribes.
- Respect and elevate Indigenous knowledge, including cultural norms and practices relevant to such consultations.
- Identify processes for notifying tribes of how the tribal input influenced the decision-making after the consultation has concluded and the decision or action has been made.
- All these principles should be applied to the extent practicable and permitted by law.

Best Practices for Tribal Consultations and Meetings⁶⁰

- **Begin Early.** Consultations and meetings should begin early in the project development phase, during the initial planning stages.

- **Establish Protocols.** Consultation and meeting protocols should be developed ahead of time.
- **When to Consult.** If there is any question whether a proposed project, policy, or other action planning will have tribal implications, then a consultation or meeting should be conducted.
- **Identify the Appropriate Decision-Makers.** Given the subject matter, all parties should ensure that representatives with appropriate expertise and, to the extent practicable, decision-making authority regarding the proposed policy or action planning are present at a consultation or meeting.
- **Creating Effective Notice.** Consultations and meetings should be scheduled such that the tribes are given timely, adequate notice, absent extenuating circumstances.
- **Read-Ahead Materials with Notice.** All pertinent information about a topic, including an agenda, should be made available at a reasonable time before the consultation or meeting.
- **Access.** Consultation or meeting opportunities should be accessible and convenient to tribal participants.
- **Notifying Appropriate Tribal Participants.** All potentially affected tribes should be notified and invited to consult or meet.
- **Meaningfully Consider Information Received from Tribal Nations.** Information obtained from tribes should be given meaningful consideration; this can happen only if tribes are both properly informed and tribal input is solicited early enough in the planning process that it may influence the decision to be made.
- **Information from a Nontribal Government Official.** When transit agencies receive information during consultation from a source other than a tribal government official, the agency should ascertain whether that source has appropriate tribal authority to speak for the tribal government by communicating with tribal government officials.
- **Considering Multiple Perspectives.** The existence, nature, or scope of an asserted tribal right may not be clearly established or may be disputed by other tribes, third parties, or others. In these instances, all parties should carefully consider information and views provided during the consultation process and, as appropriate, engage with others with related interests and responsibilities that may be affected by the decision.
- **Record of Consultation.** Maintain a record of the consultation or meeting process that includes: a summary of tribal input received; an explanation of how tribal input influenced or was incorporated into the transit agency action; and the reasoning for why tribal suggestions were not incorporated into the proposed policy or action, or why consensus could not be attained.

Operations and Planning

This section focuses on the day-to-day operational issues that may be exclusive to tribal transit systems.

Operations

Operations Management Models. There are many factors to consider when deciding whether to manage operations in-house or to use a third-party contractor for transit system management and/or operations. In addition to tribal, state, or local laws, factors to consider may include the cost associated with a particular program, availability and expertise of staff, planning strategies, control over operations, customer service, and training standards. Tribal transit agencies may also choose to enter into management contracts (MOUs) with tribal or nontribal organizations. Using a third-party operator does not relieve an FTA grantee from compliance with federal

requirements. How the tribe has chosen to access tribal transit funds (e.g., the TTP and TTSGP) may also be a factor. Contracting models are described briefly as follows:

- **Turnkey Contract.** A company that takes care of everything from providing the management team and staff members to owning and maintaining all necessary capital, including vehicles, equipment, and operating/administrative facilities. The responsibilities of a tribal transit agency as an FTA grantee would include planning, applying for, and administering the grant, establishing policies, and overseeing the contract.
- **Executive Employment Contract.** Hiring high-profile executives for upper management transit professionals, while all other employees and capital equipment would be an in-house responsibility.
- **Full-Time or Part-Time Employment Contract.** Provides all transit staff positions for the complete administration and operation of the transit agency. The contractor can be a single individual or a contracting firm. The tribal transit agency owns all the capital equipment.
- **Service Contract.** A tribal transit agency may work with one or several contractors to provide service for fixed-route, demand response, and human service transportation. The agency might also contract out for vehicle maintenance, or other support functions associated with the service.

Human Resources. Many tribal transit agencies have developed effective ways of managing their human resources. It should be noted that, depending on how a tribe receives its transit funding (federal or from other sources), there may be policies and procedures that apply only to transit and to nothing else in the tribe's services to members. An example might be the FTA drug and alcohol requirements. The following list includes links to a sampling of tribal transit human resource policies:

- Comanche Nation Transit⁶¹ has policies in place to ensure that they are following all guidelines with respect to people's and employees' rights.
- The Colville Confederated Tribes Department of Transportation (CCT/DOT) Transit⁶² has a Tribal Council⁶³ and a policy manual for the Confederated Tribes of the Colville Reservation (<https://www.colvilletribes.com/>).⁶⁴
- The Cherokee Nation⁶⁵ uses its Tribal Employment Rights Office⁶⁶ where applicable.
- Muscogee (Creek) Nation Transit⁶⁷ has a Tribal Human Resources department.⁶⁸ It is a great resource for assistance with maintaining records and employee management needs.
- The Mississippi Band of Choctaw Indians⁶⁹ uses the Mississippi Band of Choctaw Indians Human Resource Department to assist and maintain records as well as employee management needs (<https://www.choctaw.org/>).

Tribal Employment Rights Ordinance. A Tribal Employment Rights Ordinance (TERO) is sovereignty-based legislation enacted by a tribal government. "The TERO program is to enforce tribally enacted Indian Preference law to ensure that Indian/Alaska Native people gain their rightful share to employment, training, contracting, subcontracting, and business opportunities on and near reservations and native villages."⁷⁰ TERO programs derive their authority from an individual tribe's inherent sovereign status and its authority to enact and enforce Indian/Native employment preference law.

Driver Recruitment, Training, and Retention. It can be challenging for tribal transit systems to recruit and retain drivers, a challenge faced across the transit industry. One strategy is to recruit standby and part-time vehicle operators. Because commercial driver's license (CDL) training is not required for smaller vehicles, the strategy of using a fleet of smaller vehicles (when there are no capacity constraints) is sometimes beneficial to recruitment, training, and retention. Big Woods Transit (BWT)⁷¹ is a small Tribal system operated by the Bois Forte Band of Chippewa

(also referred to as Ojibwe) in Northern Minnesota. Like many other transit systems, BWT had difficulties hiring and retaining vehicle operators, resulting in its transit managers filling in as drivers. BWT was able to hire part-time workers from other tribal government departments. CDL training was not required, as most of their vehicles are smaller. Some of the new vehicle operators were hired on regular schedules, and some were on call.

Training. A selection of tribal transit-specific training resources is listed:

- National RTAP maintains a Tribal Transit Topic Guide⁷² that includes training and technical assistance specifically geared toward tribal transit managers, drivers, and riders.
- State RTAP funds are often used to provide training and technical assistance for rural and tribal transit agency personnel and to fund scholarships to attend transit-related conferences and training. Each state and U.S. territory has an RTAP manager. Available state RTAP websites⁷³ and a Directory of State RTAP Managers⁷⁴ are listed on the National RTAP website.
- The Native Learning Center (NLC)⁷⁵ offers free training and technical assistance to Native Americans and those working within Indian Country. The types of training and technical assistance offered by NLC include multi-session programs in grants management for tribes and tribal entities and grant writing training. The NLC website also contains a page dedicated to grant opportunities for tribes.
- The National Transportation in Indian Country Conference (NTICC)⁷⁶ describes itself as the “one-stop-shop transportation conference FOR TRIBES, BY TRIBES.” NTICC is an annual event that offers access to relevant training and an opportunity for tribal transportation professionals to engage with federal officials and their peers regarding Tribal transportation issues and challenges. Past conference tracks have included Safety, Workforce Development, TTP 101, Planning, Transit, Leadership/Infrastructure Bill, Project Management, and Hot Topics.
- FHWA’s Office of Tribal Transportation (OTT website)⁷⁷ has links to up-to-date resources and training opportunities.
- The FHWA Office of Transportation Workforce Development and Technology Deployment manages a Tribal Technical Assistance Program (TTAP)⁷⁸ to provide technical assistance and training to tribal governments.
- The Intertribal Transportation Association (ITA)⁷⁹ provides training and information and facilitates meetings between tribes and local, state, and federal agencies. ITA annual meetings are usually held in December.

Some tribal transit agencies have also developed their own internal training programs. For example, the Chickasaw Nation Transportation Services Program⁸⁰ purchased transit safety software and uses it as an orientation tool for new hires. They use the tool at monthly safety meetings to keep all driving staff updated and trained.

Additional Training Resources. As part of Human Trafficking Awareness Month, January 2023, FTA hosted a webinar, Human Trafficking Prevention in Tribal Communities,⁸¹ that explores how human trafficking impacts Native American/Alaskan Native populations, the role transportation and public transit has in human trafficking, how transit agencies can prevent human trafficking, and how drivers and other frontline transit employees can intervene if trafficking is identified.

The Rolling Oklahoma Classroom (ROC)⁸² is an FTA-funded grant initiative that provides hands-on mobile training to transit agencies and transit awareness for the public. Training opportunities include human trafficking awareness, understanding mobility issues, wheelchair securement training, transit availability options, and career opportunities.

Safety, Security, and Emergency Management. If a tribe’s transit program is to build and model a culture of safety, it must be combined with tribal government and community experience.

The following is a list of available resources that are specific to tribal safety, security, and emergency management needs:

- Tribal Declaration Process⁸³ Tribal governments may still choose to seek assistance, as they have historically, under a state declaration request. However, federally recognized Native American Tribal governments have the option to request a Presidential emergency or major disaster declaration independent of a state (<https://www.fema.gov/disaster/how-declared-tribal-nations>).
- The Federal Emergency Management Agency (FEMA) Tribal Affairs⁸⁴ website contains many tribal-specific resources intended to assist tribal governments with establishing and implementing emergency management processes and practices, including considerations for tribal transit. FEMA's Tribal Affairs Hub includes the following:
 - Regional Contacts. FEMA recognizes that Tribal Nations preexist state governments and that tribal lands may not follow state boundary lines. FEMA's Regional Contacts web page⁸⁵ is helpful when tribal lands cross multiple FEMA regions.
 - Tribal Consultations. Upcoming and past consultations are listed along with FEMA's Tribal Consultation Policy (2019) and Tribal Consultation Policy Instructions (2021).⁸⁶
 - Tribal Funding, Mitigation, and Planning Resources.⁸⁷ This page includes the following tribal resources:
 - Grants for Tribal Nations provides specific instructions for tribes wishing to apply for the Tribal Homeland Security Grant Program, the Staffing for Adequate Fire and Emergency Response, and other funding.
 - Tribal Mitigation Planning Resources provides access to a wide variety of resources related to the development of a hazard mitigation plan that includes transportation considerations. A Tribal Mitigation Planning Handbook and an Introduction to Tribal Mitigation Planning (YouTube) video are among the guidance. The FEMA Fact Sheet: Tribal Hazard Mitigation Planning Training, which provides an overview of FEMA's independent study course, IS-350: Tribal Hazard Mitigation Planning Training, is also included.
 - Preparedness - Training Resources for Tribal Nations⁸⁸ offers links to training opportunities, grant programs, and preparedness tips for tribes. The Center for Domestic Preparedness hosts the Annual Tribal Nations Training Week.⁸⁹
- Other examples of publications available on the FEMA website⁹⁰ include the agency's 2022–2026 FEMA National Tribal Strategy (August 2022), a Pre-Disaster Recovery Planning Guide for Tribal Governments (2019), and a Disaster Financial Management Guide: Guidance for State, Local, Tribal & Territorial Partners (April 2020). Is Your Tribe Ready? Prepare for Emergencies in Tribal Nations⁹¹ is an online brochure from FEMA's Ready Indian Country campaign to encourage Tribal communities and Alaska Native villages to prepare for disasters.
- Barriers to Collaboration Between Tribal and County Governments: Planning for Major Disasters and Other Emergencies, Health Policy Brief (January 2016).⁹² This brief discusses the complexity and offers guidance about negotiating formal cross-jurisdictional-sharing arrangements to identify how both governments (tribal and county) may work together to respond to and recover from emergencies across many jurisdictions.
- Local Tribal NIMS Integration: Integrating the National Incident Management System into Local and Tribal Emergency Operations Plans and Standard Operating Procedures Version 1.0 (2004).⁹³ This document was developed as a joint initiative between the National Incident Management System (NIMS) Integration Center and the Office of Grants and Training. It contains an Emergency Operations Procedures Checklist that Tribal Systems may find helpful.

Planning

This section highlights features of transit planning that tribal transit systems need to be aware of and identifies resources for specific planning topics.

Strategic Planning. Strategic planning activities provide transit agencies with a roadmap with guidance as their communities change over time. This roadmap provides a structure for service planning activities. Many tribal transit agencies (e.g., Oneida Indian Nation Transit System) have received funding through the FTA's Tribal Transit Competitive Program to develop strategic plans.

Service Planning. Service planning involves taking inventory of the transit agency's available resources and the transit needs of tribal members. Once the tribal transit manager or planner knows what funding, vehicles, facilities, equipment, and staff are available, and the services that are currently provided, those resources and services can be matched with the transportation needs of tribal members and other stakeholders, as appropriate. The reader is directed to the Tribal and Nontribal Systems Working Together section of this chapter for additional ideas on how to effectively engage tribal members and other stakeholders (e.g., public involvement, passenger surveys, community surveys, focus groups, and public meetings). All stakeholders will need to know who has the authority to speak and act for the tribe as a whole and to represent it in negotiations with federal, state, and local governments.

Required Planning. The type of planning required of a tribal transit agency depends on how the tribe has chosen to draw down or access TTP funds (e.g., TTP, TTS GP). If the tribe is a grantee under Section 5310 funding, the project would be required to be included in a locally developed public transit human services transportation plan. See also the Funding and Compliance section of this chapter.

Service Options. Like nontribal transit systems, tribal transit systems use a mix of service types: paratransit, fixed-route, flex-route, and microtransit. They transport residents to and from small communities, low-density areas, and extremely isolated areas. Tribal transit can include service areas that include communities 10 miles apart or 50 miles apart, communities on the fringes of urbanized areas, or communities that are hours away from an urban area. Along with this array of service types and service areas comes an assortment of customer transportation needs. For example, some riders commute to work in an urban area, some need access to key destinations such as local and specialty health care, shopping, and school, and some need a connection to an intercity bus for longer, regional travel.

A 2023 survey of 12 tribal transit agencies in seven states⁹⁴ illustrates the wide variety of services that tribal transit systems provide:

- The system size ranged from two vehicles to over 30 vehicles.
- There was an even mix of service areas, including remote rural to small urban/suburban settings.
- Most systems operated paratransit (including some microtransit). Others operated fixed- or flex-route services.
- Ridership included systems with 3,000 annual trips or close to 100,000 annual trips.

Decisions about what types of services a tribal transit agency should provide should consider information gathered during the inventory of resources and public outreach (see the sections on Strategic Planning and Service Planning). See also the Service Options section of the National RTAP's Transit Manager's Toolkit, with descriptions of the basic service types—fixed-route, flexible route, and demand response service, including microtransit and guidance about when each service type should be used. Information about service types and when and where they are appropriate is also provided in Chapter 2, Service and Design Strategies and Practices, in Part I of the Guide.

Choosing Vehicles. It is important to match the correct vehicle for the needs of the community and the environment. This is true for any transit agency, and tribal transit managers will

find guidance and resources to help with choosing the right vehicles in Chapter 6, Management and Operations Considerations, in Part II of the Guide.

Communications and Technology. While the area of communications and technology in public transportation is continually evolving, the most basic communication and technology need in tribal transit is making sure customers have a way to contact the transit agency, and tribal transit staff have a way to reach drivers while they are on the road (and vice versa). Tribal transit agencies have been implementing new technologies at a rapid pace. The following are examples:

- Cheyenne River Sioux Transit⁹⁵ uses software to schedule and route rides as well as provide an electronic fare system.
- Oneida Public Transit⁹⁶ has dispatching software and tablets and uses the Oneida Nation⁹⁷ website for communication purposes. NCRTD⁹⁸ in Espanola, New Mexico, operates in a large, mostly rural geographical area and provides service to four counties, several cities, many small towns, eight Pueblos, and the Jicarilla Apache Nation. NCRTD offers a Blue Bus Tracker⁹⁹ for its riders and provides its schedule and geographic data (GTFS feed)¹⁰⁰ online for public use to assist with transit and travel-related information.

Facilities Planning. Tribal transit systems will, at a minimum, require an office and a secure location to park vehicles. The following resources are available to tribal transit agencies who are involved in facility or infrastructure planning and implementation as they begin to work with stakeholders, both tribal and nontribal, on projects that may affect historic properties and the human environment.

- Integrating NEPA and Section 106.¹⁰¹ The National Historic Preservation Act (NHPA)¹⁰² and the National Environmental Policy Act (NEPA)¹⁰³ are two separate laws that require federal agencies to “stop, look, and listen” before making decisions that impact historic properties and the human environment.
 - NEPA ensures that environmental issues are considered alongside social, economic, and other competing concerns that arise when planning a project. For rural/tribal transit agencies, the NEPA review process is most likely to be required when planning for a project involving new construction, major rehabilitation, or renovation of a facility. Other types of FTA-funded rural/tribal transit projects are generally considered “categorical exclusions” under the NEPA requirements.
 - NHPA, as amended, is a public law to protect, rehabilitate, restore, and reuse districts, sites, buildings, structures, and objects significant in American architecture, archaeology, and culture. Section 106 of the NHPA¹⁰⁴ ensures that historic preservation is considered in federal project planning.
- Advisory Council on Historical Preservation (ACHP).¹⁰⁵ ACHP administers NHPA and addresses critical issues like NHPA’s relationship to NEPA review processes. The Office of Tribal and Indigenous Peoples (OTIP)¹⁰⁶ oversees ACHP’s Indigenous initiatives. ACHP’s web page for Indian Tribes, Native Hawaiians, and Indigenous Peoples has several resources available to assist federal agencies, applicants, Native American Tribes, and Native Hawaiian organizations in navigating the Section 106 process and its consultation requirements. Tribal Coordination and Consultation for Infrastructure Projects¹⁰⁷ speaks to issues around the timing of Section 106 consultations; consultations with tribes are often started late in the project process and are separate from the NEPA review process. Other ACHP web pages and publications of interest include: Whom to Contact if You Have Section 106 Issues: Guidance for Indian Tribes and Native Hawaiian Organizations (NHOs),¹⁰⁸ Indigenous Knowledge – Understanding and Implementation,¹⁰⁹ and Traditional Knowledge and the Section 106 Process: Information for Federal Agencies and Other Participants.¹¹⁰

- Under the U.S. Constitution, treaties with tribes are the law. In Tribal Treaty Rights in the Section 106 Process,¹¹¹ the U.S. Advisory Council on Historic Preservation (ACHP) notes that it is not uncommon for tribes to raise treaty rights concerns during Section 106 consultations. ACHP recommends that, early in project planning, applicants for federal funding or approvals determine if a proposed project might affect treaty-protected rights and provide additional guidance on how to make such a determination.
- Tribal Historic Preservation Office Program (THPO).¹¹² THPO is available to advise federal, state, and local agencies on the management of tribal historic properties and instruct municipalities on Section 106 reviews to represent Tribal interests.

There have been numerous successes in facility planning for tribal transit agencies. As of 2024, Oneida Public Transit began construction of a new facility that will store all vehicles and provide the agency with a wash bay and office space. Big Woods Transit began constructing a transit facility for the Bois Forte Band of Chippewa using a design that includes parallel parking to maximize the use of the space, which will include a bus wash, a mechanics work bay with a lift, and a solar power generation system.

Sustainability. Sustainability refers to the ability of a transit program to continue and maintain its ability to provide service. Sustainability for tribal transit systems depends on the system's ability to operate as efficiently as possible and to secure funding. Tribal transit systems may find the following resources helpful:

- Planning, Funding, Developing, Starting and Sustaining a Tribal Transit System (2020) by South West Transit Association.¹¹³
- *TCRP Report 154: Developing, Enhancing, and Sustaining Tribal Transit Services—A Guidebook*¹¹⁴ addresses the various steps for planning and implementing a tribal transit system.
- Native Americans on the Move: Challenges and Successes¹¹⁵ brochure was developed to accompany *TCRP Report 154*.

Fare Policies. Like rural transit systems, tribal transit programs should develop fare policies based on the needs of their community and comprehensive planning. For example, Muckleshoot Tribal Transit can offer fare-free service with grant funding allocated from both the FTA and BIA.

Monitoring and Evaluation

A tribal community's needs and resources change over time, as happens in other communities. To ensure that transit services are as appropriate today as they were yesterday, it is important to have a system for monitoring and evaluation. The following examples and resources are recommended reading for tribal and nontribal transit stakeholders:

- The NCRTD Tribal Transit Service Plan (May 2023)¹¹⁶ is an excellent example of a tribal transit service plan that documents existing conditions, outreach to tribal entities, transportation needs, and challenges, and identifies a "strategic mobility vision for Tribal entities served by NCRTD."
- *TCRP Report 154: Developing, Enhancing, and Sustaining Tribal Transit Services—A Guidebook*¹¹⁷ offers guidance on the various steps for planning and implementing a tribal transit system. The steps may be used for planning a new transit system, enhancing an existing service, or taking action to sustain services.



CHAPTER 15

Compliance with Federal and State Requirements

As a condition of their federal and state transit grants, rural and tribal transit systems have the primary responsibility for managing their projects. Specifically, the transit service must be operated in compliance with federal and state regulations. This includes ensuring the transit service meets local needs, is safe, is operated effectively and efficiently, meets performance standards, is coordinated with other agencies, and is marketed to the public and human service agencies.¹¹⁸

Rural and tribal transit systems that receive federal or state transit funds are undoubtedly familiar with the myriad regulations they need to comply with and agree to comply with them through the grant application and agreement process. A brief reminder of the underpinnings of these regulations includes the following:

1. **Code of Federal Regulations (CFR).** The requirements start with the CFRs that contain the general and permanent rules published by federal agencies/departments. While the regulations established by the U.S. DOT are found in Title 49 of the CFR and the CFRs for FTA are found in Parts 600-699, there are parts of Title 49 that apply to all DOT agencies (e.g., 49 CFR Part 37 includes the ADA rules that apply to transit providers).
2. **FTA Circulars.** FTA provides guidance on program regulations for how the CFR rules are implemented or applied through circulars. The FTA circulars provide grantees with program-specific requirements. FTA grantees are required to comply with all circulars and agree to do so by signing annual FTA certifications and assurances. Since entities receiving Section 5311 funds are subrecipients of the state, states pass through these requirements as part of their grant application process, and they are included in grant agreements between the state and rural and tribal transit providers. The unique circumstances for tribal governments that are direct recipients of FTA funding are addressed here and in Chapter 14, Roles of Tribal Governments.
3. **Other Federal Circulars.** FTA circulars incorporate other federal requirements such as the Office of Management and Budget's uniform requirements, cost principles, and audit requirements for federal awards, often commonly referred to as "the Super Circular," which applies to any organization receiving federal funding. It consolidated eight circulars into uniform regulations in Title 2 of the CFR, specifically 2 CFR Part 200. The requirements in 2 CFR Part 200 are reflected in FTA Circular C.5010.1F Award Management Requirements. It is important to note that the FTA circulars are occasionally revised, so it is important to use the most recent version that can be found on the FTA website. For example, in 2024, FTA updated the following circulars with an implementation date of November 1, 2024:
 - Award Management Requirements (C 5010.1F).
 - Urbanized Areas Formula Grant Programs Guidance (C 9050.1A).
 - Rural Areas Formula Grant Programs Guidance (C 9040.1H).
 - Enhanced Mobility of Seniors and Individuals with Disabilities Program Guidance (C 9070.1H).

Final versions of the circulars are available on FTA's website (<https://www.transit.dot.gov/regulations-and-programs/fta-circulars/circulars>).

Compliance Monitoring by FTA and State DOTs

An important role played by FTA and state DOTs is ensuring that grantees comply with the federal and state regulations and requirements associated with the grants they receive.

Federal Compliance Monitoring

In its role of providing financial assistance to improve, maintain, and operate public transit systems nationwide, FTA oversees thousands of grants provided to states, tribes, and local public transit agencies. FTA is responsible for conducting oversight activities to ensure that recipients of grants use the funds in a manner consistent with their intended purpose and in compliance with regulatory and statutory requirements.

Several FTA oversight review programs exist¹¹⁹ to accomplish this goal, but the primary programs that most rural and tribal transit providers will encounter are state management reviews (SMRs) to monitor transit funding that states administer and, possibly, Triennial Reviews (for funding in urban areas). Unless a rural system also receives S.5307 funding or is a tribal government with direct grants from FTA, their experience with FTA monitoring will probably be confined to possible involvement in their state's SMR.

The SMR of their state DOT includes a review of how well the states oversee their subrecipients, and the process includes an in-depth review of selected subrecipients to assess state management practices and program implementation of FTA grants. The reviews are conducted every 3 years and encompass 17 areas of compliance, including grant administration, procurement, and civil rights.

The annual Comprehensive Review Contractor's Manual is an excellent resource when attempting to understand most federal regulations since it conveys FTA's expectations for assessing compliance of FTA recipients. This manual is valuable to transit operators because it includes the minimum requirements to which all applicable recipients are expected to comply, directly tied to a law, regulation, circular, master agreement, or other binding guidance.

State Compliance Monitoring

In addition to ensuring that subrecipients of the states comply with federal requirements, states actively monitor whether subrecipients comply with state rules and regulations. Monitoring activities can take many forms, but the most visible monitoring activities are periodic compliance reviews conducted by the state DOT on their grantees that receive federal transit funds, including, in some states, urban S.5307 grant subrecipients where the state administers the federal S.5307 funds.

These reviews are to ensure that transit agencies/subrecipients can manage the FTA- and state-funded programs in accordance with the grant application, the grant agreement, the FTA-approved State Management Plan, and all applicable laws and regulations using sound management practices. The reviews are intended to look behind the signed certifications and assurances submitted by grant subrecipients to ensure that subrecipients are adhering to all statutory and program requirements when expending federal and/or state funds on local projects.

It should be noted that state monitoring programs can vary significantly depending on policy or programmatic differences. For example, while some requirements are not imposed by the FTA on Section 5311 grantees or grantees that only receive state funds, the state can require that all public transit grantees meet these compliance regulations. Although some of the FTA requirements do not "flow through" to state grantees (since the state is the direct recipient of

federal funds), some states require that all public transit grantees meet FTA program management requirements as detailed in the last section of this chapter.

Uniqueness of Tribal Governments

A federally recognized tribe may choose to access its share of the Public Transportation on Indian Reservations (Tribal Transit) Program (TTP) federal funds directly from the federal government. It may also elect to receive Section 5311 through state programs.

If the tribe receives funding through the Section 5311 program and the tribe is a subrecipient of the state, the tribe has the same grant program administration requirements and responsibilities as other tribal and nontribal Section 5311 subrecipients.

On the other hand, federally recognized Native American Tribes or Alaska Native villages, groups, or communities are eligible direct recipients under the Public Transportation on Indian Reservations Program (Tribal Transit Program or TTP) in Section 5311(c)(1). Federally recognized tribes may also elect to become direct recipients under Section 5311 in addition to being subrecipients under a state 5311 program. For direct TTP recipients, tribal requirements vary in specific ways from those under the Section 5311 program. FTA monitors their direct recipients directly, while states monitor the state 5311 program subrecipients.

Impact on New Service Strategies

Federal requirements have an impact on some items that could affect the implementation of new or enhanced transit services by rural and tribal communities. For example, the requirements of the ADA and Title VI both need to be considered when designing services. ADA regulations may dictate the need for accessible vehicles and facilities and the need for complementary paratransit services. And, Title VI requires transit providers to consider the impact of service changes on minorities or low-income populations to ensure they are not disproportionately negatively affected or are not receiving the benefits of a program proportionately.

Rural and tribal transit providers must comply with the following state and federal (FTA) requirements. The following materials are based on the FTA's Comprehensive Review Contractor's Manual, which conveys FTA's expectations of our oversight contractors for assessing compliance of FTA recipients¹²⁰:

1. **Legal.** Rural and tribal transit operators must be eligible and authorized under state and local law to request, receive, and spend FTA funds and to execute and administer FTA-funded projects. They must be organized and governed in a manner that allows the transit system to comply with federal regulations. Implementing new services or practices may require adjustments to how the system is controlled or governed. One issue that may arise is in labor projections. Section 5311 subrecipients must have signed the Special Section 5333(b) Warranty that addresses labor issues such as collective bargaining, employee displacement, and dismissal. Changes to services that affect employment practices for staff should be examined in this context.
2. **Financial Management.** Rural and tribal transit providers must demonstrate the ability to match and manage FTA and state grant funds, cover cost increases and operating deficits, cover maintenance and operational costs for FTA-funded facilities and equipment, and conduct and respond to applicable audits. They also must have the financial management systems to account for and report on FTA and state grant assistance. As they implement new or enhanced services, rural and tribal transit systems must have sufficient local resources to provide the required match and carry out the proposed project.

3. **Technical Capacity.** Tribal and rural transit providers must be able to implement any new services and FTA- and state-funded projects using sound management practices and in accordance with FTA and state requirements. This includes the ability of transit providers to administer the grant program and to report status and progress to the state. Of particular importance when implementing new services or strategies, transit providers must have an adequate number of staff (who possess the necessary knowledge, skills, and abilities) and maintain adequate documentation of key policies. If implementation involves contracting for services, providers may need to develop new procedures for managing transit service contractors to ensure that contractors comply with federal requirements and that quality service is provided.
4. **Maintenance.** Transit providers must keep FTA- and state-funded vehicles, equipment, and facilities in good working order, including the ADA accessibility features on all vehicles and facilities. Subrecipients must maintain project equipment and facilities at a high level of cleanliness, safety, and mechanical soundness. To accomplish this, transit providers must have a written maintenance plan for vehicles, facilities, and facility-related equipment, and procedures to track when preventive maintenance inspections are due and to schedule preventive maintenance inspections promptly. If the implementation of new service strategies requires a change in vehicle types and facilities, these plans and procedures may need to be revised.
5. **The Americans with Disabilities Act (ADA).** Titles II and III of the ADA provide that no entity shall discriminate against an individual with a disability in connection with the provision of transportation service. The law sets forth specific requirements for vehicle and facility accessibility and the provision of ADA complementary paratransit service. Chapter 12 in Part II of this Guide provides in-depth information on the ADA implications associated with the implementation of new service strategies. For example, systems providing fixed-route service must also provide complementary paratransit service that is comparable to the level of service provided to individuals without disabilities who use the fixed-route system. For systems that operate route-deviation service, if deviations are available to all members of the public (not just persons with disabilities), the service is considered demand response, and the system does not need to operate ADA complementary paratransit services. If deviations are only available for persons with disabilities, complementary paratransit requirements must be met as well. In very rural areas where demand is low, transit agencies might be able to meet the complementary paratransit requirements through off-route deviations. It is cautioned that, when operating complementary paratransit through route-deviation services (services that do not deviate for all riders), fixed-route operators must provide complementary paratransit service in full compliance with Part 37 Subpart F. This includes administering an eligibility determination process to determine who is ADA paratransit eligible.
6. **Title VI (TVI).** Transit grantees must ensure that no person is, on the grounds of race, color, or national origin, excluded from participating in, denied the benefits of, or subject to discrimination under any program or activity receiving federal financial assistance. FTA also prohibits discrimination on the grounds of low-income status. Rural and tribal transit providers are required to have a TVI plan—most will only have to include the elements required of nonurban transit systems (TVI plans for systems that operate 50+ fixed-route vehicles in peak service and are in UZAs with a population of 200,000 are more complicated). The TVI plan must be adopted by the governing board and updated every 3 years. Depending on the state, TVI plans may be submitted to the state. If systems are implementing new fixed routes, one implication is the need to update their TVI plan to establish the required fixed-route TVI service standards (on vehicle load, on-time performance, headways, and service availability) and service policies (vehicle assignment and distribution of transit amenities).
7. **Procurement.** When procuring supplies, equipment, or services, tribal and rural transit providers must comply with the requirements and regulations of the FTA Circular 4220.1G, 2 CFR Part 200/1201, and any state and local procurement rules. The transit providers can

follow the same policies and procedures they use for procurements with nonfederal funds, but must comply with the following statutory and administrative requirements:

- Conduct all procurements in a manner that provides full and open competition.
- Exclude in-state or local geographical preferences.
- Not enter into any contract for rolling stock with a period of performance exceeding 5 years.
- Ensure that every purchase order and contract includes all applicable clauses required by federal statutes and executive orders.
- Use competitive proposal procedures based on the Brooks Act when contracting for architectural and engineering services if the state has not adopted a statute governing the procurement of such services.
- Award to only responsible contractors.
- Comply with Buy America, Build America, Buy America Act, and vehicle manufacturer requirements, and debarments/suspensions and lobbying restrictions.

Often, procurements will be reviewed and approved by the state DOT to ensure compliance with FTA and state procurement regulations, especially if they are expected to exceed a certain dollar limit. Complying with procurement regulations is particularly important as providers implement new services that involve purchasing new equipment/vehicles or contracting with other service providers. For example, if the provider has generally purchased vehicles through a state contract that does not offer the type of vehicle needed for the new service, this may be its first experience with vehicle procurement (not an easy process). Similarly, if the transit provider has not contracted with a taxi or TNC in the past, dealing with these contracts may be challenging.

1. **Disadvantaged Business Enterprise (DBE).** Rural and tribal transit providers must not discriminate on the basis of race, color, national origin, or sex in the award and performance of FTA-assisted contracts. All providers must give DBEs the maximum opportunity to compete for and perform contracts and subcontracts financed in whole or in part with federal funds. If the implementation of new service strategies involves new purchases, DBE requirements must be considered.
2. **Satisfactory Continuing Control.** Tribal and rural transit providers must maintain control over real property, facilities, and equipment that were funded with FTA and/or state transit dollars and ensure that they are used for their originally authorized purpose of providing public transportation. As part of this requirement, transit operators must participate in an FTA-compliant transit asset management (TAM) plan either by developing their own or participating in a group TAM plan. The number of spare vehicles must be appropriate to the size and age of the fleet, the amount of peak demand, and the projected ridership growth. It must carry enough insurance to replace any FTA- and/or state-funded facility or equipment, such as bus washers, bus lifts, and so forth. Implementation of new service strategies may require changes in vehicles and the facility, necessitating adjustments to the system's TAM plan and capital program. Insurance coverage may need to be adjusted if equipment and facilities are modified.
3. **Planning and Coordination.** Transit providers must participate in the transportation planning process in accordance with FTA requirements and the metropolitan and statewide planning regulations. Chapter 5 in Part II of the Guide provides details on effective planning for new service strategies. For example, as providers implement new service strategies, they must coordinate to the maximum extent feasible with transportation from other federal sources and must involve the public in service planning. An effective public involvement program has an educational and outreach component and addresses transit riders, the non-riding public, special populations, community leaders, and civic groups. Many states require that applicants for FTA and state transit funding conduct public hearings annually as part of the grant application process and also document adequate outreach to the community.

4. **Public Comment Process on Fare Increases and Service Reductions.** FTA expects Section 5311 subrecipients to have a written, locally developed process for soliciting and considering public comments before raising a fare or carrying out a major transportation service reduction. Some states extend this requirement to all subrecipients that receive transit funds through the state, including Section 5311. Depending on their funding and state requirements, some transit providers may need to have and follow this process when raising fares or reducing various services to make way for new service implementation.
5. **Charter Bus.** Tribal and rural transit providers are prohibited from using federally funded equipment and facilities to provide charter service except in accordance with allowable exemptions or exceptions. Some states require that subrecipients have a written charter bus policy, and most require that they keep records of charter services provided. Subrecipients are allowed to operate community-based charter services excepted under the regulations. One key exemption allows Section 5311 and Section 5310 subrecipients to provide services to older adults and individuals with disabilities for program purposes. There are also exceptions for some services, such as government officials (80 hours annually). None of the new service strategies should meet the definition of charter bus.
6. **School Bus.** Rural and tribal transit providers are prohibited from providing exclusive school bus service. In no case can federally funded vehicles, equipment, or facilities be used to provide exclusive school bus service. A school tripper service that operates and looks like a regular service is allowed. Head Start transportation is considered human service transportation, not school bus service. None of the new service strategies should meet the definition of an exclusive school bus.
7. **Safety and Security.** Section 5307 subrecipients must comply with the federal Public Transportation Agency Safety Plan regulation to ensure public transit providers develop and implement an Agency Safety Plan. When implementing new service strategies, rural and tribal transit providers, regardless of funding, should achieve the highest practical level of safety and security to protect passengers, employees, revenues, and property. For example, they must document that drivers have a valid operator's license, have a safe driving record, and have been trained to ensure transit systems are operating safely. If contracting for new services, oversight over safety and security by transit providers is essential.
8. **Drug-Free Workplace/Drug and Alcohol Testing Program, Drug and Alcohol Policy.** Tribal and rural transit providers, and any contractors, must maintain a drug-free workplace for all employees and have an ongoing drug-awareness program. Further, they are required to have a drug- and alcohol-testing program in place for all safety-sensitive employees. Implementation of new service strategies could have significant implications in this area if new concepts or entities are involved.¹²¹ For example, for volunteer driver programs, the volunteers are not subject to testing unless the volunteer is required to hold a CDL or receives reimbursement in excess of expenses incurred while engaged in the safety-sensitive function. Maintenance contractors for providers in areas with populations under 200,000 are not required to have a drug- and alcohol-testing program. Taxis and TNCs are not required to have a testing program if riders can randomly choose among taxi companies/TNCs. However, if a transit provider contracts exclusively with one taxi or TNC, then that company needs to have a compliant drug- and alcohol-testing program, which the transit operator will need to oversee. This includes the independent drivers of a TNC/ridesourcing company contracting with a public transit agency.
9. **Equal Employment Opportunity (EEO).** Transit providers must ensure that no person shall, on the grounds of race, color, religion, national origin, sex, age, or disability, be subject to discrimination in employment. They must post notices setting forth an EEO policy in conspicuous and accessible places and make these available to employees and applicants for employment. Reasonable accommodations must be made for persons with disabilities on request. It is unlikely that many rural and tribal transit providers will need to prepare

a formal EEO program (since only transit agencies with 50 or more transit employees are required to prepare and maintain an EEO program). However, if the provider contracts with a private company that meets this threshold, they must ensure that the company has a compliant EEO program.

10. **Other Service Requirements.** There are several other requirements for the provision of services when using FTA transit funds that may affect service implementation of new or enhanced service strategies:
 - **Services Must Be Open to the General Public.** Services may be designed to maximize use by older adults and persons with disabilities. Coordinated human service transportation, which primarily serves older adults and persons with disabilities, but is not restricted from carrying other members of the public, is considered available to the general public if it is marketed as a public transit service.
 - **Use of S.5311 Funds in Urbanized Areas.** Section 5311 funds can only be used for public transportation projects and intercity projects in nonurbanized areas. If urbanized area recipients of Section 5307 funds also receive Section 5311 funds to carry out projects in surrounding rural areas, these subrecipients should use Section 5311 funds only to assist the rural portion of those localities. FTA expects the subrecipient to develop a reasonable basis related to the service provided for allocating operating costs between the two FTA funding sources.
 - **Incidental Use of Vehicles and Property Funded with Transit Grants.** Incidental use of a Section 5311 vehicle for nonpassenger transportation on an occasional or regular basis, such as meal delivery, must not result in a reduction of public transit service quality or availability. Incidental services must cover the operating costs associated with providing the services. Transit operators may provide incidental service with FTA-funded vehicles and facilities, but the service must not interfere with the provision of transit service and must bear the costs of providing the service.

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PART IV

Case Studies

CHAPTER 16

Introduction to Part IV

Part IV of the Guide presents the nine case studies completed for the research project. These case studies provided the opportunity to delve into examples of new and enhanced service strategies implemented by rural and tribal transit providers that improved mobility for their communities. They also serve to illustrate service strategies and practices presented in Part I of the Guide.

Following is a snapshot of the nine case studies organized by the categories of strategies used in Part I. However, many of the case studies involved multiple new service strategies and so do not fit neatly into only one category.

The detailed case studies are available on the National Academies Press website (nationalacademies.org/publications) by searching for *NCHRP Research Report 1144/TCRP Research Report 251: Public Transportation in Rural Areas and Tribal Communities: Guide for Service Improvements* and looking under “Resources.”

Service Strategies

Microtransit

CARTS NOW—Microtransit as a Fundamental Change for Rural Transit

The Capital Area Rural Transportation System (CARTS), serving the rural areas across nine counties in central Texas, successfully piloted microtransit in the small community of Bastrop. Ridership on the new microtransit service, CARTS NOW, quickly took off as riders liked the on-demand nature and dependability of the service. The CARTS NOW model was then replicated, just as successfully, in other small communities within CARTS’ service area.



SOURCE: Capital Area Rural Transit System.

High Valley Transit—Expanded Fixed-Route and Consolidated Microtransit and ADA Paratransit Service

High Valley Transit (HVT) serves the populated areas of western Summit County and Wasatch County, Utah, while also connecting these areas with Park City and Salt Lake, 30 miles away. HVT is an independent transit district created in 2020, which evolved from earlier transit endeavors in the region. HVT now operates an expanded fixed-route service and a general public, shared-ride microtransit service that is successfully operated with the same fleet of vehicles and drivers as the ADA paratransit service.



SOURCE: High Valley Transit.

Mountain Empire Transit—METGo! Microtransit Providing Better Quality Service for Rural Riders in Southwestern Virginia

Mountain Empire Transit (MET), an arm of Mountain Empire Older Citizens Inc. (MEOC), has provided advanced reservation demand response public transportation in rural southwestern



SOURCE: Mountain Empire Older Citizens Inc.

Virginia for more than 35 years. With a grant from the Virginia Department of Rail and Public Transportation to pilot microtransit service, MET launched METGo! in 2021 to explore how microtransit technology can improve service efficiency and the rider experience in rural areas. Service began in an 11-square-mile zone that included two important trip destinations. Rider-ship data and experience show that METGo! has been a success, and MET is considering the feasibility of introducing microtransit service in other portions of its large service area.

Route Design Improvements

Okanogan County Transit Authority “TranGO”—Nontraditional Service Design



SOURCE: Okanogan County Transit Authority.

The Okanogan County Transit Authority (OCTA), located in north central Washington on the Canadian border, provides transit service under the brand name “TranGO.” The transit agency chose a service design based on where people in its region need to go, identified through a public survey. Instead of a more traditional hub-and-spoke design, TranGo operates nine interconnecting routes, pulsing at five different transfer points; surrounding the transfer points are circles with 5-mile radii where general public dial-a-ride service is provided. Within the dial-a-ride zones, the dial-a-ride vehicles also provide ADA paratransit service within the ¾-mile corridors of the fixed routes. ADA paratransit trips outside the dial-a-ride zones are provided by deviations from the fixed routes.

North Central Regional Transit District—Meeting Needs in a Diverse Multijurisdictional Service Area



SOURCE: North Central Regional Transit District.

North Central Regional Transit District (NCRTD) operates in a large, mostly rural service area in north central New Mexico that includes four counties; seven cities, towns, and villages; eight Pueblos; and one Tribal Nation. NCRTD’s strategy of ensuring representation from all its regional stakeholders, including the tribal stakeholders, has been particularly important to NCRTD’s success. With a strong regional foundation, NCRTD has implemented service design innovations, such as a revised service design that increased ridership by 45% and a new micro-transit service in two towns branded as MyBlue™.

Combining Multiple Modes

Cascades East Transit—Leveraging Funding and Partnerships to Maximize Mobility in Central Oregon



SOURCE: Cascades East Transit.

Cascades East Transit (CET), a department of the Central Oregon Intergovernmental Council, operates service in three rural counties, one urbanized area, and a tribal reservation. Through proactive short- and long-term planning, CET uses a range of funding sources (capital and operating) to provide diverse transit services to meet the service area’s diverse transportation needs. It’s a “family of service” concept, with services that include, for example, a regional city-to-city connector service designed to meet residents’ trip patterns; a general public dial-a-ride that brings the more rural riders into nearby communities’ seasonal recreation shuttles; and a van-pool program.

Mobility Management and Coordination

Open Plains Transit—Rural Demand Response and Intercity Bus Service



SOURCE: Open Plains Transit.

Open Plains Transit, operated by the nonprofit Senior Services Inc., provides service in the far western counties of Nebraska known as the panhandle. Transit services include demand response service in three of the panhandle’s 11 counties and, more recently, implemented intercity bus

routes serving eight of the 11 counties. The intercity routes provide service that connects to regional transportation, such as Burlington Trailways and the Western Nebraska Regional Airport. Open Plains' services follow the organization's mission to connect individuals with needed services and other modes of transportation for regional travel.

PICK Transportation—Consortium of Public Transit Agencies Uses Microtransit to Fill Gaps in Traditional Service

PICK Transportation, the name for a partnership of four rural public transit agencies operating in eastern Oklahoma, was awarded an Integrated Mobility Innovation FTA grant to operate microtransit service during evening and Saturday hours, extending service beyond the traditional service provided by the transit agencies. The four transit agencies each operate the new microtransit service; however, to the public, it is branded as one service: PICK Transportation. The single branding and marketing of the new service is intended to offer an innovative model for regional trip coordination and integrated regional payments.



SOURCE: PICK Transportation.

Connections to National and Regional Networks

South Dakota's Rural Intercity Bus Feeder Services

Six rural transit agencies in South Dakota along with a major private for-profit regional intercity bus carrier and the South Dakota Department of Transportation worked together to create a statewide network of feeder service, linking small towns in rural areas with the national intercity bus network, providing linkages to the state's largest cities as well as destinations in other states. This coordinated effort has addressed a need common to many rural areas and small towns that have lost transit connectivity to the national intercity bus network and larger population centers.



SOURCE: South Dakota Department of Transportation.



A P P E N D I X

Literature and Resources: General Guidebooks and Literature by Topic Area

This appendix contains the literature review conducted for this project. It is available on the National Academies Press website (nationalacademies.org/publications) by searching for *NCHRP Research Report 1144/TCRP Research Report 251: Public Transportation in Rural Areas and Tribal Communities: Guide for Service Improvements* and looking under “Resources.”

Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI-NA	Airports Council International-North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
GHSA	Governors Highway Safety Association
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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ISBN 978-0-309-59990-0

